

Pension Reform Survey 2026: Everyone knows reform is needed, few expect it to happen



Content

Page 3

Executive Summary

Page 4-7

Most agree reform is needed – few believe it will be delivered

Page 8-10

Pension reform is not on the top of the list

Page 11-13

The reform options are known – but there is no one size fits all

Page 14-19

Retirement preparedness starts with knowing the numbers

Page 20-21

Financial literacy, a growing imperative for pension policy

Page 22-23

More capital-funded pension provision is a win-win

Page 24

Appendix

Executive Summary



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- Citizens recognize the need for social security reforms, but have far less confidence that reforms will actually be delivered.** Our survey of more than 8,000 respondents across Austria, France, Germany, Italy, Poland, Spain, the UK and the US explores public attitudes towards social security reform, pension sustainability and retirement preparedness. Recognition of the need for reform is overwhelming, exceeding 80% in most countries. Yet confidence in governments' ability to implement the reforms needed to ensure long-term sustainability is much weaker. The latter ranges from just 36% in Italy and 42% in Germany to 63% in Poland and 66% in the US. Views diverge more sharply when it comes to how the adjustment burden should be shared. Taking greater personal responsibility through additional saving is the most widely accepted option, while higher taxes and social contributions receive little support. Importantly, while respondents differ on how adjustment should be achieved, only a minority rejects all proposed measures outright. The challenge for policymakers is therefore no longer convincing citizens that reform is needed, but demonstrating that reforms can be delivered and designing politically viable reform packages.
- Older generations are not blocking reform.** Recognition of the need for reform increases with age, but trust that governments will successfully implement the necessary reforms declines. While 61% of respondents aged 18 to 34 express confidence that the necessary reforms will be implemented, this falls to 54% among those aged 35 to 49 and to just 44% and 43% among respondents aged 50 to 64 and 65 to 79, respectively. The result is a striking paradox: the older cohorts are most convinced that reform is necessary, while younger respondents attach less urgency to the issue, suggesting that pensions suffer from a salience gap despite their long-term importance. Contrary to common stereotypes, respondents aged 50 to 64 are also among the most willing to accept measures such as working longer, postponing retirement or receiving lower public benefits.
- Responsibility for retirement is increasingly shifting onto individuals, yet many lack the knowledge to prepare for it.** Only half of respondents expect the public pension system to provide the majority of their retirement income. As occupational and private pension provisions gain importance, pension and financial literacy are becoming essential pillars of retirement income adequacy and pension system sustainability. Yet only one in six respondents shows a high level of financial literacy (18%), while one in four scores low (26%). The knowledge gaps have concrete implications: only one-third of respondents correctly identified how long they are likely to spend in retirement, and only half state that they have a clear picture of their expected financial situation in retirement. These findings suggest that the financial literacy gap is also a pension literacy gap.
- More funded pension provision – as envisaged in Germany – would be a win-win for households and the wider economy.** Stronger occupational and private pension pillars would improve retirement security through greater private wealth accumulation while simultaneously providing long-term capital to financial markets. This could support investment in innovation, infrastructure and the green transition. Asset-backed pension arrangements amount to more than 200% of GDP in Denmark, 145% in the US and 78% in the UK, compared with just 6% in Germany and 13% in France. If Austria, France, Germany, Italy, Poland and Spain had reached the UK's level of pension assets relative to GDP in 2025, an additional EUR8.8trn of long-term investment capital would have been available.



Most agree reform is needed – few believe it will be delivered

Ageing populations are putting social security systems under growing pressure. With government spending on social protection already amounting to more than 25% of GDP in many countries and demographic change pushing old-age dependency ratios even higher, public pension and healthcare systems urgently need to be reformed¹. Against this backdrop, we surveyed more than 8,000 respondents across Austria, France, Germany, Italy, Poland, Spain, the UK and the US to explore public attitudes towards social security reform, pension sustainability and retirement preparedness.

How citizens view the need for reform, the likelihood that reforms will be delivered and their own role in the adjustment process matters greatly for reform success. Public support can facilitate reform, while a lack of trust or perceived unfairness can undermine even well-designed policy proposals. The survey therefore examines attitudes towards social security reform and pension sustainability from several perspectives, including perceptions of reform need, trust in implementation, retirement preparedness, pension and financial literacy as well as individual preferences regarding different reform options.

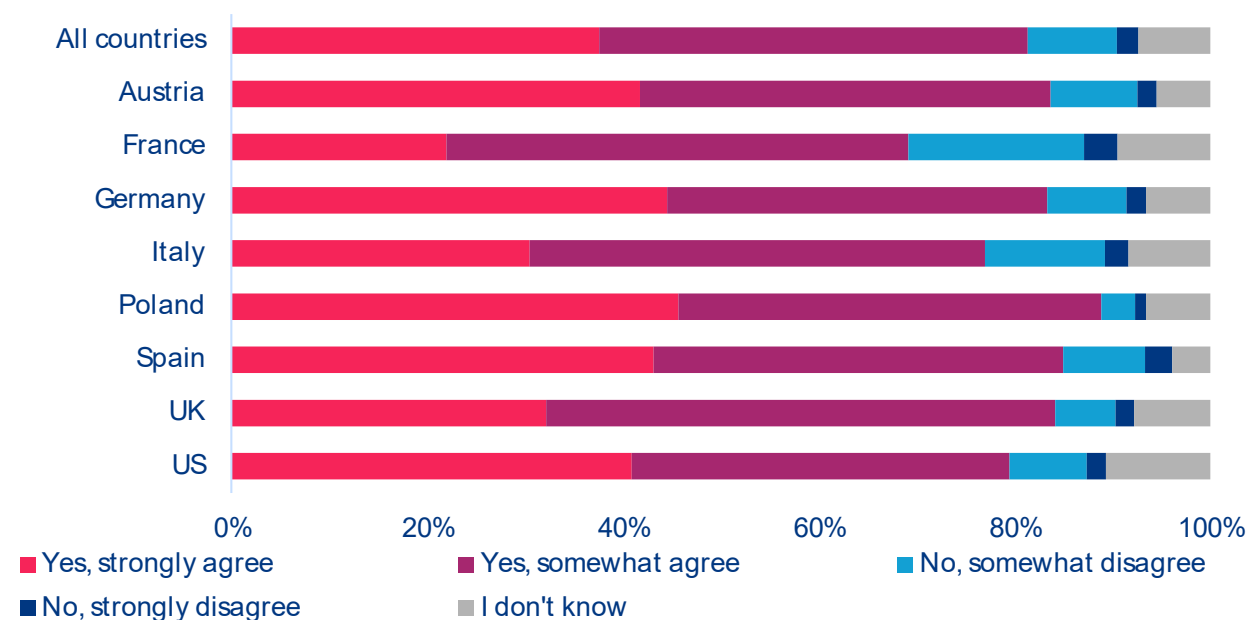
Nearly half of respondents in Poland and Germany recognize the need for reform, compared to only 22% in France. Across all eight countries, more than 80% of respondents agree or strongly agree that social systems must be reformed to remain sustainable. The share of respondents who strongly agree is highest in Poland (46%)

and Germany (45%), followed by Spain (43%), Austria (42%) and the US (41%). In the UK and Italy, around one-third of respondents express a strong need for reform. France is the clear outlier, with only 22% strongly agreeing and 21% either somewhat or strongly disagreeing (Figure 1).

Support for reform is broad across all age groups, although older respondents express slightly stronger conviction. The share of respondents who strongly agree that reform is needed ranges from 36% among young adults aged 18 to 34 to 39% among those aged 35 to 49. The relatively limited sense of urgency among younger respondents may reflect a salience and financial-literacy gap. For many younger people, topics such as pensions and healthcare remain too distant to register as immediate concerns, despite the fact that they will be most affected if governments fail to improve the sustainability of social security systems. The survey also reveals gender differences: Men express stronger conviction that reform is needed than women, with 39% strongly agreeing compared with 36% of women (Figure 2).

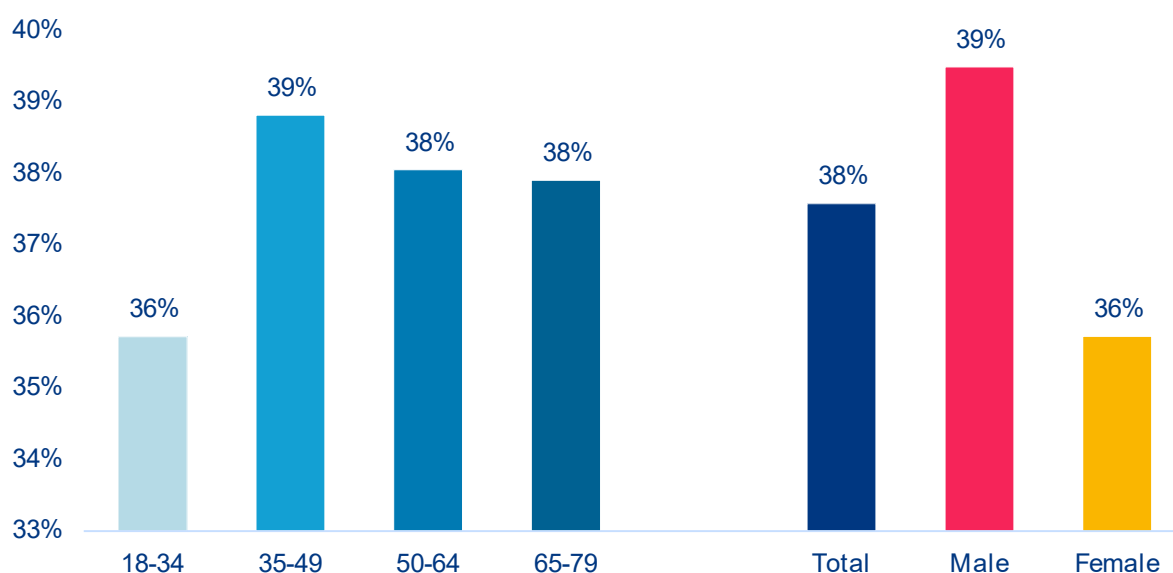
¹ In the eight countries covered by our survey, public social expenditure as a share of GDP in 2024 stood at: Austria 31.6%, France 30.6%, Germany 27.9%, Italy 27.6%, Spain 25.9%, Poland 23.1%, UK 23.0% (2023), US 19.8%. Source: OECD, Public social spending, [Social spending | OECD](#).

Figure 1: To what extent do you agree with the following statement: “Current social systems (e.g., social security, pensions, healthcare) will need to be reformed to remain sustainable.” Share of answers, in percent of all answers, by country



Source: Allianz Research.

Figure 2: To what extent do you agree with the following statement: “Current social systems (e.g., social security, pensions, healthcare) will need to be reformed to remain sustainable.”, share of respondents who “strongly agree”, by age group and gender (in percent of respective population group)



Source: Allianz Research.

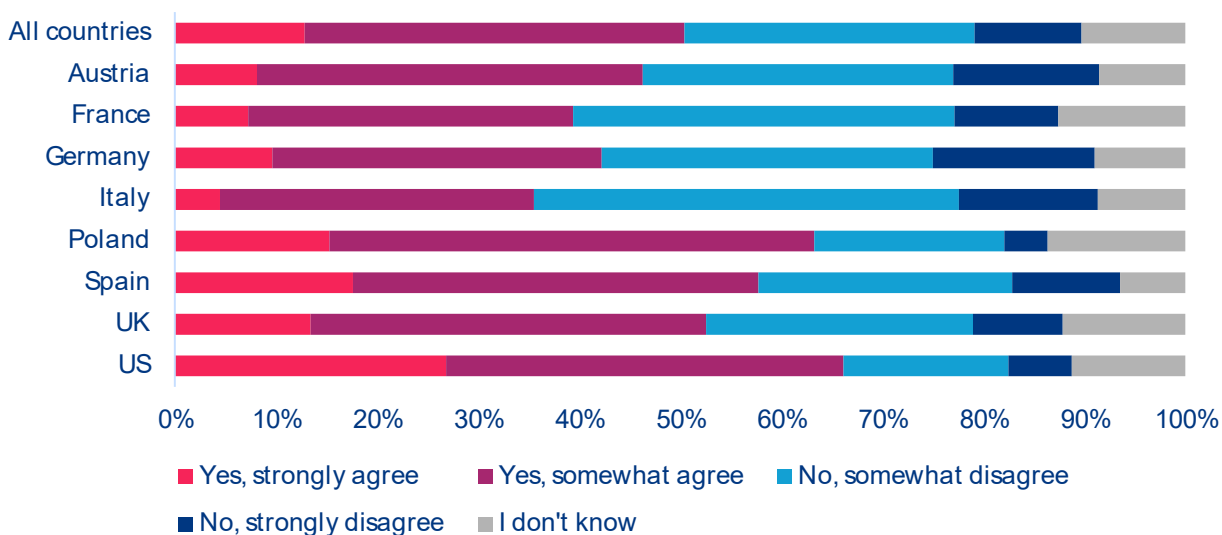
Surprisingly, nearly four in five retirees believe reform is necessary even though they are already drawing benefits from the current system. While the intensity of reform demand varies somewhat across subgroups, support remains broad throughout the sample. It is strongest among university graduates (85%) and lowest, though still substantial, among respondents with lower educational attainment (71%). Employment status shows a similar pattern. Recognition of reform need is highest among the employed (85%) and self-employed (82%), suggesting that those who bear social contributions most directly are particularly aware of the pressures facing the system. The more striking finding, however, comes from retirees. While the level of support is at 78% somewhat lower than among the workforce, nearly four in five still agree that reform is necessary despite already drawing benefits from the current system. Recognition of reform need is therefore not confined to future contributors; it is also shared by many who already depend on the system today.

While more than 80% of respondents in most countries agree that social systems require reform, confidence that governments will follow through on the reforms needed is far lower. Across all eight countries, only around 50% of respondents believe that the necessary reforms will

actually be implemented. Confidence is highest in the US, where around two-thirds of respondents believe reforms will be delivered, and relatively strong in Poland (63%). By contrast, confidence is markedly lower in Germany at 42% and lowest in Italy, where only 36% believe the necessary reforms will be implemented (Figure 3). The challenge therefore appears less one of recognizing the need for reform than of building confidence that policymakers will follow through on reform commitments.

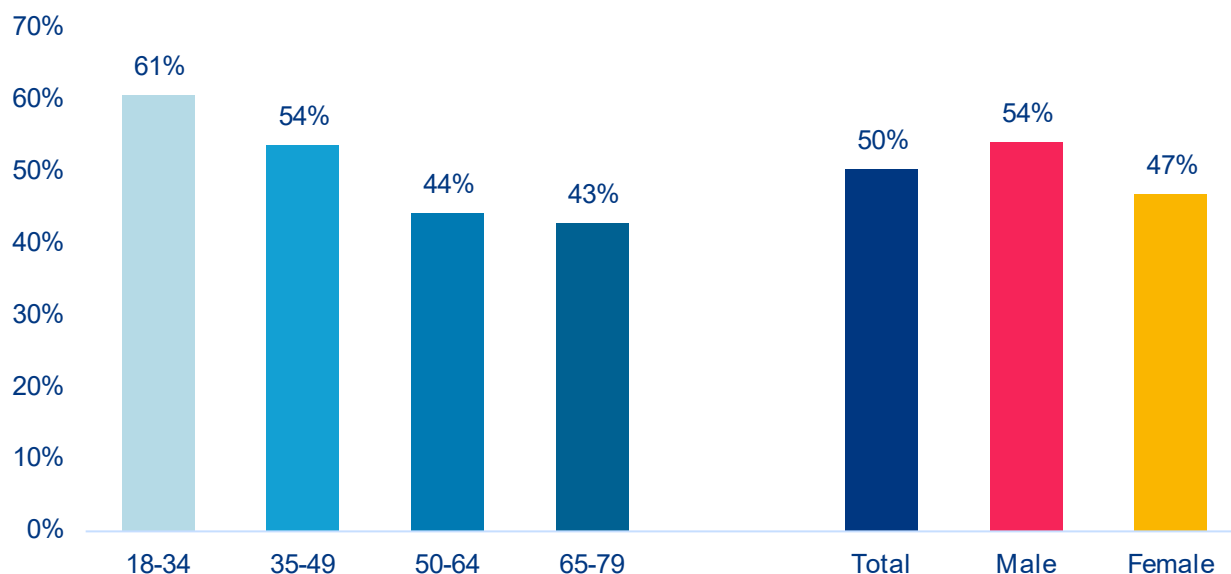
The generational pattern is the mirror image of reform need. While recognition of reform need tends to rise with age, confidence that reforms will actually be implemented declines. Younger respondents remain considerably more optimistic that governments will follow through on reform commitments, whereas older respondents combine the strongest support for reform with the deepest skepticism that it will happen. This divergence between reform demand and reform confidence is one of the most striking findings of the survey and helps explain why broad support for reform so often fails to translate into expectations of action (Figure 4).

Figure 3: Do you think your country is capable of implementing the reforms needed to ensure long-term economic sustainability? Share of answers in percent of all answers, by country



Source: Allianz Research

Figure 4: Do you think your country is capable of implementing the reforms needed to ensure long-term economic sustainability?, approvers by age group and gender (in percent)

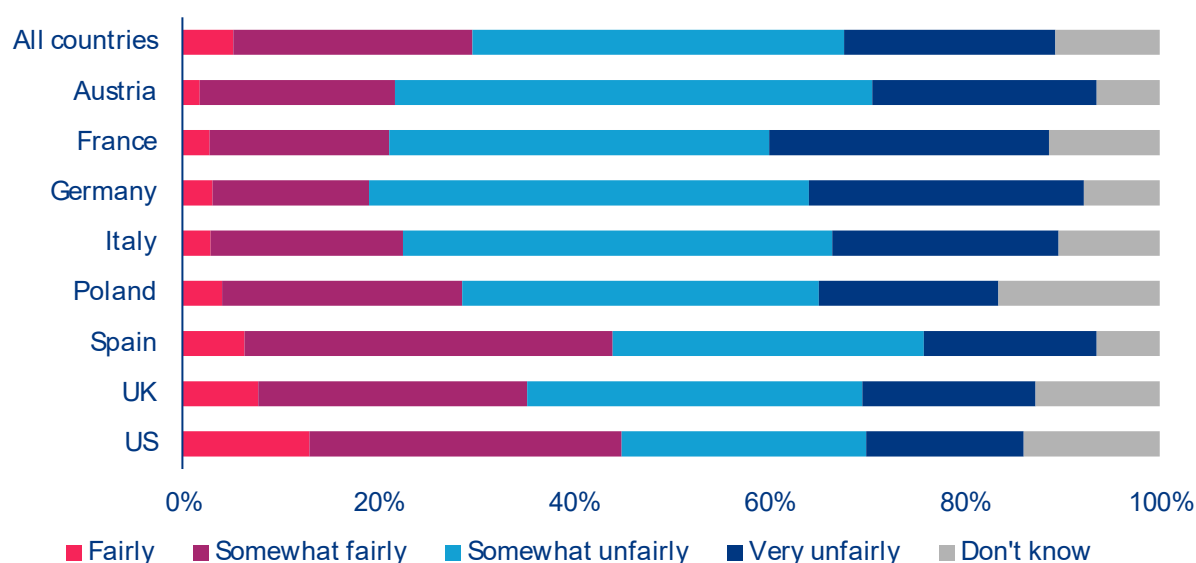


Source: Allianz Research.

Many citizens doubt not only that reforms will happen, but also that they will be fair. Across the eight countries, **only** around 30% of respondents expect the costs and benefits of reform to be distributed fairly, while almost 60% expect an unfair outcome (Figure 5). This skepticism is particularly pronounced in continental Europe and may

help explain the broader trust deficit identified above. Citizens do not merely question whether governments will implement reform; many also question whether they will do so in a fair and balanced way. Reform credibility is therefore as much a question of legitimacy as of policy design.

Figure 5: How fairly do you think the costs and benefits of reforms will be distributed across society?, share of answers in percent of all answers, by country



Source: Allianz Research



Pension reform is not on the top of the list

Pension reform is widely recognized as necessary, but it is not at the top of the public's reform agenda.

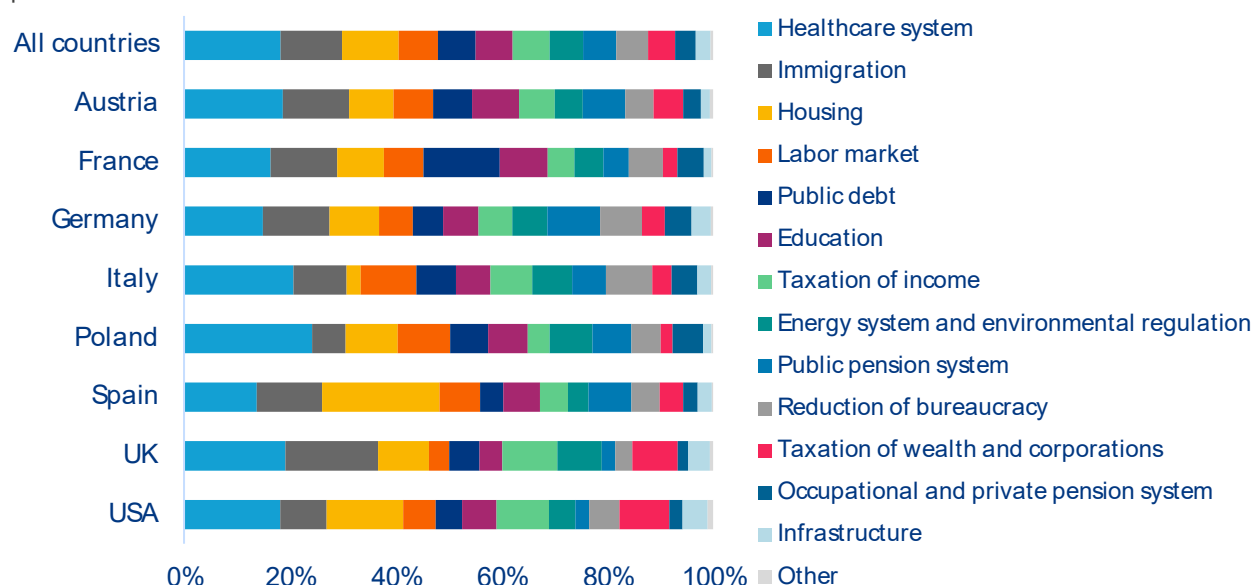
When asked where reforms are most urgently needed, respondents consistently rank healthcare, immigration and housing at the top, far ahead of public, occupational and private pension systems (Figure 6). This may help explain why pension reform is so frequently postponed despite broad recognition of the underlying challenges. The issue is therefore not a lack of awareness, but a lack of urgency.

the broader survey finding that perceptions of reform need increase with age. The pattern points to a salience effect: as retirement moves from a distant prospect to an imminent reality, concern about the sustainability of pension systems rises markedly. Younger generations, despite having the greatest long-term stake in the sustainability of pension systems, tend to view retirement as a distant concern.

The older the wiser: the strongest demand for pension reform comes from those closest to retirement.

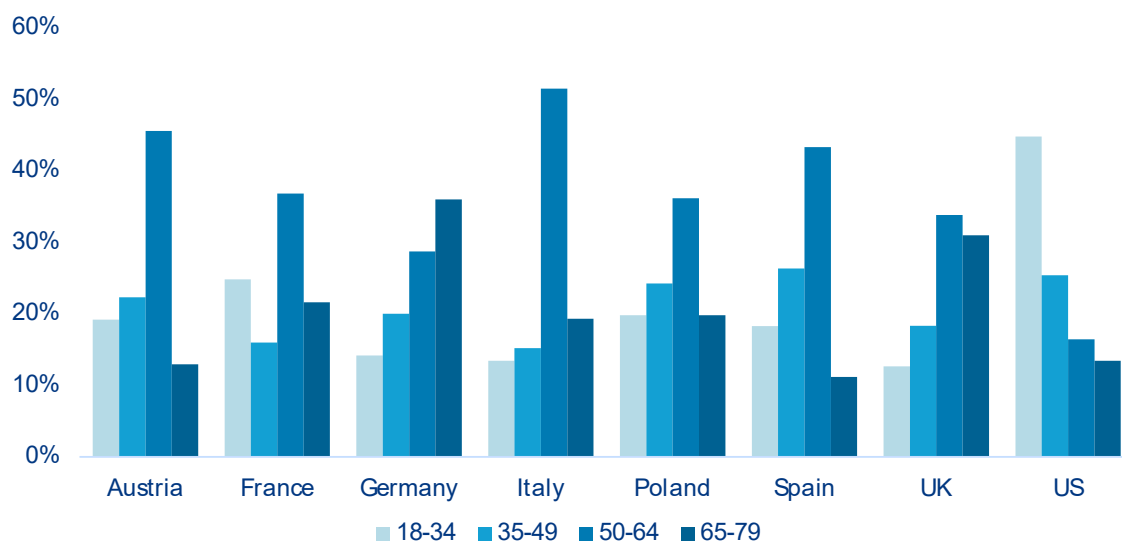
Respondents aged 50 to 64 are most likely to identify pension reform as a policy priority (Figure 7), mirroring

Figure 6: In which areas do you think reforms are most needed in your country? Select max. 3, share of answers in percent of all answers



Source: Allianz Research

Figure 7: In which areas do you think reforms are most needed in your country? Select max 3. - Public pension system, approvers by age group (in percent)



Source: Allianz Research

Demographic pressures are accelerating and the window for gradual reform is narrowing. With the baby-boomer generation reaching retirement age, the increase of the old-age dependency ratio (OADR) – the number of people aged 65 and over relative to every 100 people of working age – is set to accelerate sharply across most advanced economies over the next decade before levelling out on a comparatively high level. The countries covered in our survey are no exception. By mid-century, in Italy and Spain, there are projected to be around 75 retirees

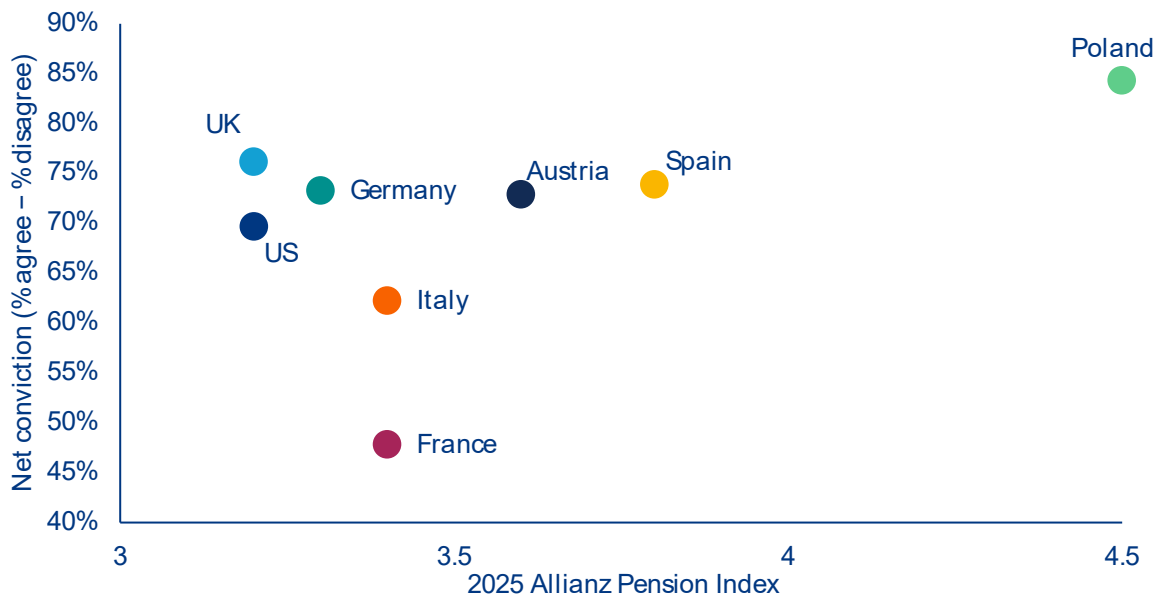
per 100 people of working age², while in Poland the ratio will exceed 60%. In Austria and Germany it is expected to stabilize at around 58%, and even in France it is projected to surpass 50%. Only in the UK and the US is it expected to remain below 40%.

Figure 8: Allianz Pension Index 2025

	API 2025	Basic Conditions	Sustainability	Adequacy
<u>Surveyed Countries:</u>				
UK	3.17	3.40	3.54	2.70
US	3.24	3.10	3.45	3.11
Germany	3.26	3.80	3.58	2.67
Italy	3.35	4.34	3.32	2.89
France	3.39	3.47	3.70	3.05
Austria	3.57	4.14	4.08	2.78
Spain	3.77	4.18	4.43	2.90
Poland	4.51	4.39	4.92	4.16
<u>Reference Countries:</u>				
Denmark	2.34	3.17	2.29	1.98
Netherlands	2.60	2.98	3.33	1.68
Sweden	2.62	2.93	2.94	2.15

1 = no need for reforms 7 = high reform pressure

Source: Allianz Research

Figure 9: Net convictions about reform need vs. Allianz Pension Index 2025, by country

Source: Allianz Research

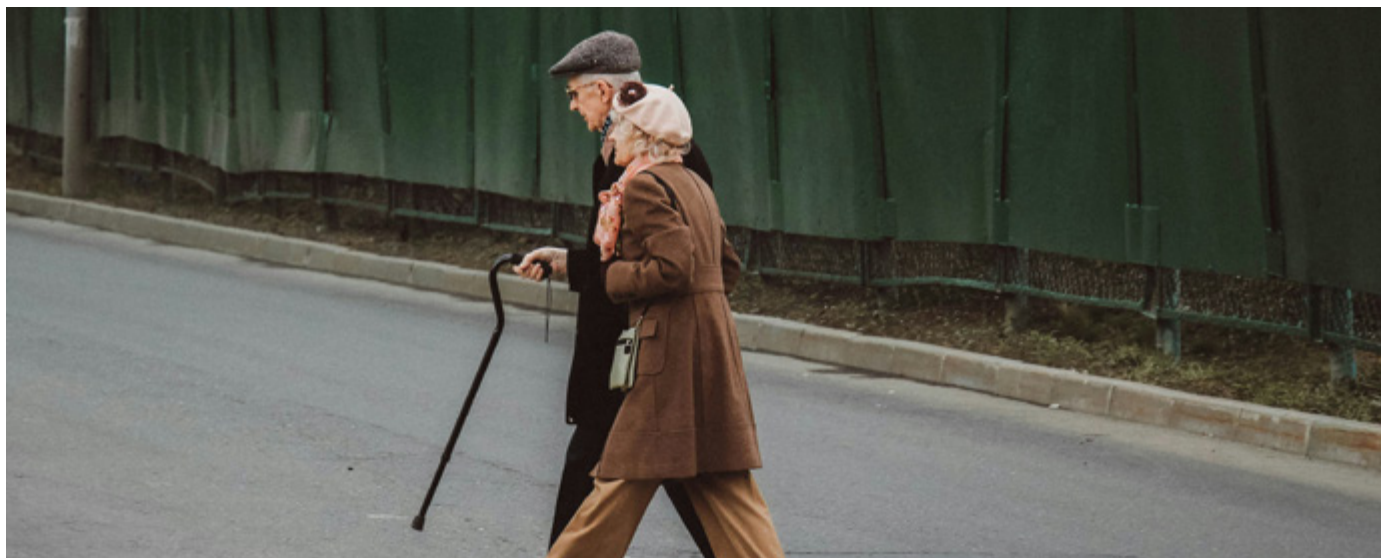
Demographic pressures are accelerating and the window for gradual reform is narrowing. With the baby-boomer generation reaching retirement age, the increase of the old-age dependency ratio (OADR) – the number of people aged 65 and over relative to every 100 people of working age – is set to accelerate sharply across most advanced economies over the next decade before levelling out on a comparatively high level. The countries covered in our survey are no exception. By mid-century, in Italy and Spain, there are projected to be around 75 retirees per 100 people of working age², while in Poland the ratio will exceed 60%. In Austria and Germany it is expected to stabilize at around 58%, and even in France it is projected to surpass 50%. Only in the UK and the US is it expected to remain below 40%.

Public perceptions of reform need broadly mirror the structural pressures facing pension systems. The stronger the reform need identified by the Allianz Pension Index³ (API), which assesses the adequacy and sustainability of pension systems across countries, the stronger the perceived need for reform revealed in our survey (Figure 8). Citizens appear remarkably adept at recognizing where pension systems face the greatest strain, suggesting that

awareness is not the main obstacle to reform. Poland tops both rankings (API: 4.5; net conviction: 84%), with Spain (3.8; 74%) and Austria (3.6; 73%) not far behind. At the other end, the UK (3.2; 76%) and the US (3.2; 70%) combine comparatively good API scores with somewhat lower reform urgency. The deeper divide, however, is not national but institutional. The perceived and actual need for reform tends to be strongest in countries that rely heavily on pay-as-you-go pension systems, while countries with stronger funded pension pillars are less exposed to demographic pressures (Figure 9). France (3.4; 48%) and Italy (3.4; 62%) remain notable exceptions, where the perceived need for reform appears more subdued than the underlying structural pressures would suggest.

² Here aged between 20 and 64. Source United Nations, Department of Economic and Social Affairs, Population Division (2024).

³ The Allianz Pension Index (API) is based on three sub-indices – starting points, sustainability and adequacy – and takes into account 40 parameters. Each parameter is rated on a scale of 1 to 7, with grade 1 signaling no need for reforms and 7 a high reform pressure.



The reform options are known — but there is no one size fits all

The inventory of possible reforms is well known, implementation remains the real challenge. The broad contours of pension reform look remarkably similar across countries: increasing the retirement age, encouraging longer working lives, strengthening funded occupational and private pensions, improving financial literacy and adjusting contribution rates and benefit levels. What differs across countries is the mix. Pension systems vary widely in their reliance on public transfers, occupational pensions and private savings, as well as in their demographic pressures and labor-market conditions. Successful reform therefore requires adaptation rather than imitation. The challenge is not a lack of solutions, but turning well-known reform options into policy action and implementing the right combination of measures before demographic pressures force more abrupt and costly adjustments.

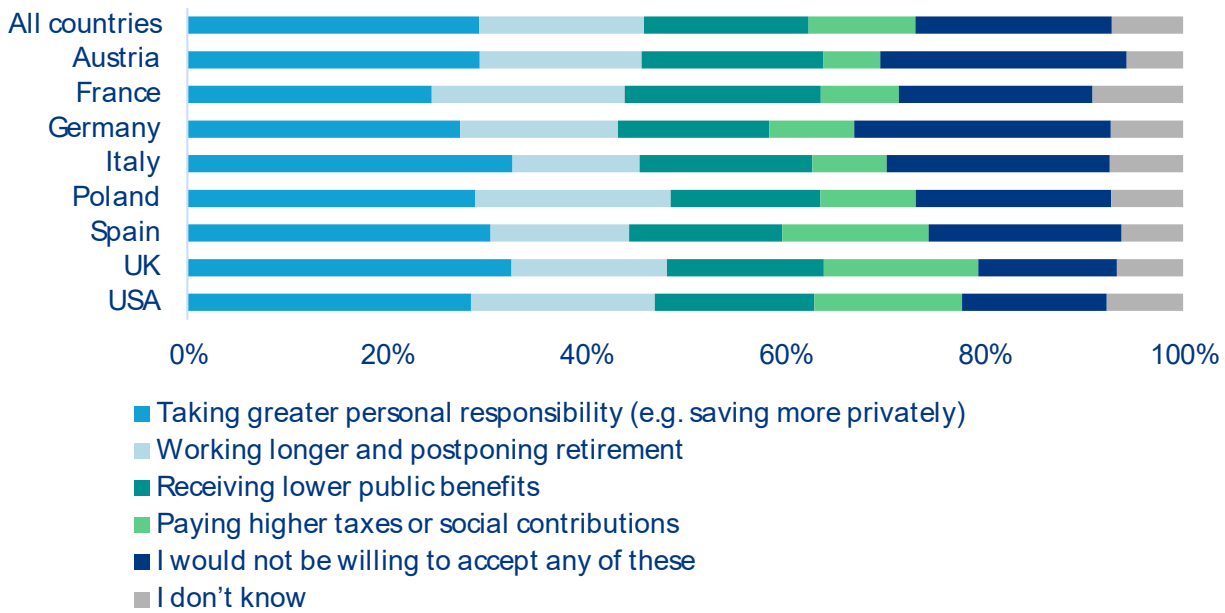
Support for pension reform is broad, but there is much less agreement on which reform measures should be implemented. When asked which measures they would personally be willing to accept to improve the long-term sustainability of public finances, respondents paint a more nuanced picture than their broad recognition of the need for reform might suggest. The most widely supported option is taking greater personal responsibility through additional private saving (29% of all responses). However, the second most common response is a refusal to accept any of the proposed measures (22%). Working longer and postponing retirement ranks third, followed by accepting lower public benefits. The results highlight a central

challenge of pension reform: while many agree that change is necessary, there is far less consensus on how the adjustment burden should be shared (Figure 10).

Taking greater personal responsibility is the most widely accepted reform option. In most countries, support for additional private saving is strongest among older age groups (Figure 11). This likely reflects greater financial flexibility later in life, when major household expenditures such as mortgages and child-related costs have often declined. Across all countries, however, support for this measure is consistently higher among men than among women.

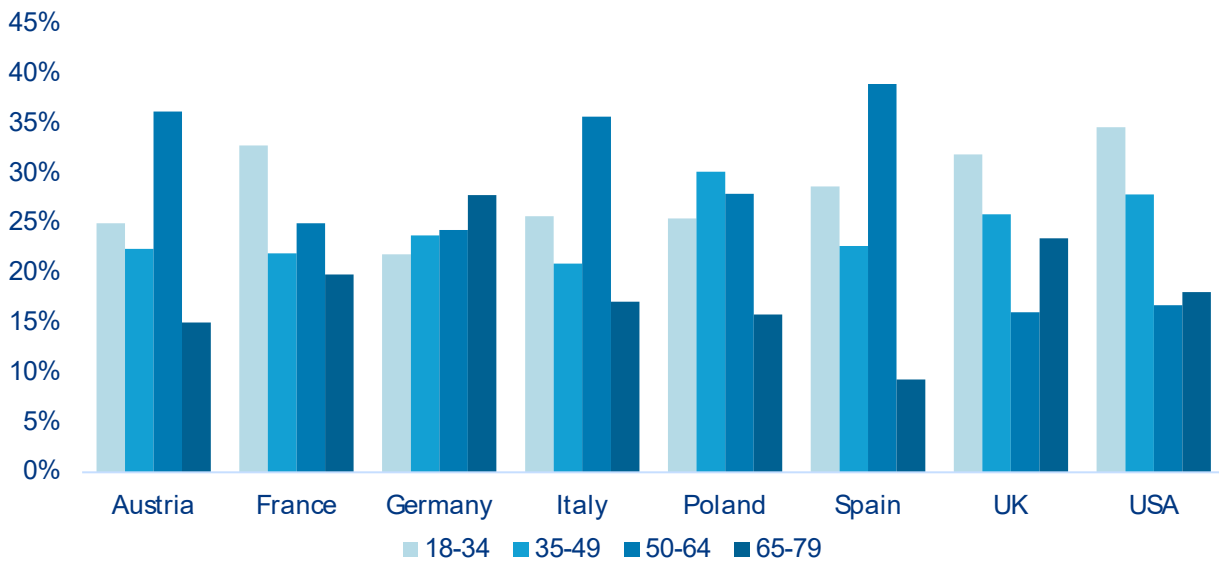
The stereotype that older workers are unwilling to work longer finds little support in the data. In Austria, Italy and Spain, more than one-third of respondents supporting later retirement are aged between 50 and 64. In France, support is distributed relatively evenly across age groups. Germany presents a more nuanced picture: only 21% of supporters are aged between 50 and 64, compared with 25% aged 65 to 79. One possible explanation is that extending working lives may appear less realistic in an environment characterized by economic weakness, corporate restructuring and relatively high long-term unemployment among older workers. By contrast, support for working longer is more broadly distributed in the UK and the US, where labor-force participation at older ages is already higher and funded pension systems play a larger role (Figure 12).

Figure 10: Which of the following would you personally be willing to accept to ensure long-term sustainability of public finances? share of answers in percent of all answers



Source: Allianz Research

Figure 11: Which of the following would you personally be willing to accept to ensure long-term sustainability of public finances? Taking greater personal responsibility, approvers by age group (in percent)

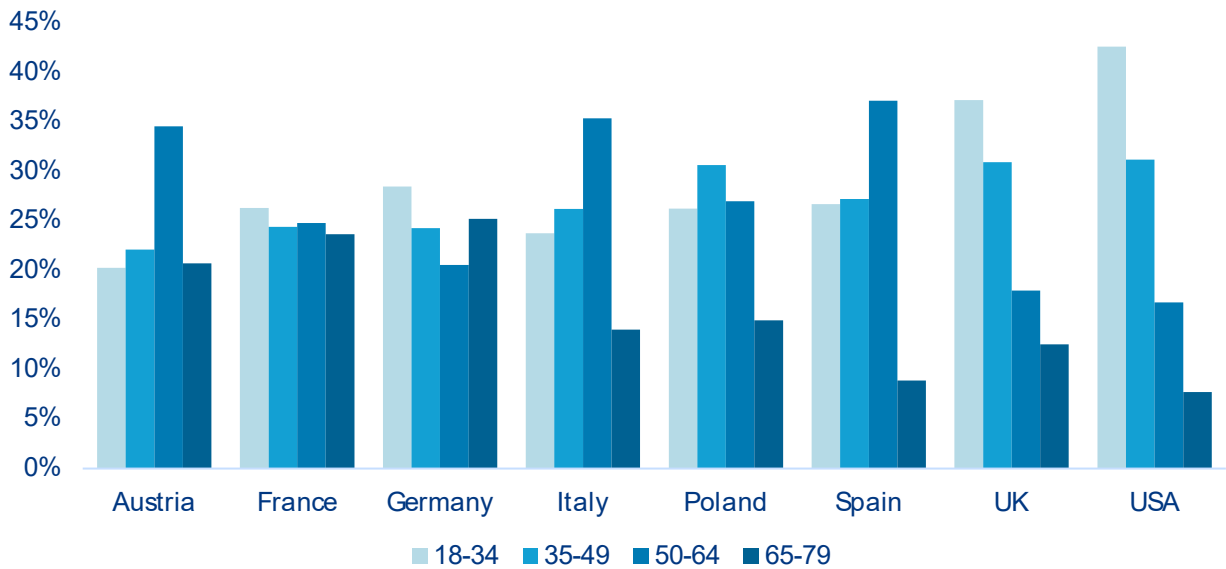


Source: Allianz Research

Support for lower public benefits remains limited, but is not confined to younger generations. Around one-quarter of respondents would consider lower public benefits as part of a reform package. In Austria, Italy and Spain, where public pension replacement rates remain comparatively high, more than one-third of supporters are aged between 50 and 64. In most countries, however, support is lower among those approaching or already in retirement than among younger cohorts, reflecting the shrinking time horizon available to compensate for benefit reductions through additional private saving (Figure 13).

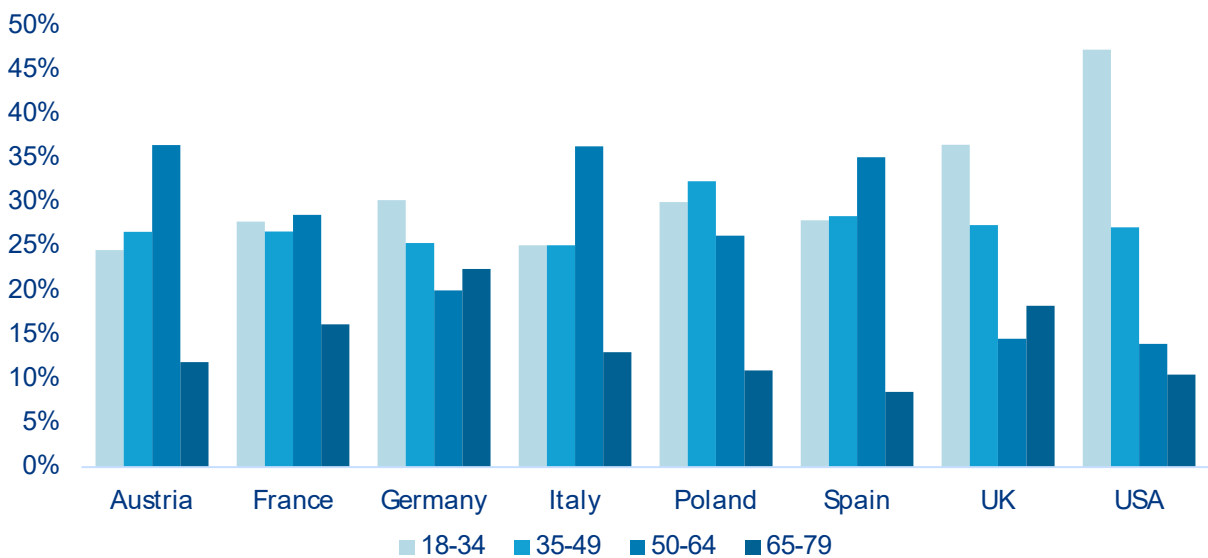
Disagreement over reform measures should not be confused with opposition to reform. While respondents differ on how adjustment should be achieved, only a minority reject all proposed measures outright. The challenge for policymakers is therefore less one of overcoming universal resistance than of combining measures in a way that distributes the adjustment burden credibly and fairly. The survey results suggest that citizens may disagree on the preferred path, but many recognize that some form of adjustment will be necessary to ensure the long-term sustainability of pension systems.

Figure 12: Which of the following would you personally be willing to accept to ensure long-term sustainability of public finances? Working longer and postponing retirement, approvers by age group (in percent)



Source: Allianz Research

Figure 13: Which of the following would you personally be willing to accept to ensure long-term sustainability of public finances? Receiving lower public benefits, approvers by age group (in percent)



Source: Allianz Research



Retirement preparedness starts with knowing the numbers

Limited knowledge of key pension parameters makes retirement planning more difficult. Pension reform ranks below healthcare, immigration and housing on the public policy agenda, and this lower salience is reflected in respondents' understanding of pension systems. Across countries, many respondents struggle to estimate key parameters such as the statutory retirement age, the number of years they are likely to spend in retirement and the level of income they can expect from the public pension system. Yet these are precisely the variables individuals need to understand in order to assess how much additional saving toward retirement will be required.

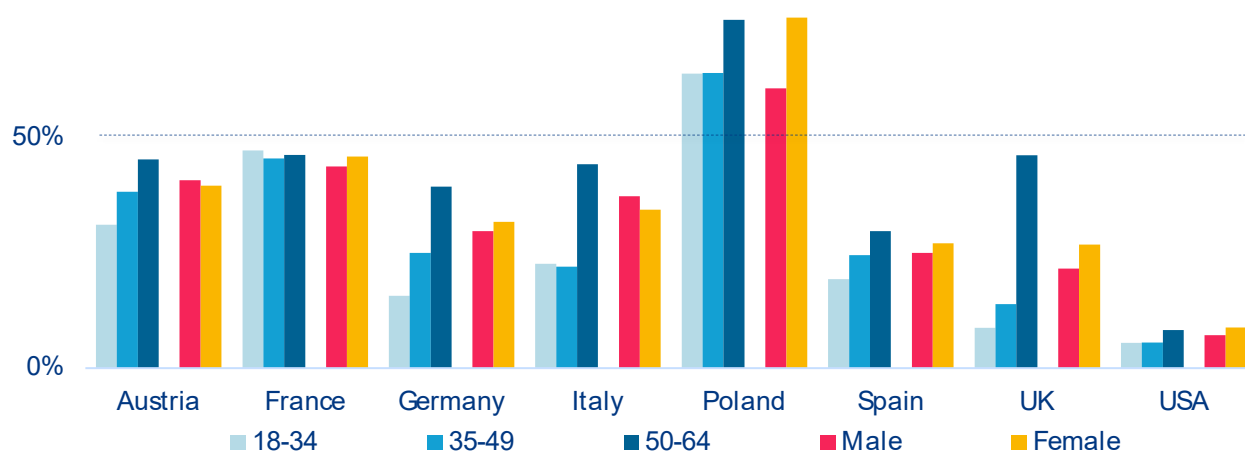
The knowledge on retirement options increases with age. More than half of respondents in every country except Poland plan to retire before the current or already legislated retirement age. However, as respondents approach retirement age and become more familiar with the options and financial implications of retiring early or

later, their expected retirement age converges towards the statutory retirement age. Furthermore, the answers reflect debates about or actual reforms of the retirement age in recent years. Respondents aged 50 to 64 were generally the most likely to plan retirement in line with the statutory retirement age, with the exception of France. Only in Poland, more than 50% of respondents plan to retire at the statutory retirement age, which has remained unchanged over the past decade (Figure 14).

Most respondents expect to retire between the ages of 65 and 69. France is the notable exception, where many respondents expect to retire as early as possible by making use of the possibility to retire at age 63, or age 64 in the future, once the required contribution periods have been met (Figure 15).

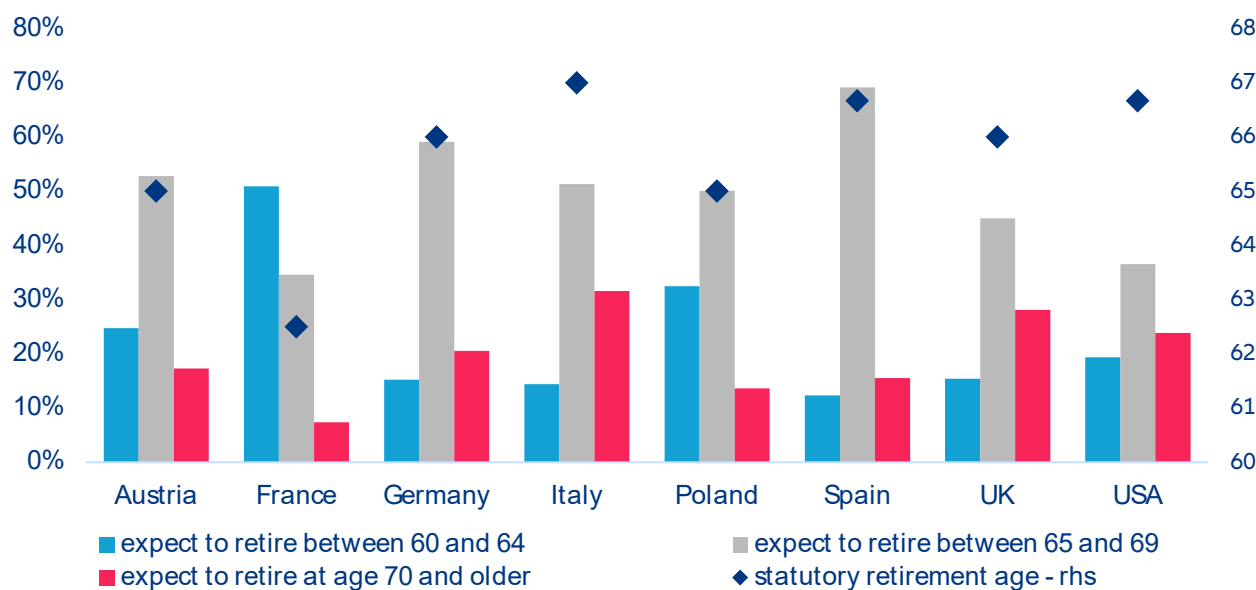
Figure 14: At what age do you realistically expect to retire?, share of respondents who gave the actual statutory current or after reform statutory retirement age (in percent of respective population group).

100%



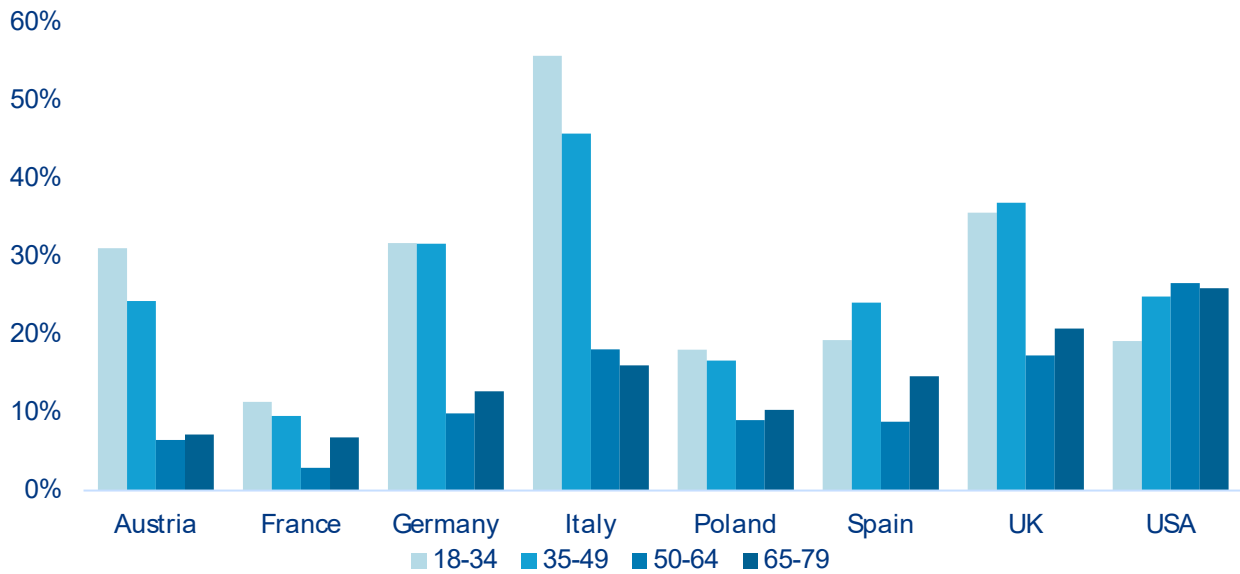
Source: Allianz Research

Figure 15: At what age do you realistically expect to retire?, age brackets (share of respondents in percent)



Source: Allianz Research

Figure 16: At what age do you realistically expect to retire? – At the age of 70 and older (in percent of the respondents in the respective age class)

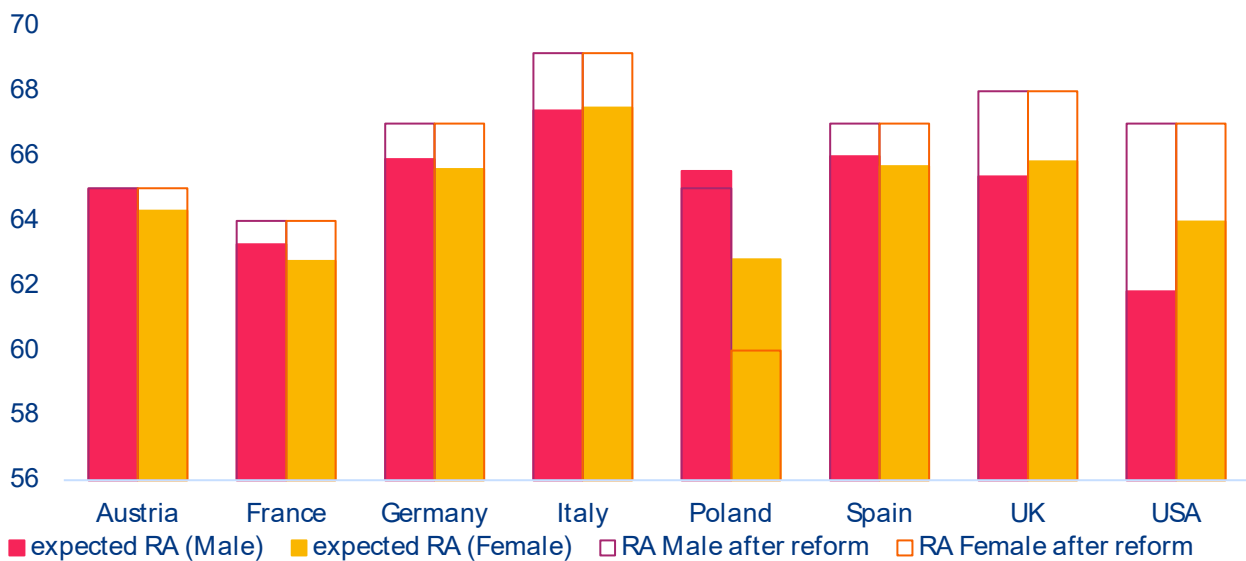


Source: Allianz Research.

Younger generations in particular expect to work beyond the age of 70. In most countries, fewer than 25% of respondents expect to retire at age 70 or later. Only in Italy and the UK does this share reach around one-third. In both countries, future increases in the retirement age have already been linked to rising life expectancy. Since these changes are being implemented gradually, it is particularly younger respondents aged 18–34 and 35–49 who expect to retire at age 70 or beyond (Figure 16).

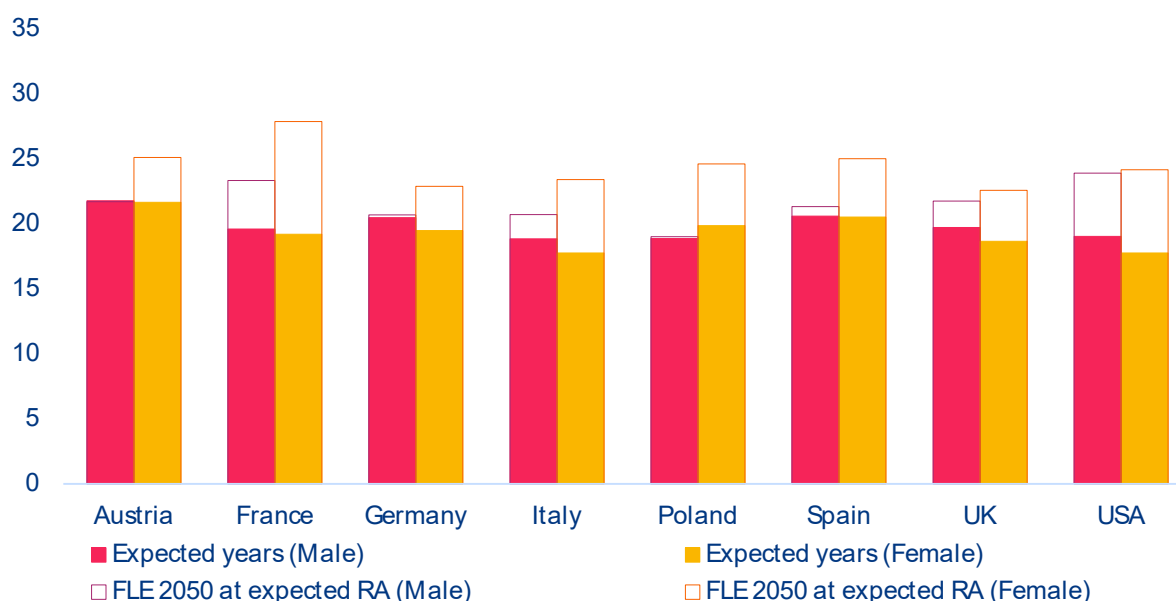
Many respondents expect to retire up to two years earlier. Ranging from 62 years in the US to 67 years in Italy, the average expected retirement age remains between one and two years below the future statutory retirement age already agreed upon in legislation (Figure 17) in most countries. Poland is the main exception, partly because many women choose to work beyond the statutory retirement age of 60 in order to receive a higher pension. Another explanation may be the expectation that the retirement age will be increased in the long run.

Figure 17: At what age do you realistically expect to retire?, average expected retirement age compared to the statutory retirement age after reforms (in years)



Source: Allianz Research.

Figure 18: How many years do you expect to spend in retirement?, expected average time spend in retirement compared to the average further life expectancy at retirement age (in years)



Sources: UN Population Division (2024), Allianz Research.

Respondents are even less informed about how long they are likely to spend in retirement. Of course, it depends on the individual situation, but the average further life expectancy at the age of retirement is a rather good proxy for the time one can expect to spend in retirement. In most surveyed countries, only around one-third of respondents correctly estimated average life expectancy at their expected retirement age. Women in particular tend to underestimate the number of years they are likely to spend in retirement. The largest gap emerges in France, where female respondents expect to spend around 19 years in retirement on average, while the average further life expectancy of a woman retiring at age 63 is currently around 26 years and is projected to increase to 28 years by 2050 (Figure 18).

Underestimating longevity can lead to a significant underestimation of retirement savings needs. Rising life expectancy poses a challenge not only for public pay-as-you-go pension systems but also for individual retirement planning. Linking retirement ages more closely to life expectancy could help preserve intergenerational fairness by maintaining a more stable relationship between working life and retirement duration across generations. The implications for individual saving behavior are equally important. Taking the example of an average French woman aiming for a retirement income equal to 70% of her final wage, annual savings of 3.7% of earnings would

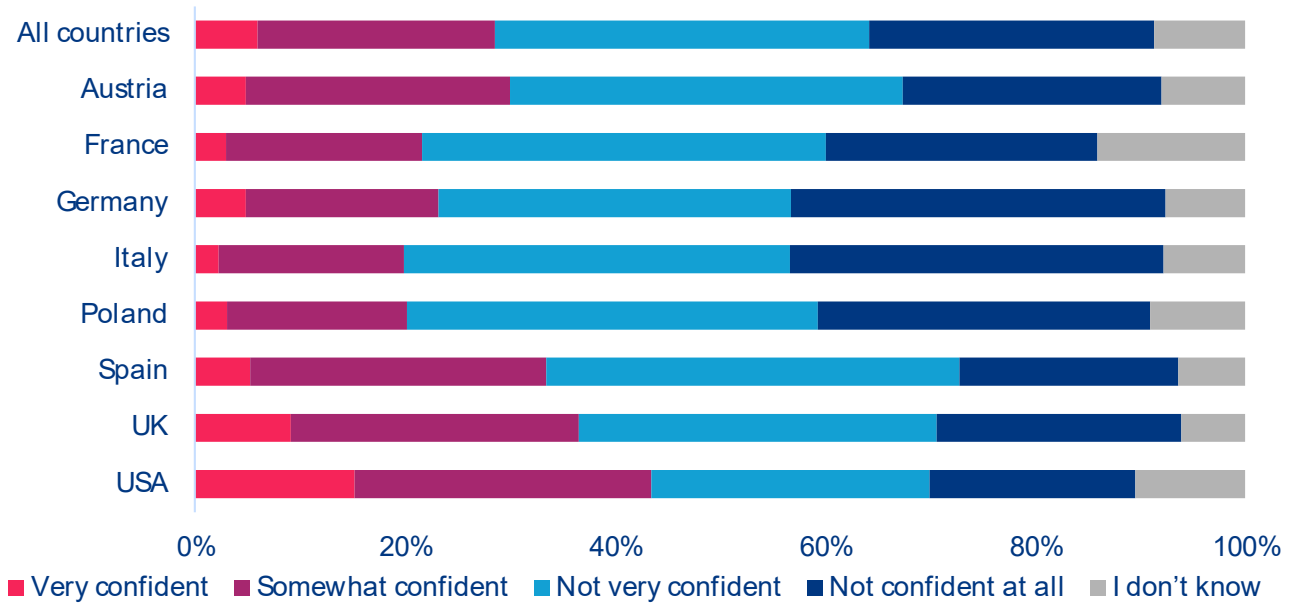
be sufficient if retirement lasted only 19 years. However, if retirement lasts closer to 28 years, the required savings rate rises to 5.0% of earnings, or alternatively requires accepting a lower retirement income of around 67%⁴.

The pension and financial literacy gap therefore has direct consequences for retirement preparedness.

Understanding retirement age, longevity and expected pension income is essential for estimating future pension gaps and determining how much supplementary saving will be required. Yet confidence in the ability of public pension systems to provide an adequate retirement income is already limited. Only in the US do more than 40% of respondents report being very or somewhat confident that the public pension system will provide a decent standard of living in retirement. In Spain this share falls to 37%, in the UK to around one-third, while in France, Germany, Italy and Poland fewer than 25% of respondents express such confidence (Figure 19). The challenge is therefore not a lack of awareness that additional retirement provision may be needed, but a limited understanding of how large the resulting savings gap may be.

⁴ Furthermore, we assumed an average inflation of 2%, an annual increase of the nominal wage of 2%, an average annual yield of 5.0% during the savings phase and of 3.0% during the payout period.

Figure 19: How confident are you that the public pension system in your country will provide you with an adequate income when you retire? share of answers in percent of all answers, by country

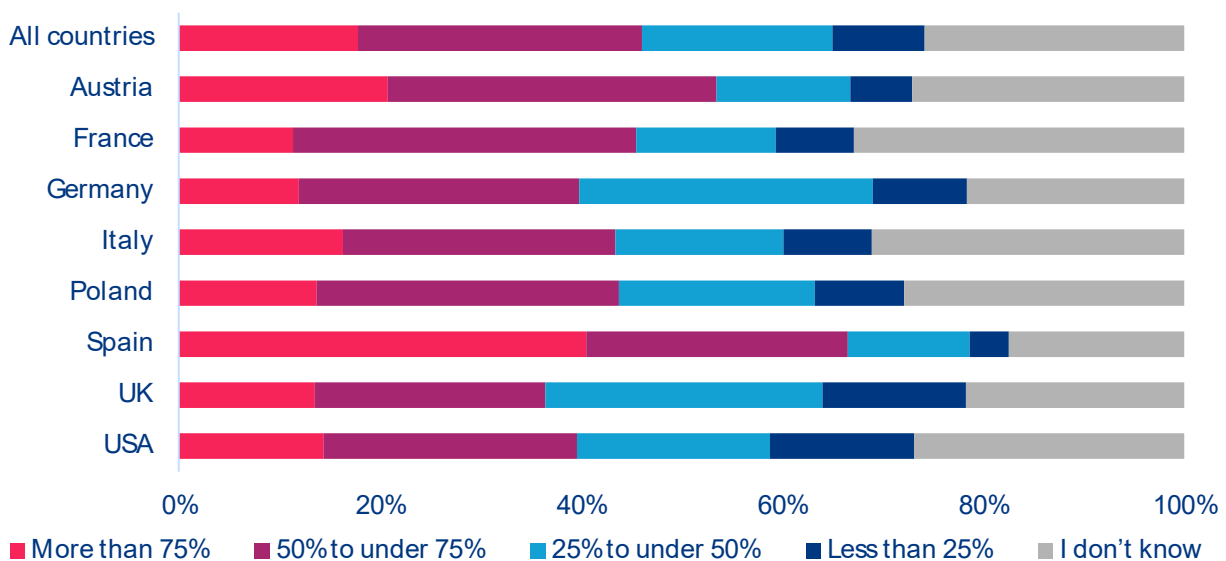


Source: Allianz Research

Many respondents expect public pensions to remain their main source of retirement income despite limited confidence in their adequacy. In Austria and Spain, 53% and 64% of respondents respectively expect the public pension system to provide more than half of their retirement income. In the remaining countries, the share ranges from less than 40% in the UK and the US to 45% in Italy (Figure 20). The results reveal a striking tension: many respondents expect to rely heavily on public pensions, yet relatively few are confident that these systems will provide an adequate standard of living in retirement.

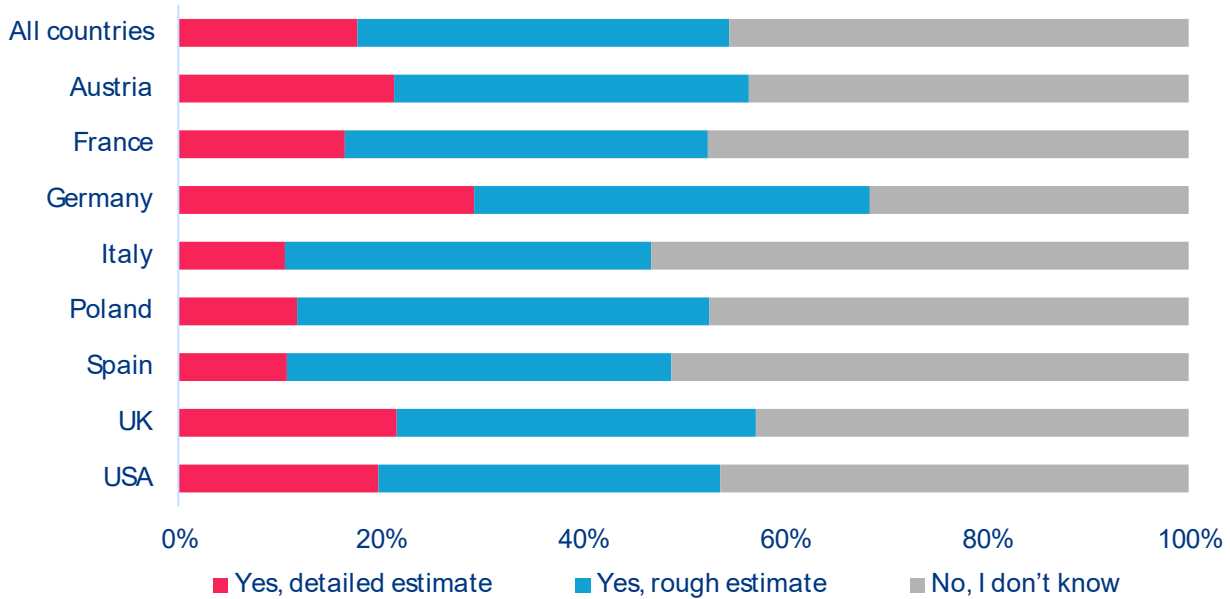
Knowledge of expected pension income remains limited. In most countries, fewer than half of respondents report having a detailed or rough estimate of their future public pension income. Across all countries, only 11% of respondents have a detailed estimate of their expected public pension income. Germany is the exception at 29%, likely reflecting the annual pension statements provided by the public pension provider. Between 34% of respondents in the US and 41% in Poland report having at least a rough estimate. At the same time, an average of 45% do not know how high their future public pension income will

Figure 20: What percentage of your retirement income do you expect to come from the public pension system? share of answers in percent of all answers, by country



Source: Allianz Research

Figure 21: Do you know what your expected public pension income will be?, share of answers in percent of all answers, by country



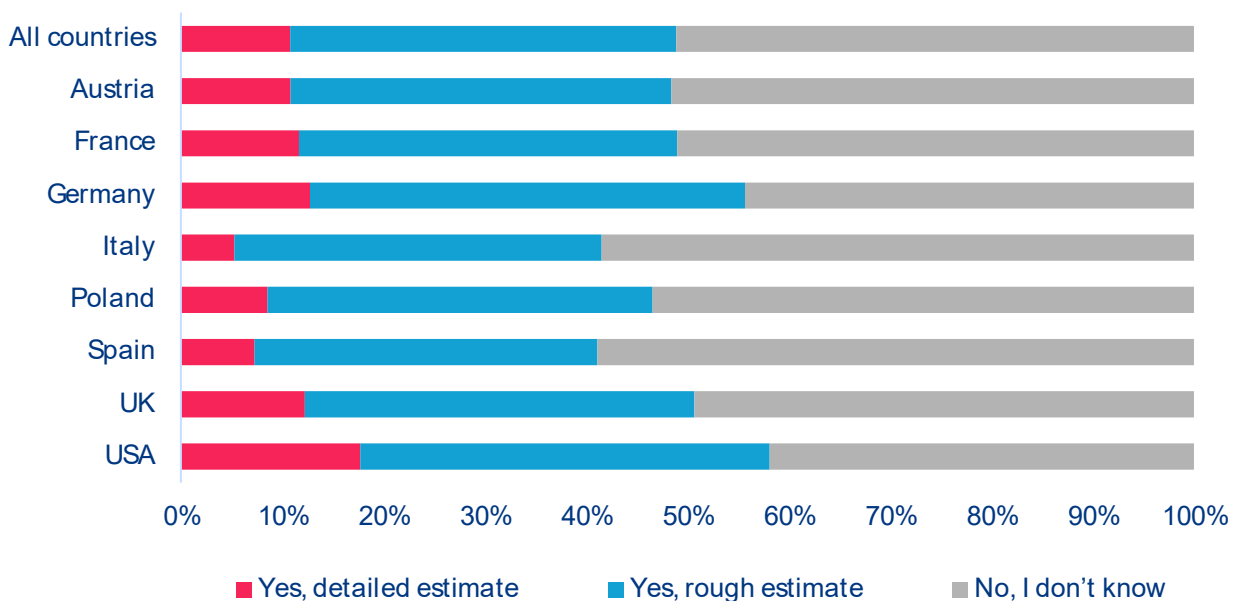
Sources: Allianz Research

be, ranging from 32% in Germany to 53% in Italy (Figure 21). For many households, retirement planning therefore begins without a clear understanding of expected retirement income.

Many respondents have not estimated how much income they will need in retirement. With the exception of the US and Germany, where the share is close to 60%, around half of respondents across countries have never

estimated their required retirement income (Figure 22). Women are particularly likely to report not knowing how much income they will need. Surprisingly, the largest share of respondents without such an estimate is often found among those aged 50 to 64, i.e. the group approaching retirement most rapidly. The results suggest that retirement planning remains incomplete even among individuals nearing the end of their working lives.

Figure 22: Have you ever estimated how much income you will need in retirement?, share of answers in percent of all answers, by country



Source: Allianz Research

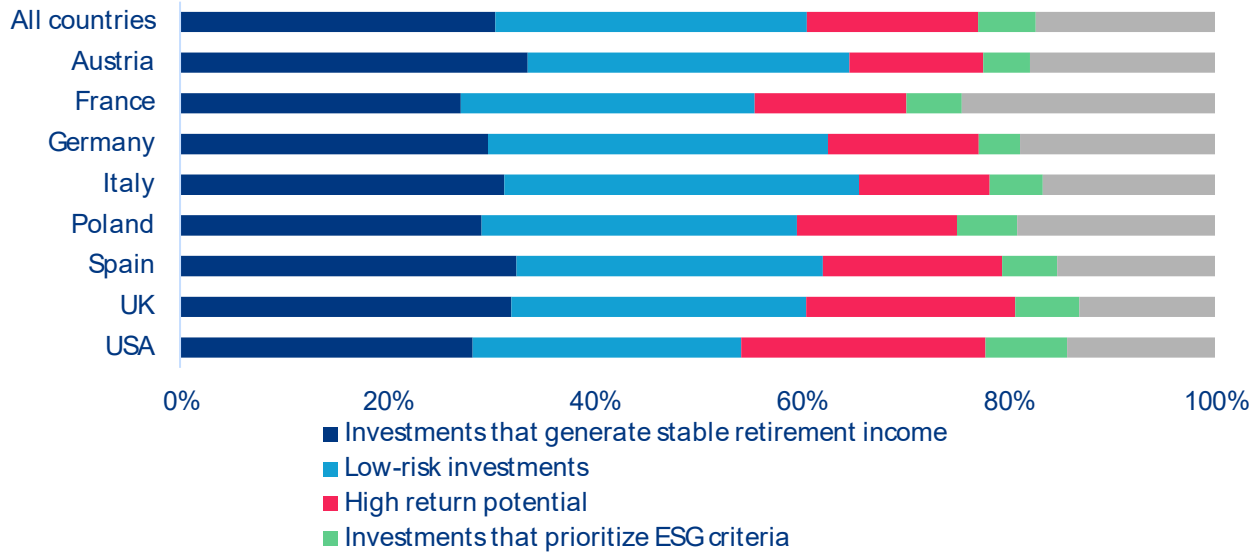


Financial literacy, a growing imperative for pension policy

Financial literacy is becoming an increasingly important pillar of retirement planning. As trust in public pension systems remains limited and funded occupational and private pensions become more important, individuals will need to take greater responsibility for retirement planning. A solid financial education becomes essential to ensuring the maintenance of the desired living standard in old-age. Across the eight countries surveyed, only one in six respondents shows a high level of financial literacy (18%), compared with 57% of respondents with medium and 26% with low financial literacy. The low levels of financial literacy translate into savings and investment behaviors that favor suboptimal low-risk investments. When asked about their investment preferences, most respondents favor low-risk investments and products that provide a stable retirement income. By contrast, fewer than 10% identify ESG characteristics as their primary consideration when saving for retirement (Figure 23).

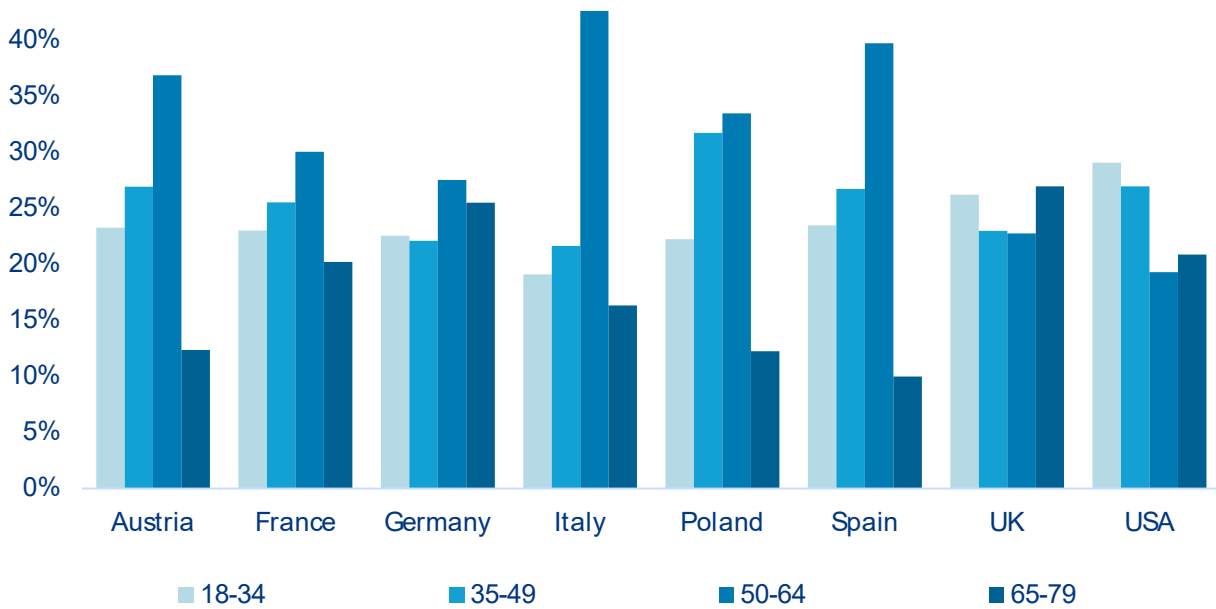
Investment preferences vary systematically across age groups. Respondents preferring low-risk investments are predominantly aged 50 to 64 (Figure 24), reflecting a shorter investment horizon and greater sensitivity to market fluctuations. By contrast, ESG-related preferences are most common among respondents aged 18 to 34.

Figure 23: When saving for retirement, which characteristics are most important to you?, share of answers in percent of all answers, by country



Source: Allianz Research

Figure 24: When saving for retirement, which characteristics are most important to you? Low-risk investments, respondents by age group



Source: Allianz Research



More capital-funded pension provision is a win-win

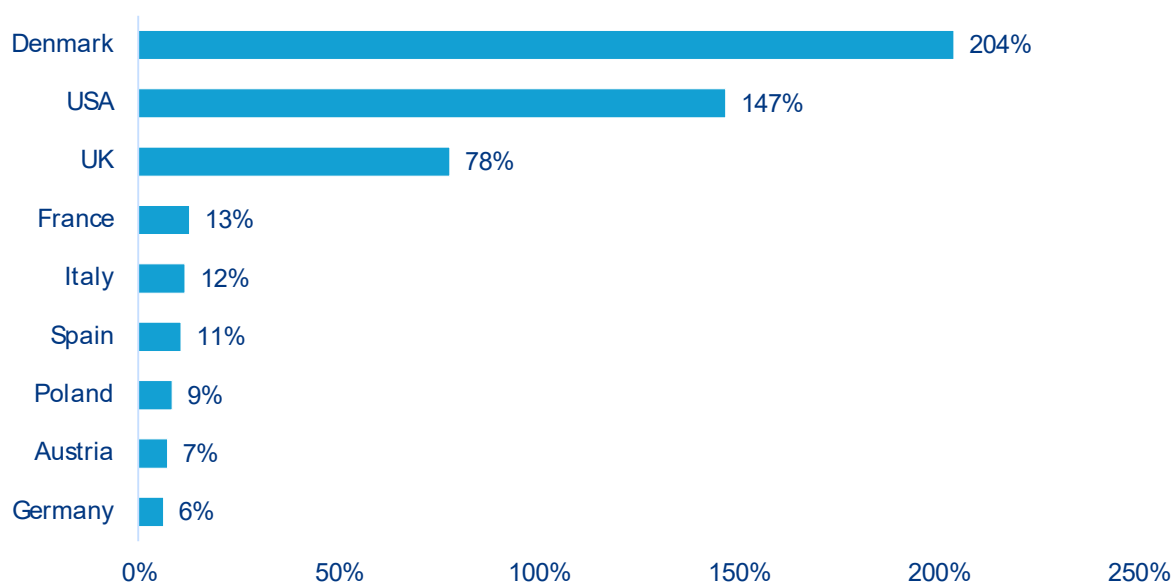
More capital-funded pension provision could create benefits beyond retirement income. Strengthening funded pension pillars could be a win-win for households and the wider economy. It would strengthen retirement security by increasing private wealth accumulation while simultaneously providing long-term capital to financial markets, supporting investment in innovation, infrastructure and projects aimed at mitigating climate change. Countries with strong funded pension pillars tend to exhibit significantly higher levels of household financial wealth. In Denmark, net financial assets per capita amount to EUR191,560, more than twice the level observed in Germany (EUR86,800)⁵. At the same time, pension assets correspond to more than 200% of GDP in Denmark, compared with just 6% in Germany. Comparable figures amount to 147% in the US and 78% in the UK (Figure 25).

To illustrate the effect of strengthening capital-funded pension pillars with respect to availability of long-term investment capital, we conduct the following thought experiment: If pension assets in Austria, France, Germany, Italy, Poland and Spain were to reach the same share of GDP as in the UK today, additional long-term investment capital of around EUR8.8trn could become available, equivalent to approximately twice the free-float market capitalization of the Euro Stoxx 50⁶ (Figure 26).

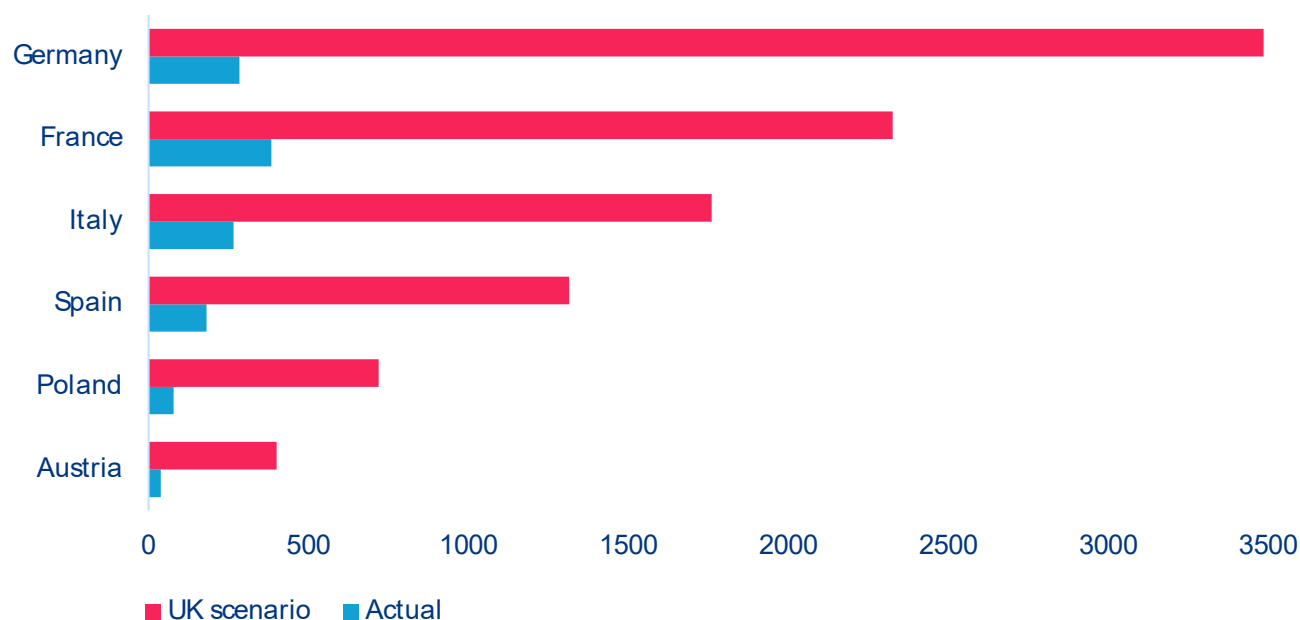
How pension assets are invested is as important as how much is accumulated. Significant differences remain in asset allocation across countries, reflecting differences in regulation and pension-system design. US pension funds invest only 12.5% of assets in bills and bonds and 1.5% in cash and deposits, whereas German pension providers allocate 42.8% to bills and bonds and 2.6% to cash. These differences have important implications for long-term returns, retirement outcomes and the availability of capital for financing economic transformation and innovation. The potential benefits of stronger funded pension pillars therefore depend not only on the amount of assets accumulated, but also on how effectively these assets are invested. Pension reform is thus not only about strengthening retirement security, it is also about mobilizing long-term capital, deepening capital markets and supporting future growth.

⁵ See Allianz Global Wealth Report 2025.

⁶ On May 29, 2026, the free-float market capitalization of the Euro Stoxx 50 was EUR 4.3trn, see Euro Stoxx (2026) factsheet, <https://stoxx.com/index/sx5e/?factsheet=true>

Figure 25: Pension assets (in percent of GDP 2024)

Source: OECD Pension Outlook 2025.

Figure 26: Actual pension assets vs UK scenario (in bn EUR)

Sources: IMF, OECD, Allianz Research.

Appendix: About the survey

The survey covered more than 8,000 respondents across eight countries: Austria, France, Germany, Italy, Poland, Spain, the United Kingdom and the United States. It was conducted as a web-based survey between 22 April and 25 May 2026. Overall responsibility lies with Allianz Research, Allianz Investment Management SE. Sample planning and fieldwork were carried out by Qualtrics.

Respondents were selected using representative quota sampling. Qualtrics was given quotas specifying how many people to survey in each country and which selection criteria to apply, with quotas set in line with official statistics for gender and age groups. A comparison with official statistics confirms that the resulting sample broadly corresponds to the population aged 18 and over in each of the eight countries.

Table 1: Sample statistics

	Total	Austria	France	Germany	Italy	Poland	Spain	UK	USA
Number of respondents	8048	1006	1006	1006	1006	1006	1006	1006	1006
Gender, in % of respondents									
Female	50.6%	50.0%	51.4%	49.9%	50.7%	50.9%	50.4%	50.9%	51.0%
Male	49.2%	50.0%	48.2%	49.9%	49.1%	49.0%	49.4%	48.9%	48.9%
Age groups, in % of respondents									
18-34	24.4%	25.0%	24.9%	23.1%	20.8%	22.7%	22.7%	27.4%	29.1%
35-49	25.0%	25.0%	23.9%	23.9%	21.3%	29.8%	24.8%	25.9%	25.1%
50-64	31.0%	36.7%	30.3%	26.0%	40.1%	30.8%	40.8%	21.8%	21.7%
65-79	18.4%	12.4%	20.4%	25.7%	16.8%	16.3%	11.5%	23.0%	21.1%
Education, in % of respondents									
Primary (less than high school)	4.8%	2.2%	3.7%	2.6%	14.1%	4.0%	2.8%	2.0%	7.2%
Secondary (high school)	52.5%	75.2%	47.9%	69.2%	52.5%	55.2%	36.0%	43.6%	40.1%
Tertiary (university or comparable)	42.7%	22.6%	48.4%	28.2%	33.4%	40.9%	61.2%	54.4%	52.8%

Source: Allianz Research

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The semiconductor premium: EM equity and the concentration risk within

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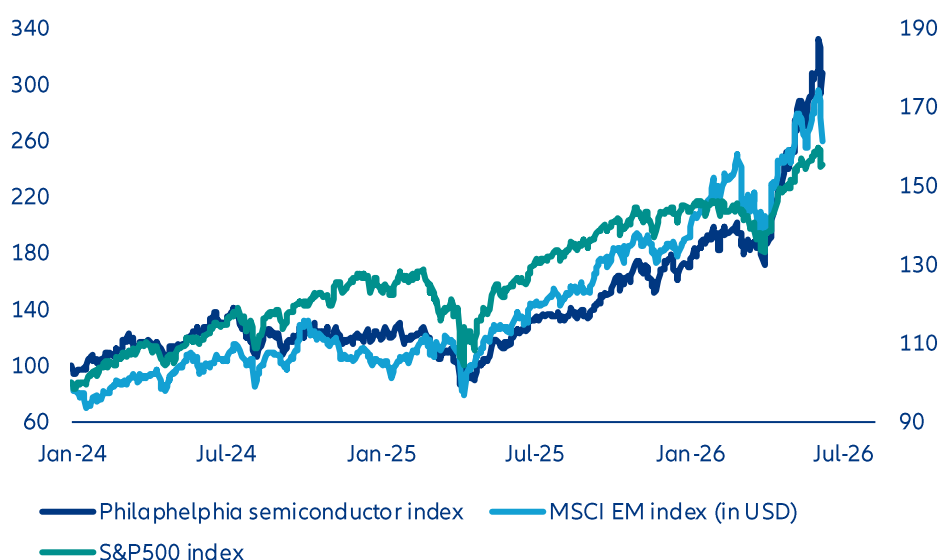
In Summary

- **Semiconductors are the backbone of the 2026 AI rally.** Capital investment is overwhelmingly directed at AI infrastructure, and the data centers doing the heavy lifting run entirely on chips. The numbers speak for themselves: the Philadelphia semiconductor index is up +90% YTD – twelve times the S&P 500's +7.5% - despite a couple of recent sell-off episodes in June. In emerging markets, which concentrate most of the supply chain, markets like South Korea and Taiwan are up +110% and +60% this year.
- **Concentration is the flip side of the boom but fundamentals are strong.** Concentration is most acute in EMs, where dependence on a single industry is now extreme. If the chips sector disappoints, the re-rating could be brutal. But the earnings are real and lopsidedly so: chips deliver 13% of S&P 500 profits on just 5% of sales, half of South Korea's profits on a fifth of its sales and over 70% of Taiwan's earnings and revenue alike. Ferocious demand colliding with supply bottlenecks from Middle East tensions are keeping prices sky-high.
- **The bull case still stands on five pillars.** First, hyperscalers show no sign of easing off. Upward capex guidance in Q1 points to a still-aggressive stance that should lift Asian chipmakers' profits by a remarkable +70% CAGR through 2026-2029. Second, the demand pipeline looks structurally solid: Broader AI diffusion, the corporate shift to agentic platforms and the memory-hungry new chip generation all reinforce one another. Third, the moat is real: capital and expertise are needed to build a modern foundry, keep competition at bay and pricing power intact. Fourth, valuations are compelling as based on solid earnings trajectory, and recent sell-offs offer new entry point on top picks. And fifth, in the over-concentrated EM space, semiconductors ensure geography and supply-chain position diversification, with an attractive trade-off compared to developed markets.
- **But strong convictions do not mean complacency: Four risks could unsettle the rally over the next 12-24 months.** First, a Middle East re-escalation could disrupt energy routed to Asia, forcing production halts – temporary or not. Second, bottlenecks in the Asian supply chain for inputs critical to data-center fleets could lift costs and lengthen lead times, throttling AI capex momentum. Third, a timing mismatch looms: New supply could come online just as capital allocation decelerates, should a fresh rate cycle take hold in the US or Europe. Fourth, competition is sharpening fast – Beijing is piling on funds and pressure to drive chip self-sufficiency (70% target in AI chips in 2026) and a bigger global footprint for its champions. Contrary to what market volatility suggests, we do not see lower inference cost to be a negative driver for memory chip business, but rather a guarantee of a sustainable book order if it comes with stronger AI diffusion and usage.

Semiconductors are driving the 2026 AI equity rally – and concentrating its risk

Equity markets are roaring back in 2026, with semiconductors as the primary engine. A brutal tech sell-off erupted last autumn, fueled by a rotation out of hyperscalers as the "AI bubble" narrative gained traction. The pain extended well into this year as frontier AI models – armed with native Customer Relationship Management (CRM), cybersecurity and data-processing capabilities – triggered a sharp de-rating of traditional software names suddenly fighting for relevance. Yet through the storm, one segment barely flinched: semiconductors. Quarter after quarter, chip makers delivered earnings that kept beating expectations, rewarding investors who held their nerve with relentless outperformance. The Philadelphia Semiconductor Index tells the story plainly, up 90% ytd, dwarfing the S&P 500's modest +7.5% gain and cementing semis as the single biggest beneficiary of the AI equity rally. But the most eye-catching trade of this cycle is unfolding not in the US but across emerging markets: The MSCI EM index has surged +23.2% YTD, and the engine behind that re-rating is, again, semiconductors as Asian chip and advanced packaging names ride the same AI capex supercycle. The investment wave is global and equity markets are pricing it that way.

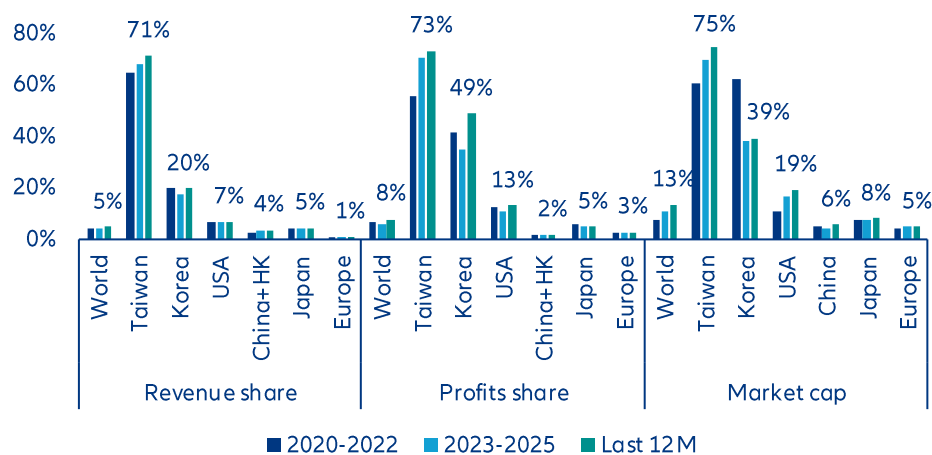
Figure 1: Equity index performance, rebased to 100 on January 2024



Sources: LSEG Datastream, Allianz Research

Semiconductors have quietly become the backbone of global equity markets – and a structural vulnerability at their core. Taiwan and South Korea stand in a league of their own. Between them, they concentrate the world's most critical foundry capacity and their stock markets reflect it starkly – Taiwan's semiconductor sector commands a 75% market cap share of its equity universe, South Korea's a still-commanding 39%, dwarfing every other major index. Yet the paradox lies in the revenue line: Semiconductor revenue shares across most exchanges remain surprisingly moderate, exposing an industry structured around a handful of dominant Asian foundries towering over a fragmented ecosystem of smaller players. Europe and the US have built their digitalization ambitions – and increasingly their AI infrastructure – on an Asian supply chain they do not control, a dependency that geopolitical stress-testing has already proven fragile.

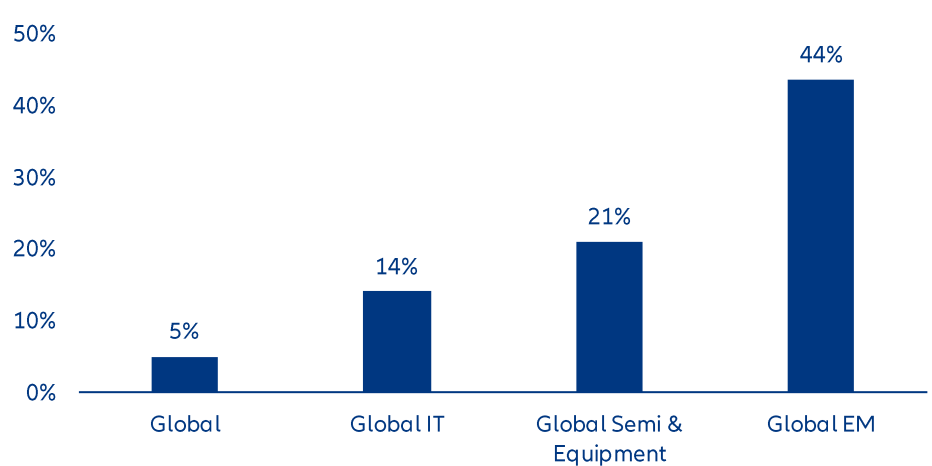
Figure 2: Semiconductor industry performance and footprint in the EM equity universe



Sources: LSEG Datastream, Allianz Research

Concentration is extreme – and the risk is double-layered. Taiwan's semiconductor profit share within its equity universe has vaulted from 55% in 2020-2022 to nearly 75% over the last 12 months. South Korea has mirrored that trajectory, climbing from 40% to close to 50% over the same period. This is not a gradual drift; it is a structural repricing of an entire industry's earnings power, driven by AI-fueled demand and ferocious pricing discipline from a handful of dominant foundries. But that concentration cuts both ways, and for EM investors, the vulnerability is not single-sided. MSCI EM already carries a 37% sector weight in IT and a 45% combined geographic exposure to Taiwan and South Korea alone, meaning the index is structurally pre-loaded with semiconductor risk before a single stock is picked. Stack on top of that the surging profit dominance of semis within those very markets and MSCI EM has effectively become a leveraged proxy on the AI capex cycle – a winning position today but a precarious one if sentiment toward the AI theme softens, earnings surprise cycles turn or geopolitical disruption bites into Asian supply chains. Any de-rating of the semiconductor complex would hit EMs through two channels simultaneously – sectoral and geographic – amplifying drawdown risk well beyond what headline index diversification would suggest. The US tells a more nuanced story: Semiconductor profit share is also at a cycle peak – approaching 20% – yet its broader and deeper equity market provides a natural buffer, diluting the concentration risk that EM investors simply do not have access to. The same industry driving the most exciting equity returns of this cycle is simultaneously the single biggest source of unpriced fragility within it.

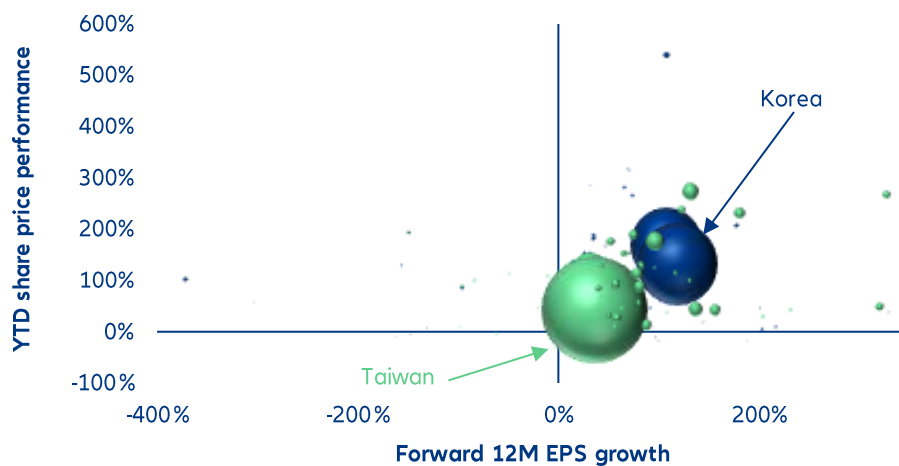
Figure 3: Aggregated market share of South Korea and Taiwan in MSCI index (as of April 2026)



Sources: MSCI, Allianz Research

Hyperconcentration of profits does not mean an absence of opportunity. South Korea dominates High Bandwidth Memory (HBM), the memory architecture at the heart of every AI accelerator, stacking dies vertically to deliver the massive low-latency bandwidth that large language models cannot function without. Taiwan holds what is effectively an unchallengeable near-term monopoly on advanced logic at 2-3nm node, where compute performance, miniaturization and energy efficiency converge – a process gap that Intel and SMIC are years from closing. The positive externalities from these twin centers of excellence ripple across equipment makers, packaging specialists and testing firms, generating an investment universe that extends well beyond the obvious names. That breadth is visible in the numbers: Median forward earnings growth recorded mid-May stood at 52% in South Korea and 42% in Taiwan, up sharply from 35% and 26% at the start of the year, a broad-based acceleration confirming that AI demand is diffusing through the entire supply chain, not just concentrating at its peak.

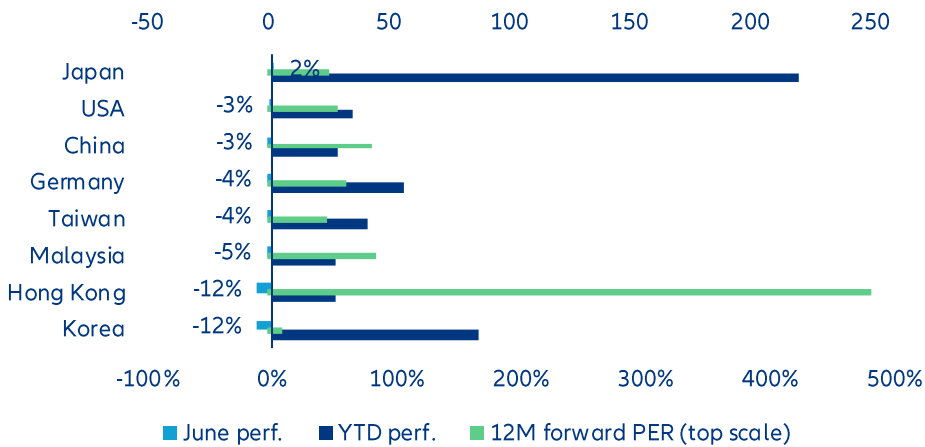
Figure 4: Forward earnings growth vs. 2026 share performance from South Korean and Taiwanese semiconductor firms



Bubble size refers to market capitalization weight within national semiconductor universe.
Sources: LSEG Datastream, Allianz Research

Volatility increases when you are under spotlights: go beyond day-to-day analysis. The -12% correction in June in global semiconductor equities was a positioning shock, not a fundamental one. It was caused by a sequence of negative signals that offer a reason for investors to cut their positions and take back their profits after an outstanding run (+82% YTD even after the sell-off). Among main downside catalysts, the disappointment caused by the non-revision of earnings trajectory from a top US chipmakers coupled with rising odds of a rate hike in USA and new concerns toward a weaker demand for HBM chips caused by the release of new Anthropic's Claude 4.8 opus that displays a more efficient inference cost model participate to pinch investor's sentiment toward semiconductor stocks. In reality, the damage was narrow rather than broad-based, concentrated in the two highest-beta markets — Korea, where the KOSPI fell 8.3% and tripped a circuit breaker, and Taiwan, where the largest foundry worldwide posted its largest-ever intraday drop. Korea bore the brunt for structural reasons: index hyper concentration in Samsung and SK hynix, heavy leverage, foreign outflows and a won past 1,560/USD, a mix that turns global tech wobbles into forced-deleveraging cascades. Yet the rally's anchors were not impaired — supply-chain demand kept accelerating and the megacaps rebounded fast — and valuations stay undemanding, with Korea near 10–11x forward earnings, the cheapest of the major Asian markets and North Asia broadly at a discount to the US.

Figure 5: Performance of semiconductor market and forward valuation per region breakdown

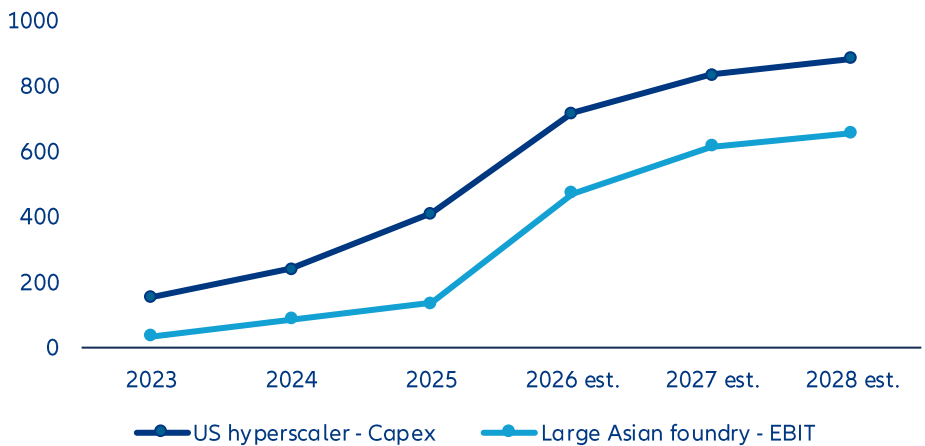


Sources: LSEG Datastream, Allianz Research

Semiconductors remain the purest expression of the AI capex cycle

The current phase of AI development is overwhelmingly infrastructural. Nations, hyperscalers and corporates alike are locked in a race to accumulate the largest possible compute capacity, with digital sovereignty increasingly measured by the volume of cloud space owned and the raw processing power available to run frontier models. Every data center announcement is, in essence, a chip order. US hyperscaler capex has surged from USD150bn in 2023 to over USD400bn in 2025, with guidance pointing to nearly USD900bn by 2028 – a near-sixfold expansion in five years. Large Asian foundry EBIT has tracked that trajectory with striking fidelity, albeit with a one-to-two-year lag that reflects the natural conversion time from capex commitment to wafer demand to earnings delivery: From a negligible USD35bn in 2023, foundry profits are projected to reach USD650bn by 2028. The lag is a feature, not a friction: It means the earnings wave is still building. With no credible sign of hyperscaler capex fatigue, the foundry profit outlook remains extraordinary. The AI-powered world being built is, above all, a chip-powered world.

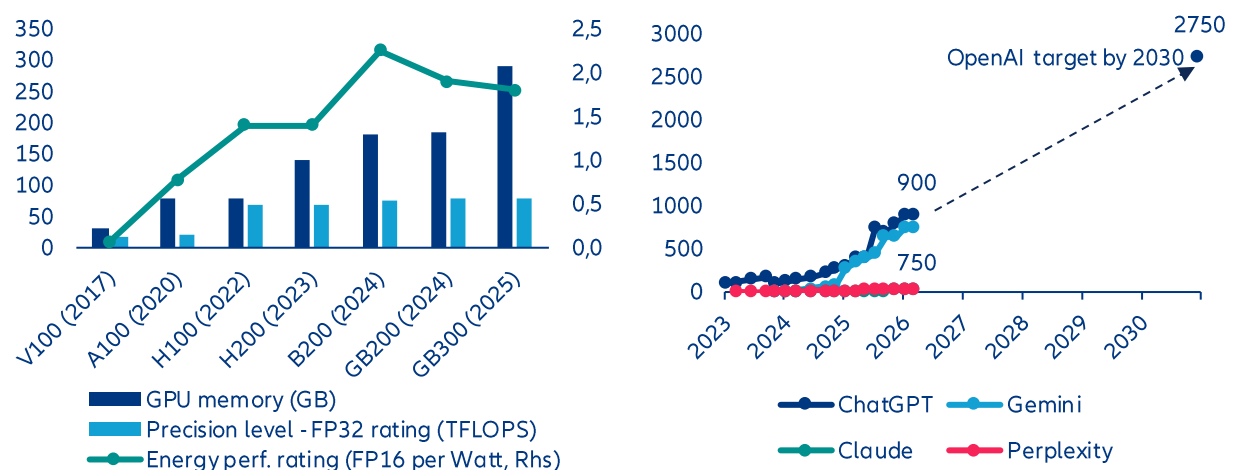
Figure 6: Guidance of US hyperscaler capital expenditure and large Asian foundry EBIT



Sources: LSEG Datastream, Allianz Research

AI diffusion is deepening, not plateauing – and memory demand follows. Each successive Nvidia GPU generation has delivered a step-change in memory capacity: from 32GB on the V100 in 2017 to 280GB on the GB300 in 2025, while energy performance per watt has expanded nearly twentyfold over the same period – a relentless hardware escalation that has made every frontier model generation structurally more chip-hungry than the last. The shift in AI workloads from training toward inference – often cited as a potential headwind for memory demand – is a false concern at the system level: what inference loses in per-query intensity, it more than recovers through sheer volume explosion. OpenAI alone is targeting 2.75bn users by 2030, implying that nearly one in four people on the planet will be regularly querying a frontier model – a demand profile that makes aggregate memory chips Dynamic Random Access Memory (DRAM) and Not-And (NAND) consumption essentially non-discretionary infrastructure spending. Unlike software incumbents facing direct substitution from AI-native capabilities, semiconductor and memory suppliers sit upstream of every model, every query and every data-center expansion – insulated from the application-layer disruption playing out below them. The democratization of AI access is not a risk to the chip cycle; it is its most powerful demand multiplier. The efficiency-bear case that used to come back at our ears at each release of new frontier AI model assumes demand is fixed, but it rarely is: cheaper, leaner inference is precisely what pulls AI into more users, more always-on agentic and multimodal workloads, and more applications that were previously uneconomic — so falling cost per query tends to expand aggregate query volume faster than it compresses the memory intensity of each one. Crucially, the memory call is driven by the scale of deployed infrastructure rather than the elegance of any single model: every incremental accelerator shipped is loaded with HBM and DRAM, and the shift toward inference and long-context, agentic workloads actually raises KV-cache memory requirements, meaning a broader, more efficient installed base de facto consumes more chips in aggregate, not fewer.

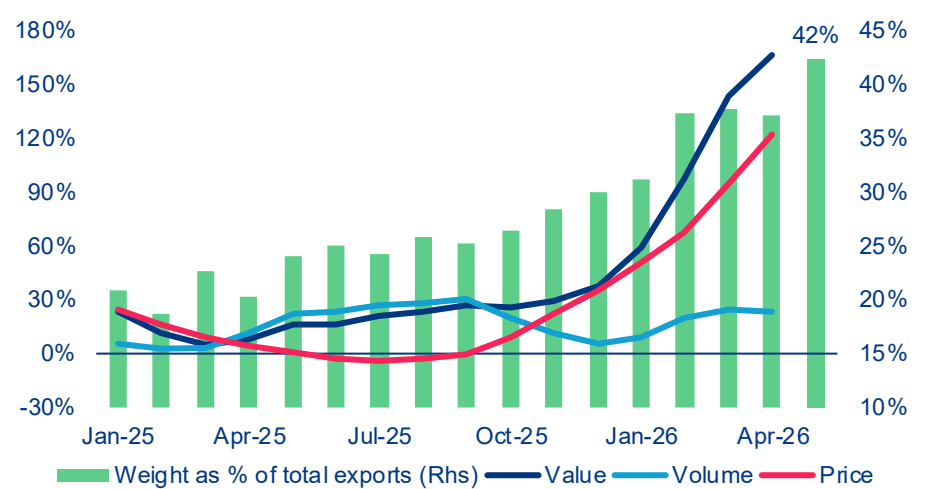
Figure 7 : Memory capacity, performance and power efficiency of GPU (based on Nvidia models) / Number of users of main LLM models, current and estimated (in mn of users)



Sources: Nvidia corporate brochure, Allianz Research

Supply tightness is extending pricing power – and acting as a macro hedge. Supply was already structurally tight before geopolitical stress entered the equation. The Middle East conflict has added an acute dimension: disruption to helium supply chains – a critical purification gas sourced partly from Qatar and indispensable to wafer fabrication – triggered precautionary front-loading across South Korean and Taiwanese buyers, compressing available inventory precisely when AI-driven demand is at its most insatiable. The result is visible in spot prices with brutal clarity: Flash NAND 64GB spot prices have nearly tripled in 2026 alone, surging from below USD5 to nearly USD20 per unit – a pass-through dynamic that semiconductor firms have executed without friction, reflecting a pricing power that few industries can match. When the end-demand justification is AI productivity gains worth trillions in enterprise value, customers absorb chip price inflation with remarkably little resistance. South Korea's semiconductor export value growth has accordingly exploded to +150% year-on-year – a magnitude that dwarfs volume growth and confirms that this is a margin and pricing story as much as a volume one, with the spread between value and volume growth at its widest on record. The slow progress in new capacity build-up postpone in time breakeven point between supply and demand, leading to a likely extension of memory shortage until 2027. As a result, the mix of supply imbalances coupled with extremely-high concentration of the segment ensure a strong pricing power Asian chipmakers that should help them keeping a high margin profile for long.

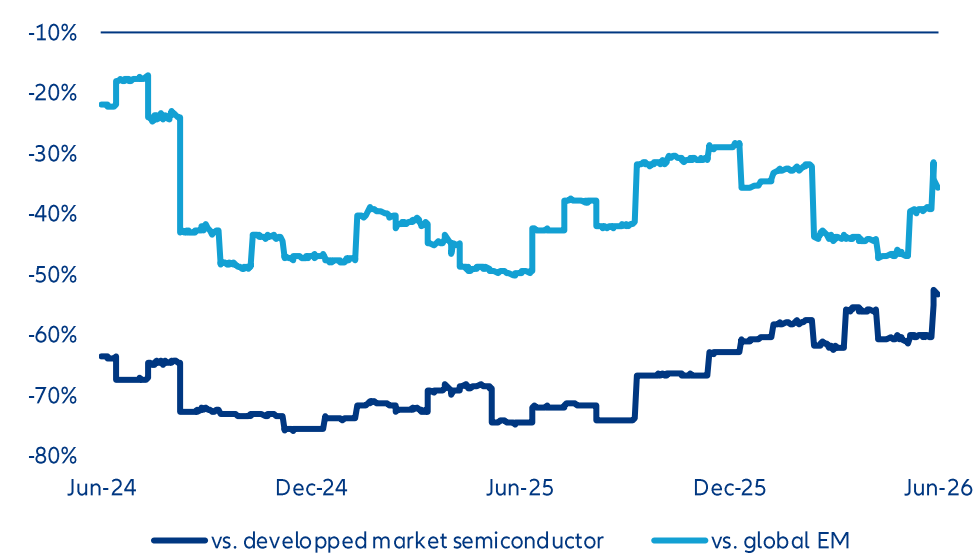
Figure 8: South Korea semiconductor exports trend (price, value and volume) and weight



Sources: Bank of Korea, LSEG Datastream, Allianz Research

Valuations remain compelling despite rally. Even after the year-to-date rally, EM semiconductors remain conspicuously cheap on a market-weighted basis: as of June-26 the industry still trades at roughly a 55% forward-P/E discount to developed market (DM) semiconductor peers and around a third below broad global emerging market benchmark (EM). The rally has admittedly started to dilute that gap — both discounts have compressed off their late-2024 troughs (circa -75% versus global semis, -50% versus global EM), with the narrowing accelerating sharply in the most recent leg, particularly against global EM. Yet the residual discount to DM names sits oddly with the reality that EM players now occupy the most critical, highest-value nodes of the chain — leading-edge foundry and HBM — a positioning that arguably warrants convergence rather than a persistent markdown. Part of the optical cheapness is also a concentration artefact: the aggregate multiple is dragged down by a handful of mega-caps whose steep forward-earnings trajectories compress the denominator, so the headline discount reflects those few names rather than uniform value across the sector. Beyond the pure valuation call, that same skew opens an arbitrage-and-diversification angle — the better risk-reward may sit further along the supply chain in equipment, materials, advanced packaging and testing, rather than in the crowded foundry and pure-memory leaders.

Figure 9: Valuation premium/discount of EM semiconductor industry vs. global EM and semiconductor industry (based on forward 12M price to earnings ratio)



Sources: LSEG Datastream, Allianz Research

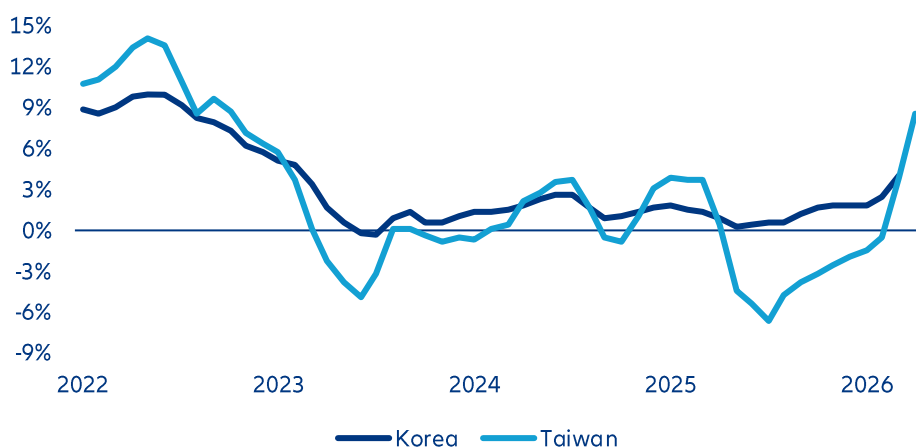
Four risks that warrant active monitoring

Despite the compelling semiconductor narrative, complacency would be a mistake. A clear-eyed view of the next 12-18 months reveals four factors that deserve active monitoring before the next leg of the rally is taken for granted.

Risk 1 - Middle East disruptions: when scarcity becomes a double-edged sword

The same supply tightness that has gifted semiconductor firms extraordinary pricing power now carries a sting in the tail. Helium storage levels have dropped to uncomfortable lows as tensions persist. Compounding the picture, wafer component costs and freight rates have moved meaningfully higher, inflating production costs at a moment when delivery pipelines are already stretched. The structural constraint is unforgiving: New fabrication lines outside Asia take years to reach operational relevance, leaving current capacity with little slack to absorb sustained disruption. Q1 earnings provided some relief – margin guidance pointed only to modest downward adjustments rather than any structural deterioration – and the industry is not defenseless: three to six months of secured helium inventory, combined with recycling protocols put in place as far back as 2022, provide a meaningful buffer against near-term shocks. But that buffer has a clock on it. Should Middle East tensions persist and logistical disruption deepen, the scarcity narrative that has been a profit tailwind could rapidly invert into a production bottleneck – with spot price spikes masking volume shortfalls that ultimately disappoint revenue expectations. The PPI chart captures the inflection already: Taiwan's production cost index has snapped from -7% year-on-year in late 2025 to nearly +9% by mid-2026 – the sharpest reversal in the dataset – while South Korea's is also accelerating, signaling that cost pressures are landing on balance sheets now, not as a future risk. For an industry priced for perfection, that reversal is worth watching with precision.

Figure 10: Production cost index in South Korea and Taiwan (PPI, annual change in %)



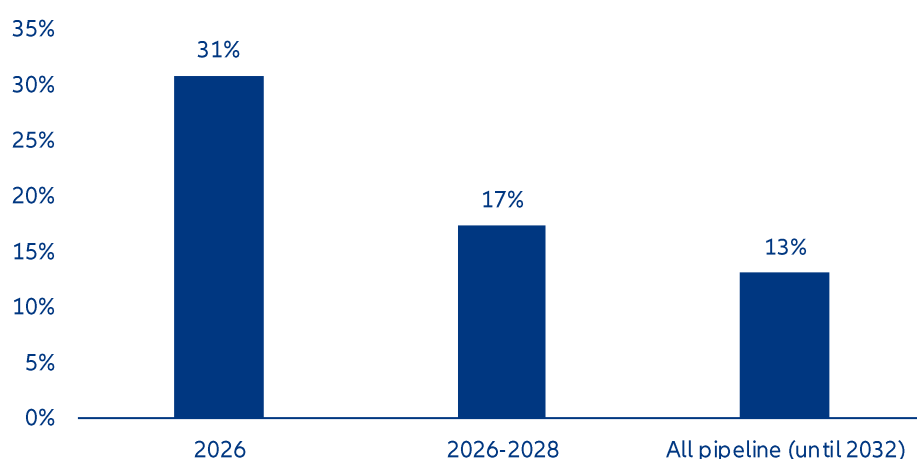
Sources: LSEG Datastream, Allianz Research

Risk 2 - AI capex execution gap: when the infrastructure pipeline meets real-world friction

The AI capex supercycle is the single most powerful demand driver underpinning the semiconductor bull case – which makes any credible threat to its execution timeline a risk that cannot be dismissed. As of end-January 2026, only 31% of US data-center capacity committed for delivery in 2026 was officially under construction – a figure that drops to 17% for the 2026-2028 window and a sobering 13% across the full pipeline to 2032. Behind the relentless drumbeat of hyperscaler capex announcements lies an execution gap that is widening, not closing, driven by planning bottlenecks, grid connection delays, construction labor shortages and permitting friction that no balance sheet commitment can shortcut. A data center that is announced but not built does not pull forward a chip order.

The risk compounds if the macroeconomic backdrop shifts. A renewed inflationary impulse – entirely plausible given rising freight costs, energy price pressure from Middle East tensions and sticky services inflation – could force another monetary tightening cycle, pushing long rates higher at precisely the moment when capital-intensive infrastructure projects are most rate-sensitive. Higher rates do not just raise discount rates on future cash flows, they make banks materially more restrictive in extending project finance to private data-center developers, the fastest-growing segment of the pipeline. Given the near-perfect correlation between hyperscaler capex trajectories and Asian foundry profit guidance, any sustained flattening of investment commitments would transmit rapidly into semiconductor earnings downgrades, eroding pricing power simultaneously as the demand urgency that has allowed cost pass-through begins to dissipate. For a sector priced at a meaningful premium, the margin for disappointment is narrow.

Figure 11: Percentage of the US data-center pipeline officially under construction



Sources: Sightline Climate, Allianz Research

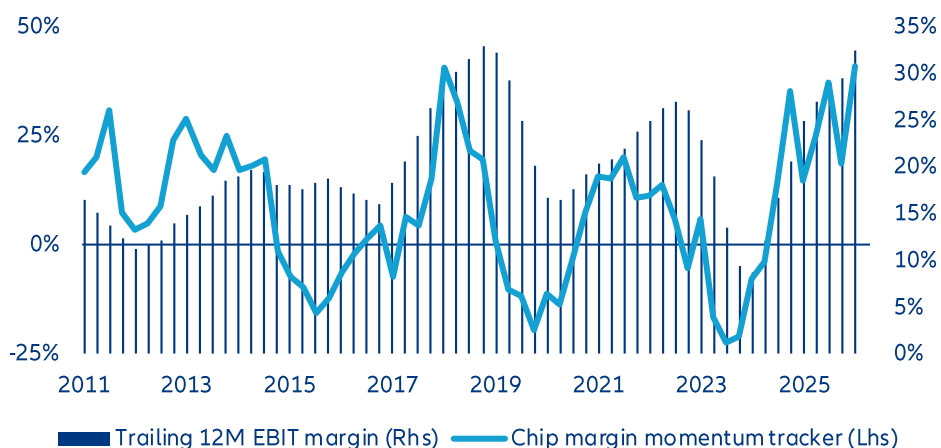
Risk 3 - The capex timing trap: will history rhyme?

The semiconductor industry has a well-documented and costly habit: it invests in new capacity at the top of the cycle, and the production lines come online just as demand rolls over. The pattern is visible with painful regularity in Figure 11: EBIT margins and chip margin momentum have oscillated in violent cycles, with the most brutal compression episodes – 2015, 2019, 2023 – coinciding precisely with waves of new supply hitting markets into weakening end-demand. The post-2022 episode is the most instructive: A surge of capacity investment triggered by pandemic-era shortage hysteria delivered production lines into a demand vacuum, crushing margins to near zero in 2023 and generating inventory write-downs across the industry. The question is whether 2027-2028 will rhyme. The timing risk is concrete and quantifiable. TSMC's Arizona Fab 21 Phase 2 – targeting 3nm and 2nm production – is scheduled for completion by 2027-2028, alongside its ongoing Japan facility expansion in Kumamoto targeting mature nodes. Samsung's P3 and P4 lines in Pyeongtaek – among the world's largest semiconductor complexes – are expected to reach full operational capacity over the same window, adding substantial HBM and advanced logic output to global supply. SK Hynix's M15X facility in Cheongju, purpose-built for next-generation HBM production, is similarly slated for ramp-up through 2027. Collectively, these projects represent the largest synchronized capacity expansion in the industry's history, hitting markets simultaneously, with limited ability to delay or redirect output once fab construction reaches completion.

Trailing 12-month EBIT margins are approaching 30-32%, levels last seen fleetingly around 2018 and never sustained beyond a few quarters. Chip margin momentum, the leading indicator, is already showing the first signs of oscillation at elevated levels, a pattern that historically has preceded compression. The bull case argument – that unprecedented data-center pipeline demand will absorb the incoming supply wave – is plausible but carries the execution uncertainty documented in Risk #2: only 13% of the US data-center pipeline to 2032 is currently under construction. If the demand realization timeline slips by even 12-18 months relative to supply delivery, the industry faces the same inventory build and margin compression dynamic it has experienced in every prior cycle.

Semiconductor profits have been the fuel of the EM equity rally; a cyclical mean-reversion in foundry margins would not just slow the trade, it would reverse it.

Figure 12: EBIT margin and margin momentum index of large Asian foundries



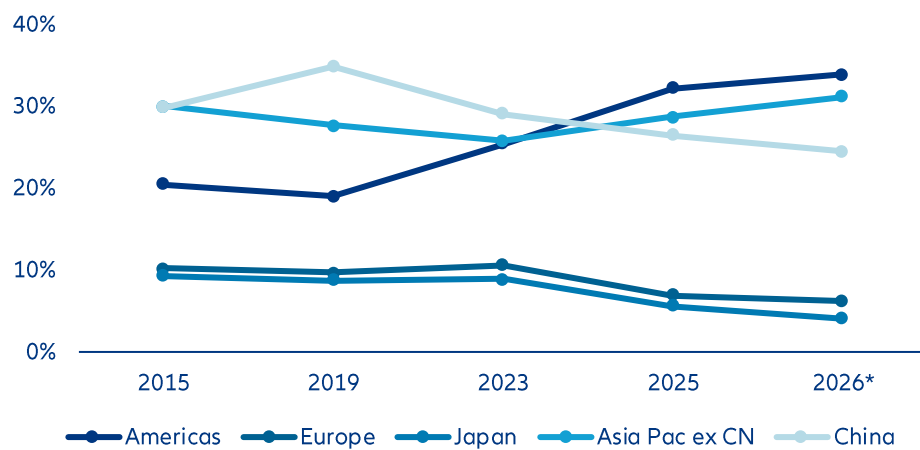
Sources: LSEG Datastream, Allianz Research

Risk 4 - The China wildcard: a competitor in construction, not yet in contention

Figure 13 below frames the competitive landscape with striking clarity: Americas' share of global semiconductor shipments by value has surged from 20% in 2015 to 34% in 2026, while Asia Pacific ex-China has climbed to 31% – the two dominant and widening poles of the industry. China, meanwhile, has seen its share compress from 30% to 24% over the same period, a trajectory that reflects both export controls tightening its access to advanced process technology and a domestic industry still years from self-sufficiency. Europe and Japan have faded to near-irrelevance at 4-6%, a structural marginalization that makes their dependence on Asian supply chains all the more acute. But the direction of Chinese policy ambition is unambiguous and dismissing it entirely would be a mistake. Beijing has set an explicit target of covering 70% of domestic semiconductor needs through indigenous production – a goal backed by the most aggressive state financing program in the industry's history. The Big Fund Phase III alone mobilizes an estimated USD30-50bn, part of a total USD70bn incentive envelope that dwarfs both the US CHIPS Act (USD53bn) and the EU Chips Act (EUR43bn) in raw commitment size. The political signal is equally unambiguous: During President Donald Trump's recent visit, Chinese officials made clear they had no interest in dismantling semiconductor trade barriers. The strategic priority is building domestic capacity, not negotiating market access. Semiconductors have become a theatre of technological sovereignty, not commercial competition.

Yet the gap between ambition and execution remains vast across the three most critical chokepoints. On EUV lithography – the indispensable tool for sub-7nm node production – China has made headline progress, with reports of a prototype developed under a project drawing on expertise from former ASML personnel. But scaling from prototype to mass production is an entirely different challenge: The official target of EUV mass production by 2028 is viewed skeptically by industry experts, with more credible timelines pointing to post-2030 at the earliest for volume output. On High Bandwidth Memory, the process know-how required to compete with SK Hynix and Samsung at HBM3/HBM4 levels demands both multi-billion dollar capex and years of yield optimization that state funding alone cannot shortcut. And on EDA software – the design automation tools that underpin every advanced chip architecture – building competitive capability requires deep, accumulated human expertise that neither subsidies nor industrial policy can manufacture quickly. China is a competitor in construction, not yet in contention. But the 2028-2030 window, when domestic capacity begins reaching meaningful scale, is precisely when Asian foundry supply is also expected to surge. The collision of those two dynamics is a risk the market has not yet begun to price.

Figure 13: Global semiconductor shipments by value - geographic market share



Sources: SEMI, Allianz Research

These assessments are, as always, subject to the disclaimer provided below.

FORWARD-LOOKING STATEMENTS

The statements contained herein may include prospects, statements of future expectations and other forward-looking statements that are based on management's current views and assumptions and involve known and unknown risks and uncertainties. Actual results, performance or events may differ materially from those expressed

or implied in such forward-looking statements.

Such deviations may arise due to, without limitation, (i) changes of the general economic conditions and competitive situation, particularly in the Allianz Group's core business and core markets, (ii) performance of financial markets (particularly market volatility, liquidity and credit events), (iii) frequency and severity of insured loss events, including from natural catastrophes, and the development of loss expenses, (iv) mortality and morbidity levels and trends,

(v) persistency levels, (vi) particularly in the banking business, the extent of credit defaults, (vii) interest rate levels, (viii) currency exchange rates including the EUR/USD exchange rate, (ix) changes in laws and regulations, including tax regulations, (x) the impact of acquisitions, including related integration issues, and reorganization measures,

and (xi) general competitive factors, in each case on a local, regional, national and/or global basis. Many of these factors may be more likely to occur, or more pronounced, as a result of terrorist activities and their consequences.

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Ten Years After Brexit: Resilience Without Revival

22 June 2026
Allianz Research



Content

Page 3-5

Executive Summary

Page 5-12

10 years since the Brexit vote: an evolving economic and energy landscape

Page 13-17

UK economic and energy policies in need of a revamp

Page 18-25

The UK's trade performance: goods imports substitution, services trade powering ahead

Page 26-29

Financial markets: unofficial dollarization and eroding confidence but the City of London keeps (most of) its shine

Page 30-17

UK private equity market has lost its magic, low tech exposure prevails

Executive Summary



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- **Brexit 10 years on: Neither collapse, nor renaissance.** As the UK looks for its 7th Prime Minister in 10 years, this week also marks a decade since Brexit roiled markets and divided forecasters. Predictions ranged from economic collapse to a renaissance. A decade later, we take a hard look at which projections came through and which failed to pass. The outcome has been more nuanced than many expected, with indicators pointing to both economic strengths and weaknesses — not all of them Brexit related.
- **Britain's Economic Backbone: The knowledge economy, tech and clean energy.** UK ICT exports to the EU have almost doubled since Brexit, demonstrating the continued competitiveness of Britain's knowledge economy. The UK remains the world's second-largest exporter of financial services, accounting for 21% of global exports, while deepening its provision of financial services to EU markets. In private markets, the UK continues to attract more venture-capital funding than any European competitor. Between 2020 and 2026, it raised roughly USD164bn. Since then, London has retained much of its global financial importance. The UK still accounts for nearly 50% of global OTC interest-rate derivatives trading and almost 38% of global foreign-exchange turnover. In clean energy, the UK has emerged a leader within Europe. Wind generation has increased by 130% since 2016, reducing the UK's fossil dependence while enabling a successfully phased out coal in 2024.
- **Burned by Brexit: Dragging growth, trade frictions and lagging Leavers.** Independent studies estimate that GDP would have been 2 to 4% larger without the political instability and trade friction triggered by Brexit. Since 2016, "Leave" regions have generally underperformed compared to the UK as a whole: 59% of the population of the "Leave" areas have seen their regions fall further behind national income per capita average. Since the vote and the formal exit from the bloc's economic structures on 1 January 2021, growth has relied increasingly on foreign-born workers, which has fueled more than half of GDP expansion. Brexit has increased frictions and reduced trade flows. While the EU remains the UK's largest trading partner, structural estimates suggest UK-EU trade is around 21% lower for goods than it would have been without Brexit. New trade agreements and diversified supply chains with the US, China and Commonwealth countries have failed to match the scale of the economic ties previously enjoyed within the EU. UK assets continue to trade at a discount relative to international peers.

- A lasting lesson of the 2022 mini-budget crisis is that fiscal credibility matters.** Investors now demand a structurally higher risk premium on UK assets against a backdrop of rising fiscal imbalances. Equity markets, where UK stocks have underperformed both US and European peers in the past decade, reflect this premium. In private markets, the magic has faded into a priced-in discount. A decade on, Brexit's imprint reads through the UK's two largest private-capital engines, private equity and venture. UK investments were 12–18% below a no-Brexit path as of 2025, a one-off level shift now cleared into a structurally higher cost of capital. Private equity has proved resilient, but increasingly as a US funded value trade rather than home grown momentum with compressed valuations turning UK companies into take-private targets for US capital.
- Beyond Brexit: Identifying and implementing solutions for the future.** The UK scores high on business creation, labor-market flexibility, higher education and research, compared with many other advanced economies. This indicates that domestic bottlenecks — exposed by Brexit but not caused by it — are the root of the country's disappointing growth performance. The government has identified many of these issues, but needs to push further. Priority actions should include:
 - Accelerating planning reform,
 - Increasing investment in housing and energy infrastructure,
 - Strengthening support for innovation and strategic industries,
 - Treating NHS reform as an economic priority, and
 - Addressing the slow diffusion of new technologies from frontier firms to the broader economy.
- A revamp of fiscal rules, combined with reforms to pensions spending and property taxation as well as broadening the VAT base, would help redirect scarce public resources toward deficit reduction and investment, with an eye to defense.** Unlike the EU-27, Britain has staged no distinct post-2022 rearmament inflection. The UK government has committed to increase defense spending to 2.5% of GDP by April 2027, and 3.0% target by 2030, but they are not set in stone. The cooperation map looks busy — AUKUS with the US and Australia, GCAP with Japan and Italy, a privileged-partner track toward the EU's SAFE instrument — yet it reads as a catalog of commitments rather than tangible delivery. Government investment would help attract investors and domestic defense companies to commit capital to UK capacity. Reintegrating more closely with European energy markets would generate meaningful economic benefits. High energy prices caused by the UK leaving the EU Internal Electricity Market, successive energy supply crises and grid bottlenecks linked to the rapid expansion of renewable energy have continued to undermine industrial competitiveness. Divergence between UK and EU carbon markets also risks creating new trade barriers for energy-intensive industries. Reintegrating energy and carbon markets could help reduce friction costs, stabilize power prices and improve supply security.

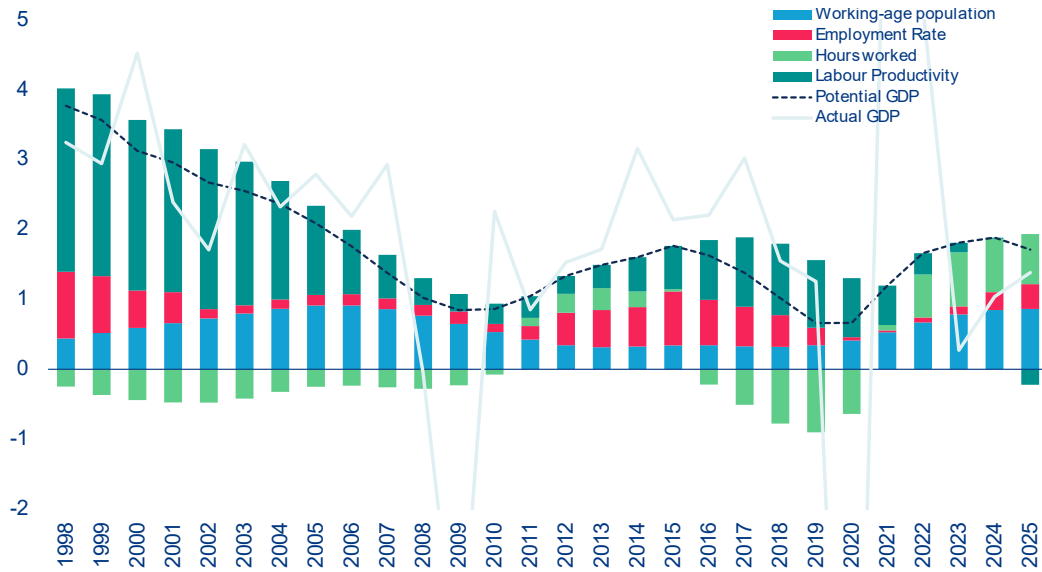


10 years since the Brexit vote: an evolving economic and energy landscape

This month marks 10 years since the Brexit vote and the promises of greater democratic accountability, reduced immigration and independent trade policy. The Brexit referendum took place on 23 June 2016, gathering a high turnout of 72.2%, with 51.9% (17.4mn) votes for leaving the EU against 48.1% (16.1mn) for remaining. The Leave campaign was built around several core themes and promises. First, sovereignty, or "Take Back Control": restoring the authority of the UK Parliament by removing the influence of EU institutions and courts. Second, immigration control: ending the free movement of people from the EU, and allowing the UK to set its own immigration policy. For most Brexiters, this meant reducing the flows of immigration drastically, including from non-EU countries. Third, trade and regulation: gaining the freedom to negotiate independent trade agreements and reducing or removing EU-derived regulations. Fourth, budget contributions: stopping contributions to the EU budget and redirecting some of those resources to domestic priorities (notably providing additional funding to the NHS). Finally, democratic accountability: decisions affecting the UK would be made by elected British politicians rather than EU bodies. In all, Leave supporters generally argued that Brexit would increase political autonomy and economic flexibility, while opponents warned about trade barriers, lower investment and weaker economic growth. The UK effectively left the EU on 31 December 2020.

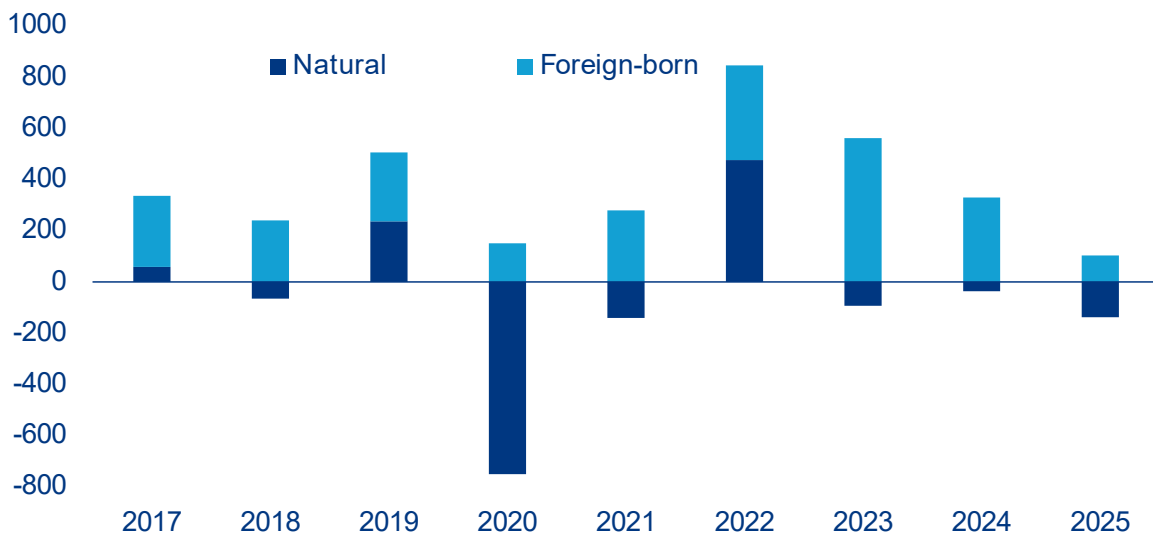
UK politics has been shaken since the 2016 vote. Debates about closer ties with the EU, or even potential re-entry into the EU, have gained momentum recently amid a Labour party leadership contest. The post-Brexit period has been one of unusually high political turnover in the UK. Since the referendum, the UK has had six Prime Ministers (PM), compared with only four PMs during the entire 1997-2016 period. Much of this high turnover was directly linked to Brexit negotiations, disagreements over the UK's future relationship with the EU and subsequent political pressures. Poor fiscal decisions against a backdrop of rising budget imbalances added to the political instability, with the infamous "mini Budget" debacle of unfunded tax cuts by PM Liz Truss and Chancellor Kwarteng in September 2022 spooking financial markets. The recent challenge of PM Starmer's leadership with the ruling Labour Party has sparked a renewed debate over the opportunity to deepen trade and economic ties with the EU. The current government's stance is to rule out rejoining the customs union or the single market or bringing back freedom of movement.

Figure 1: Potential GDP growth decomposition (% y/y)



Sources: LSEG Workspace, Allianz Research

Figure 2: Employment growth decomposition (000s)



Sources: LSEG Workspace, OECD, Allianz Research

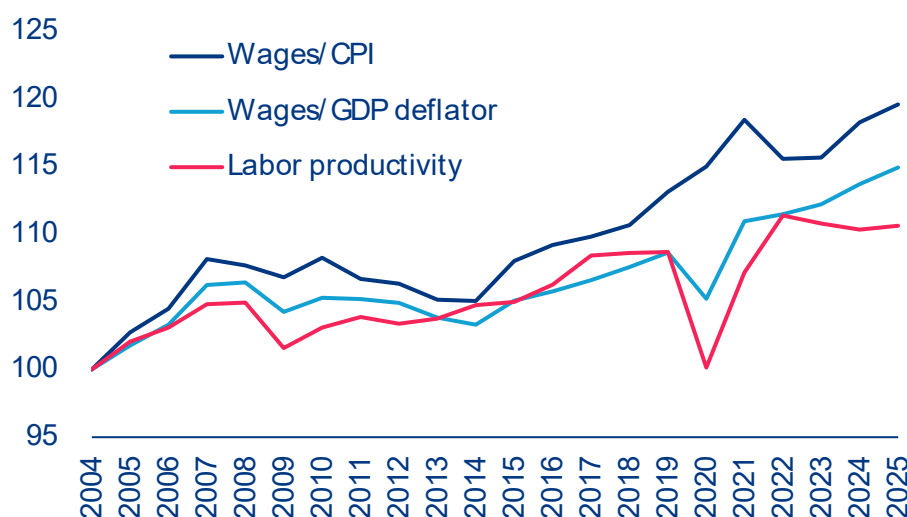
Note: we take into account only permanent migration, which is defined by legal status, not duration of stay by the OECD. It includes migrants with a pathway to remain indefinitely such as refugees, family migrants and some labor migrants, while those without such rights (e.g. students or seasonal workers) are not considered. As per the OECD estimate, we assume that roughly 75% of new immigrants to the UK become employed.

Reduction of regional economic disparities has been disappointing. While Scotland and London strongly supported remaining within the EU, other regions such as the Midlands and North East of England voted firmly to Leave. Reducing inequalities across regions, was a key talking point both during and after the Brexit referendum. ONS data indicates that convergence has generally failed to materialize. We look at the sample of 332 areas in Great Britain representing over 61 million people. 227 areas voted Leave and 105 voted Remain. 59% of the population of “Leave” areas have seen their regions fall further behind national income per capita average since 2016, although some important leave-voting cities such as Blackpool, Doncaster and Wolverhampton have seen an improvement compared to the UK average.

GDP growth has held up since Brexit, but shifted towards more government consumption and less business investment. Higher immigration flows and working hours per worker have offset worsening productivity trends. Despite the challenging environment since Brexit, including the Leave vote, “Truss” moment, political instability, Covid pandemic and energy shock of 2022, UK GDP growth has held up well. Annual GDP growth averaged +1.4% since 2016 — exactly the same

average pace than the previous 10 years (2005-2015). The composition of growth changed, however, with higher growth of government consumption (+2% vs +1.3%) and residential investment (+3.6% vs -1.8%) offsetting lower business investment growth (+2.4% vs +3.7%). Export and import volumes both decelerated at roughly the same pace (-0.7pp to +2.5% for imports, -0.5pp to +2.4% for exports). Household consumption growth was unchanged (+1.3%). From a supply-side perspective, the underlying drivers of potential/sustainable GDP growth¹ changed even more dramatically. The working-age population re-accelerated after 2016 (Figure 1), supporting job creations and GDP. Job creation remained concentrated among foreign-born workers (newcomers) (Figure 2), amid worsening demographics for native and longer-term residents. In all, foreign born workers’ jobs have grown +0.8% per year on average since 2016, contributing to more than 50% of UK GDP growth². Meanwhile, trend growth of hours worked per worker has also accelerated, whilst trend productivity growth faltered. In all, while UK GDP growth has been resilient since Brexit, studies generally estimate that in a counterfactual were Brexit not to happen, GDP would have been between +2% and +4% larger³.

Figure 3: Labor productivity & real wages (indexed 100 in 2004)



Sources: ONS, Allianz Research

¹ We look at potential GDP to strip out cyclical and financial effects that could blur the analysis.

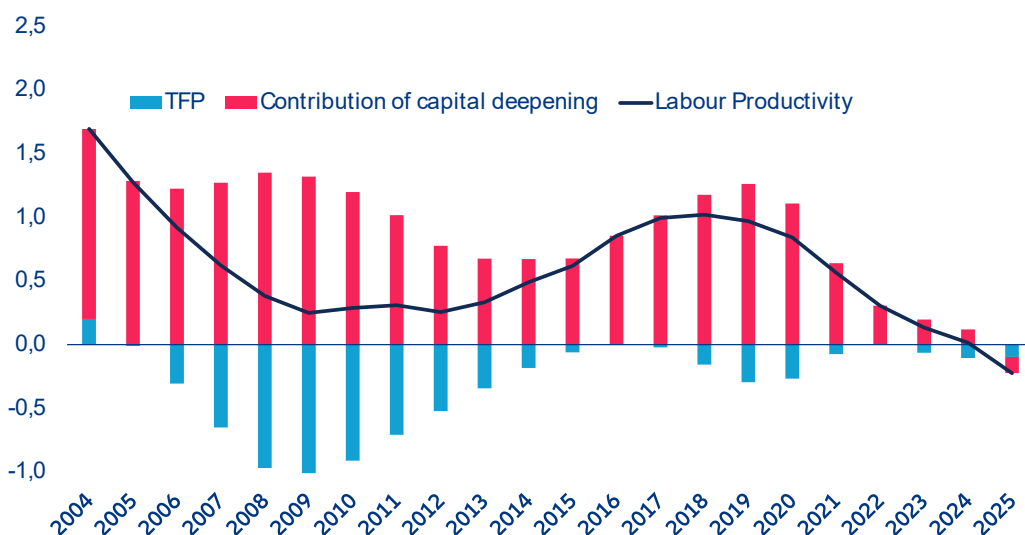
² Immigrants may generally have lower productivity levels than the rest of the population in aggregate (because they tend to work in lower-paid sectors), and therefore weigh on productivity dynamics and GDP growth by this channel. As a result, we take a conservative estimate (50%) lower than headline employment numbers would imply (57%).

³ For instance the Bank of England estimates a cost of -3.5% GDP.

Real wages and income growth has been decent despite an inflation surge, but poor productivity is likely to increasingly weigh on prospects.

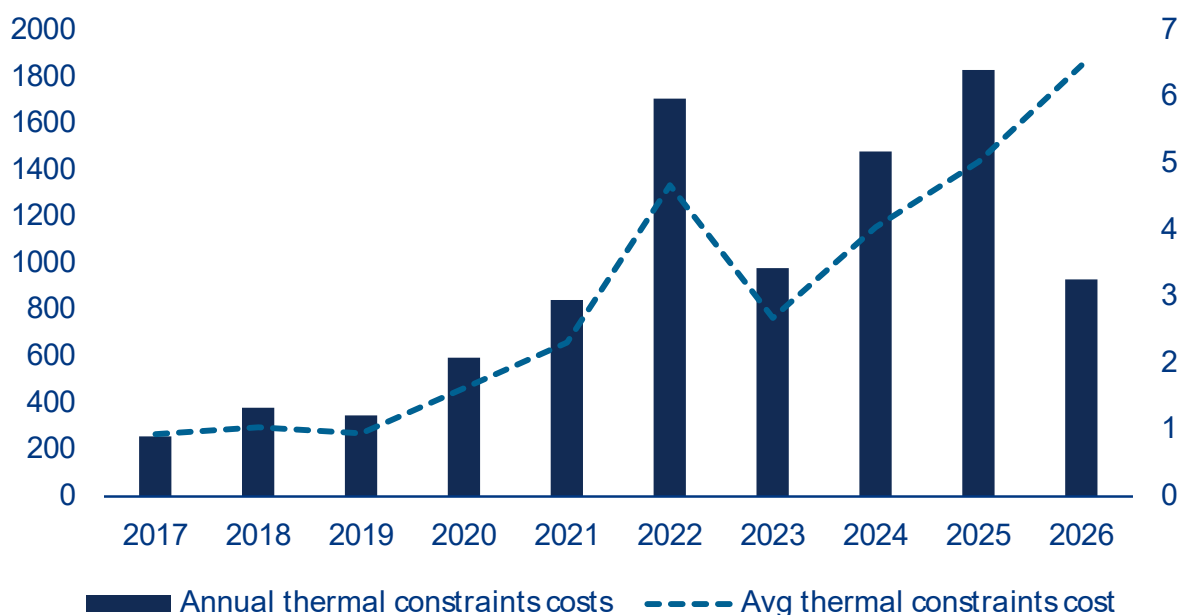
Real income per capita and wage growth have picked up pace since Brexit, despite continued poor productivity performance and elevated inflation. Real income per capita has grown by +0.9% per year on average since 2026, against +0.5% in the ten years prior. Real wage (adjusted by CPI inflation) trends showed even better improvements (+0.9% vs. +0.1%). But this occurred against a backdrop of lingering productivity weakness, which has increased in importance since the pandemic (Figure 3), signaling increasing margin compression for corporates. In other words, decent real wage (and income) growth is unlikely to last unless productivity picks up and/or corporates continue to take the hit on their margins. Very slow investment and capital accumulation has intensified since the pandemic, increasingly weighing on trend productivity growth (Figure 4). The persistent weakness of total factor productivity (TFP), i.e. the underlying efficiency of the economy, reflects many factors that are common to many other developed nations. Recent analysis suggests the slower diffusion of new technologies from frontier firms to the broader economy has been a key impediment to TFP⁴.

Figure 4: Trend productivity growth decomposition (% y/y)



Sources: LSEG Workspace, Allianz Research

⁴ The basic idea comes from a series of OECD papers by Dan Andrews, Chiara Criscuolo and co-authors. They show that productivity growth at the global frontier remained relatively strong during the 2000s and 2010s, but that the gap between frontier firms and the rest of the economy widened. In other words, innovation did not stop; the problem was that innovations spread more slowly to ordinary firms.

Figure 5: Annual thermal grid constraints/congestion costs in the UK power grid (in EUR mn, left) and average daily costs

Sources: LSEG Workspace, Allianz Research

Sector-wise, some services activity (transport and storage, ICT, support services activities, professional services) and public administration output have done well, Manufacturing flatlined but with some bright spots. Structural changes since the pandemic, including global ones, such as increasing Chinese competition on manufacturing, have likely played more of a role in shaping economic outcomes than Brexit itself. The UK manufacturing sector was -1.4% lower in 2025 relative to 2016. However, this masks dramatic shifts within the industrial sector, with energy-intensive sectors shrinking and less energy-intensive sectors doing well. Transportation equipment (+25% between 2025 and 2016), pharmaceuticals (+43%), textiles (+8%), agrifood (+31%), electrical equipment (+20%) and computer & electronics (+20%) saw a sharp increase of production. Meanwhile food/paper (-11.7%), coke/refined petroleum products (-38.4%), chemicals (-10%) and gas steam/electric output (-54%) shrank dramatically.

On energy markets, the UK's departure from the EU Internal Electricity Market (IEM) on 1 January 2021 is one of the most direct and measurable Brexit-related costs for UK energy consumers. Under IEM market coupling, interconnectors used implicit capacity allocation to converge prices efficiently across borders. After Brexit, the shift to explicit auctions reduced trading efficiency and increased transaction costs, resulting in an estimated upward pressure on wholesale prices of between 0.25-0.7% (GBP90-250mn) in 2021⁵. The UK also lost its previous role in EU electricity-market governance and ENTSO-E decision-making structures, although operational cooperation between British and continental system operators remains extensive.

⁵ UK-EU Energy and Climate Cooperation: Why heightened engagement is imperative for Net Zero (Energy UK)

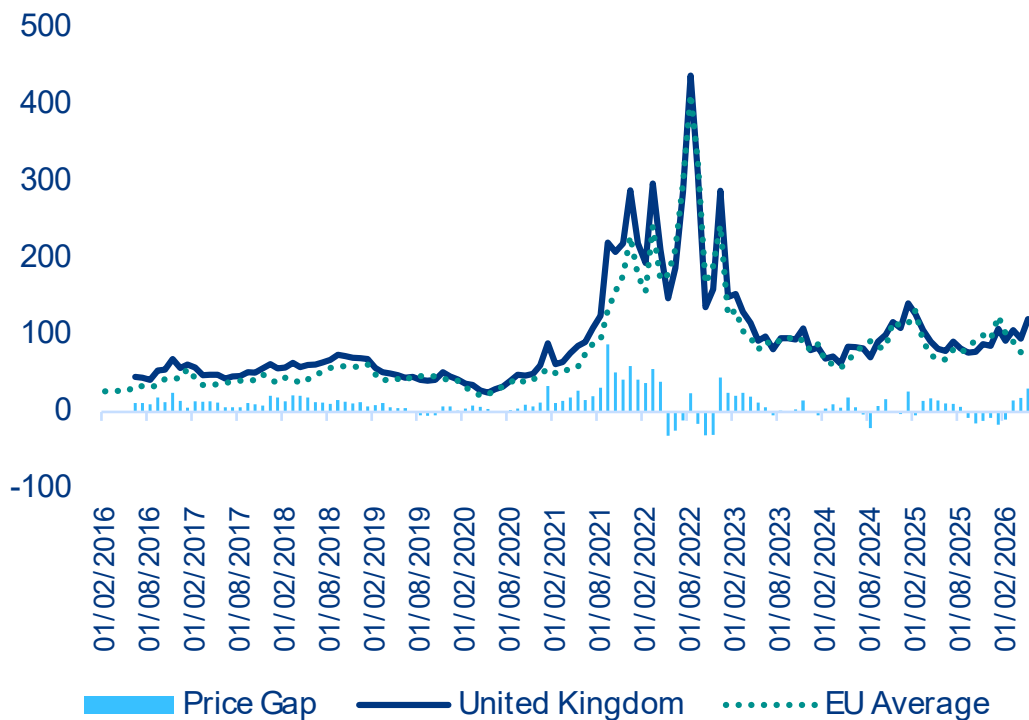
These institutional changes coincided with a broader structural evolution of the UK electricity system. Wind generation has more than doubled since 2016 (+130%), while solar output has increased by 85%, marking a rapid expansion of variable renewable capacity. This transformation enabled a landmark milestone: the UK completed its coal phase-out in September 2024, becoming one of the first major economies to eliminate coal from its electricity system entirely.

However, the rapid expansion of renewable generation has been highly geographically concentrated, particularly in northern Scotland, creating a growing imbalance between generation and demand centers. This has resulted in a severe north–south transmission bottleneck with rapidly escalating system costs. Thermal constraint costs, the cost of managing congestion in the electricity grid when power cannot be transported from where it is generated to where it is needed, rose from GBP258mn in 2017 to GBP1.83bn in 2025 (Figure 5). Based on the H1 2026 run-rate of GBP6.5mn per day, full-year 2026 costs are on track for approximately GBP2.4bn (+29% year-on-year). While these costs are primarily driven by domestic grid constraints and the

spatial distribution of renewable generation, rather than Brexit itself, Brexit remains relevant in a second-order sense. Reduced integration with neighboring electricity markets and slower progress towards coordinated North Sea infrastructure may have limited the system’s ability to alleviate congestion through cross-border balancing and joint offshore grid development.

The UK has materially benefited from cross-border electricity trade, with interconnectors enabling imports of lower-priced continental electricity and reducing the need for domestic redispatch. This is reflected in a close relationship between UK–EU price differentials and physical flows, with the UK typically acting as a net importer when domestic wholesale prices are below those in neighboring EU markets (Figure 6). This mechanism also helps reduce price volatility by allowing continuous arbitrage across markets. In 2022, however, this relationship temporarily reversed: the European energy crisis and reduced French nuclear availability tightened continental supply conditions, leading the UK to become a net exporter as flows adjusted to extreme price differentials.

Figure 6: UK vs EU power price development and gap (USD/MWh)

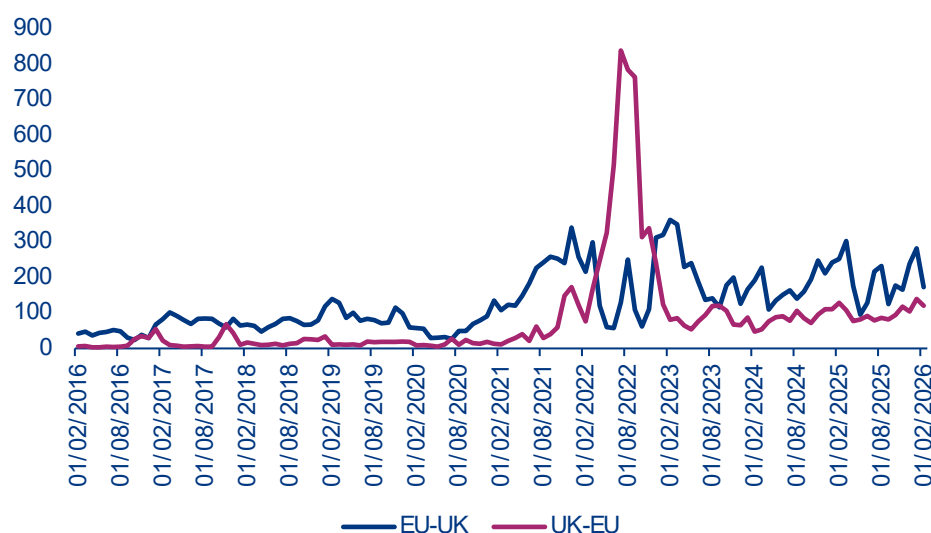


Sources: EMBER, Eurostat, Allianz Research

By contrast, the widening of the UK–EU electricity price gap in 2021–22 reflected a period in which both domestic flexibility and cross-border trading efficiency were reduced. This included the IFA1 interconnector outage, an unusually low wind year in Great Britain, increasing reliance on gas-fired generation and reduced interconnector efficiency following the post-Brexit shift from implicit to explicit capacity allocation. Together, these factors increased the decoupling of UK and EU power prices during the initial post-Brexit period and

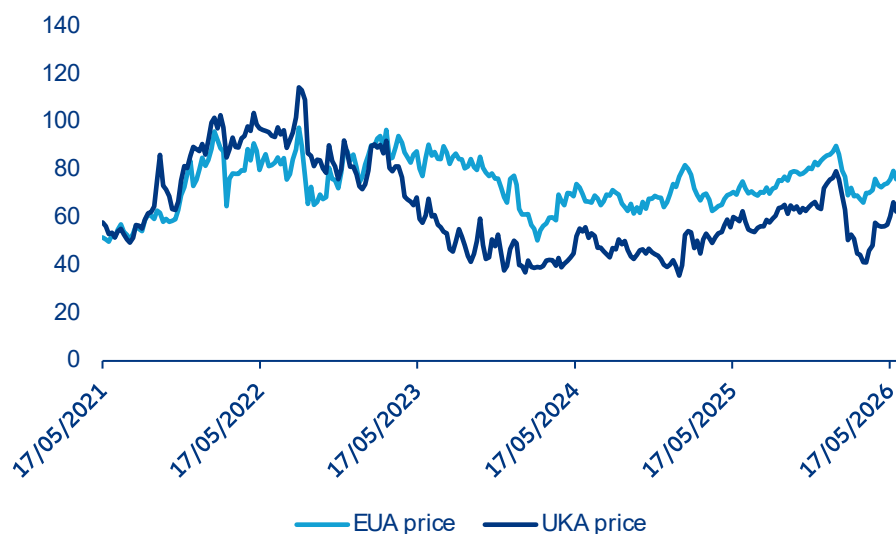
subsequent energy shock. As a result, the post-Brexit UK electricity price premium over the EU rose by 23.6%, peaking between September 2021 and May 2022, when UK wholesale prices averaged EUR52/MWh above the EU average. Although this premium has partially normalized, it remains structurally elevated at EUR15–25/MWh in 2024–25, reflecting persistent integration frictions and a higher reliance on domestic gas-fired generation.

Figure 7: Cross-region power trade (EUR mn, right)



Sources: EMBER, Eurostat, Allianz Research

Figure 8: Co-development of UK and EU ETS allowance prices (in EUR/tCO₂)



Sources: LSEG Workspace, Allianz Research

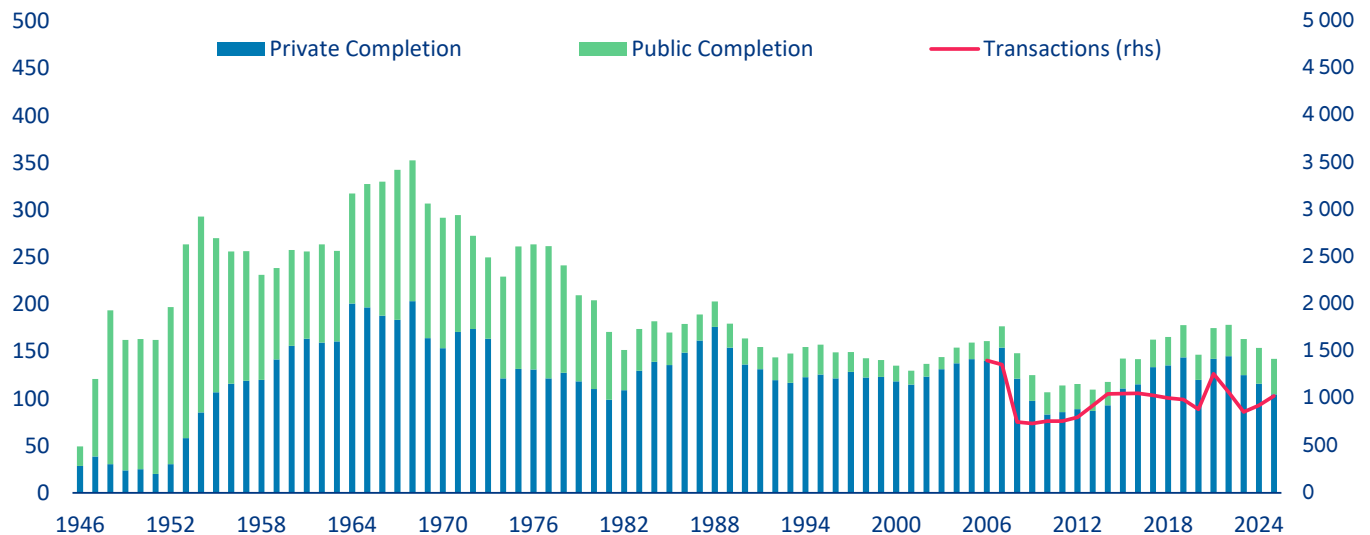
While post-Brexit divergence in energy markets has remained relatively limited, carbon pricing is emerging as a potentially more material channel of separation through its impact on trade-exposed industrial costs.

The UK ETS has traded at a persistent discount to the EU ETS since 2023 (Figure 8). This reflects weaker allowance demand in the UK power sector following rapid decarbonization and greater policy uncertainty around the future tightening path relative to the more clearly signaled trajectory of the EU system. While this reduces compliance costs for UK emitters, it weakens the relative carbon price signal and creates a growing competitiveness gap for energy-intensive sectors exposed to European markets.

This gap becomes more consequential in the context of the EU Carbon Border Adjustment Mechanism (CBAM), which from 2026 will require importers of carbon-intensive goods such as steel, cement, aluminum, fertilizers and electricity to pay the difference between EU and domestic

carbon prices. As a result, the existing UK–EU ETS price differential is increasingly translated into an effective cost wedge on UK exports to the EU, rather than remaining a purely domestic compliance gap. As free allocation in the EU ETS is phased out through 2034, this transmission of carbon price differences into border-adjusted costs is set to strengthen further. Taken together, ETS divergence and CBAM therefore increasingly embed carbon price differentials into trade frictions for energy-intensive UK exports to the EU.

Figure 9: Co-development of UK and EU ETS allowance prices (in EUR/tCO₂)



Sources: LSEG Workspace, Allianz Research



UK economic and energy policies in need of a revamp

A decade after the Brexit referendum, the UK economy remains constrained by a few, but big, structural growth drags: elevated electricity costs and infrastructure bottlenecks.

At the same time, the UK continues to boast many strengths. Productivity growth has been among the weakest in the developed world since the GFC, business investment remains low relative to peer economies, housing supply is chronically insufficient and infrastructure bottlenecks continue to constrain economic activity. Table 1 looks at the performance of the UK on key structural metrics relative to “well performing” developed economies which display elevated levels of income per capita: the US, the Netherlands, Denmark and Sweden. In terms of “output”, the UK is clearly lagging behind in terms of productivity, but hours worked per working person and the employment rate are not particularly low, although the latter has improved less than peers since Brexit. In terms of education proficiency, the UK boasts good scores, particularly in science, and has even improved in country rankings in mathematics and reading proficiency. The UK devotes the largest budget to education (6.1% GDP). In terms of business environment and financial markets integration, the UK also scores well. High electricity costs and a low AI infrastructure index are where the UK really stands out as lagging behind.

In this regard, the current Labour government has correctly identified the economy’s core challenges. Its emphasis on raising public and private investment and accelerating infrastructure delivery reflect a recognition that the UK’s growth problem is fundamentally a supply-side problem. The government’s reform agenda has also correctly highlighted the role played by excessive planning restrictions and regulatory barriers. According to the OECD, the UK’s planning system imposes unusually

high costs and uncertainty on investment projects. Lengthy and unpredictable permitting processes have become an important constraint on housing construction, infrastructure deployment and private investment, contributing to the country’s weak capital accumulation and productivity performance. The consequences are visible in persistently high housing costs, lengthy approval processes and chronic underinvestment in transport, energy and industrial projects. The government’s efforts to streamline planning procedures therefore target one of the most important barriers to long-term growth.

Similarly, the government’s attempts to provide greater strategic direction and higher funding to industrial policy goes into the right direction. The Industrial Strategy identifies priority sectors – including advanced manufacturing, clean energy, defense, digital technologies and life sciences – in which the UK enjoys strong comparative advantages. The creation of the National Wealth Fund and additional support for clean-energy investment represent useful steps towards mobilizing private capital. Recently, the step-up of public funding toward AI hardware companies (GBP1.1bn) represents a welcome development.

Efforts to improve pay and working conditions in essential public services, particularly the NHS, are sound policies to be pursued. The government has well recognized that persistent recruitment and retention difficulties have weighed on labor participation, healthcare outcomes and overall economic performance. Restoring the attractiveness of these occupations is a necessary condition for improving public service delivery and raising the economy’s productive capacity.

Table 1: Economic outcomes and performance of UK economy versus peers and change since Brexit

	Output			Education and Labour				Business		Innovation and AI						Energy	
	Productivity per hour	Hours worked per worker	Employment Rate	PISA Maths Scores Rank	PISA Reading Scores Rank	PISA Science Scores Rank	Educational Institutions Spending (% GDP)	Financial Markets Depth Index	Product Market Regulation	R&D (% GDP)	AI R&D (% GDP)	Total AI Investment (% GDP)	AI R&D Index Rank (per capita)	AI Infrastructure Index Rank (per capita)	AI Investment Index Rank (per capita)	Electricity Cost (\$ per kWh)	Wholesale Electricity Prices (€/Mwh) - Average last 5 years
UK	74,0	1512	74,8	14	13	15	6,1	0,99	0,9	2,7	0,2	0,9	13	27	10	0,4	128,2
US	84,1	1796	71,9	34	9	16	5,8	0,99	1,6	3,5	0,4	1,2	5	7	2	0,2	
Denmark	92,3	1379	77,2	13	15	20	5,3	0,61	1,1	3,1	0,3	1,0	15	12	9	0,4	107,4
Netherlands	82,1	1445	82,3	10	35	25	5,0	0,96	0,9	2,3	0,2	1,5	16	9	13	0,3	117,5
Sweden	81,9	1431	76,7	22	18	21	5,3	0,93	0,8	3,6	0,3	0,8	12	5	3	0,2	66,1
Change since 2015	%	%	%	2018 Rank GDP p.p.				GDP p.p.		GDP p.p.						% change since 2015-2019	
UK	5,0	-0,1	1,9	27	22	15	-0,1	0,4		0,4						135,5	
US	14,9	-1,9	4,6	40	24	25	-0,3	0,7									
Denmark	8,5	-2,0	6,8	12	18	21		0,0								223,2	
Netherlands	0,5	-0,3	8,7	11	15	17	-0,4	0,1								186,1	
Sweden	5,0	-2,4	1,6	24	17	28	0,0	0,4								99,0	

Sources: various, Allianz Research

Restoring the attractiveness of these occupations is a necessary condition for improving public service delivery and raising the economy's productive capacity.

However, while the government's diagnosis is broadly correct, implementation has often fallen short of its ambitions.

Despite the government's emphasis on industrial policy, funding commitments remain modest relative to the scale of the challenge. Public support for strategic sectors remains fragmented across numerous programs and falls short of the large-scale industrial policies pursued by competitors such as the US, China and increasingly the EU. The result is a mismatch between ambitious rhetoric and limited fiscal resources. Similarly, some of the initial ambition of planning reform has been diluted in response to political opposition, raising concerns that housing supply and infrastructure delivery may not improve sufficiently to alter the UK's long-term growth trajectory.

Questions also arise regarding the credibility of the fiscal strategy. The increase in employer National Insurance (NI) Contributions in 2025 raises the cost of labor at a time when the economy requires stronger hiring, investment and labor demand. From an efficiency perspective, greater reliance on broad-based consumption taxes such as VAT would likely have generated similar revenues while imposing fewer distortions on employment decisions. The government has also relied heavily on relatively narrow

tax measures targeting capital gains, carried interest and non-domiciled taxpayers, despite the fact that such revenue sources are often uncertain and susceptible to behavioral responses. Table 2 depicts the key public expenditures and revenues ratios for the UK against the "well performing" peer countries. Figures from the OECD (for cross-country comparisons purposes) are only from 2024, but highlight that the UK has a relatively low tax take on goods & services consumption (essentially VAT), and this take has decreased since Brexit (at 10.2% GDP vs over 11% in Sweden and Denmark). VAT is one of the least distortive form of taxation (ie with limited negative effect on potential growth⁶). Social security contributions, while starting from a relatively low level, have increased, and the NI contributions hike in 2025 would have increased the ratio further. Meanwhile, a substantial share of the planned fiscal consolidation is backloaded towards the latter years of the Parliament. This reduces confidence that the measures will ultimately be implemented. Markets typically discount future fiscal tightening when political incentives may change before implementation occurs. This contributes to uncertainty regarding the medium-term fiscal outlook and may increase the risk premium demanded by investors, leading eventually to higher funding costs for the private sector, weighing on business investment – a key constraint on growth.

⁶ VAT has several appeals over some other forms of taxation: it does not directly discourage work, it is difficult to avoid, it raises a lot revenue

Table 2: Selected public expenditures & revenues ratios

	Fiscal Policy				Healthcare and Social Spending		
	Firms' Income, Profits and Capital Gains Tax	Goods and Services Taxes	Social Security Contributions (% GDP)	Personal Income and Capital Gains Taxes (% GDP)	Healthcare Spending (% GDP)	Public social expenditure (% GDP)	Public expenditure on pensions (% GDP)
UK	3,5	10,2	6,1	10,8	11,1	23,0	7,1
US	2,2	4,1	6,0	10,3	17,2	19,8	7,3
DK	4,2	12,3	0,1	25,2	9,4	26,4	7,5
NL	4,3	11,1	12,0	9,8	10,0	18,9	6,4
SW	3,9	11,2	9,0	11,3	11,3	26,1	8,0
Change since 2015	GDP p.p.	GDP p.p.	GDP p.p.	GDP p.p.	GDP p.p.	GDP p.p.	GDP p.p.
UK	1,19	-0,71	0,11	1,96	1,2	-3,7	-1,2
US	0,14	-0,20	-0,17	-0,23	0,8	0,3	0,4
DK	1,15	-2,23	-0,02	-0,07	-1	-1,3	-0,4
NL	1,63	-0,05	-1,74	2,33	-0,2	-0,7	-0,1
SW	0,93	-0,98	-0,61	-1,33	0,4	1,3	-0,3

Sources: OECD, Allianz Research

A durable growth strategy requires a fundamental overhaul of the UK's fiscal rules and fiscal framework.

On top of poor fiscal policy choices, credibility is also undermined by poorly designed fiscal rules — a subject often overlooked. The current fiscal framework relies on two principal fiscal rules: public sector net debt must be falling as a share of GDP in the fifth year of the forecast horizon, while the current budget must be in balance over the same period. These targets are problematic because both depend heavily on forecasts produced by the Office for Budget Responsibility (OBR) and are assessed at a moving five-year horizon. As a result, governments can often satisfy the rules through relatively small forecast revisions rather than genuine improvements in underlying fiscal sustainability. The framework is therefore complex, opaque and poorly understood outside specialist circles. A simpler framework combining a medium-term expenditure growth rule with a deficit ceiling would offer substantial advantages. Spending rules are transparent, easier to monitor and less vulnerable to forecast manipulation. They also allow automatic stabilizers to operate during downturns, thereby supporting macroeconomic stabilization without requiring repeated discretionary interventions. Such frameworks have generally delivered stronger fiscal discipline and greater credibility in several northern European economies.

Reducing deficits and creating additional fiscal space will require confronting the rapid growth of age-related spending.

Although UK public pension spending remains moderate by international standards (at around 7% GDP, see Table 2), the triple lock creates a source of spending growth that is difficult to justify in the context of mounting demographic pressures. By guaranteeing annual increases based on the highest of inflation, earnings growth or 2.5%, the mechanism tends to raise pension incomes faster than required to preserve either purchasing power or relative living standards. As the population ages, this implies a steadily rising claim on public resources. Reforming the triple lock would therefore not address an immediate pension crisis, but rather prevent future spending pressures from crowding out public investment in areas such as housing, infrastructure, innovation and healthcare that are more likely to support long-term growth.

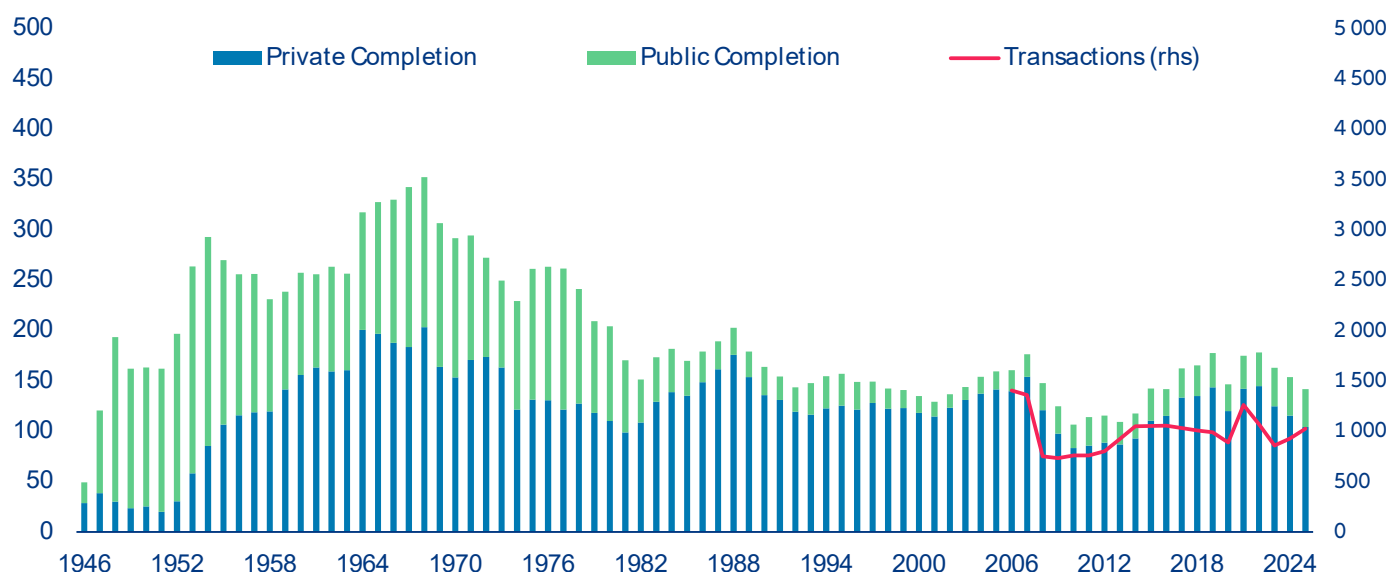
Fiscal consolidation should rely on a more efficient tax mix. A broader VAT base (reduced exemptions and zero rates), continued fiscal drag through temporarily frozen income-tax thresholds and reform of the pension triple lock would raise significant revenues while imposing relatively limited costs on growth. Part of the proceeds could be directed towards lower-income households. At the same time, the government should eliminate distortions within the tax system, notably the effective 60% marginal tax rate faced by individuals earning between GBP100 000 and GBP125 000. Additional savings could be generated through improvements in public-sector productivity, procurement reform and greater administrative efficiency. The government should consider reforming property taxation, which remains based on decades-old valuations and imposes fewer distortions on growth than taxes on labor or business investment⁷. The resulting fiscal space could be directed toward defense, housing, infrastructure, healthcare and innovation, strengthening both the sustainability of the public finances and the economy's long-term growth potential.

Doubling down on housing policy and the NHS should occupy a central place in the government's growth strategy. While planning reform is essential, it is unlikely on its own to generate the scale of construction required to close decades of underbuilding. The public sector therefore has an important role to play both through direct investment and by supporting local authorities and housing associations. From 1950 to 1970, the building of public housing was behind the surge in housing construction, while the private sector's role never really took off. Increasing housing supply would generate benefits extending far beyond the construction sector. Lower housing costs would raise real disposable incomes, reduce welfare expenditures linked to housing support and improve labor mobility by allowing workers to relocate more easily to areas with stronger employment opportunities. More affordable housing would also reduce cost pressures on employers and improve the allocation of labor across the economy. Meanwhile, addressing the NHS' core challenges should be a top priority, rather than focusing on cutting disability benefits, where the current government faced stiff opposition from backbench Labour MPs on its welfare reform. The government's objective

should not simply be to reduce benefit expenditure, but to address the underlying drivers of inactivity through improved healthcare provision, faster treatment, stronger occupational health support and more effective incentives to remain connected to the labor market. Long waiting times for treatment delay the return of workers to employment, increase absenteeism and weaken productivity. Improving healthcare capacity, investing in preventative care and addressing workforce shortages within the NHS would therefore yield economic returns beyond the healthcare sector itself. A healthier workforce is more productive, participates more actively in the labor market and imposes lower fiscal costs on the state. In an ageing society facing persistent labor shortages, improving population health may represent one of the most effective ways of expanding the economy's productive capacity.

One often overlooked feature of the UK's productivity problem is that it does not primarily stem from a lack of innovation. The UK hosts world-class universities, a vibrant start-up ecosystem and globally competitive firms in sectors such as finance, life sciences and artificial intelligence. Rather, the challenge lies in the diffusion of innovation across the wider economy. OECD research shows that productivity growth at frontier firms has remained robust, while the gap between leading and lagging firms has widened over time. In the UK, a relatively small number of highly productive firms and regions have continued to innovate, but many smaller firms have struggled to adopt new technologies, invest in digitalization and upgrade workforce skills. As a result, the benefits of innovation have not spread sufficiently across sectors and regions. Improving technology diffusion through stronger investment incentives, skills policies, business support programs and closer collaboration between universities and firms could therefore represent one of the most effective ways to raise economy-wide productivity growth

⁷ The UK's main recurrent residential property tax, Council Tax, is still based on property values assessed in England in 1991 (2003 in Wales). As a result: a £500,000 property and a £5 million property may face similar tax bills, effective tax rates fall as property values rise. The system is therefore regressive with respect to property wealth. Moreover, the UK taxes transactions heavily instead of ownership.

Figure 9: Housing completions & housing transactions (000s per year)

Sources: LSEG Workspace, Capital Economics, Allianz Research

On energy policy, the Brexit-related energy costs point to three priorities, two of which are already the subject of active negotiations. The most immediate is completing IEM reintegration: the post-Brexit shift to explicit capacity auctions has contributed to a persistent UK wholesale price premium over continental markets, with efficiency losses that grow as electricity import dependence increases. Following the May 2025 Lancaster House Summit, the EU Council authorized the Commission in March 2026 to open negotiations on UK participation in the Internal Electricity Market. These talks are time-sensitive: delays prolong consumer costs and investment uncertainty around interconnector capacity, while the potential efficiency gains remain material at system scale.

A second priority is linking the UK and EU ETS. The UK carbon price discount has evolved from a domestic pricing divergence into a direct trade cost via CBAM for energy-intensive exports, an effect that strengthens as EU free allocation is phased out through 2034. The May 2025 summit established a political agreement in principle to pursue linkage, with CBAM exemption forming the central commercial rationale. Translating this into a binding agreement is the most direct near-term lever to limit carbon-related trade frictions.

The third focus should lie on deepening North Sea infrastructure cooperation. The “Great Grid Upgrade” remains essential to address the UK’s internal north–south transmission bottleneck by connecting Scottish renewable generation to demand centers in the south. In parallel, deeper coordination under the North Seas Energy

Cooperation framework offers scope to reduce system costs through shared offshore grid planning and hybrid interconnector development, particularly in the southern North Sea where congestion pressures and offshore wind expansion most directly overlap. While these initiatives would not substitute for domestic grid investment, they can partially ease balancing constraints and improve cross-border optimization in an increasingly interconnected regional system.

The UK’s productivity challenge ultimately requires a broader investment strategy. Energy costs remain significantly higher than in many competing economies, reducing the attractiveness of the UK as a destination for manufacturing and capital-intensive investment. Accelerating grid expansion, nuclear deployment and domestic energy production should therefore be viewed as industrial policy as much as energy policy. At the same time, the UK should expand support for innovation through a combination of enhanced R&D tax incentives and mission-oriented public investment. While institutions such as ARIA, Innovate UK and the National Wealth Fund move in this direction, they remain considerably smaller than programs such as France 2030 or the industrial initiatives introduced under the US Inflation Reduction Act and CHIPS Act. A more ambitious innovation strategy, combined with housing and infrastructure investment, would help address the UK’s longstanding productivity shortfall and strengthen the economy’s capacity to generate sustainable growth in the post-Brexit era.

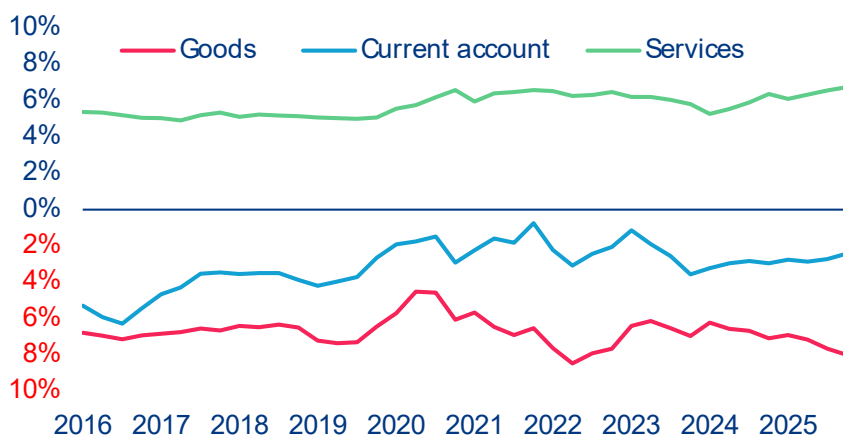


The UK's trade performance: goods imports substitution, services trade powering ahead

The UK's goods trade deficit has worsened since Brexit but rising services surplus have offset this rise. The UK's trade deficit in goods increased by +1.3pp of GDP (and by +1.5pp incl. gold) between 2016 and 2025, rising from 6.7% of GDP to 8.0% of GDP (Figure 10). The UK runs a goods trade deficit with most of its major trading partners. Around one-third of the total deficit is attributable to trade with the EU-27 (USD110bn), which represents a decline compared with 2016, when 70% of the deficit came from the EU-27). Commonwealth countries account for 8% of the deficit (22% incl. Gold), while China accounts for 28% and

the US 7%. On services, the UK's substantial surplus helps offset much this widening goods trade deficit. As a result, the current account balance has remained relatively stable over time and has even shown a modest improvement, reaching a deficit of USD97bn (2.4% GDP) by the end of 2025. This highlights the continuing importance of the UK's services sector – particularly financial, professional and business services – in mitigating the deterioration of its goods trade balance.

Figure 10: UK current account balance breakdown (four-quarter moving sum, % GDP)

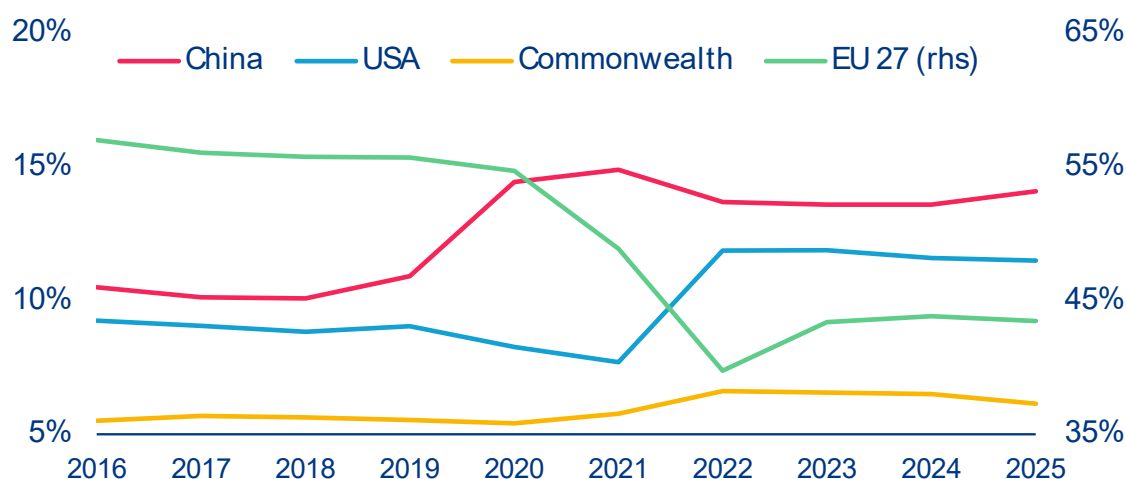


Sources: IMF BoP, Allianz research

Since the Brexit vote in 2016, the structure of UK goods exports toward foreign partners has not changed much. However, on the import side, major economic blocs have benefited from the reorientation of UK trade away from the EU. UK goods exports have decreased by 2.9pp of GDP since the Brexit vote, reaching 11.2% of GDP in 2025. The EU remains the UK's largest export market, accounting for 48% of total goods exports, a share that has remained stable from 50% in 2016. The shares of other major partners such as the US, the Commonwealth countries and China have also remained relatively stable. On the import side, however, changes have been much more significant. Total imports increased by 2pps of GDP between 2016 and 2025, reaching 18.7% of GDP. At the same time, the composition of imports has shifted substantially (Figure 11). The EU now accounts for 43% of the UK's total goods imports, compared with 57% a decade ago. Meanwhile, the importance of China and the US has grown, representing 14% (vs 10%) and 11% (vs 9%), of total imports, respectively. Two main factors explain these developments: First, imports from China surged since the pandemic, driven by strong demand for technology goods, and second, imports from the US increased significantly following the outbreak of the war in Ukraine, largely due to energy purchases.

The UK has diversified both its sources of supply and the types of goods it imports, notably through increased reliance on Chinese manufactured goods and US energy exports. Today, UK imports are dominated by machinery, chemicals, agrifood and automotive products. Compared with 2016, the composition of total imports across these products has become more balanced. Energy imports also gained importance, while chemicals, textiles, transport equipment and automotive products experienced a decline in their relative shares of total imports. A closer look at the changing composition of the UK's suppliers reveals substantial shifts between 2016 and 2025. China has made the most significant gains, particularly in electronics and electrical equipment (the UK is the top destination, with 35% total gross imports of this category), a trend similar to that observed in the EU's growing dependence on Chinese technology goods. China has also increased its presence in automotive products, machinery, chemicals and transport equipment. In short, it has expanded its market share across virtually all manufacturing sectors. Meanwhile, the US has strengthened its position in energy imports, increasing its share by 16pps of the UK's total energy imports. It has also gained ground in machinery and chemicals. Finally, Commonwealth countries have expanded their presence across nearly all sectors, particularly in textiles and chemicals, but the overall share is stable (+0.6pp).

Figure 11: Share of global gross goods imports of the UK by main partner (excluding gold)



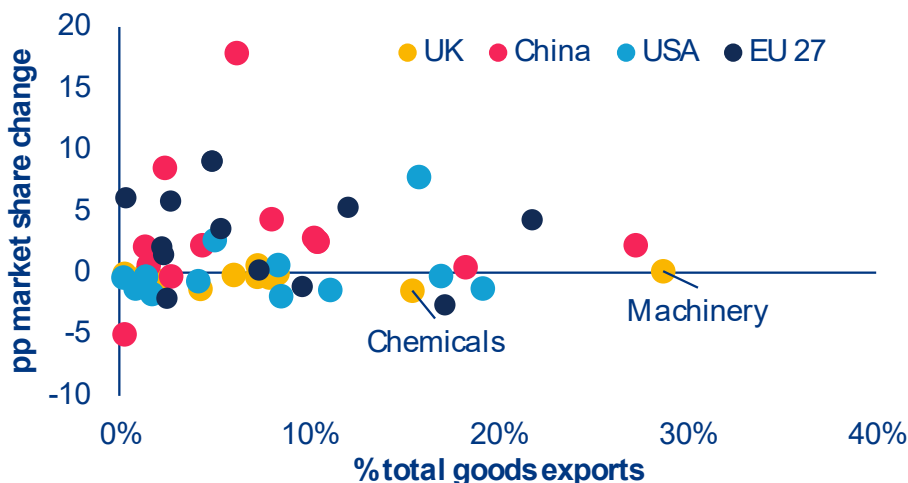
Sources: Trade Map, Allianz research

UK exports have reduced their market shares. Exports are more concentrated in machinery, with a loss of market shares in some traditional strengths (autos, chemicals). Machinery exports are primarily composed of turbojets, turbopropellers, other gas turbines used in the aerospace industry and specialized medical equipment. They now account for almost 30% of total gross exports. By contrast, the importance of the automotive and chemicals sectors has declined significantly. The share of chemicals in total goods exports fell from 20% to 15%, while the automotive sector's share decreased from 13% to 9%, a trend also reflected in broader transport equipment exports. These declines are visible not only in relative terms but also in absolute values, with automotive exports falling by 20% and chemical exports by 8% between 2016 and 2025. At the product level, the UK exports less to the EU across most products than it did in 2016 (as a share of total gross exports), with the notable exceptions of transport equipment and energy. However, these declines have not been offset by another economic bloc in relative share. Although the UK's export structure appears relatively stable in relative terms, with the same trading partners and some rebalancing across products, this masks significant shift towards machinery and energy exports, alongside a decline in sectors that have traditionally been among the

UK's strengths, namely automotive and chemicals. Figure 12 shows that overall, the UK has experienced a decline in its global market share across most of the products it exports compared to the other major economic blocks.

This analysis of UK trade excludes gold exports and imports, a category in which the UK has experienced strong growth in market share and which has had a significant impact on its trade structure as a result of the gold price rally in recent years. Including gold could distort the assessment of the underlying state of British manufacturing capabilities and the structural shifts currently taking place in global trade, such as the growing footprint of Asian economies — particularly China — across most goods categories, and the increasing role of the US in energy supply. When gold is included, the UK's degree of openness remains broadly unchanged at around 38% between 2016 and 2025. Excluding gold, however, the indicator points to a weakening integration of UK into global trades, with a share that decreased from 35% in 2016 to 30% in 2025.

Figure 12: Market share changes 2025 vs 2016 (vertical axis) and share of total gross exports of the country by product in 2025



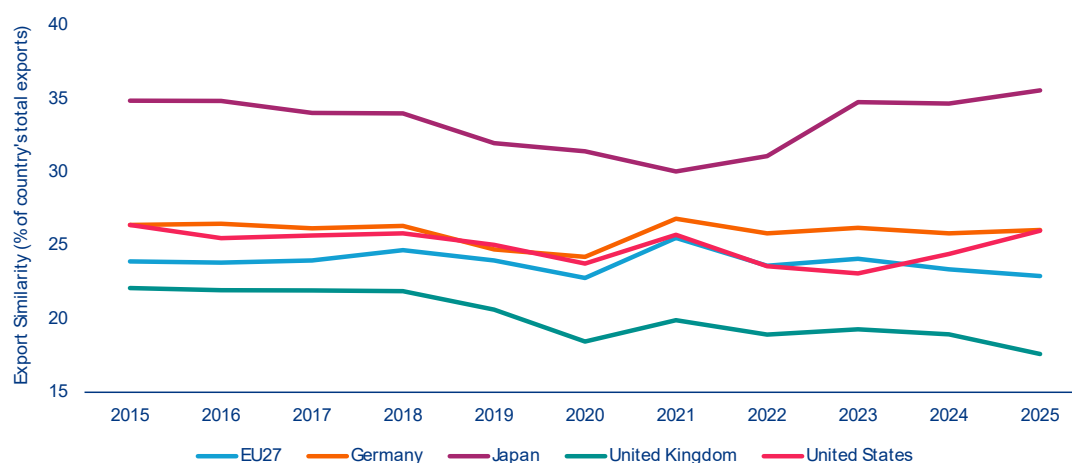
Sources: Trade Map (H2 classification), Allianz research

With regard to Chinese competition, the UK is less exposed. In 2025, about 15% of UK exports were in products among the top 25 of China's exports. In contrast, 25% of US exports and 26% of German exports compete directly with the Asian giant's top exports. As China has moved towards high-value manufacturing, its top exports now consist of telecommunication devices, integrated circuits and computers. The UK, historically a smaller exporter of electronics, is thus less exposed to Beijing's shift. Pharmaceuticals and aerospace, a large source of the UK's competitiveness advantage, are less in the crosshairs. Since the pandemic, a key area of competition is also the automobile industry. Cars represent the UK's second largest source of exports at over USD30bn per year. A falling similarity index could be a cause for concern. Indeed, a lack of competitiveness that erodes a sector's share in total exports will mechanically decrease the index. For the UK, car exports have increased by +1.1% in nominal value since 2022 while total exports (excl. gold) increased by +11.2%, highlighting a flailing automotive sector and thus decreasing the similarity index.

UK trade agreements have expanded market access, but they have not matched the scale of the economic relationship with the EU or fully offset Brexit-related trade losses. Although UK-EU tariffs remain relatively low, with an average applied tariff of 1.5% (reaching up to 19%

for certain products), the principal barriers to trade are non-tariff measures (NTMs). Since leaving the EU in 2020, the UK has introduced 58 trade-related measures affecting EU countries, including 25 export-related measures, 20 subsidies, seven tariff measures and five price controls. Over the same period, the EU has introduced 337 measures affecting the UK, predominantly subsidies (275), alongside export restrictions (19), tariff measures (15), licensing and quota measures (13), price controls (six), trade-protective measures (four) and one foreign direct investment (FDI) measure. These policy changes have contributed to a significant decline in bilateral trade integration. Structural gravity estimates suggest that Brexit reduced UK-EU trade by -21.3% for goods and -49.7% for services. This weaker bilateral performance reflects Brexit-related trade and supply-chain disruptions arising primarily from higher NTMs and, to a lesser extent, tariffs.

Figure 13: Similarity index to Chinese goods exports



Sources: ITC Trade Map, GTA, Allianz Research

Note: Using GTA methodology, similarity is calculated for each country as the sum of the export value of all their products appearing in China's top 25 HS4 codes, divided by their total exports. The HS code for cars is included for all years.

7 The UK's main recurrent residential property tax, Council Tax, is still based on property values assessed in England in 1991 (2003 in Wales). As a result: a £500,000 property and a £5 million property may face similar tax bills, effective tax rates fall as property values rise. The system is therefore regressive with respect to property wealth. Moreover, the UK taxes transactions heavily instead of ownership.

In parallel, the UK sought to offset some of the economic costs of Brexit through an independent trade policy. Since Brexit, the UK's free trade agreement (FTA) strategy has evolved from initially rolling over agreements previously negotiated through EU membership to pursuing new bilateral and plurilateral arrangements. Major milestones include agreements with Japan (2020), Australia (2021), New Zealand (2022), the EEA EFTA states (Norway, Iceland and Liechtenstein), digital trade agreements with Singapore and Ukraine, accession to the Comprehensive and Progressive Agreement for Trans-Pacific Partnership (CPTPP) in 2024, the India FTA signed in 2025, an upgraded agreement with South Korea and the Gulf Co-operation Council (GCC) agreement concluded in 2026 but not yet in force (Table 3). Collectively, these post-Brexit agreements account for approximately 21% of UK trade in 2025 (without the US, which is a prosperity deal to offset the impact of imposed tariffs rather than a FTA). The UK is also negotiating or upgrading agreements with Canada, Switzerland and Türkiye. Despite expanding the UK's global trade network, the scale of these agreements remains substantially smaller than the UK's economic relationship with the EU, which continues to account for the largest share of UK trade at 37% in 2025.

Deeper UK–EU trade ties could unlock billions in economic gains. Looking ahead, the UK and EU may find scope for greater alignment in goods trade, while the UK and US are more likely to deepen cooperation in services, technology, finance and regulation. However, only a

limited share of the trade frictions created by Brexit is likely to be reversed. Computable general equilibrium (CGE) simulations indicate that a deep UK–EU trade agreement could generate annual gains of USD25.2bn for the UK, comprising USD19.6bn in trade creation and USD5.6bn in trade diversion. This would increase the value of UK exports to the EU by +23.1%, with the largest proportional gains in sugars, food preparations and cereals, followed by vehicles, alcoholic beverages and textiles. EU countries could gain USD31.9bn annually (+17.4%), driven primarily by vehicles — reflecting the restoration of previously integrated value chains — as well as food preparations, dairy products and chocolate.

Table 3: UK Free Trade Agreements since Brexit

Partner	Agreement	Year	Key provisions	UK bilateral trade (share of total UK trade in 2025)
European Union	EU–UK Trade and Cooperation Agreement	2020 (effective 2021)	Necessary replacement for EU membership. Zero tariffs and quotas on qualifying goods, customs cooperation, services framework, fisheries, transport, energy, law-enforcement cooperation	37% with EU remaining the UK's largest trade partner
Japan	UK–Japan CEPA	2020	Goods market access, digital trade, financial services, data flows, investment protections.	1,50%
Australia	UK–Australia FTA	2021 (effective 2023)	Tariff elimination, services access, mobility provisions, digital trade, procurement.	1,30%
New Zealand	UK–New Zealand FTA	2022 (effective 2023)	Near-total tariff elimination, services liberalization, digital trade, investment access.	0,10%
Norway, Iceland, Liechtenstein	UK–EEA EFTA FTA	2021	Goods, services, investment, digital trade and mobility provisions.	2,30%
Singapore	Digital Trade Agreement	2022	Digital identities, e-commerce, fintech cooperation, cross-border data flows.	1,00%
Ukraine	Digital Trade Agreement	2023	Digital trade rules, e-signatures, cybersecurity cooperation, data flows.	0,20%
Comprehensive and Progressive Agreement for Trans-Pacific Partnership	UK accession to CPTPP	2023 (effective 2024)	Tariffs, services, investment, procurement, digital trade across 11 Pacific economies.	9,20%
India	UK–India FTA	2025 (not yet in force)	Tariff reductions, services access, professional mobility, customs simplification, investment provisions. Expected to increase bilateral trade substantially.	1,70%
South Korea	UK–South Korea FTA	2025 (updated from earlier EU agreement)	Replaces the continuity agreement inherited from EU membership. Modernizes rules of origin, services and digital trade, maintains tariff-free access on most goods, and strengthens access for automotive, financial services and technology sectors.	0,90%
United States*	Economic Prosperity Deal	2025 (partially implemented)	A non-binding framework intended to reduce the impact of US tariffs and improve market access. Includes preferential treatment for UK autos, aerospace, steel and aluminum, and a framework for future cooperation on pharmaceuticals and supply-chain security.	13,40%
Gulf Co-operation Council (Bahrain, Kuwait, Oman, Qatar, Saudi Arabia, UAE)	UK–GCC FTA	2026 (not yet in force)	The UK's first FTA with the GCC and the GCC's first FTA with a G7 country. Covers tariff reductions, customs facilitation, services, investment and market access.	2,60%

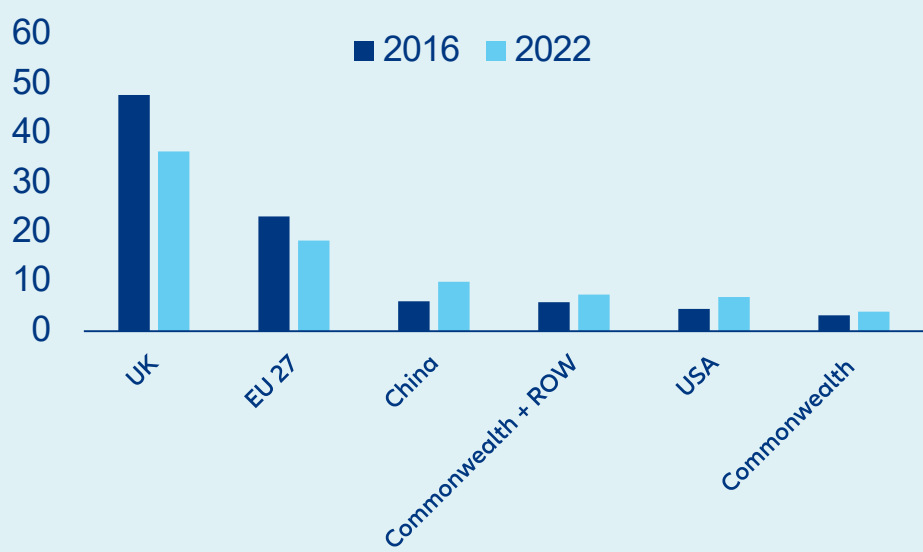
Sources: UNComtrade, UK House of Commons, Allianz Research. Notes: *The US Economic Prosperity Deal is not a FTA, but a deal to lower the impact of tariffs

The evolution of UK trade in global supply chains

From a supply-chain perspective, the UK's domestic demand has become increasingly dependent on industrial foreign suppliers' inputs, while the country has struggled to expand its export footprint in global supply chains. The OECD Trade in Value Added (TiVA) dataset has data only available up to 2022, yet it provides valuable insights into the changes to supply-chains. It looks at the foreign and domestic value added embodied in final demand. For the UK, the data indicates a gradual reorientation of domestic demand away from EU products. Although the EU remains the UK's largest trading partner in valued added, the older EU member states are no longer its preferred trading counterparts in supply chains as the relative importance of partners such as China, the US and Commonwealth countries has increased significantly over the past decade.

Indeed, between 2016 and 2022, the share of industrial foreign value added in UK final demand originating from the EU declined from 52% to 39% in the manufacturing sector. Over the same period, China's share increased by 8pps, making it the UK's largest single-country partner in 2022. The US gained 2pps and became the second-largest partner, while Germany's share fell by 8pps, causing it to drop from first to third place. Within the EU, only Ireland and Poland increased their shares, whereas most Western European countries experienced a decline. This shift is particularly pronounced in the computer, electronics, and optical products sector, where China's share has overtaken that of the EU. Similarly, in the motor vehicles industry, the US gained around 5pps, again at the expense of both EU and UK manufacturing value added. These trends suggest a gradual reconfiguration of the UK's supply chains away from its historical European partners and towards a more globally diversified network of suppliers. In terms of UK exports, between 2016 and 2022, the UK has lost a bit of market share in the trade in value added embodied in other countries' final demand for manufacturing (-0.4pp at 1.6% at the global level), visible for the main UK's industries (chemicals, electronics, machinery, automotive) .

Figure 14: Insurance penetration, selected countries (gross written premiums in percent of GDP)

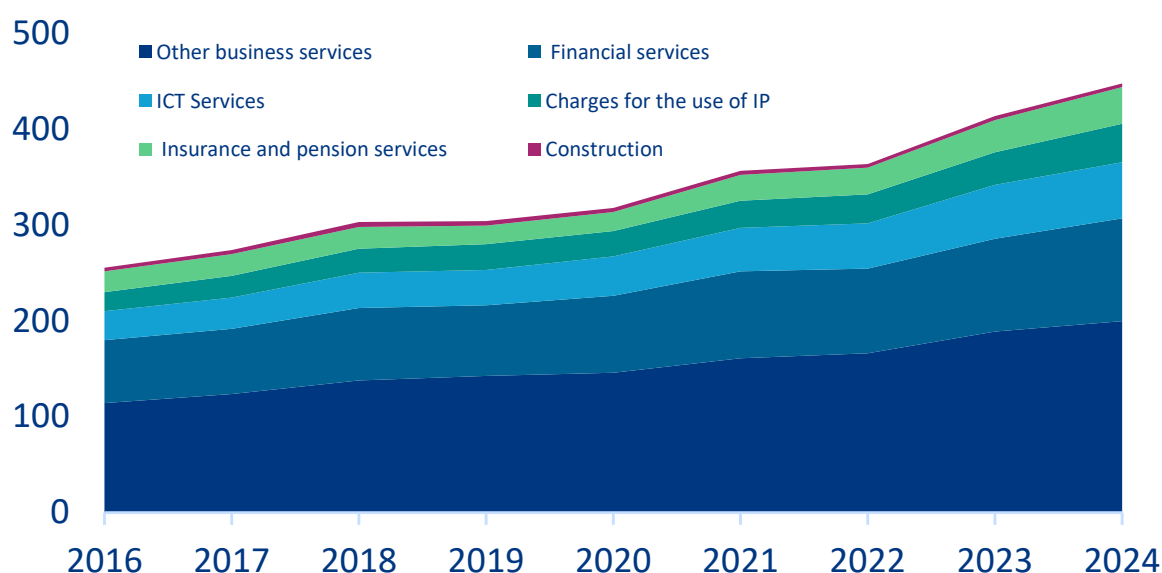


Source: OECD TiVA, Allianz research

While Brexit introduced additional compliance requirements and regulatory uncertainty, the services sector has largely avoided the trade frictions experienced in goods markets. UK services exports to the EU increased by USD96bn between 2026 and 2024, representing growth of +63%, only slightly below the pace of growth in UK services exports globally (+65%). Financial services have retained much of their international competitiveness despite the loss of preferential access to EU markets. While some trading activity, assets and jobs shifted to Amsterdam, Dublin and Paris, London remains Europe's leading center for foreign exchange, derivatives, insurance and international banking. Financial services exports to the EU grew by 58% between 2026 and 2024, compared with 63% globally. Moreover, while the UK's share of value added embodied in global final demand for financial services decreased by 0.3 percentage points between 2016 and 2022, its share in EU-27 final demand increased by 0.6 percentage points, suggesting a deeper integration of the UK financial sector into European value chains. The share of financial services in total UK service exports has remained broadly stable at around 17% in 2024⁹. The UK's trade surplus in financial services has increased over the period and the export destinations remain heavily concentrated in the US, which accounts for around 27% of UK financial services exports, followed by Ireland, Luxembourg, other EU

member states and a range of offshore financial centers. In the global financial services market, the UK's position has remained remarkably resilient. It continues to rank second worldwide, accounting for approximately 21% of global financial services exports, outperforming the EU and behind the US, which retain the first place. The main changes in the global ranking have occurred among the following positions, with Singapore notably strengthening its role and consolidating its position as the fourth-largest exporter of financial services. These trends suggest that while Brexit has not significantly enhanced the UK's dominance in financial services, neither has it undermined London's status as one of the world's leading financial centers. Actually, the UK is exporting mostly "other business services" more than financial services + insurance and pension services, including "R&D services", "consulting services" and "technical, trade-related services" (Figure 15). Finally, ICT services have done even better. Supported by digital trade provisions in the EU–UK Trade and Cooperation Agreement and continued EU data adequacy arrangements, UK ICT services exports to the EU grew by +94% since Brexit, outpacing growth to the rest of the world (+76%).

Figure 15: Distribution of UK's services exports (USD Bn)



Sources : WTO-OECD, Allianz research

⁹ From a value-added perspective, financial service activities accounted for 4.4% of total UK value added in 2023, down from 6.5% in 2016

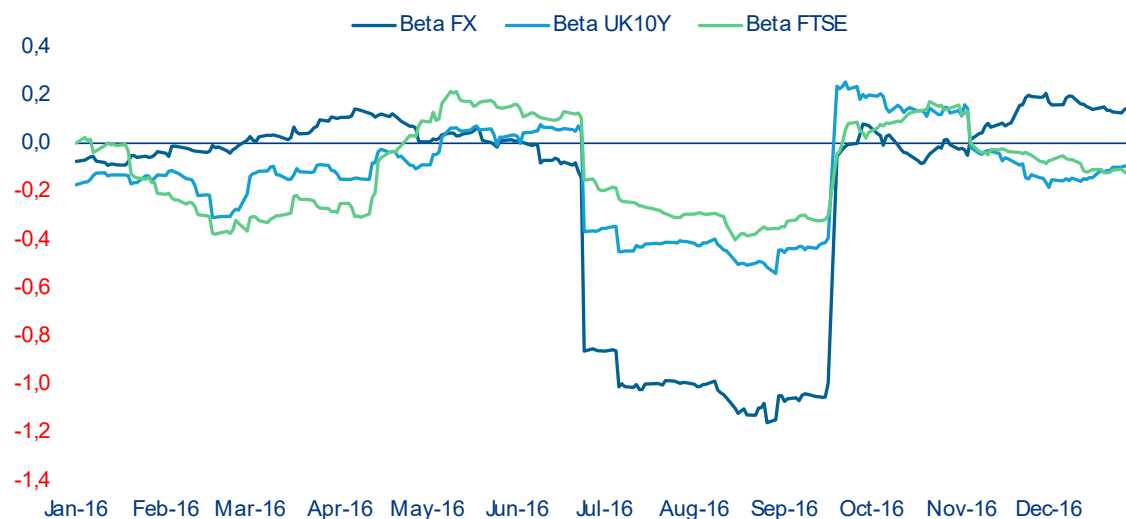


Financial markets: unofficial dollarization and eroding confidence but the City of London keeps (most of) its shine

From a capital market perspective, Brexit set in motion a sequence of structural changes — monetary, institutional and commercial — whose cumulative effects only became visible after the central bank's protective cover was removed.

The immediate market impact of the referendum was absorbed almost entirely by the exchange rate. From the moment Leave probabilities began rising in late 2015, sterling traded as the primary signal of UK political and growth uncertainty. The Gilt market was insulated: the Bank of England cut rates and relaunched asset purchases within weeks of the result, compressing term premia and anchoring the long end of the curve. The primary driver of Gilt yields throughout this period was the global rates cycle — Fed signaling above all — rather than any UK-specific risk premium. Quantitative easing masked the Brexit effect on rates; the FX market was the honest signal, the bond

market the managed one. On the day the referendum result was announced, 24 June 2016, the GBP/USD fell ~8.1% in a single day — a 14.8-sigma move relative to its normal daily volatility, by far the most extreme reaction of any asset. By contrast, the UK 10Y yield moved -28.8bp ($z = -5.3$) and the US 10Y just -16bp ($z = -3.1$), while the FTSE 100 fell -3.1% ($z = -3.3$). In standardized terms the currency shock was roughly three to five times larger than the rates and equity shocks. The Gilt had, in effect, become a UST proxy at a discount, its terminal rate tracking US conditions more closely than domestic fundamentals.

Figure 16: Rolling 61-day beta (z-score returns) vs policy index – Zoom 2016, Brexit 24/06

Sources : LSEG Datastream, Allianz Research

Beneath this surface stability, Brexit was accelerating a structural reorientation of the UK's external sector.

The share of UK exports invoiced in GBP fell from approximately 63% before 2015 to 38% by 2022, with USD invoicing rising from one-third to 45% over the same period. This transition was directly triggered by the currency-mismatch valuation shock of the 2016 depreciation and proved persistent. As the UK economy became progressively coupled with US financial conditions, the dependency on Fed policy transmission deepened. This dollarization of trade was the most consequential — and least visible — legacy of the referendum years. The risk premium currently visible in GILT was not observable right after Brexit.

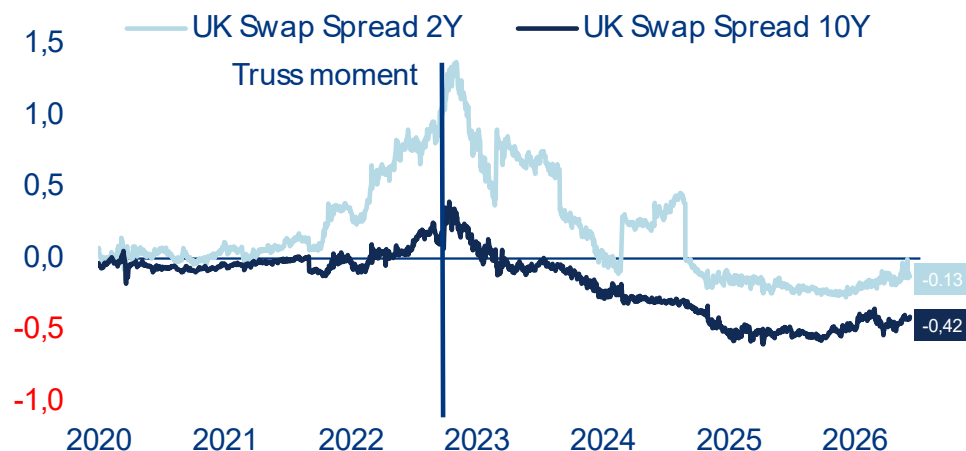
The City's resilience post-Brexit has proved more robust than pre-referendum predictions suggested, but headline stability conceals a meaningful structural shift at the margin. London's pre-eminence was built on two foundations: its historical role as the birthplace of the Eurodollar system and its subsequent emergence as the natural bridge between dollar and euro financial markets. It hosted the clearing infrastructure for euro-denominated derivatives on LCH, the primary market for rates and FX instruments across both currencies and the center of international banking for non-domestic entities operating in Europe.

The aggregate BIS data tells a story of remarkable persistence.

In April 2025, the UK accounted for 49.6% of global OTC interest rate derivatives turnover – up from 42.9% in 2022 – and handled 37.8% of global FX turnover, broadly stable versus the pre-Brexit baseline (BIS Triennial Survey, 2025). These figures point not only to an expansion in gross activity but also underline the UK's persistent global leadership. The depth of London's liquidity pool and the network externalities that make migration costly have proved more durable than political pressure alone could dismantle.

The euro clearing franchise is where nuance matters.

LCH retained approximately 81% of global euro IRS clearing as of end-2023, with Eurex holding the remaining 19% – a shift that occurred almost entirely in the 2019–2021 window driven by Brexit uncertainty and Eurex incentive programs, and which has not materially progressed since (ECB Financial Integration Report, June 2024). Around 95% of euro-denominated interest rate swaps were still cleared through LCH in 2024, and the European Commission's renewal of UK CCP equivalence until June 2028 is a tacit acknowledgement that the infrastructure cannot be relocated by political fiat. The exception is euro repo, the one segment where migration has been documented and sustained: London's banking ties to the Eurozone have loosened in part owing to the shift of euro repo clearing from London to Paris (BIS Bulletin No. 65, 2022; ICMA repo survey data, 2024).

Figure 17: Swap spread widening after the Truss moment.

Sources: ILSEG Datastream, Allianz Research

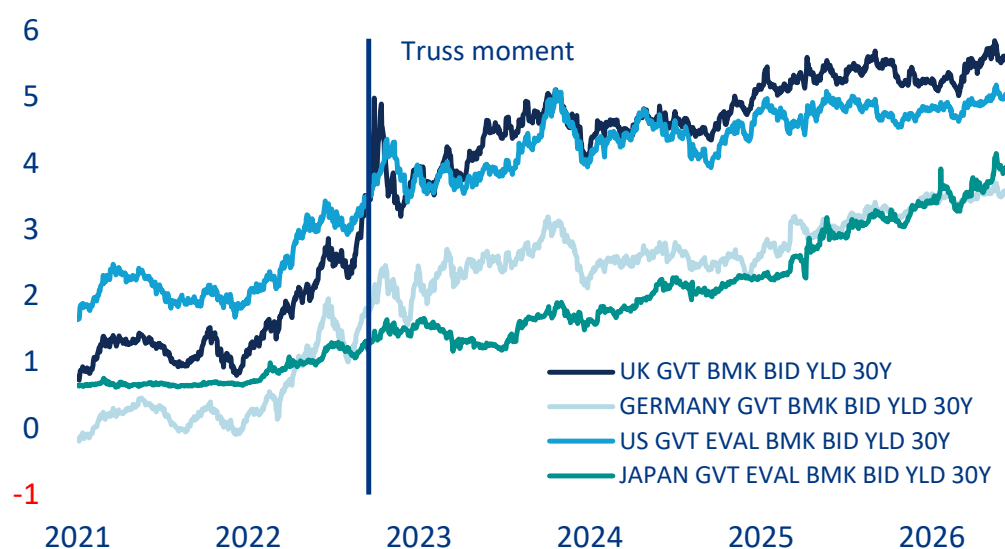
The honest assessment is that London's clearing and derivatives franchise stopped growing into its natural trajectory rather than declined outright.

Euro-denominated IRD trading surpassed US dollar-denominated instruments to account for the largest share of global IRD turnover in 2025 — growth that, absent Brexit, would have accrued disproportionately to London. The counterfactual loss is real even if absolute volumes remain strong. For the Gilt market, this matters through a second-order channel: a City that is less central to euro financial intermediation generates less natural structural demand for UK sovereign paper as collateral, reinforcing the shift toward a thinner, more price-sensitive foreign investor base.

The September 2022 fiscal episode is, in retrospect, the most consequential single event for the structural repricing of UK sovereign risk, more so than the referendum itself. It exposed the inherent fragility of a market dominated by concentrated, leveraged duration holders. The high-weighted average maturity of the Gilt stock (UK WAM is touching 13.5 year, against 7 year for Germany, or 6 year for US, at the end of 2025 according to the OCDE)) combined with the structural role of liability-driven investors and pension funds as dominant holders of long-dated paper, created a high vulnerability to duration shock and a self-reinforcing feedback loop that the Bank of England had to interrupt with emergency purchases of

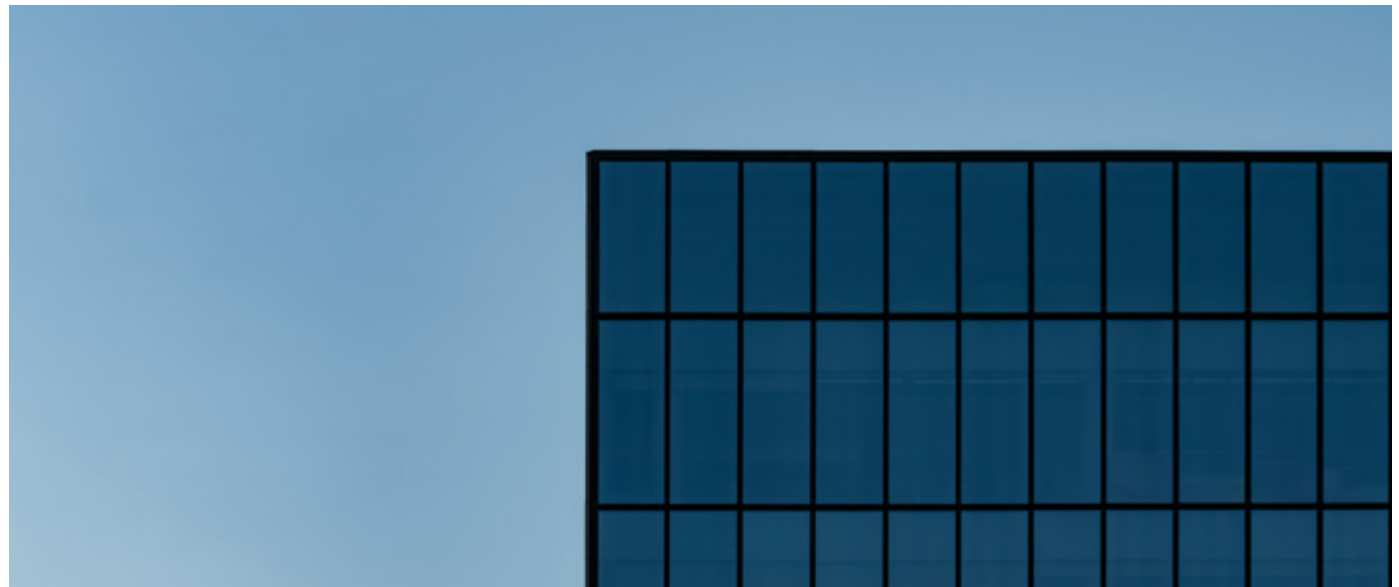
GBP65bn over 13 trading days. The episode established one structural fact that had not previously been priced: the central bank backstop for Gilts is conditional on fiscal credibility. Swap spreads, which had traded near zero prior to September 2022, moved to persistently negative territory, reflecting a banking sector constrained in its capacity to absorb newly issued sovereign paper, and a durable liquidity premium now embedded in the yield curve.

The transition from QE to quantitative tightening has compounded the vulnerabilities exposed in September 2022, but the causality is not purely mechanical. The Bank of England has pursued QT aggressively, in part as a deliberate signal to sovereign bond markets: a demonstration that institutional discipline had been restored after Truss. That signal has had a cost. The regime of scarce collateral and abundant reserves has given way to one of elevated Gilt supply and tighter bank funding conditions — GC rates now sit modestly above the BoE policy rate, consistent with a structural shift in money market. Banks depend more heavily on money market financing, reducing their capacity to absorb issuance and amplifying market sensitivity in stress episodes.

Figure 18: DM 30Y rates and GILT discount since the Truss moment

Sources: LSEG Datastream, Allianz Research

The BoE has credibly anchored the short end, what it has not recovered is control over the long end. Long-dated yields are increasingly set by supply dynamics, global uncertainty and term premia rather than by forward guidance or domestic macro anchors. US macro announcements — Fed decisions, Treasury supply, CPI prints — exert a more immediate and measurable effect on Gilt pricing than domestic budget decisions. This is the EMification dynamic in its precise formulation: not a loss of formal monetary sovereignty, but a structural compression of the market's capacity to price UK fundamentals independently of the global financial cycle. While this is a global phenomenon, the magnitude is strongly pronounced in the GILT market that exhibit a higher rate than the other DMs.



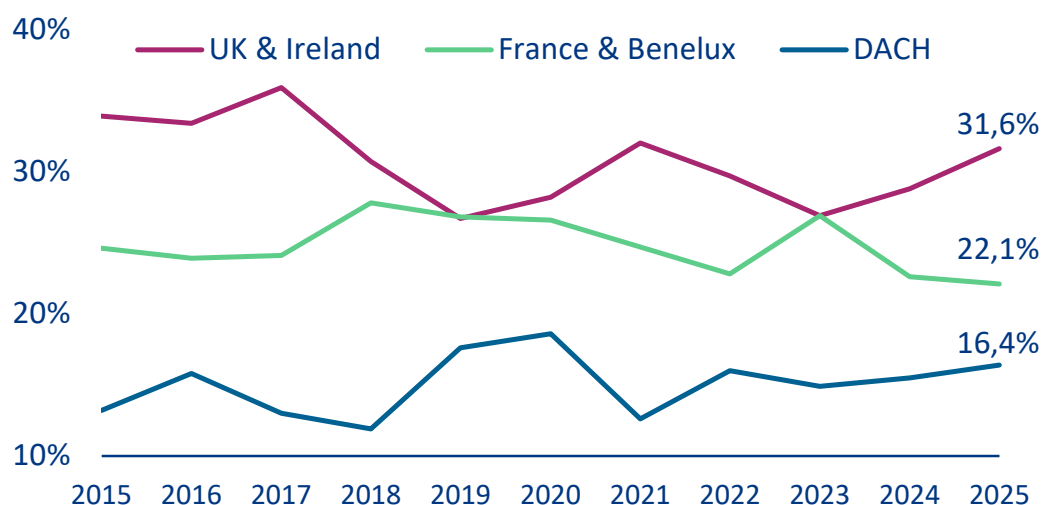
UK private equity market has lost its magic, low tech exposure prevails

A decade after the referendum, Brexit's imprint on UK private markets reads as a level shift that has now largely cleared the price. The UK is still Europe's dominant private capital hub, accounting for roughly one in five European deals and close to one in three fund closes, but it has stopped pulling away from the continent. The damage is real, concentrated and uneven: a one-off downshift in trend growth and relative attractiveness, since layered over by the pandemic, the energy shock and a higher-rate cycle that have blurred attribution. What remains is a market trading at a structural discount, increasingly financed from abroad, and now leaning on domestic policy to re-rate. Within that, the clearest bifurcation is in venture capital, the generalist fundraising base has hollowed out, yet the UK retains a relative structural edge in AI and deep-tech, where it continues to anchor the continent's largest rounds.

Brexit was a level shift, and it is now largely priced in. Estimates from Bloom et al. indicate that, by 2025, Brexit had reduced overall UK investment by around 12–18% relative to a no-Brexit counterfactual. Transmission ran through trade and labor frictions, the loss of passporting and equivalence, which pushed UK managers onto national private-placement regimes and EU-domiciled affiliates at higher operating cost, and through uncertainty itself, which depressed UK private equity volumes via a real-options "value of waiting" channel. Hundreds of financial firms have shifted staff, balance sheets and legal

entities into the EU, re-anchoring fund servicing across Dublin, Luxembourg, Paris and Frankfurt and embedding a more multipolar European capital map. The discrete adjustment is behind us; what persists is a structurally higher cost of capital for UK risk assets.

Private equity proved resilient, increasingly as a US-funded value trade. The UK retains roughly a third of European private equity deal value, a position held through the post-referendum period, yet its deal value growth over the decade since the vote has materially lagged Germany and France, and Germany has displaced France as the continent's second market. The structural mechanism behind the UK's resilience is partly a discount story: compressed equity valuations relative to US peers have turned listed UK companies into prime take-private targets, drawing a sustained and disproportionately large share of US capital into the market, a pattern consistent across the post-Brexit cycle and the highest rate of US participation of any major European PE market. Buyout activity has increasingly defined the deal mix, reflecting both the take-private dynamic and the relative underrepresentation of growth equity in the UK versus continental peers.

Figure 19: The UK retains roughly a third of European private equity deal value.

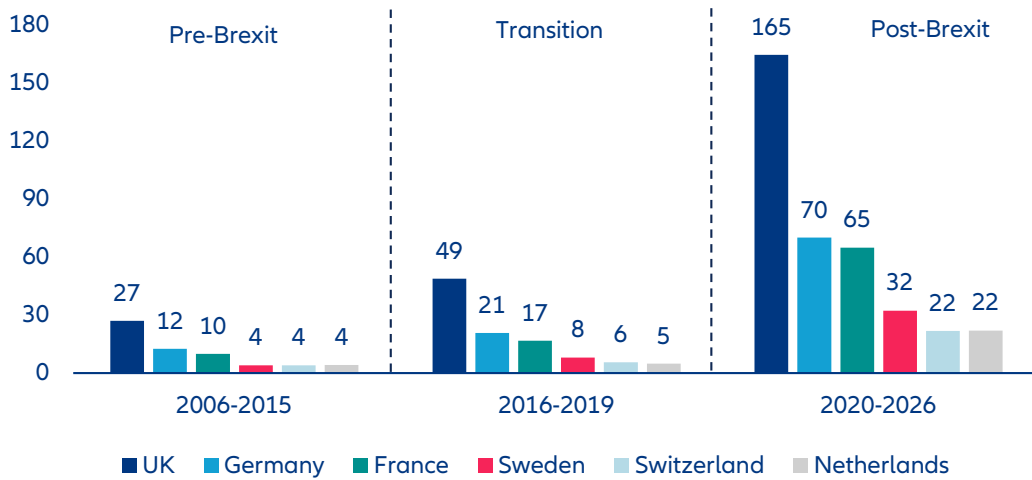
Sources : Pitchbook, Allianz Research

Notes: European PE deal value as a share of annual total by region.

The exit channel is the binding constraint, and policy has to be what clears it. Exit values have contracted sharply from the 2021-2022 peak and have been slow to recover, with disposals lagging deal formation and the ratio of PE-backed companies to exits widening to historically elevated levels, a backlog that accumulated across the rate-shock years and has yet to clear. The public listings market has been a structural drag, LSE has shed net listings across the post-Brexit decade, PE-backed IPO activity has been episodic rather than sustained and the equity re-rating visible in the FTSE 100 has yet to translate into a reliable exit route for private sponsors. Tax and domicile changes have added a directional headwind to manager incentives, reinforcing the pull of competing jurisdictions. Against that, the policy response is the most substantive in a decade, the Mansion House Accord targets GBP50bn of pension capital into private markets by 2030, PISCES provides an intermittent secondary venue short of a full listing and FCA deregulation is progressively lowering the post-Brexit compliance burden. Tentative green shoots are already visible in the exit data, PE exit value reached GBP20.9bn across 131 transactions in the first four months of 2026, pacing toward a record year by count, but the recovery is being led by buyouts rather than public listings, so the IPO channel most central to London's re-rating has yet to reassert itself.

The UK remains the most dynamic start-up ecosystem in Europe but the gap has narrowed since 2020. As Figure 21 shows, it still towers over Europe, drawing some USD 164bn of venture capital over 2020–2026, about two-and-a-half times Germany (68) or France (65) and roughly 44% of the six-country pool. Yet the UK's relative margin has quietly thinned since the referendum: its share of that pool eased from a 2016–2019 peak near 46% back to about 44%, and its lead over France narrowed from roughly 2.7x to 2.5x, as faster-growing rivals erupted around it. France and Germany remain the two heavyweight challengers, but the more telling shift is the rise of a second tier: Sweden, Switzerland and the Netherlands each pulled in between five and eight times more capital after 2020 than across the entire pre-Brexit decade (2006–2015) — Sweden, up eightfold to EUR32bn, has now overtaken both Switzerland and the Netherlands (22 each) to become Europe's third-largest hub. In short, Brexit left British innovation magnetic but no longer singular: the Continent is building its own pools of attraction, and the UK's edge, while still commanding, is slowly compressing.

Figure 20: Venture capital investment in UK and top European peers between 2006 and 2026



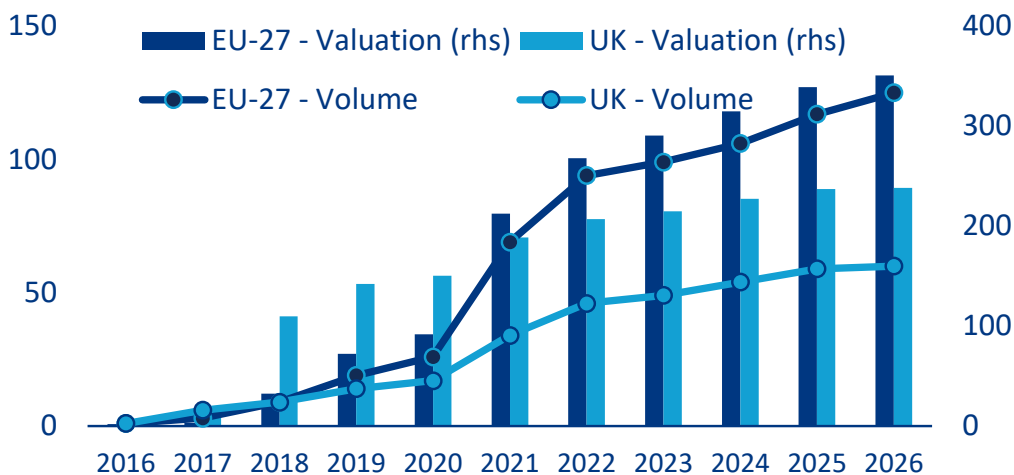
Sources : Dealrooms, Allianz Research

UK market dynamism hides a double concentration bias.

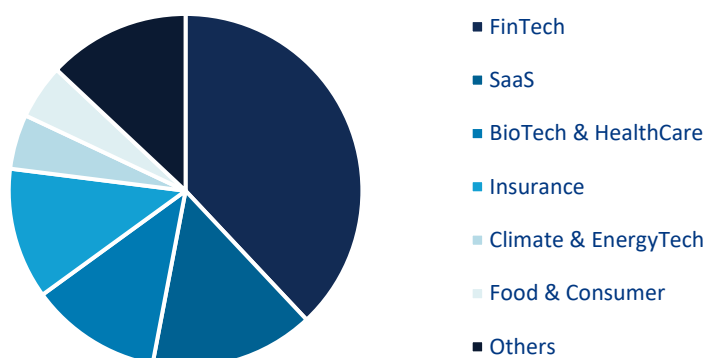
With 60 active unicorns, the UK is the fourth largest market worldwide behind the US, China and India. However that fleet leans on a handful of sectors. Fintech alone makes up roughly 40% of the pack, with software, health and insurtech filling most of the rest; the deep-tech and frontier layers that build lasting advantage barely register. And value clusters even tighter than companies do – three names carry around 25% of capital raised over the past decade and initially 40% of the entire pool valuation. That

fragility showed when the post-2021 sell-off hit. Having bet so hard on fintech and software, the very sectors that re-rated most brutally, the UK took the heaviest blow in Europe, unicorn fleet valuation being estimated in Q1 2026 at just 0.77x its last-round value, the most stressed major market, against a sturdy 0.97x in Sweden and 0.87x in Germany.

Figure 21: Venture capital investment in UK and top European peers between 2006 and 2026



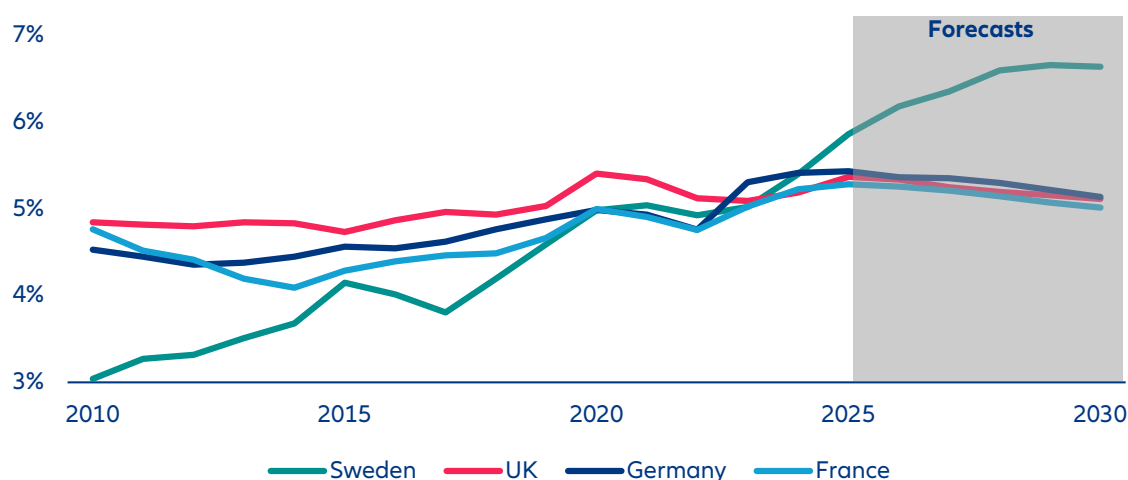
Sources : CBInsights Allianz Research

Figure 22: Sector breakdown of current unicorn ecosystem in UK (per volume)

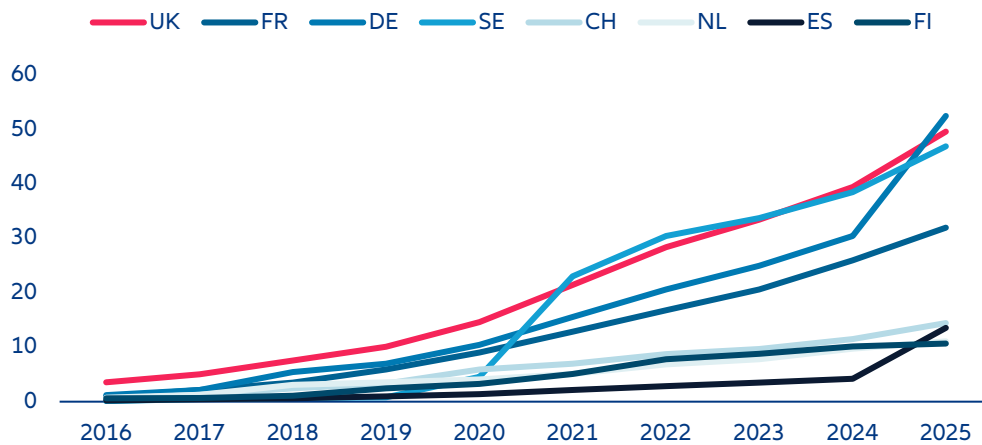
Sources : CBInsights Allianz Research

Diverging pattern: technology footprint is edging down in UK. ICT has grown into the UK's output more slowly than across the Channel — roughly 5.6% a year against 6% in France and 7% in Germany and Sweden — and the gap is widening, with Sweden breaking clear toward 6.5% of output by 2030 as Britain flattens near 5%. The causes compound. At home, finance, insurance and healthcare soak up the capital, starving the hardware and deep-tech that actually lift ICT's weight; the 2022 energy-and-inflation shock bit harder than on the Continent; and, cut off from EU structural funds, the UK now leans almost entirely on a private VC market that bankrolls software, not silicon. The deeper drag is structural: Europe's engineering schools feed a denser pipeline into

mechanical and frontier-AI work, while the UK's own innovation watchdog warns its science-and-engineering skills shortage is now structural rather than That is the crux, as the Cambridge Industrial Innovation Policy's UK Innovation Report concludes, closing the country's scale gap demands complementary capabilities like skills, supply-chain depth and manufacturing capacity, not simply more capital. And with no homegrown ICT champion — its highest-ranked tech firm sits just 389th in global R&D spending — there is no locomotive to pull a cluster along behind it.

Figure 23: Evolution of the weight of ICT sector into some European economies (based on gross output)

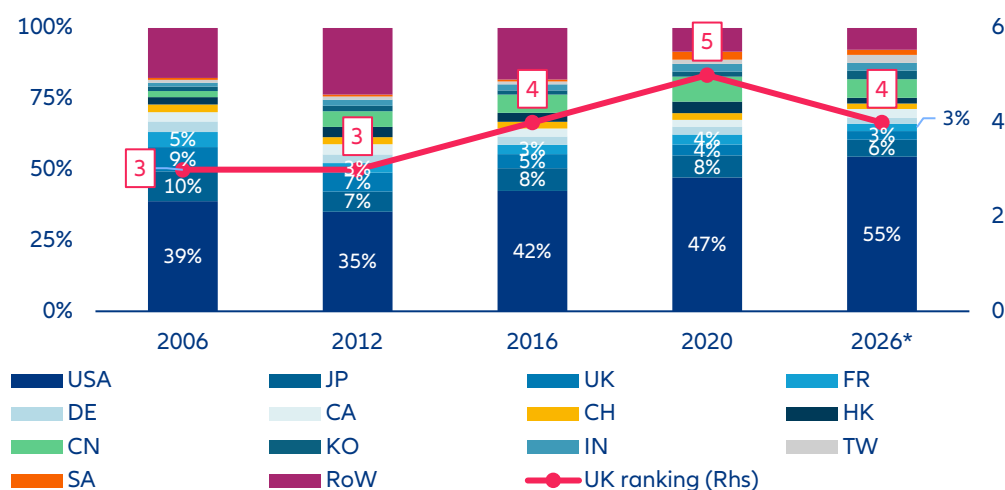
Sources : Oxford economics, Allianz Research

Figure 24: Cumulative private investment into AI firms in European countries since 2016

Sources :Center for Security and Emerging Technology (2026), Allianz Research

And the proof lands in AI itself. Narrow the lens to AI-specific capital and Britain's lead evaporates: Across 2020–2025 it drew around USD39bn, behind both Germany (USD45bn) and Sweden (USD46bn), with inflows growing just +21% a year lately against +24% in France and +37% in Germany – and a fast-rising Spain at +70%. The gap exposes where the UK sits in the value chain: strong in the materials layer – compound-semiconductor substrates, specialty chemicals – but absent from the parts that compound as comparatively to its European peers, the UK does not produce lithography machines or provide sovereign frontier-model and cloud services. London hosts among Europe's largest data-center fleets, but renting out the racks is not the same as owning the chips, models and clouds that fill them. That is the real verdict on Brexit-as-liberation: Far from a break-free moment to slip Europe's regulatory harness and chase the frontier, leaving the EU turned the UK inward — doubling down on the finance, insurance and software it already knew rather than the new domains where France, the Netherlands and Sweden are planting flags. The pandemic and the energy crisis weighed on the effort, but they cannot be the whole story, because the same storms crossed the Continent and its rivals emerged building, not retreating. The magnet still works – it is simply drawing capital into a narrower, more familiar field, while the frontier is mapped elsewhere.

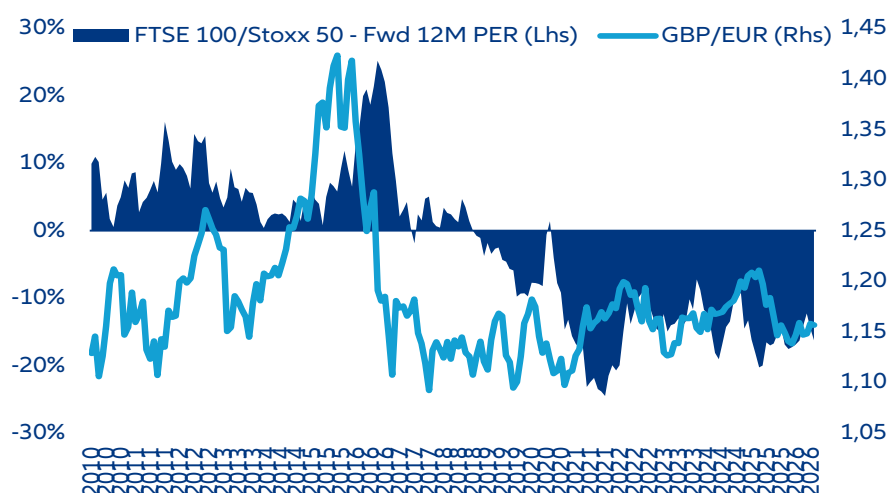
The UK stock market has also lost its shine. The same loss of altitude shows up in public markets. London remains the world's fourth-largest equity venue by market capitalization — behind the US, China and Japan — but that perch is now crowded, with South Korea, Canada and Taiwan each circling at roughly 3% share. Trading itself hasn't dried up: daily turnover has actually risen since the 2016 vote, to about GBP4.8bn from GBP3.8bn, though that merely tracks a global pick-up in equity activity that ran hotter still in the US and parts of Europe. What has unmistakably shrunk is relative weight. The UK's share of global market cap nearly halved from around 8.5% in 2006 to 5% by 2016, then slid again to roughly 3% by 2026 — even as the US vaulted from 42.5% to 54.5% in the past decade alone. And British equities now trade at a 15–20% discount to their Eurozone peers, a gap that has barely budged in years.

Figure 25: Market share of largest stock exchange place per market capitalization

Sources : LSEG Datastream, Allianz Research

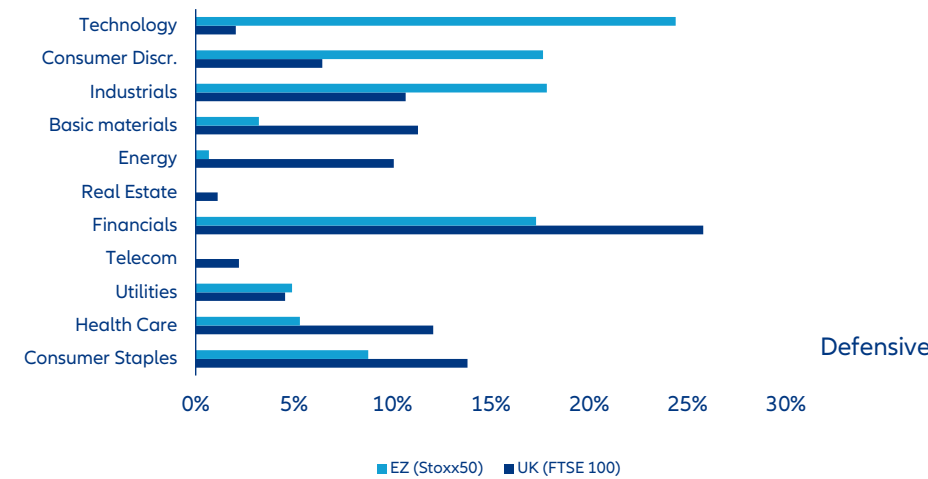
Persisting post-Brexit risk premia and low-technology profile burden stock attractiveness. The roots run deeper than Brexit — the slide was underway well before 2016 — but the divorce plainly didn't help. That lingering discount sits alongside a pound that never fully recovered against the euro or dollar, and a standing political-risk premium: global investors repriced UK assets downward after the referendum and never repriced them back. The bigger drag, though, is composition. The London index is structurally defensive — heavy in financials, energy, healthcare and staples, light on cyclicals — at the very moment markets have been relentlessly risk-on, powered by the AI trade. With technology barely 2% of the index, the UK simply lacks the exposure to ride that rally: its

make-up cushions the blows in volatile, risk-off spells but caps the upside precisely when, as now, the world is reaching for growth. The equity market, in short, mirrors the innovation economy above it – safe, familiar and increasingly on the wrong side of where the frontier is moving.

Figure 26: Relative valuation of UK equity toward Eurozone peers

Sources : Dealrooms, Allianz Research

Figure 27: Sector classification differences

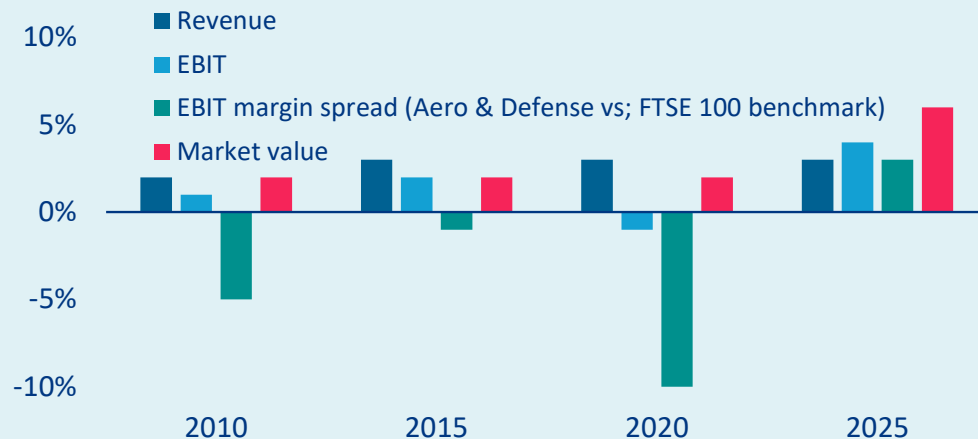


Sources :Center for Security and Emerging Technology (2026), Allianz Research

Box: Defense – Act or Lag

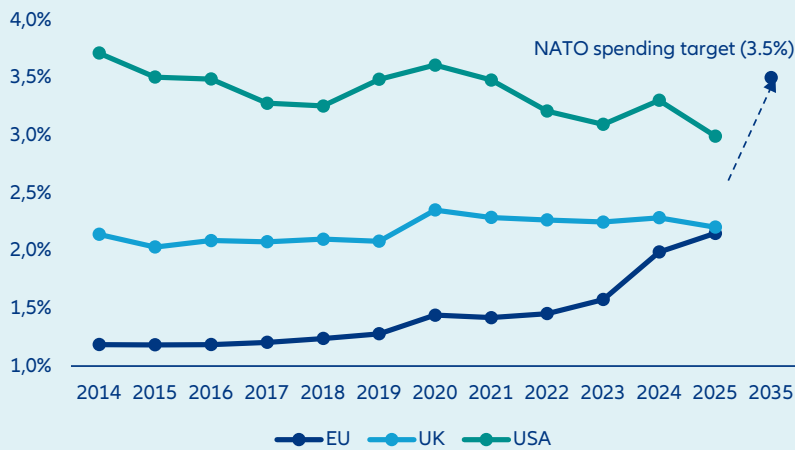
The Russia–Ukraine war handed European equity one of its scarcer assets — a genuine European growth story in which UK looks a lackluster against its peers. Continental names had a blockbuster 2025: the STOXX Europe Total Market Aerospace & Defence index returned roughly 68% over the twelve months into early 2026, taking the cumulative rally since Russia's February 2022 invasion to about 260%, with Rheinmetall — the rearmament proxy — up roughly 150% in 2025 alone and more than 1,000% above pre-war levels. Momentum has since cooled: the STOXX Europe Aerospace & Defence index is down ~1% year-to-date against ~5% for the STOXX 600 (late May 2026), and Rheinmetall sits about 38% below its January peak – a consolidation, not a reversal. The UK captures only a thin slice of that externality: core defense names in UK have roughly tripled their FTSE 100 market-cap weight since 2020, well ahead of the benchmark, but from such a low base that A&D stays a near-negligible index line. The US case is different again, and easily overstated as a Ukraine dividend: The S&P Aerospace & Defense Select index returned ~54% over the trailing year, comfortably ahead of the S&P 500 – but on an equal-weight basis that flattered smaller space and drone names) as much as giant domestic aerospace & aircraft. The marquee primes lagged: Lockheed fell ~13% over the trailing year and underperformed the S&P 500, leaving US primes at a discount to richly-rated European peers – a domestic budget-cycle trade (a ~USD1trn FY26 outlay) more than a war premium.

Figure 28: Weight of aerospace & defense core companies into FTSE 100 index benchmark (labeled in term of end fiscal 12-month trailing revenue, EBIT and market value)



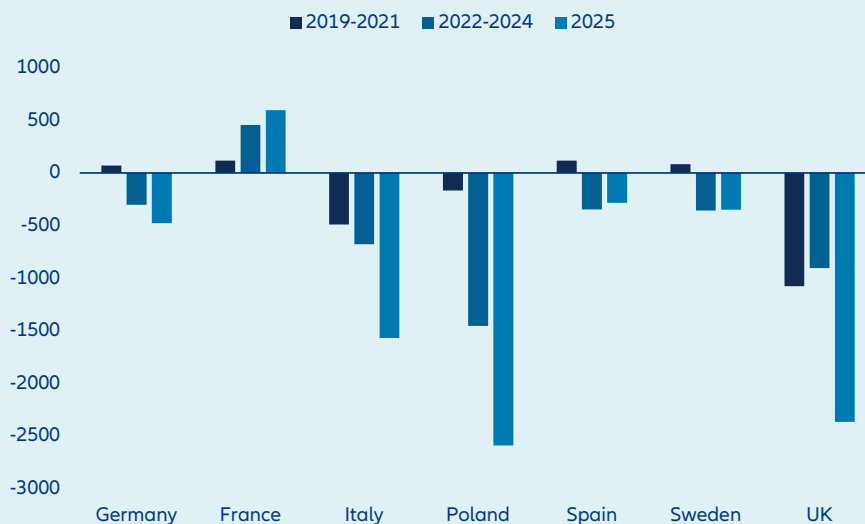
Source: LSEG Datatsream, Allianz Research

Figure 29: Evolution of military spending as % of GDP in EU, UK &



Source: NATO, Allianz Research

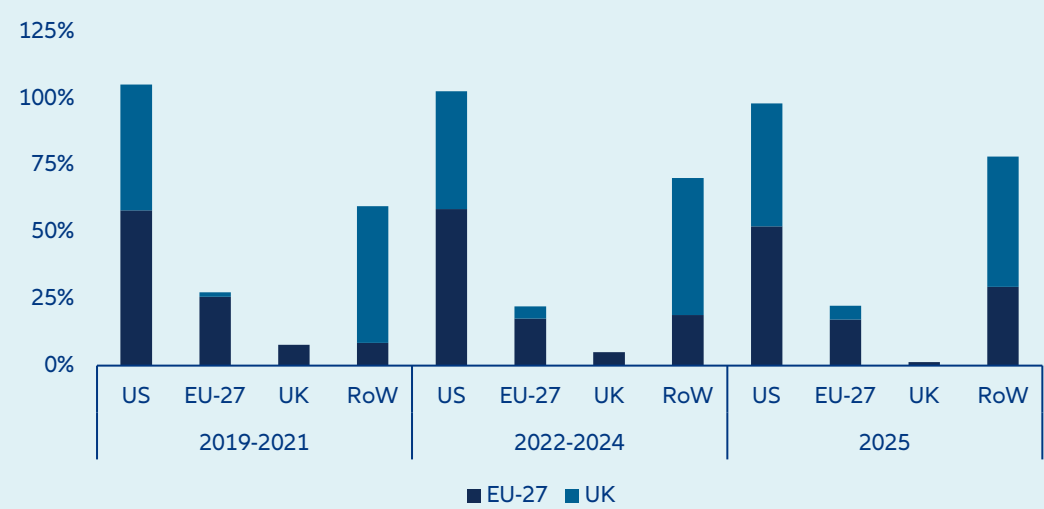
Figure 30: European countries trade balance with US in term of weapons and military equipment transfer (in mn of



Source: Sipri, Allianz Research

Neither war nor an American retreat drew Britain into Europe's defense fold. The chart tells it plainly: Across every window, including the 2022–24 war years, the UK's sourcing from EU-27 suppliers stays thin and by 2025 thins further toward the floor, while Britain remains a captive customer of US equipment, combat aircraft and munitions above all, for which it is Washington's only national "Level 1" industrial partner on the F-35, with roughly 15% of each airframe built in the UK. The corollary is a squandered windfall. Having opened negotiations to access the EU's EUR150bn SAFE rearmament facility after the May 2025 security pact, London walked away in late November over the entry fee and procurement terms — Brussels' demand reportedly ran to ~EUR6.75bn, even as Canada joined for around EUR10m — relegating UK firms to "third-country" status capped at 35% of any funded contract's value. And with the EUR150bn already parceled out to member states (Poland alone ~EUR44bn), that first-round ship has sailed. Global conflict handed Britain an open door to European defense-industrial integration and a multi-year demand pipeline to underpin its primes' earnings; it stayed welded to the dollar instead.

Figure 31: Regional distribution of arms & weapons imports by supplier breakdown (data labeled into millions of SIPRI trend-indicator values (TIVs))



Source: Sipri, Allianz Research

Stalling risks: funding constraints to hurt ambitious roadmap. On paper the trajectory is bold – 2.6% of GDP by 2027 en route to NATO's new 3.5% core target by 2035, 5% all-in — and the UK is embedded in a thick layer of partnership commitments, from AUKUS to the SAFE instrument and the trilateral GCAP fighter with Japan and Italy. In practice, a roughly GBP28bn funding gap and a Defence Investment Plan delayed several times have left those ambitions hostage to a Treasury unwilling to absorb more borrowing and a government wary that the tax rises needed to close the gap would render the agenda politically toxic. GCAP is the live illustration: lacking a long-term contract, the program has been running on a ~GBP686m bridge deal funding work only through end-June 2026, and frustration is rising in Italy and Japan that press UK government to offer funding guarantee. For the domestic base the signal is corrosive: immobilism today undercuts the case for capacity investment tomorrow and erodes the very argument primes need to attract capital – just as peers accelerate. The risk is not a reversal of intent but a creeping credibility gap between what Britain pledges and what it actually funds.

A photograph showing a group of diverse hands stacked on top of each other, resting on a tree trunk. The hands are of various skin tones and some have accessories like a black wristband and a silver ring. The background is a lush green forest. The text 'Our team' is overlaid on the image, with 'Our' in white and 'team' in yellow.

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Allianz Research | 18 June 2026

Public vs. private equity: The widest equity return gap in two decades

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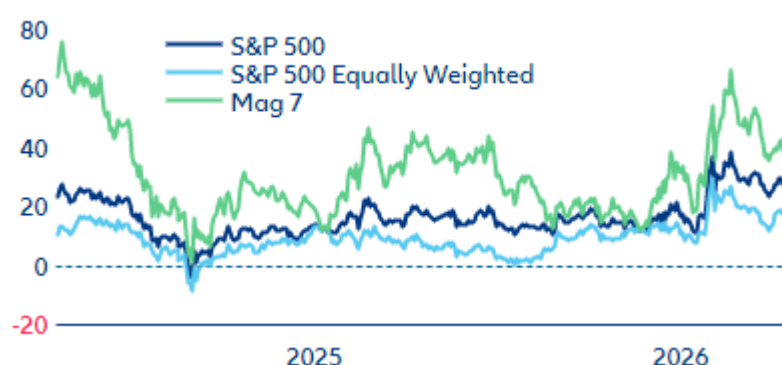
In Summary

- **Public equities have repriced the future while private equity is still monetizing the past.** Since the October 2022 trough, the S&P 500 has compounded at more than 20% a year for three straight years, while Private Equity buyout portfolios assembled at the 2020-2021 peak, and now carrying more expensive debt into a thin exit market, have lagged on every benchmark and horizon. At this point in time, the gap between liquid and illiquid equity returns seems to be the widest in two decades.
- **The rally that opened the gap is narrow and earnings-led, not a liquidity melt up.** The Magnificent Seven alone delivered more than half the S&P 500's three-year total return, strip them out and the index is an ordinary performer, and the equal weighted version has compounded at about half the pace, a gap unseen since the late 1990s. But the move rests on delivered earnings and a real AI capex super cycle rather than multiple expansion, so the premium looks more durable than the dot-com peak.
- **Yet private equity's long-run premium is real, this cycle has just interrupted it.** On a like for like basis, matched to public indices for geography, size, sector and leverage and net of fees and carry, buyout has beaten public markets over three decades, with MSCI estimating pooled direct alpha of roughly 400bps a year since 1994. The exception is the 2021-2023 vintages, which currently show a negative direct alpha of about ~800bps against the MSCI ACWI, the first consecutive run to trail public markets in the series. Consequently, the data points to rough PE performance ahead even allowing for the fact that these vintages are still young and J-curve distorted.
- **Private equity can no longer count on rising valuations and must now drive returns by growing the underlying business.** For over a decade, buyout returns came mostly from rising valuations, selling companies for a higher multiple than was paid. That is mostly gone as borrowing costs have roughly doubled and exit valuations have stopped climbing, so returns now come from growing the business. The contrast on the financing side is also large, with the largest listed companies still holding more cash than debt, so benefiting from higher rates, while buyout owned companies sit on floating rate debt due to be refinanced in 2026-2028, where higher rates only add to their interest bill.
- **As capital drains from private equity, private debt is catching the outflows.** The same jump in interest rates that made buyout deals more expensive made lending the more attractive trade. Big long term investors continue to like senior private credit because it pays a steady, contractual cash yield, its value barely moves and losses have been low, none of which buyout has delivered lately. So new money is rotating out of private equity and into private debt.
- **2026 will decide whether the public-private gap was a passing extreme or a lasting regime change.** Our central case assumes an economic mix that keeps returns positive but well below the post-Covid peaks with public equity settling in the low teens, while buyout follows in the mid-teens through 2027-2028. All in all, the way forward for private equity seems to be less about how much to own and more about how to own it. This means favoring managers and vintages with a demonstrable edge over generic exposure, treating liquidity and valuation discipline as features to be tested rather than assumed and resisting the temptation to buy the label without the underlying skill.

Public markets have sprinted ahead

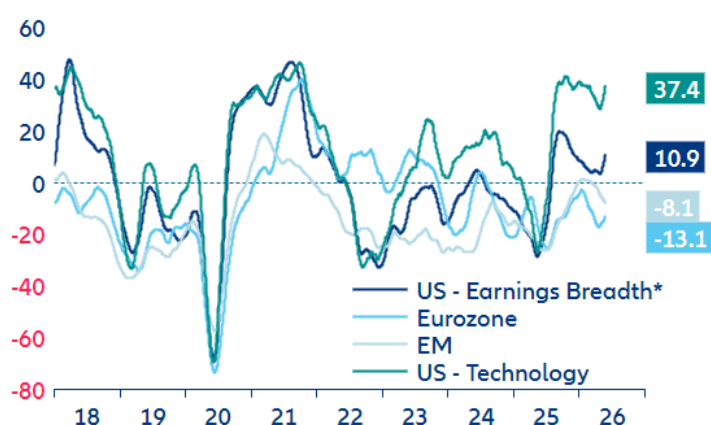
Since the October 2022 trough, public equities have staged one of the sharpest re-ratings in modern memory. The S&P 500 has compounded at more than 20% a year for three straight years, with other large-cap benchmarks close on its heels. But the headline flatters the market. The rally has been extraordinarily narrow: The Magnificent Seven (*Mag 7* - Apple, Microsoft, Alphabet (Google), Amazon, Nvidia, Meta (Facebook) and Tesla) alone delivered more than half of the S&P 500's three-year total return. Stripped of those seven names, the index is a thoroughly ordinary performer. The dispersion shows up in valuation, too. The top ten names now trade at a roughly 50% forward price/earnings premium to the rest of the index, wide by any historical yardstick. But that premium rests on an earnings path well above the dot-com peak of the 2000s, a sign that the fundamentals are more resilient now than they were then (Figure 1). Earnings breadth is mending as well (Figure 2), if grudgingly: Barely a third of S&P 500 members beat the index in 2025, up from a record low the year before. At the same time, small caps have scarcely shown up at all. The equal weighted cut of the same indices tells a more sober story still, with the S&P 500 equal weight compounding at about half the pace of its capitalization weighted parent, a gap unseen since the late 1990s. All in all, what reads at the index level as a broad bull market is, mechanically, the re-rating of a very small number of balance sheets.

Figure 1: Equity performance y/y%



Sources: LSEG Datastream, Allianz Research

Figure 2: Earnings breadth (3mma)



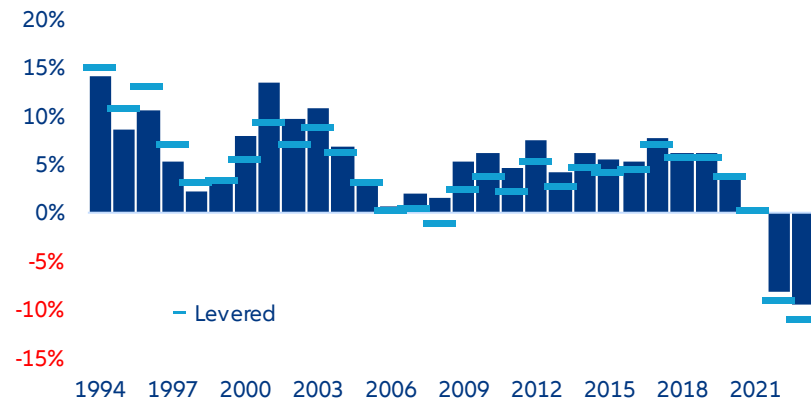
Sources: IBES, LSEG Datastream, Allianz Research

Note: Earnings breadth measures the share of index constituents receiving upward earnings estimate revisions.

The scale of the recent gap is easier to appreciate against private equity's long-run record. On a like-for-like basis, comparing each buyout fund with a public index matched for geography, size, sector and leverage, and measuring excess return net of fees and carry, private equity has, in fact, beaten public markets over the past three decades. MSCI estimates pooled buyout direct alpha of roughly 4% a year since 1994. That long-run premium is precisely what the current cycle has interrupted. Broken out by vintage, buyout funds delivered positive alpha against global equities in almost every cohort since the mid-1990s. The exception is the 2021-2023 vintages, which currently show

direct alpha of roughly -5% to -12% against the MSCI ACWI, the first run of consecutive vintages to trail public markets in the series. Of course, those readings come with an important caveat since the funds are young, and early fee drag (the J-curve) mechanically depresses returns that typically recover as portfolios mature, so those returns are provisional rather than final. But even allowing for that, the message is consistent: The vintages bought into the cheap-money peak are, so far, the worst relative performers in a generation and may still lead to some struggling years ahead (Figure 3).

Figure 3: Buyout direct alpha vs. public equity by vintage (in %)

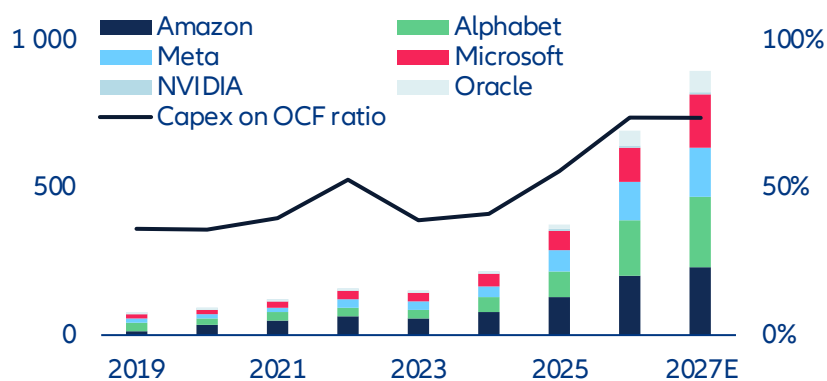


Sources: MSCI Private Capital Universe; Lester, O'Shea & Warren (2025), Allianz Research

Note: Direct alpha is the annualized excess return of buyout funds over a leverage- and sector-matched public index, net of fees and carry. 2021-2023 vintages are shaded to indicate they remain J-curve-distorted and are not yet representative of since-inception performance; 2024-2025 are omitted as too young to have generated representative exit data.

The character of the move matters as much as the magnitude. This has not been a liquidity-driven melt up of the 2020-2021 variety. It has been an earnings- and margin-led re-rating built on top of a genuine capex super cycle (Figure 4). Along these lines, forward earnings for the S&P 500 are up by roughly a third since the 2022 trough, and the bulk of the index's price return is explained by delivered earnings rather than multiple expansion. That distinction matters for how we read the gap to private markets performance (Figure 5). Public equity has not simply re-priced risk, it has also re-priced a specific, identifiable growth thesis around AI infrastructure, hyperscale compute and the operating leverage of asset-light platforms. Private equity, holding a portfolio assembled for a different regime, has had no comparable thesis to ride, and software, one of its biggest post-Covid allocations, has suffered from major potential disruptions. The second feature worth flagging is what the rally has done to the investable opportunity set itself. Free float in the largest names has shrunk as buybacks have accelerated and insider holdings have stayed concentrated, even allowing for the supply that the latest mega-IPOs have brought back. The effective supply of mega-cap equity available to incremental buyers is therefore tighter than headline market capitalization suggests (Figure 6). At the same time, the US now has roughly half as many listed companies as it did at its late 1990s peak, and the firms that do go public are older and larger than they used to be. So when a private equity sponsor comes to sell a company today, the public market it can exit into is smaller, more concentrated in a few large names and pickier about what it will absorb than the market sponsors were counting on when they bought in over 2018–2021. Consequently, the gap between public and private valuations in that light does not look like a temporary mispricing that will simply close, but a lasting result of a deeper shift in where growth capital is raised and how it is priced.

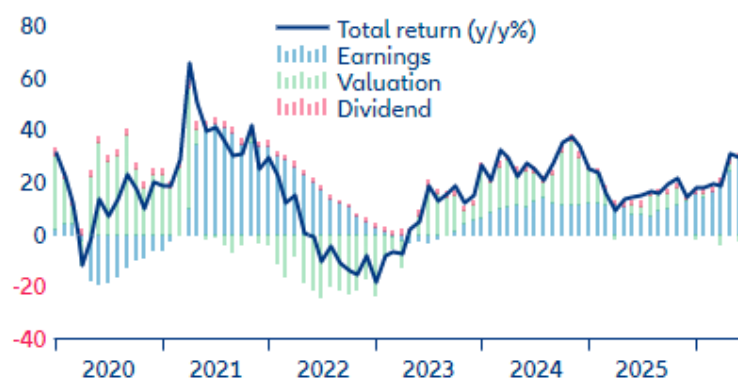
Figure 4: AI capital expenditure



Sources: Bloomberg, Allianz Research

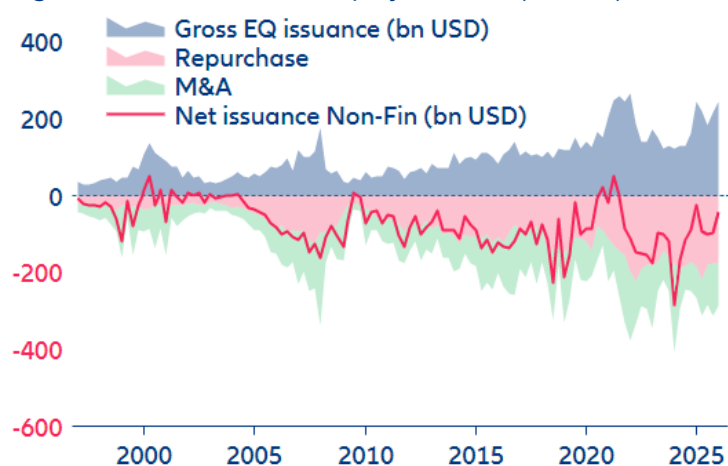
Note: OCF – Operating Cash Flow

Figure 5: S&P 500 return breakdown



Sources: LSEG Datastream, Allianz Research

Figure 6: US non-financial equity issuance (bn USD)

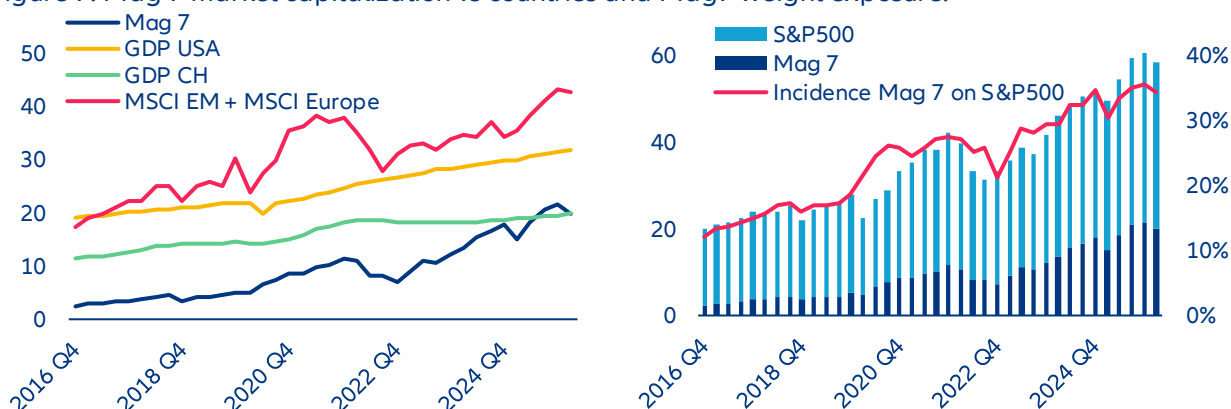


Sources: LSEG Datastream, Allianz Research

Public markets have absorbed private equity's growth engine

The narrowness of the public equity rally deserves closer inspection, because it is both the source of the divergence with private equity and a risk factor in its own right. The Mag 7 now represent roughly a third of S&P 500 market capitalization, up from around an eighth a decade ago, and their combined market value exceeds the GDP of every country except the US and China (Figure 7). Of course, the concentration is also self-reinforcing, because as these companies grow, passive and benchmark tracking flows mechanically direct more capital toward them, deepening their index weight independently of any fresh view on fundamentals.

Figure 7: Mag 7 market capitalization vs countries and Mag7 weight exposure.



Sources: Bloomberg, Allianz Research

Earnings concentration is also at extremes, and it now runs in parallel with the concentration in price. The same handful of mega-caps that dominate index weight also generate a disproportionate and rising share of S&P 500 earnings, so the benchmark's profits, not just its market value, increasingly depend on a single cohort. On valuation, the index trades well above its long-run average forward multiple, but the premium is almost entirely a mega-cap phenomenon; strip out the largest names and the rest of the market sits much closer to its historical norms, leaving a two-tier index in which the most expensive part is also the largest part (Figure 8). Because so much of the index now sits in long duration growth names, the multiple is also unusually sensitive to moves in rates and growth expectations; a small shift in either can swing the level materially. As an anchor for that sensitivity, a reversion of the cluster's valuation premium to its long-run average alone, before any change to underlying earnings, would cost the S&P 500 close to 10-20% of its value.

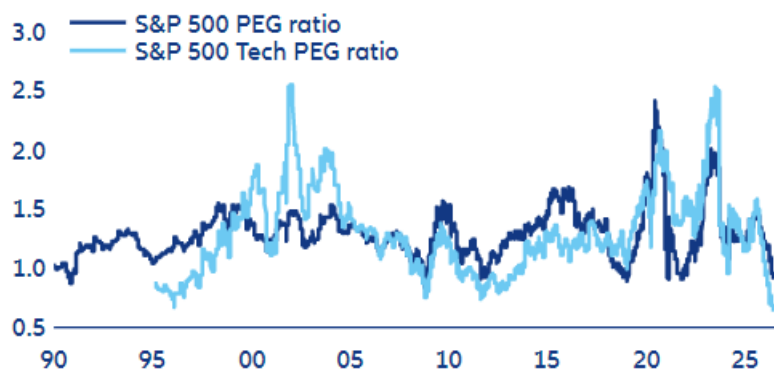
Figure 8: S&P 500 12m forward PE ratio



Sources: IBES, LSEG Datastream, Allianz Research

The same concentration invites two sharply opposing readings, and the contest between them is itself a market risk. The bullish case argues that the Mag 7 are nothing like the 2000 dot-com leaders as they generate hundreds of billions of dollars of free cash flow a year between them, sit on net cash balance sheets and dominate the decade's most powerful secular trend: AI compute, cloud infrastructure and advertising-data networks. Consensus forward earnings growth comfortably exceeds the broader market's, and on a PEG (Price/Earnings-to-Growth ratio) basis the cluster trades at parity or a discount. Nonetheless, the bearish case argues that the AI thesis rests on hyperscale capex sustaining at current levels, that enterprise AI revenue has yet to scale into the returns that capex implies and that the concentration itself is a source of fragility. An earnings disappointment from one or two names could de-rate the whole index in days. Both views can be right at once but which one prevails comes down to whichever is dominating the narrative at the time (Figure 9).

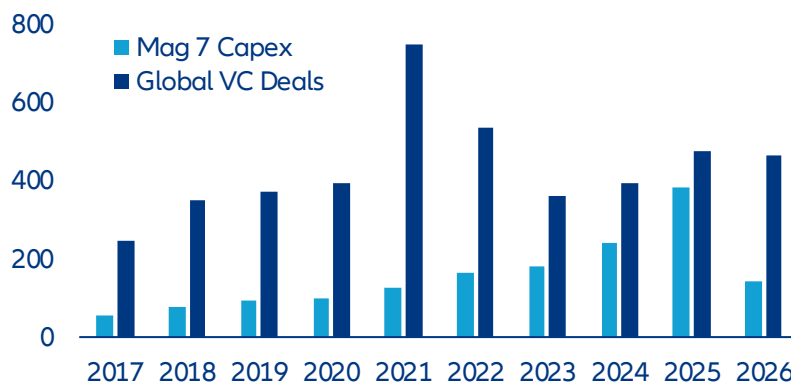
Figure 9: S&P 500 PE to growth ratio



Sources: IBES, LSEG Datastream, Allianz Research

Why the concentration matters for the public-private comparison is subtler than it first appears. The Mag 7 are themselves “long duration” growth assets, capex heavy and R&D intensive, with revenue streams that compound over a decade rather than monetize in a five-year LBO window. In effect, US public markets have absorbed the kind of growth equity exposure that used to migrate to private markets. Combined hyperscale capex now approaches the entire global venture capital (VC) deployment. Private investors who, in earlier cycles, would have funded the next generation of compute or AI infrastructure through late-stage venture rounds can now access that exposure through publicly listed mega caps, at any size and at low friction (Figure 10).

Figure 10: Mag 7 Capex vs VC deployment (USD Bn)



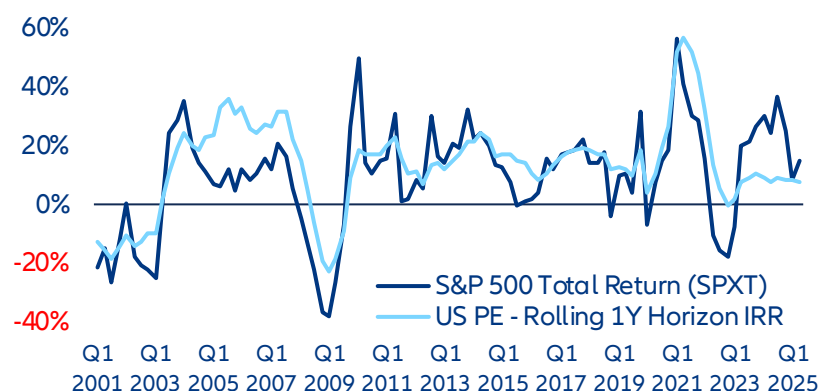
Sources: Bloomberg, PitchBook, Allianz Research

The corollary is uncomfortable for private markets. The universe of high-growth, scaled, profitable technology companies available exclusively to private investors has shrunk. At the time of writing, some of the private mega caps remain private holdings (though not for long, as SpaceX, Anthropic and OpenAI have IPO'd or filed to do so),

but the depth of comparable opportunity in private markets has narrowed precisely as fundraising and deployment has slowed.

Private equity, by contrast, is still digesting the previous cycle. The asset class entered the 2022 rate shock with peak fundraising, peak entry multiples, peak leverage and aggressive growth assumptions. The 2020-2021 vintages, in particular, were priced for a world of cheap money and continued multiple expansion, a world that ended within 18 months. Even top-quartile buyout funds returned single-digit IRRs in 2025, well behind both the S&P 500 and MSCI World, and the average vintage of the past decade barely cleared mid-single-digits (Figure 11).

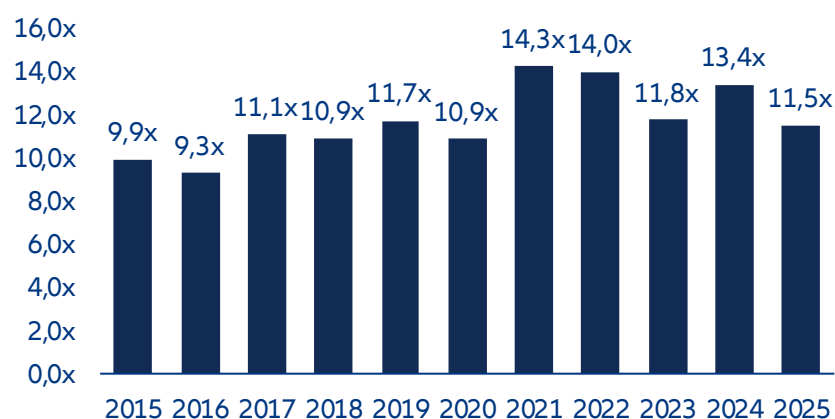
Figure 11: US Private Equity 1y IRR vs S&P 500



Sources: Bloomberg, PitchBook, Allianz Research

The lag has many causes, but the most mechanical is a reset in the cost of capital. A typical deal a decade ago cleared at around 10 times EBITDA, with half its capital in single-digit yield debt; today's clears closer to 13 times, with materially less leverage at far higher financing costs. Sponsors paid those record prices expecting to exit into further multiple expansion. That expansion has not arrived, and because thinner leverage cushions the entry, the drag on equity returns runs deeper than the change in multiples alone would suggest (Figure 12).

Figure 12: North America buyout entry multiples

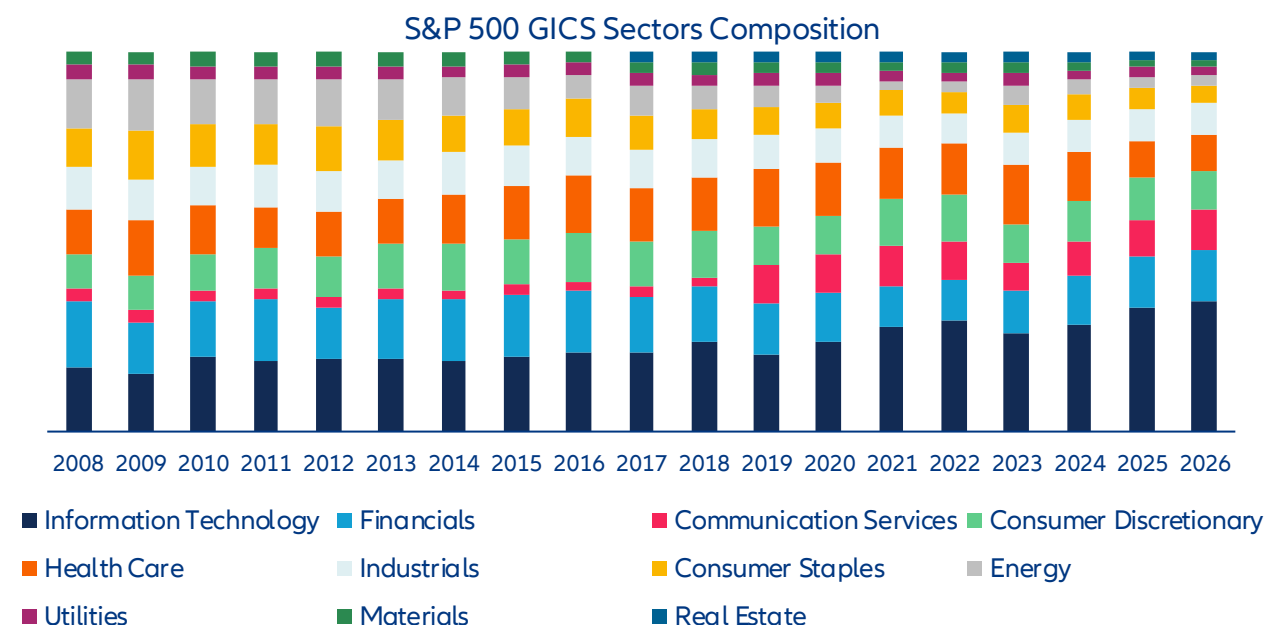


Sources: Pitchbook, Allianz Research

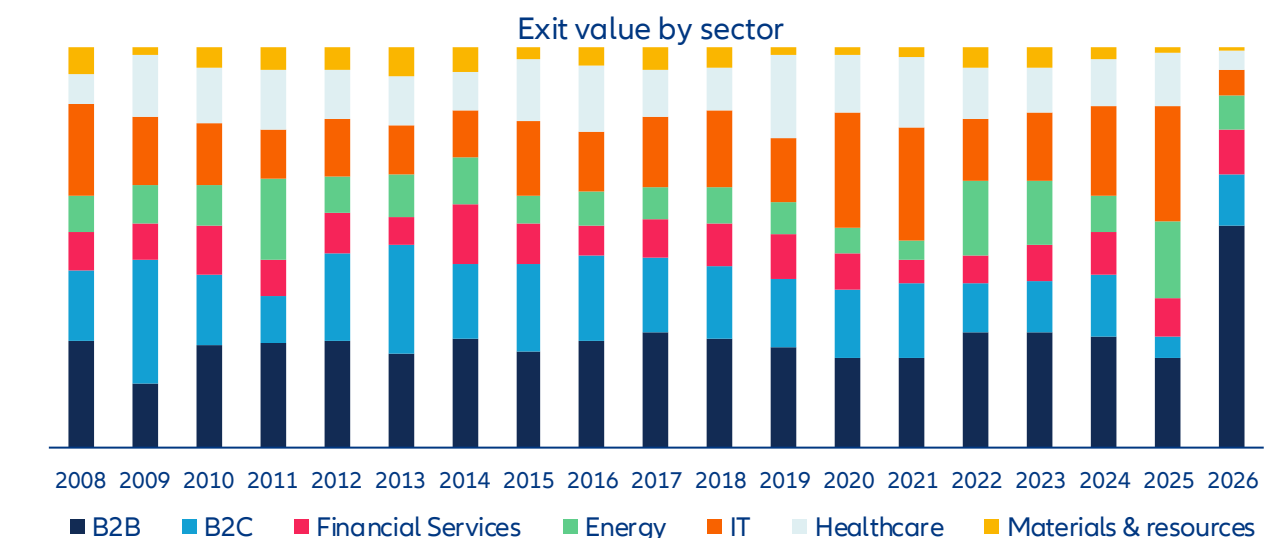
The mismatch is as much about sector mix as about timing. Public benchmarks are now dominated by mega-cap technology, semiconductors and communication services, the very cohort the AI capex cycle has re-rated, while financials, healthcare and industrials make up the balance. Private equity's deployment skews the other way, with software and technology being the single largest buyout sector for most of the past decade, peaking at roughly a third of deal value in the 2020-2021 vintages, with healthcare, business services and industrials filling out most of the remainder.

The catch is that the software the public market rewards and the software private equity owns are not the same exposure. Listed technology leadership is concentrated in hyperscale compute, AI infrastructure and platform businesses with net-cash balance sheets; the typical Private Equity-owned software asset is a mature, leverage-financed vertical or horizontal SaaS business bought on a high revenue multiple, precisely the cohort most exposed to slowing seat growth, AI-driven displacement risk and a higher discount rate. So even where the two universes appear to overlap on the word "software," they sit on opposite sides of the AI trade: public markets long the enabler, private portfolios long the potentially disrupted incumbent (Figure 13-15).

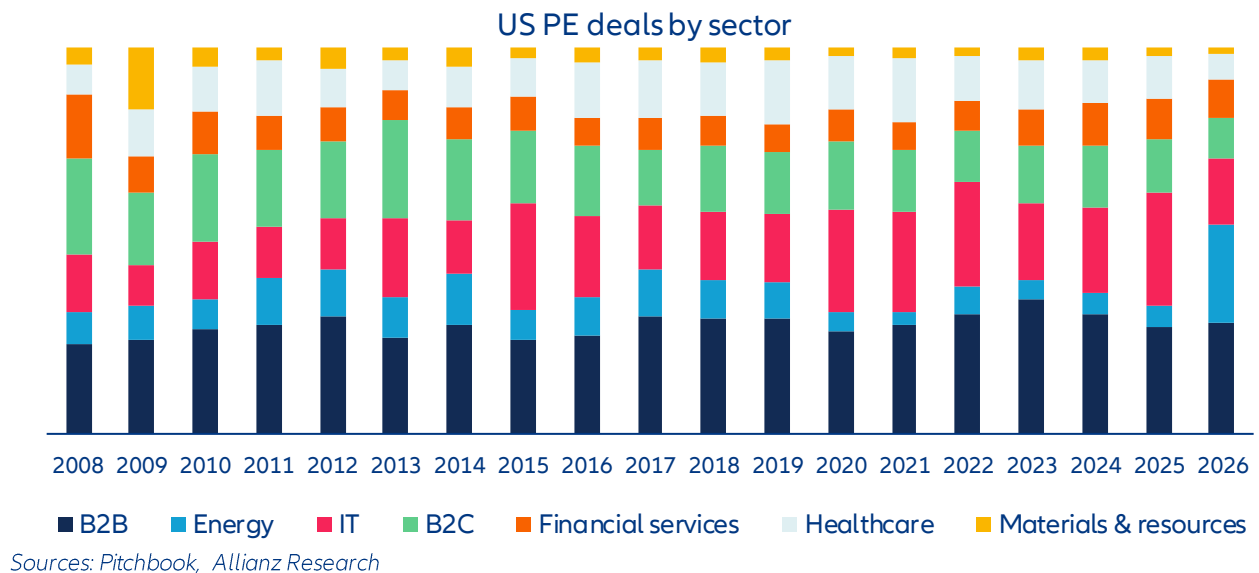
Figure 13-15: US Public vs Private EQ sector allocation (in %)



Sources: Bloomberg, Allianz Research



Sources: Pitchbook, Allianz Research



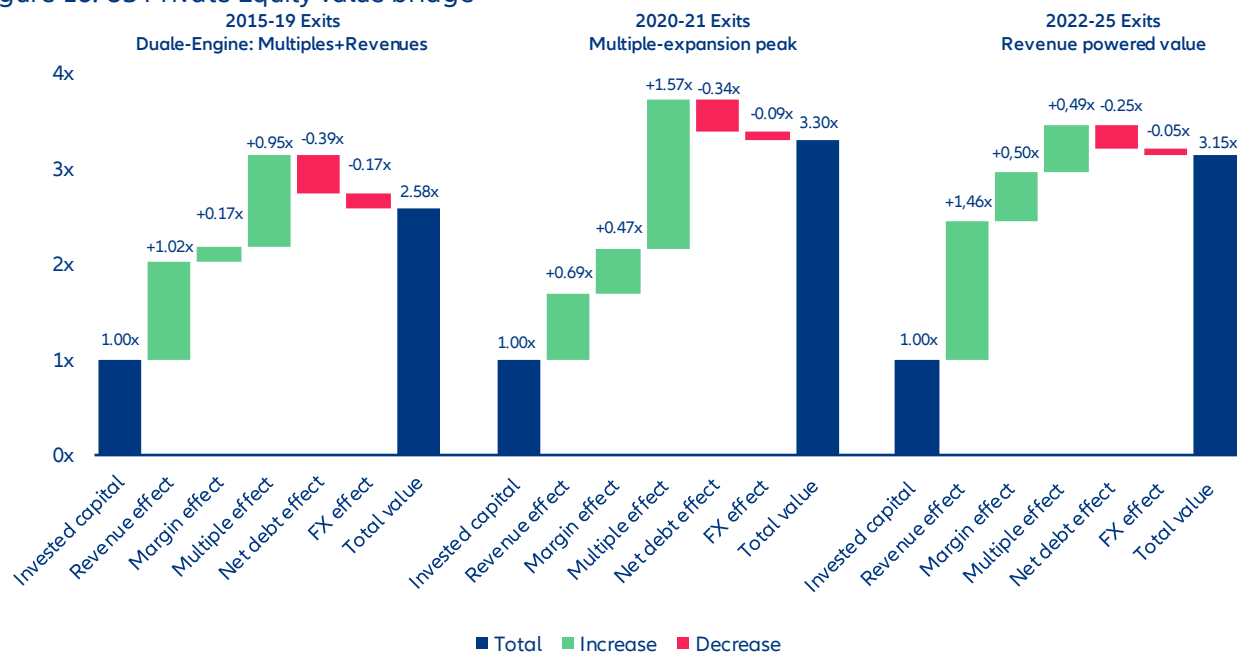
The Private Equity engine has changed, not just stalled

What ultimately separates public and private equity today is not how much they have returned, but what is driving those returns. Strip a buyout down to its parts and the value created comes from three places: the business growing its earnings, the multiple on those earnings rising between purchase and sale and the debt used to finance the deal. For most of the past decade, one of those did the heavy lifting. It was not, as is often assumed, leverage; across successive cohorts of exited deals, debt has been close to neutral and frequently a quiet drag once its rising cost is counted. The real engine was the multiple. Sponsors bought well, watched valuations climb and sold into a market that reliably paid more than they had, with growth in the underlying business playing only a supporting role.

That engine has stalled. Purchase multiples sit at or near record highs, so there is little room left for further re-rating, and exit multiples are no longer drifting up to reward patient owners. With cheap leverage gone and the multiple tapped out, only one source of value is left standing: the business itself. Revenue growth, once the junior partner, is now the dominant contributor to private equity returns, and the deals that work are the ones where the company genuinely grows rather than the ones where the multiple obligingly expands.

This is exactly where the two universes part ways. Public market returns are still, overwhelmingly, a re-rating story, but a narrow one, concentrated in a handful of net-cash mega caps that the AI trade has repriced upward. Private markets have been denied that same rerating, and the businesses sponsors own carry debt rather than cash, so their returns now have to be earned the hard way, one point of revenue growth at a time. Put simply, public equity is still being lifted by the market; private equity increasingly has to build its returns from the ground up. The bar has risen accordingly: only operationally well-versed managers in genuinely growing sectors can clear it (Figure 16).

Figure 16: US Private Equity value bridge



Data as of Q2 2025. Pooled value bridge across a subset of exited buyout holdings, weighted by invested capital.
Source: MSCI Private Capital Universe, MSCI Private Asset and Deal Metrics data

The exit market is where the Private Equity cycle's hangover shows most clearly. Buyout funds hold tens of thousands of unsold portfolio companies worth trillions of dollars, and median holding periods have stretched markedly. The starkest symptom is liquidity, the cash investors actually get back. Unlike a public equity, which can be sold on any trading day, a private holding returns capital only when the fund exits a deal, and that channel has narrowed sharply. Distributions to LPs (limited partners, the investors who fund the buyout) stayed below the 20% mark for a fourth straight year, the longest such stretch on record, and a depth last seen during the 2008-09 crisis. After three years of investors paying in more than they took out, net cash flow edged only modestly above breakeven in 2025, a change in direction rather than a true recovery.

The other exit routes tell the same uneven story. Deal and exit values rose sharply from a low base in 2025, but the rebound was narrow, driven by a handful of mega-deals while the mid-market stayed constrained. The IPO window has cracked open, but aftermarket pricing suggests public investors are still cautious on sponsor owned businesses, with Private Equity-backed listings tending to trade less buoyantly than venture backed ones. The secondary market has absorbed part of the gap, transacting a record volume in 2025 at a blended 87% of NAV (net asset value, the carrying value of the fund's holdings), with continuation vehicles taking a meaningful share, but it clears only a small slice of the unsold inventory each year. The plumbing is working again; the backlog behind it will probably take years to drain (Figure 17).

Figure 17: TTM US Private Equity buyout fund distributions as a share of beginning NAV



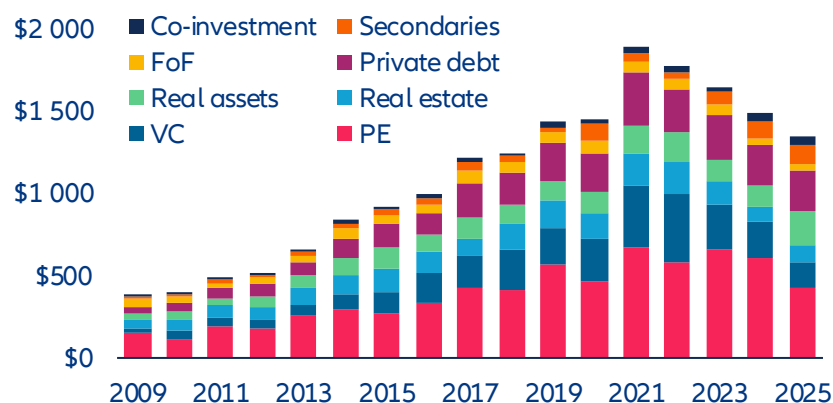
Source: Pitchbook, Allianz Research (Data as of Q3 2025)

Note: TTM = trailing twelve months. The line shows cash distributions paid to investors over the prior twelve months as a share of the fund's beginning net asset value (NAV); lower bars mean less cash is being returned.

Private credit has absorbed what private equity gave up

Private credit deserves its own section because it has been the structural beneficiary of the very forces that have pressured private equity. The same rate reset that raised LBO yields and made buyout math harder turned lending into the more attractive side of the trade, and as banks retreated from leveraged lending under regulatory pressure and sponsors still needed acquisition finance, direct lending stepped into the gap at unprecedented scale. Global direct-lending AUM has more than quadrupled since 2018 and is on track to keep growing through 2028. All-in yields on senior unitranche loans to mid-market, sponsor-backed borrowers remain in the low double digits, equity-like returns earned from a senior, secured position in the capital structure rather than the bottom of it. That combination is exactly what private equity has failed to deliver over the past three years: For long-duration insurers and pension funds, private credit has provided contractual cash yield, low mark-to-market volatility and a credit-loss experience that has stayed below long-run averages despite higher rates (Figure 18).

Figure 18: Private capital raised (USD bn) by strategy

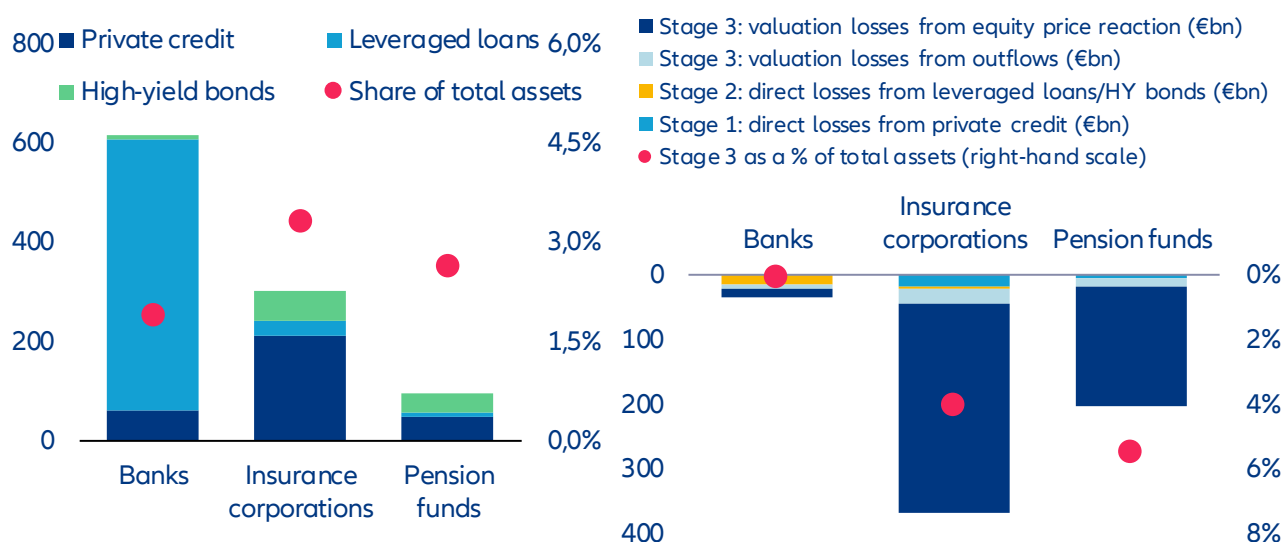


Sources: Pitchbook, Allianz Research

Two cautions are worth registering. First, private credit's default experience has been benign, with realized losses well below long-run averages, but the asset class has not yet been tested through a full credit cycle at current scale. The borrower base is also more levered than at any prior measurement point, and payment-in-kind (PIK) structures, in which coupons accrue rather than pay in cash, have grown sharply, suggesting some borrower-level stress that is being deferred rather than recognized. Second, private credit is increasingly intertwined with the insurance

balance sheets that own much of the asset class. The IMF and the ECB have flagged this as a systemic concern; when liquidity pressures emerge, insurance regulators may impose forced de-risking, creating sudden price-discovery events in a market built on stable, marked-to-model valuations. Neither concern is a near-term catalyst. Both argue for portfolio construction that favors senior secured exposure, granular underwriting and clear pricing discipline over chasing the highest-yielding mezzanine and second-lien strategies. (Figure 19-20)

Figure 19-20: Direct exposures of Eurozone entities to private credit and leveraged debt markets (left); Simulated losses for Eurozone financial entities from broad defaults in private credit and a severe shock in the software sector (right)



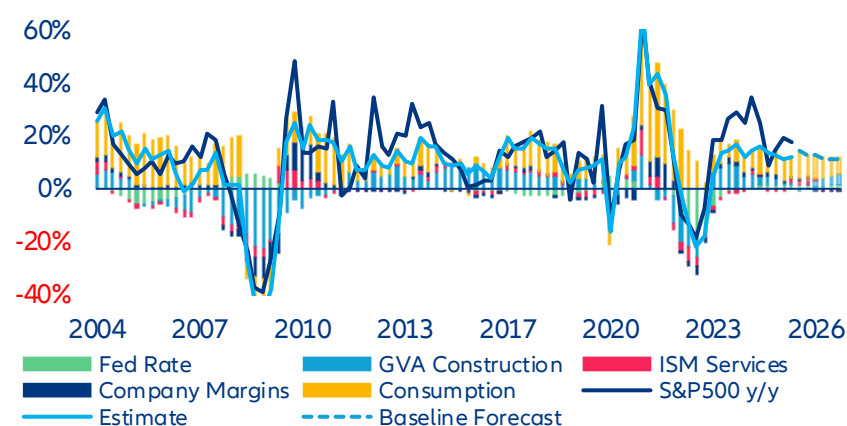
Sources: ECB, Allianz Research

The broader implication for private credit is that the asset class should be treated as a core, structural holding rather than a tactical reach for yield. The case does not depend on rates staying high. Even as financing costs normalize, the bank retreat from leveraged lending looks structural, the borrower base is largely captive and the asset's contractual cash yield and low mark to market volatility answer precisely the liquidity problem that has made private equity so hard to hold. For insurers and pension funds matching long-dated liabilities, that makes private credit a natural anchor allocation rather than a trade. But "core" tends to mean senior, not adventurous. The allocation belongs at the top of the capital structure, in senior secured exposure run by managers who can underwrite through a full cycle. Mezzanine, second-lien and opportunistic strategies still earn their place as return enhancers, but they are best treated more tactically, sized around that senior core rather than forming it. The suggested posture, in short, is a larger and more permanent allocation held more conservatively.

Outlook 2026: three drivers, three scenarios

The next 12 months will decide whether the 2022-2025 divergence between public and private market returns was a cyclical extreme or the start of a longer regime change. The central case assumes a prolonged but contained Middle East conflict, the Federal Reserve maintaining high rates and the AI capex cycle sustaining without a major guide down, a configuration that keeps returns positive across asset classes but well below the peaks of the post-Covid recovery. Under those conditions, public equity returns would be moderate into the low teens. Consumption and construction sentiment, the two biggest supports for the S&P 500 since 2010, hold up, while higher for longer rates shift from neutral to a modest drag. Private equity follows with a lag, with buyout returns settling in the mid-teens through 2027-2028 as financing conditions stabilize and the income flows that drive most of the variation in buyout returns recover from their 2023-2024 trough (Figure 21).

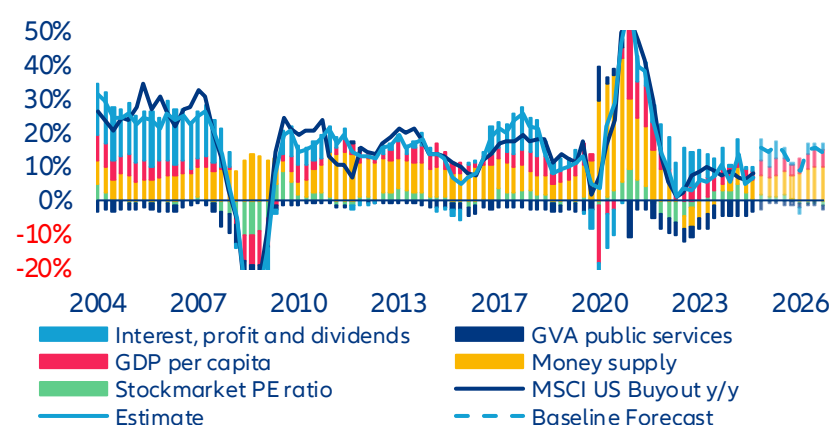
Figure 21: S&P 500 return decomposition and forecast (y/y%)



Sources: LSEG Datastream, Oxford Economics, Allianz Research

Note: Please note that these scenarios are produced using the Oxford Economics model and scenarios for comparability purposes. For official forecasts please refer to our quarterly scenario.

Figure 22: MSCI US Private Equity buyout return decomposition and forecast (y/y%)

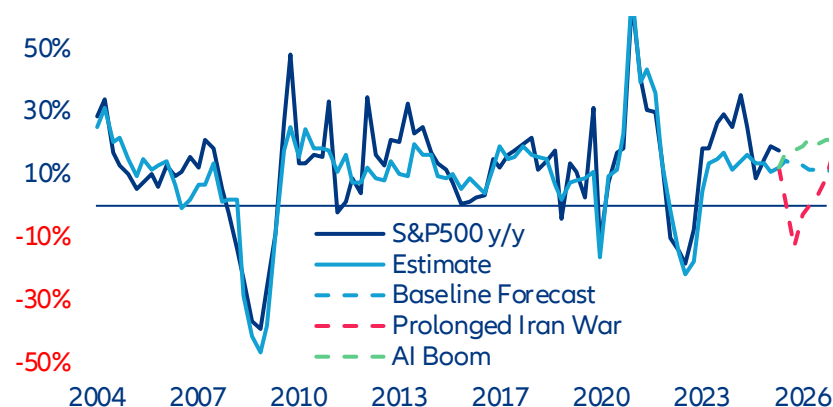


Sources: LSEG Datastream, Oxford Economics, MSCI, Allianz Research

Three drivers will decide which scenario plays out, each operating on a different horizon and through a distinct channel. In the near term, the resilience of the Middle East deal is the dominant variable. If oil and gas trade continue to normalize through the negotiated settlement, the path for both asset classes would look very different than if disruption to the Strait of Hormuz reignites beyond September. In that second case, a forced Fed tightening would turn rates from a modest drag into an active headwind for public equity. In this regard, consumption which is itself sensitive to rates through mortgage costs and credit availability, and the US consumer would enter any oil driven tightening from an already fragile position. The shock would also hit the real economy directly, with construction activity stalling as input cost visibility would collapse and commodity pass through would erode real

incomes. Under that path, public equity would turn negative in 2026, with buyouts following one to two quarters later as valuations reset with their characteristic lag (Figure 23).

Figure 23: S&P 500 scenario analysis (y/y%)

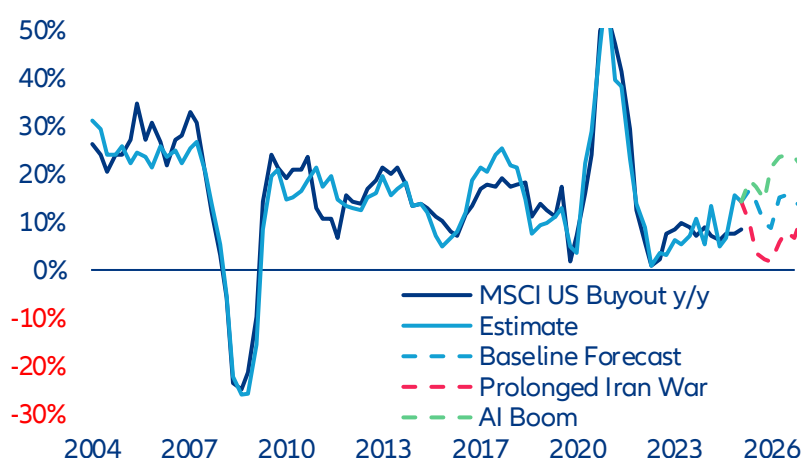


Sources: LSEG Datastream, Oxford Economics, Allianz Research

Note: Please note that these scenarios are produced using the Oxford Economics model and scenarios for comparability purposes. For official forecasts please refer to our quarterly scenario.

Over the medium term, the path of rates and the sustainability of AI capex are likely to become the binding constraints, and their interaction will determine whether the baseline recovery holds or extends. A higher for longer rate environment would delay the recovery of buyout profits, holding the downside path well below the cost of illiquidity in 2026. If AI capex were to guide down, whether through enterprise revenue disappointment, regulatory friction or balance sheet discipline, large cap tech margins would be hit first, and operating leverage means the damage would reach returns faster than the analyst revision cycle would suggest. From there it would propagate into buyout exit valuations. At that point a 10 to 15% public market de-rating is the single factor most capable of turning a baseline buyout year into the 4% downside outcome (Figure 24).

Figure 24: US Private Equity buyout scenario analysis (y/y%)



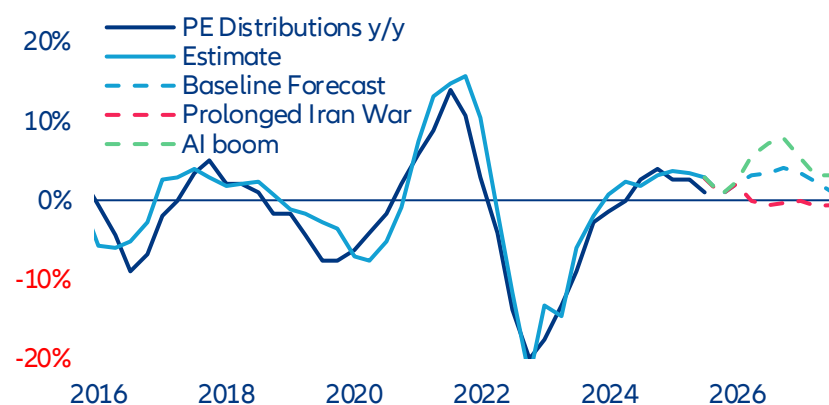
Sources: LSEG Datastream, MSCI, Oxford Economics, Allianz Research

The upside case requires both the near- and medium-term constraints to ease together, and when they do, the dynamics are mutually reinforcing in a way that produces outsized returns. In this scenario, with the Middle East conflict de-escalating after the June 2026 agreement, the Fed gains leeway and AI investment accelerates toward the 1990s capex cycle, with digital technology spending and business investment rising sharply. In that configuration the outcomes are assertive. Public equity peaks above 20% in 2027 as the consumption, margin and Fed rate drivers align at once, historically the setup that produces the sharpest single year public equity returns.

Buyout surpasses 20% shortly after as multiple expansion feeds through into exit valuations and money supply conditions ease, with the equity market PE ratio and the interest and profit rate drivers all contributing at the same time. Europe participates, but structurally less so. With the ECB's more gradual easing delivering less impulse than the Fed, Eurozone growth stays subdued and the AI preparedness gap, lower adoption capacity and smaller hyperscaler exposure mean the productivity gains accrue disproportionately to the US, compounding the transatlantic divergence of the post-2020 period.

The central message of the downside is not the magnitude of the return shock but its asymmetric transmission across asset classes. In the downside scenario, public equity would reprice immediately and sharply as margins and consumption deteriorate together under the stagflationary mix of input cost pressure and slowing growth, then would recover gradually as base effects and eventual policy normalization rebuild the factor stack. Private equity NAVs would adjust with a two to three quarter lag, technically positive in 2026 but close to the cost of illiquidity. The losses would be more contained in return space yet materially worse in liquidity terms, and nowhere is that gap more visible than in distributions. Historically, borrowing spreads and the CAPE ratio drive distribution activity, with rates the most persistent drag since 2022 and equity issuance, a proxy for IPO activity, the cyclical amplifier that has been conspicuously absent through the drought. The mega-IPO pipeline is where that factor becomes concrete. Successful mega-IPOs at premium multiples would lift the equity issuance driver, validate private AI marks across the buyout book and reopen the exit channel for sponsors holding inventory, converting years of paper gains into LP cash on a scale that dwarfs any prior exit event. Nonetheless, a collapse of the June agreement and renewed Iran escalation would break the sequence entirely, with spreads widening, the CAPE ratio deteriorating, rates staying higher and the IPO window closing before the queue clears (Figure 25).

Figure 25: US Private Equity distributions y/y abs change



Sources: LSEG Datastream, MSCI, Oxford Economics, Allianz Research

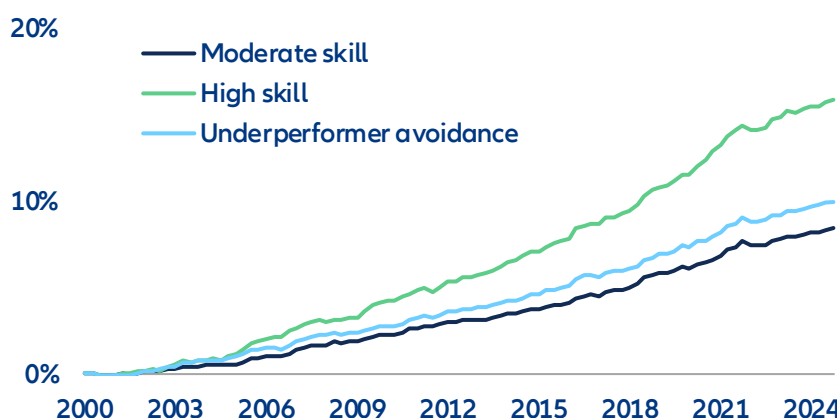
Note: Please note that these scenarios are produced using the Oxford Economics model and scenarios for comparability purposes. For official forecasts please refer to our quarterly scenario.

From beta to selection on both sides of the divide

All in all, the story is, at heart, a story about the fading power of beta. For most of the last two decades, simply being in the room was enough, public markets rode a long bull run and private equity compounded on cheap leverage and steadily expanding multiples. That era has closed on both sides of the divide. Public benchmarks now carry their returns on a narrow cohort of AI-exposed mega caps, while private equity is still working through vintages bought at the top of the cycle. What used to be a rising-tide question, how much to allocate to private markets, has become a question of which managers, which vintages, and which exposures. The center of gravity has shifted from beta to selection.

That shift is not a matter of conviction but of arithmetic. The dispersion between the best and worst private equity managers has widened to the point where the median fund tells you very little, and the returns that justify the asset class increasingly sit in its upper reaches. Skilled selection, on the evidence, is now worth more than the allocation decision itself, but it is a genuine skill, not a default setting, and an allocator without a durable edge is as likely to give that premium back as to capture it. Beta has become a commodity, the scarce input is judgement (Figure 26).

Figure 26: Average Private Equity buyout cumulative total portfolio alpha



Sources: Pitchbook, Allianz Research

For investors looking forward, that reframes the central question. The debate worth having is no longer "public or private," but "differentiated or undifferentiated" within each, because passive index beta and generic, broad-exposure buyout have quietly become the same crowded trade, priced for a tailwind that is no longer blowing. The democratization of private markets, through evergreen and semi-liquid wrappers, only sharpens the point as it widens access to an asset class precisely as access stops being the thing that pays, and the funds that actually generate the premium remain the hardest to reach. The way forward is less about how much to own and more about how to own it. Investors should favor managers and vintages with a demonstrable edge over generic exposure, treating liquidity and valuation discipline as features to be tested rather than assumed, and resisting the temptation to buy the label without the underlying skill. The next market correction will be the first real test of which private market structures were built to endure and which were built to sell, and the distinction between robust and fragile will become visible quickly. The alpha, on both sides of the liquidity divide, now belongs to those with the conviction and the capability to select it.

These assessments are, as always, subject to the disclaimer provided below.

FORWARD-LOOKING STATEMENTS

The statements contained herein may include prospects, statements of future expectations and other forward-looking statements that are based on management's current views and assumptions and involve known and unknown risks and uncertainties. Actual results, performance or events may differ materially from those expressed or implied in such forward-looking statements.

Such deviations may arise due to, without limitation, (i) changes of the general economic conditions and competitive situation, particularly in the Allianz Group's core business and core markets, (ii) performance of financial markets (particularly market volatility, liquidity and credit events), (iii) frequency and severity of insured loss events, including from natural catastrophes, and the development of loss expenses, (iv) mortality and morbidity levels and trends, (v) persistency levels, (vi) particularly in the banking business, the extent of credit defaults, (vii) interest rate levels, (viii) currency exchange rates including the EUR/USD exchange rate, (ix) changes in laws and regulations, including tax regulations, (x) the impact of acquisitions, including related integration issues, and reorganization measures, and (xi) general competitive factors, in each case on a local, regional, national and/or global basis. Many of these factors may be more likely to occur, or more pronounced, as a result of terrorist activities and their consequences.

NO DUTY TO UPDATE

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Allianz Research | 17 June 2026

Southern Europe won the last decade. The next one is less obvious

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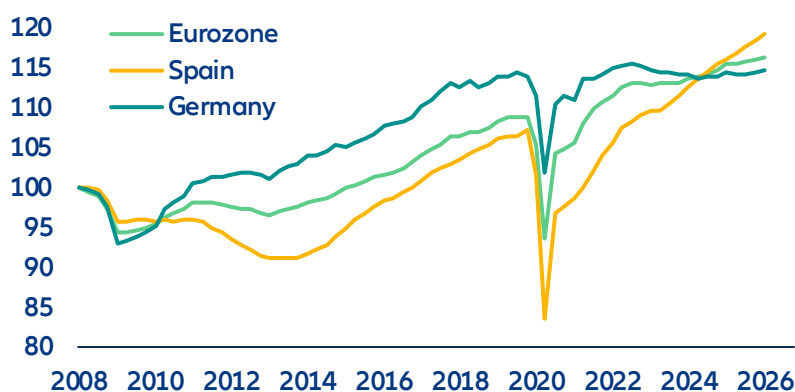
In Summary

- **Southern Europe has delivered an impressive turnaround this past decade, broader and more durable than many expected.** Spain, Portugal and Greece are running ~11% above pre-pandemic output, while Germany has stagnated for four consecutive years. Sovereign spreads that peaked above 250bps during the debt crisis have compressed to around 70. Southern banking systems, once the epicenter of Eurozone fragmentation risk, have largely completed their NPL clean-up (it is German and French banks that are now carrying rising asset quality pressures). This is not a cyclical blip but reflects an earned shift through labor market reforms, fiscal consolidation and the NGEU investment impulse – the largest coordinated public investment program the Eurozone has ever deployed. Spain's renewable buildout – now supplying more than half of its electricity – has added a further dimension: an industrial energy cost advantage that did not exist before 2021 and that has no equivalent in the North. The reversal is real but increasingly reflected in market pricing.
- **The growth model has delivered output convergence, but productivity – the engine of durable income catch-up – remains the unfinished business.** Employment-led recovery has run its course as the primary driver; Greece remains below its 2008 per capita income level. Sustaining the narrative through the next phase requires a more difficult set of structural changes: judicial efficiency, capital market deepening, R&D intensity and a completed Banking Union. The NGEU disbursements ending in 2026 sharpen that test: what phases out is not just a fiscal impulse but a reform discipline mechanism, and where implementation has been compliance-driven rather than institutionally embedded, the risk of slippage rises. The fiscal impulse is fading, with no clear replacement, and private cross-border capital flows remain too limited and episodic to fill that gap organically – the handover from public investment anchor to market finance is assumed but not yet visible in the data. These are not reasons to dismiss the story – they are the conditions that would extend it.
- **The political calendar and spread geometry call for selectivity.** France, Italy, Spain and Greece all face national elections in 2027, creating fiscal pressure points at the precise moment external conditionality fades and the ECB continues quantitative tightening. At around 70bps, BTP and Bonos spreads leave limited room for positive surprises and some room for temporary widening if credibility wavers. This is not a crisis configuration – sovereign balance sheets are in materially better shape than in 2011 and the ECB's Transmission Protection Instrument (TPI) provides a credible backstop – but it is a configuration in which episodic volatility is more likely than further compression. The region's reform record since 2012 is real and the underlying story has not exhausted itself. What comes next is more differentiated: returns will accrue to investors who discriminate by country, sector and reform trajectory rather than treating Southern Europe as a monolithic allocation.

The convergence scorecard: the wreckage of the 2010s and the post-pandemic catch-up

The Eurozone's defining fault line of the 2010s – the North–South divide – has materially narrowed over the past five years. The 2008 financial crisis and the subsequent Eurozone sovereign debt crisis produced a split of historic proportions. Between 2008 and 2013, Spanish GDP fell by roughly -9%, unemployment peaked at 28% and youth unemployment exceeded 55%. Greece suffered the most severe peacetime output contraction in a developed economy in the post-war era. Italy and Portugal followed trajectories nearly as damaging, albeit with somewhat better access to external support. In the years that followed, Mediterranean economies experienced only a sluggish recovery, while the core bloc (Germany, France, the Netherlands and Belgium) continued to expand, led by Germany's export competitiveness and manufacturing rebound, powered by surging global demand for automobiles and machinery in emerging markets.

Figure 1: GDP 2008=100



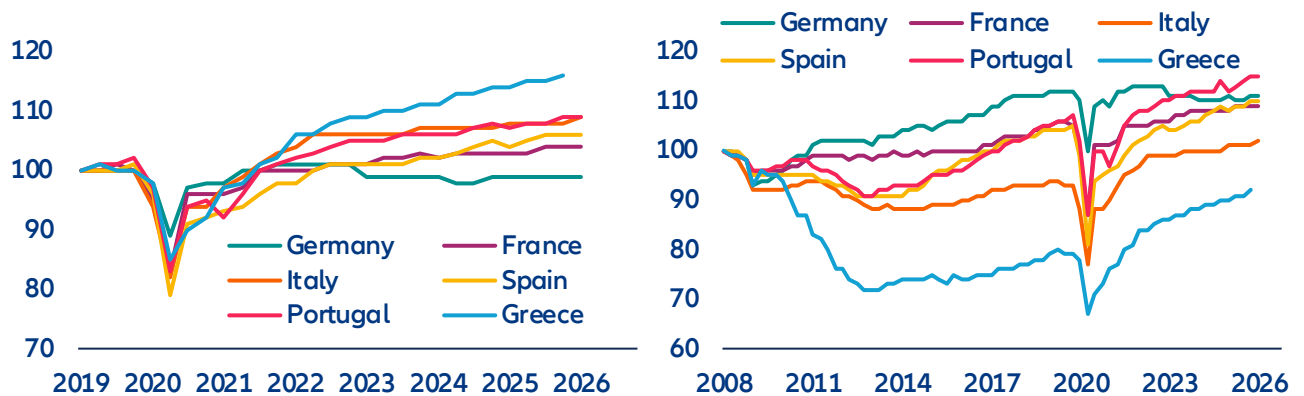
Sources: LSEG Datastream, Allianz Research

The post-2021 recovery marked a clear shift in the geography of Eurozone growth, with Southern economies moving from chronic underperformance to cyclical leadership. This shift reflects three reinforcing mechanisms. First, reforms implemented in the aftermath of the sovereign debt crisis – often under EU and IMF conditionality – improved competitiveness, labor market flexibility and external adjustment, laying the foundations for stronger employment and export performance. Second, the recovery has been strongly sector-biased, with contact-intensive services – particularly tourism – benefiting disproportionately. In Southern economies, tourism typically contributes between 10% (in Italy) and 20% of GDP (in Greece and Portugal) once indirect and induced effects are included. Third, the deployment of Next Generation EU (NGEU) funds supported public investment and credit conditions, amplifying domestic demand and accelerating structural upgrading, including through infrastructure and digitalization.

At the same time, Germany's growth model, built on cheap energy inputs and external demand, has been structurally disrupted. The loss of Russian gas, combined with weaker demand from China, has exposed vulnerabilities linked to underinvestment and slower technological adaptation, contributing to prolonged stagnation between 2022 and 2025. The resulting convergence combines catch-up and compression – a critical distinction. The narrowing of income gaps reflects not only structural improvement in the periphery, but also a rebalancing of relative growth across the Eurozone.

However, this convergence remains incomplete when assessed through a structural lens. While output growth has converged, convergence in GDP per capita – the more relevant measure of welfare – has been significantly more limited. Despite strong post-pandemic performance, longer-run comparisons point to persistent scarring effects from the sovereign debt crisis. Greece remained below 2008 per capita income levels in 2025, Spain has broadly closed the gap, while Portugal has exceeded it, reflecting sustained reform implementation and a strong tourism-led rebound.

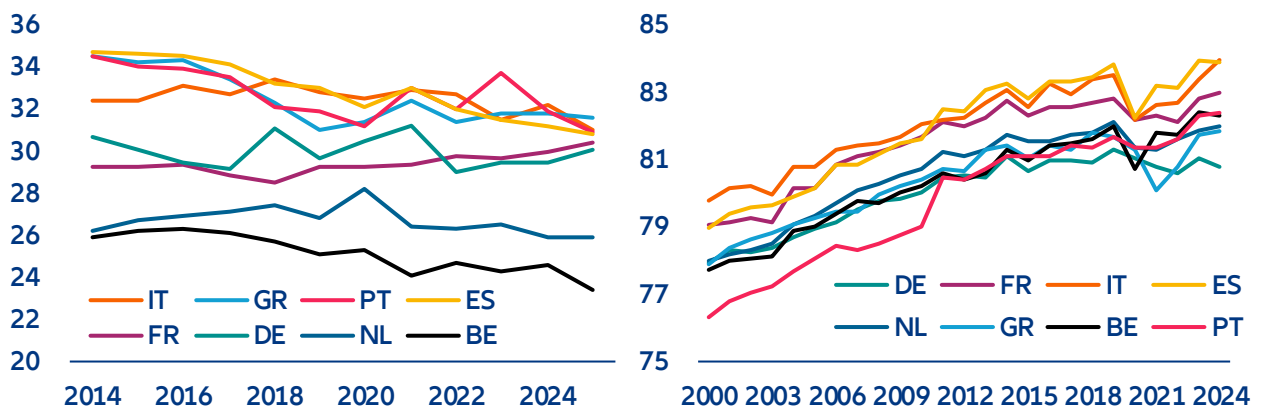
Figure 2: GDP per capita, 2019=100 and 2008=100



Sources: LSEG Datastream, Allianz Research

This divergence reflects the nature of the current growth model. Since 2021, growth has been predominantly employment-driven rather than productivity-driven. This has delivered rapid gains in output, but only partial convergence in income levels. The human dimension of the crisis decade remains visible. Southern Europe leads the EU in life expectancy yet carries persistently higher income inequality than its northern peers (Gini coefficients of 32–34 versus 25–30 in core economies), with recent gains that have not reached the bottom of the distribution. The current phase therefore risks resembling “convergence by slowdown” in the core as much as genuine structural catch-up in the periphery.

Figure 3: Gini coefficient and life expectancy



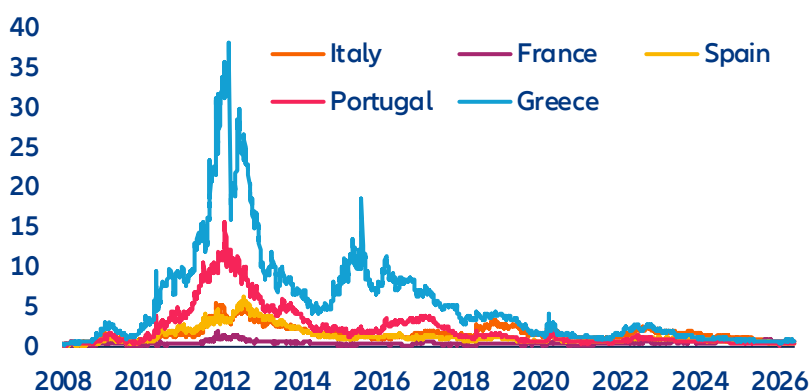
Sources: LSEG Datastream, Allianz Research

The capital market scorecard

Financial markets have broadly confirmed the convergence narrative. Italian 10-year spreads compressed from a peak of 250bps in 2022 to an average around 70bps in early 2026; Greek government bond spreads consistently trade tighter than Italy. EGB spreads are nowadays broadly back to the levels of 2008. NGEU has also been a key driver of this compression, with its combined impact on growth, fiscal balances and policy credibility accounting for a significant share of the observed convergence¹. Equity markets echo this: Italian, Spanish and Greek indices have all outpaced the Stoxx Europe 600 since 2022, with gains driven by value rotation and banking rehabilitation rather than a fundamental re-rating. Measured from the sovereign crisis period (2008=100), Southern European markets experienced a prolonged phase of underperformance followed by a gradual recovery.

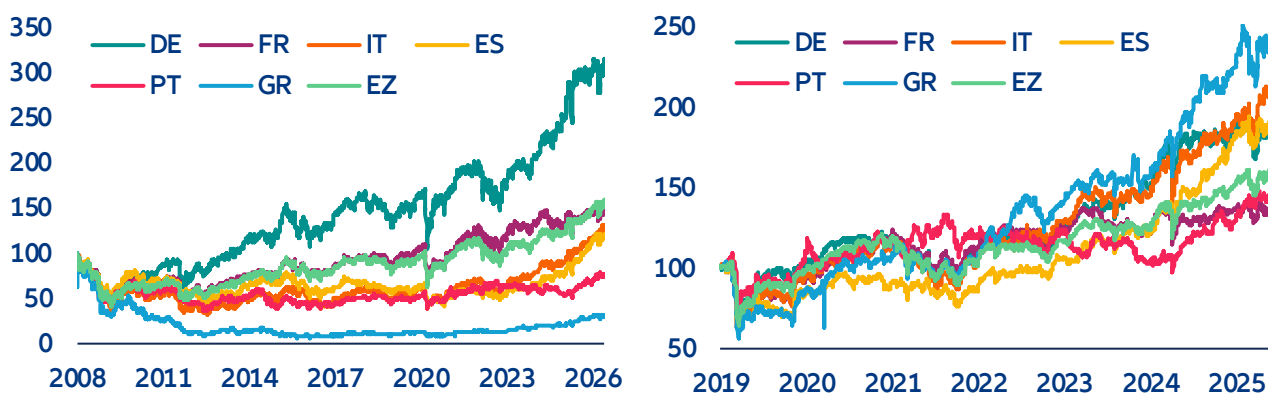
¹ [What to watch | November 14, 2025 | Allianz](#)

Figure 4: 10y government bond spreads over Bund



Sources: LSEG Datastream, Allianz Research

Figure 5: Equity index (2008=100, 31/12/2019=100)

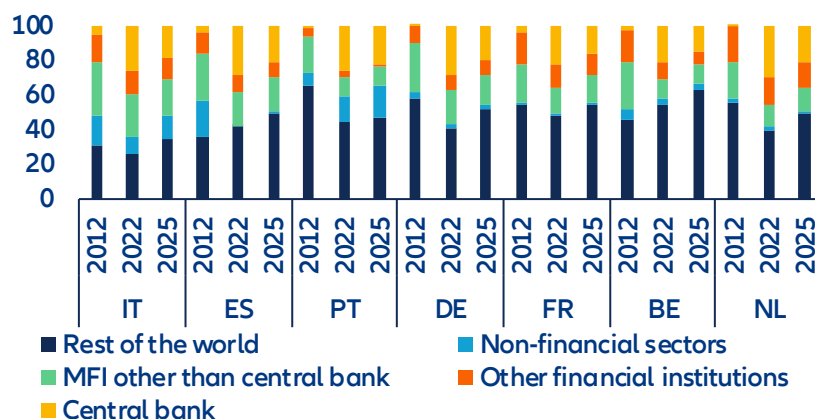


Sources: LSEG Datastream, Allianz Research

A substantial share of this re-rating reflects a catch-up from deeply discounted levels rather than a transition to premium valuation. The initial phase of financial convergence – the compression of crisis-era risk premia – has largely played out. Whether it becomes durable will depend on the transition to real convergence through productivity gains. Future performance will depend less on multiple expansion and more on the underlying drivers of earnings growth.

One important structural caveat: the ECB's gradual withdrawal as price-setter via QT means marginal sovereign funding increasingly depends on hedge funds and active managers with shorter horizons, making spreads more reactive to fiscal signals than during the QE era. For Italy this matters most – retail absorption capacity is approaching its limits, and marginal funding depends increasingly on investors with firmer fiscal conditions and shorter time horizons. The fact that spreads have compressed despite this shift is a testament to the improvement in underlying fundamentals elsewhere in the South; for Italy, it is a risk that remains latent.

Figure 6: Government debt holdings by investor type



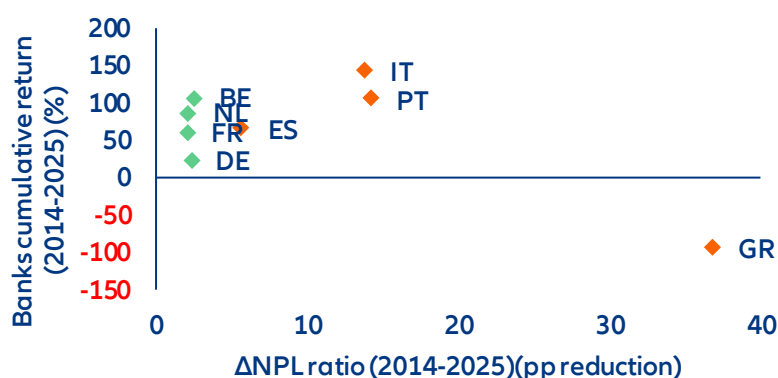
Sources: LSEG Datastream, Allianz Research

Banking sector: the structural inversion markets have not fully priced in

Banking sector repair is arguably the most underappreciated development in European finance over the past decade. A decade ago, Southern banks were the epicenter of Eurozone fragmentation risk: non-performing loan (NPL) ratios stood at 17% in Italy, 18% in Portugal and close to 50% in Greece. The subsequent clean-up – driven by ECB supervisory pressure under the Single Supervisory Mechanism, stricter provisioning and active NPL disposals – was slow and painful, but is now largely complete. By Q3 2025, all four Southern banking systems continue to reduce NPL volumes: Spain currently stands at 2.4%, Italy -2.7%, Portugal 2.6% and Greece 2.8%.

What makes this development particularly relevant is the direction of travel elsewhere. Germany and France are now experiencing rising NPL volumes, driven by commercial real estate and corporate sector stress. A decade after the sovereign debt crisis, it is core European banks – not Southern ones – that are carrying the asset quality problem. This inversion is structural, not cyclical, and is not yet fully reflected in prevailing narratives about fragmentation risk in the Eurozone.

Figure 7: NPL ratio change (%) vs Banking Equity Index Cumulative Return (2014 Q3-2025Q4)



Sources: LSEG Datastream, Allianz Research

Improved asset quality also has direct macro-financial implications. The sharp reduction in NPL stocks freed up capital, reduced provisioning needs and strengthened profitability, increasing banks' capacity to extend credit to the private sector. In that sense, banking sector repair is not just a legacy issue but a precondition for the broader economic convergence, as it restored the transmission of monetary policy and supported investment and growth.

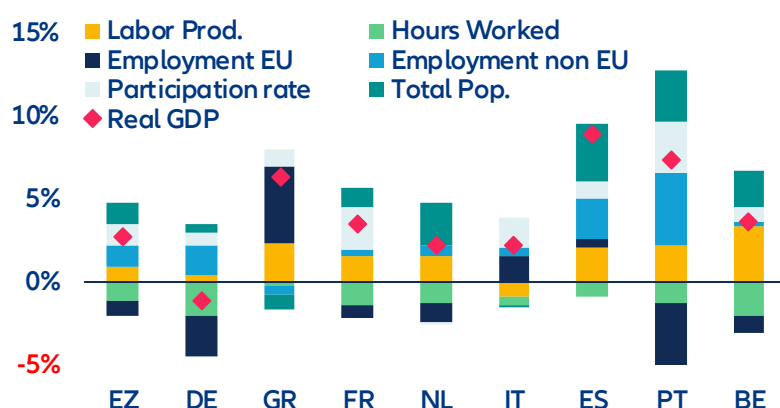
What did the South do right?

Labor market dynamics: the core driver of Southern resilience

Labor market performance is central to explaining the strength and persistence of Southern Europe's recovery. Since 2021, employment has not lagged output but has acted as the primary engine of expansion. Labor input – through both rising participation and higher employment rates – has accounted for the bulk of GDP growth, pointing to a distinctly labor-intensive recovery phase.

This dynamic reflects both cyclical demand and structural adjustment. Participation has increased across demographic groups, while migration has played a critical stabilizing role. Foreign workers accounted for more than half of labor force growth in recent years, easing labor shortages at a time when supply constraints were binding, particularly during the 2022 energy and supply shock. At the same time, earlier labor market reforms – especially in Southern Europe – have improved hiring dynamics, reduced rigidities and increased resilience to shocks.

Figure 8: Supply-side GDP decomposition, 2022-2025 cumulative growth

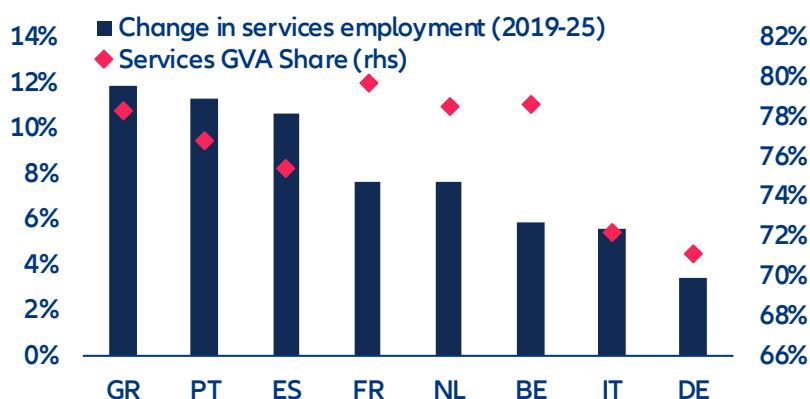


Sources: LSEG Datastream, Allianz Research

Spain provides the clearest example of a structural break. The 2021 labor market reform directly targeted the long-standing duality between temporary and permanent contracts. The impact is visible in the contractual structure: temporary employment declined by more than 10pps between Q4 2021 and Q4 2025, while hiring shifted decisively towards open-ended contracts. The share of fixed-term contracts – previously close to 90% of new hires – fell to below 60%, and youth temporary employment declined sharply, converging towards the EU average.

On the demand side, the sectoral composition of the recovery has amplified these effects. Strong services demand – particularly tourism – translated into high employment elasticity, allowing output growth to feed rapidly into job creation. Compared to the pre-crisis period, employment has responded more quickly and more proportionately to economic activity. Importantly, employment gains have been predominantly private-sector driven, pointing to robust underlying demand dynamics rather than administrative expansion.

Figure 9: Employment in services vs services GVA/ total GVA



Sources: LSEG Datastream, Allianz Research

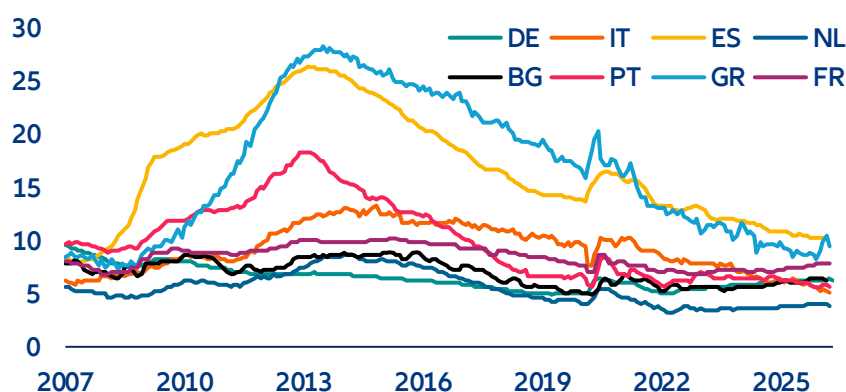
These shifts are not merely statistical. Greater contract stability alters firm incentives, making investment in training and skills more profitable, improving match quality and supporting a gradual shift towards higher-quality employment. At the same time, increased labor availability – particularly in high-demand sectors such as construction, hospitality and ICT – has allowed employment to expand without triggering proportional increases in unemployment or strong second-round inflation effects. Migration composition also matters: Non-EU workers are disproportionately represented in shortage sectors and often exhibit high participation rates.

At the aggregate Eurozone level, compositional changes have contributed to the decline in unemployment.

A larger share of older and more educated workers – groups with structurally lower unemployment rates – has mechanically compressed aggregate unemployment, while migration has expanded labor supply without raising joblessness.

These gains come with clear trade-offs. The recovery has been driven predominantly by the extensive margin, while average hours worked remain below pre-pandemic levels. This reflects both cyclical and structural factors, including the expansion of part-time employment, the shift towards services with structurally lower working hours and precautionary labor hoarding amid persistent shortages. As a result, labor input has increased mainly through employment gains rather than higher utilization per worker, dampening productivity growth and weakening the pass-through from output growth to aggregate income. Strong employment performance therefore partly masks an incomplete normalization of labor utilization. The implication is that output convergence has translated only partially into income convergence, as employment gains have not been matched by commensurate productivity growth. The key medium-term question is whether improvements in labor market structures – contract stability, participation and labor supply – will translate into a sustained reacceleration of productivity.

Figure 10: Unemployment rate (%)



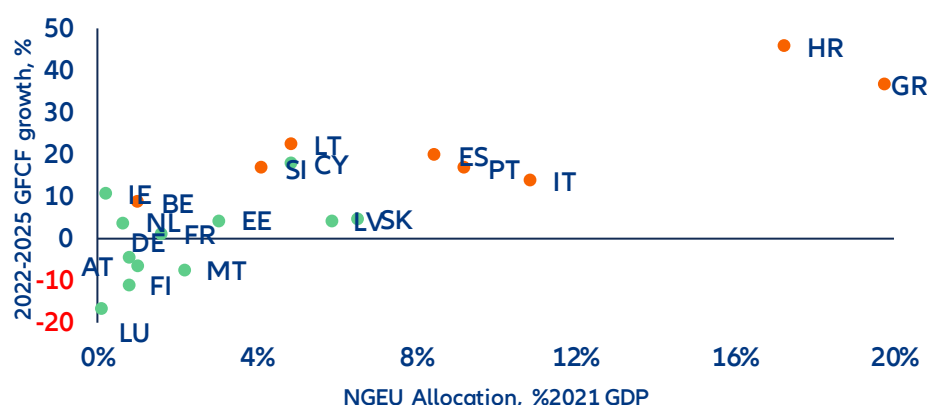
Sources: LSEG Datastream, Allianz Research

NGEU: front-loaded investment impulse with reform conditionality

The scale of the NGEU allocation to Southern Europe is easily underestimated. Greece received funds equivalent to nearly 20% of GDP, Italy close to 11%, Portugal 9.3% and Spain 8.5% – allocations that dwarfed those of core economies, with Germany receiving less than 1% of GDP and France just 1.6%. For economies where national fiscal space remained severely constrained, this represented a public investment impulse with no post-war precedent outside of crisis interventions.

The macroeconomic transmission operated through three distinct channels. First, the direct investment effect: Disbursements concentrated from 2022 onwards drove a visible recovery in public investment as a share of GDP across the South, reversing a decade of austerity-driven compression. Second, the fiscal accounting effect: Grants do not count as borrowing – they flow through the expenditure side without adding to national debt. For high-debt economies running primary surpluses, this was analytically significant: it allowed investment to expand while headline debt ratios continued to fall, contributing directly to the snowball dynamic and to the spread compression that followed. Third, the conditionality effect: By linking disbursements to predefined milestones, the NGEU transformed reform commitments into binding obligations, lowering the political cost of reform by externalizing part of the decision process. Spain's labor market reform, Portugal's digitalization agenda and Greece's administrative overhaul were all advanced under this framework.

Figure 11: NGEU allocation vs GFCF cumulative growth



Sources: Eurostat, Allianz Research. Orange dots represent country which have requested loans on top of grants.

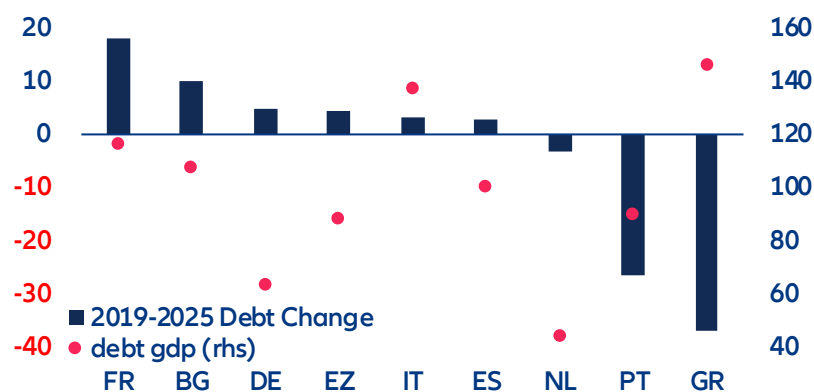
NGEU did not operate in isolation from fiscal consolidation – it amplified it. By flowing through the expenditure side without adding to national debt, grants improved headline balances and contributed directly to the debt dynamics that followed. The two pillars are complementary: NGEU created the fiscal room; Southern governments, for the most part, used it well.

Fiscal consolidation: the pillar that is cracking

Following the sovereign debt crisis, Southern economies implemented prolonged and politically costly consolidation strategies – cutting deficits, rebuilding primary balances and restoring institutional credibility at significant social cost. After the pandemic, they made the most of a favorable window: strong nominal growth and still-low effective interest rates combined to generate a powerful snowball effect, pushing debt ratios down materially. Portugal cut its debt ratio by over 40pps to below 90% of GDP since 2020. Spain delivered primary surpluses from 2022 onwards. Greece regained investment-grade status by 2025. Monetary policy set the lower bound for spreads across the board, but fiscal consolidation determined the extent of the re-rating.

Italy's picture is more nuanced. The return to a primary surplus in 2024 was a genuine improvement and should not be dismissed. However, the surplus has not been large enough to stabilize the debt ratio against a backdrop of weakening growth and past generous policy (Superbonus tax credit) – debt has continued to drift upward, leaving the fiscal position rather vulnerable. Italy's spread compression reflects ECB backstop credibility and a period of relative political stability as much as fundamentals, and it is the case that most warrants watching as financing conditions normalize and the ECB's role as marginal buyer fades.

Figure 12: Change in debt-to-GDP ratio



Sources: Eurostat, Allianz Research

Energy transition: from structural vulnerability to competitive advantage

The 2022 energy crisis triggered a structural shift in the distribution of energy exposure across the Eurozone – one that cuts against the traditional North–South framing. Southern economies – particularly Spain and Portugal – accelerated investment in renewable capacity, materially reducing their dependence on imported gas. Spain now sources more than 50% of its electricity from renewables, lowering its exposure to gas price volatility. On electricity, where the renewable build-out has been fastest, Southern Europe has moved from structurally higher-cost to lower-cost relative to the German industrial benchmark.

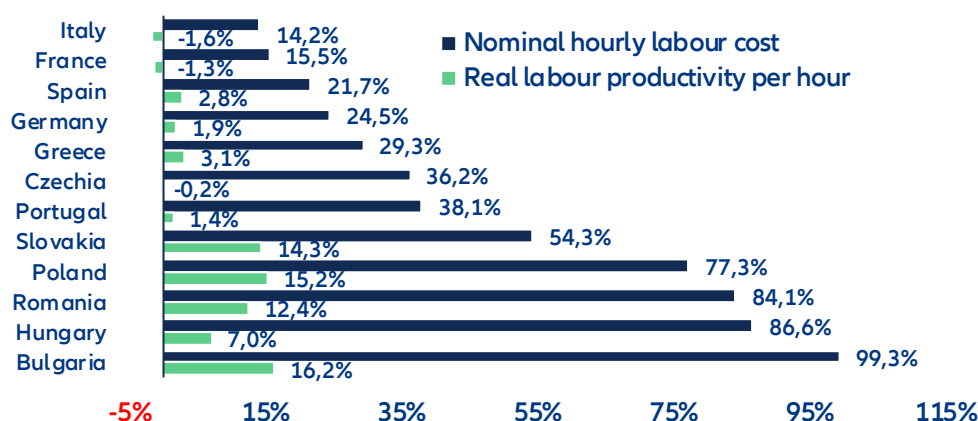
The implications extend beyond near-term growth. Energy has become a binding constraint for capital allocation in energy-intensive sectors, including digital infrastructure and AI. As a result, energy-intensive industries – chemicals, steel, aluminum and cement – are increasingly locating in energy-advantaged jurisdictions. Southern Europe, with its expanding renewable base and Atlantic LNG access, is becoming more competitive in attracting this reallocation.

However, the adjustment remains incomplete. Three residual exposures deserve explicit quantification. First, oil dependency: Southern economies remain net importers of crude, with transport fuels accounting for roughly 35–40% of final energy consumption. Second, EV adoption is uneven – the EV share of new car sales reached ~23% in Germany versus ~14% in Spain and ~8% in Italy in 2024 – leaving Southern transport sectors structurally more exposed to oil price volatility for at least the next decade. Third, energy poverty remains a distributional vulnerability: Approximately 13% of EU households in Southern member states report an inability to adequately heat their homes (Eurostat 2024), limiting the political durability of energy transition policies.

External tailwinds: the CEE rotation

Beyond the South gaining on the North, a broader reordering of comparative advantage is unfolding across the European economy. Some of the factors that previously powered Central and Eastern Europe's convergence are increasingly becoming sources of pressure, exactly as they are turning into relative advantages for Southern Europe. Since end-2019, nominal hourly labor costs rose by +99% in Bulgaria and +87% in Hungary, against productivity gains of just +16% and +7% respectively, while productivity in Czechia remained broadly flat despite strong wage growth (Figure x). The drivers are structural rather than cyclical. Labor markets in CEE have tightened to record levels, with unemployment at 3.1% in Poland and 2.8% in Czechia, while working-age populations have shrunk sharply since 2015 (Bulgaria –13.5%, Poland –11.3%, Romania –8.8%). The low-wage manufacturing model that anchored the region's integration into European supply chains is therefore becoming structurally harder to sustain as labor scarcity and rising costs erode part of the competitiveness advantage that originally underpinned it. At the same time, Southern Europe is benefiting from almost the opposite configuration: more moderate labor-cost pressures, stronger labor-market normalization and a more services-oriented growth structure supported by tourism and renewable investment.

Figure 13: Nominal labor cost and real productivity, cumulative % change since 2019



Sources: LSEG Datastream, Allianz Research

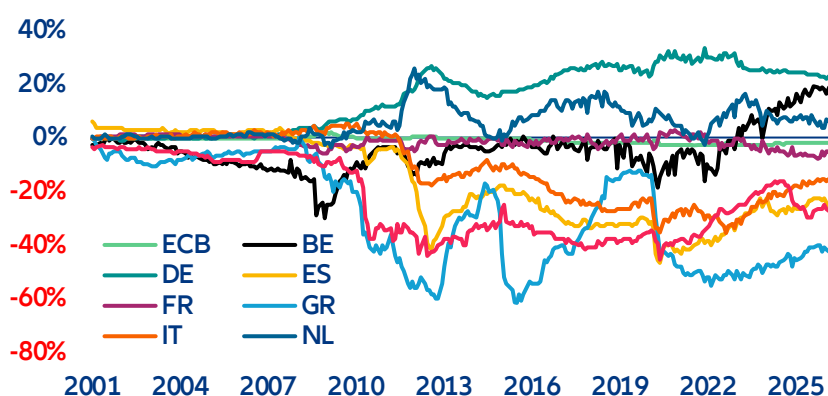
This divergence is reinforced by the changing geography of industrial competitiveness. Southern Europe's renewable build-out, together with Germany's loss of cheap Russian gas, has materially altered relative energy costs across Europe. In H2 2025, ex-tax electricity prices for medium-sized industrial users stood at EUR10.6 cents/kWh in Spain and EUR10.7 in Portugal, compared with EUR17.1 in Hungary, EUR16.0 in Slovakia, EUR15.6 in Romania and EUR14.6 in Czechia – a divergence of roughly 5–6 cents/kWh that did not exist before 2021. At the same time, CEE economies remain heavily concentrated in sectors tied to the weakening German industrial cycle, particularly autos. Motor vehicles account for 24.7% of manufacturing value added in Slovakia, 21.3% in Czechia, 15.8% in Hungary and 14.6% in Romania, all above Germany's own 19.5% and several multiples of Spain's 9.1% or Italy's 5.9%. The region is therefore simultaneously among the most exposed to high industrial energy costs and the most dependent on a German automotive model facing intensifying Chinese competition and weak external demand. Southern Europe, by contrast, carries neither exposure to the same extent. The result is a broader reordering of comparative advantage across Europe: Germany remains constrained by the structural weaknesses exposed after the energy shock, Southern Europe is increasingly gaining competitiveness in energy, services and investment attraction, while Central and Eastern Europe is confronting the limits of the very model that originally drove its convergence success.

The post 2026 cliff

2026 is the decisive test – political and institutional – as the external anchor fades. The NGEU is explicitly time-bound, with disbursements scheduled to end in 2026. Italy, Spain, Portugal and Greece together account for almost 60% of the total EU envelope, making the post-2026 transition a system-level test. The key question is whether convergence gains are becoming structurally embedded – in administrative practices, legislation and firm behavior – or remain largely compliance-driven. Where implementation remains compliance-driven, the risk of policy slippage increases sharply once the external anchor weakens.

This creates a dual risk. First, the withdrawal of policy support may expose underlying growth dynamics that remain heavily reliant on labor rather than productivity. Second, the loss of a binding reform framework could make policy slippage more likely, particularly in high-debt economies facing renewed fiscal constraints as financing conditions normalize. Convergence is currently policy-supported, not yet fully self-sustaining: Private cross-border capital flows remain limited and episodic. Capital flight has stopped, but it has not clearly reversed – Target2 imbalances have stabilized but have not structurally declined.

Figure 14: TARGET2 balances (% of nominal GDP)



Sources: LSEG Datastream, Allianz Research

The institutional gap: A completed Banking Union is the critical missing piece for the transition from public to private-led convergence. Without it, private cross-border flows remain too shallow to replace the NGEU impulse post-2026. Despite a common monetary policy, financial integration remains incomplete: Cross-border bank lending and portfolio allocation within the Eurozone still exhibit strong home bias, with domestic sovereign exposures dominating bank balance sheets. The absence of a European Deposit Insurance Scheme means risk-sharing remains incomplete: ECB backstops continue to substitute for missing institutional integration rather than operating within a fully unified financial system. Initiatives such as the Savings and Investments Union aim to deepen capital markets and reduce home bias, enabling more efficient cross-border allocation of savings, but the scale of private risk-sharing remains limited relative to the size of the Eurozone economy, and the transition from public to private investment as the primary driver of adjustment remains incomplete.

Political risk. The political sustainability of reform is not guaranteed. France, Italy, Spain and Greece all face national elections in 2027, putting fiscal discipline and reform commitment to a simultaneous political test across Eurozone economies. Populist movements in Italy and Greece remain a latent risk to reform continuity. Spain's political fragmentation – governing with a minority coalition dependent on regional nationalist parties – adds procedural uncertainty even where the policy direction is broadly reform-positive. Reform gains have not come without a social price: fiscal consolidation in the early phase involved material cuts to healthcare, education and social transfers that compressed living standards for the bottom half of the income distribution, and that political legacy has not fully faded.

From policy supported convergence to structural test. The first phase of Southern rebalancing – compressing crisis-era risk premia, rehabilitating banking systems, restoring fiscal credibility – has largely run its course, driven by a combination of domestic policy effort and EU-level architecture. From a macro perspective, the catch-up is real but unfinished: employment-led growth has delivered output convergence without closing the income gap, and the post-2026 withdrawal of the NGEU anchor will test how much of the improvement is structurally embedded. For markets, spreads at decade lows and equity valuations normalized leave little room for further re-rating on recovery alone: Outperformance now requires a differentiated view on which economies can convert institutional gains into productivity-led growth.

These assessments are, as always, subject to the disclaimer provided below.

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(v) persistency levels, (vi) particularly in the banking business, the extent of credit defaults, (vii) interest rate levels, (viii) currency exchange rates including the EUR/USD exchange rate, (ix) changes in laws and regulations, including tax regulations, (x) the impact of acquisitions, including related integration issues, and reorganization measures,

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US-Iran Deal: Markets Price Peace, Economies Still Pay for War

In Summary

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- **Markets have celebrated the end of the Strait of Hormuz conflict. The economy has not yet earned that relief - and for most households, corporates, and governments, things may get worse before they get better. The physical reopening is a multi-month process, not a switch.** The US-Iran MoU and 60-day ceasefire mark a meaningful turning point, but they are a de-escalation framework, not a final settlement. Normalization will be slow, frictional, and reversible, and markets have frontloaded the good news before the transmission lags have even peaked. Mine-clearing alone takes 30 to 50 days. After that, commercial traffic returns only as fast as shipowner confidence allows - the 1988 Iran-Iraq precedent shows normal conditions took more than three months to restore even with US naval escorts. Our base case restores 65% of the disrupted 4-5 mb/d within three months and 80% within four, with full normalization by year-end. The corridor reopens under managed, not free, navigation. Any stumble in nuclear negotiations or ceasefire adherence on Lebanon restarts the clock.

- **Energy prices ease, but inflation peaks later than markets think.** Brent stabilizes around USD 80/bbl in Q3 before easing to USD 75/bbl in Q4 and USD 67/bbl by end-2027. Past energy increases are still passing through supply chains, utility bills, and rents. US headline CPI averages 3.3% in 2026 and core CPI peaks at 3.1% in Q4; Eurozone inflation peaks around 3.4% in Q4 before averaging 3.1% in 2026. Real wage growth turns positive only in Q1 2027. Markets pricing disinflation from today are running at least two quarters ahead of the data.

- **The shock hits every balance sheet - but Europe bears the deepest scar.** The US absorbs the blow with structural advantages: a net energy exporter, it captures a terms-of-trade windfall through higher mining revenues and fiscal receipts, while tax rebates cushion households and AI-driven investment sustains corporate capex. Europe has no such offset. Governments across the Eurozone have deployed just EUR12bn YTD - 0.1% of GDP - with Germany's defense and infrastructure stimulus the sole demand anchor. Households face a sharper and more persistent squeeze: energy bills represent a larger share of disposable income than in the US, floating-rate mortgage exposure is higher across the UK and Netherlands, consumer confidence has not recovered to pre-war levels, and real purchasing power will remain compressed through most of 2026. For corporates, the energy cost relief, most visible in transport and petrochemicals where fuel runs 25-40% of operating costs, runs directly into this demand void. Pricing power is the dividing line: airlines and branded pharma can defend margins through surcharges; automotive OEMs and generic drugmakers cannot. Input costs are easing but wage bills remain elevated into 2027, and listed-company cash has already fallen 3% to EUR 36.7trn. Any ceasefire relapse would recouple all three pressures simultaneously - reigniting input costs while households are still digesting the first wave and fiscal space is exhausted - with Europe more exposed throughout given higher energy intensity, greater trade openness, and no room left for transfers.

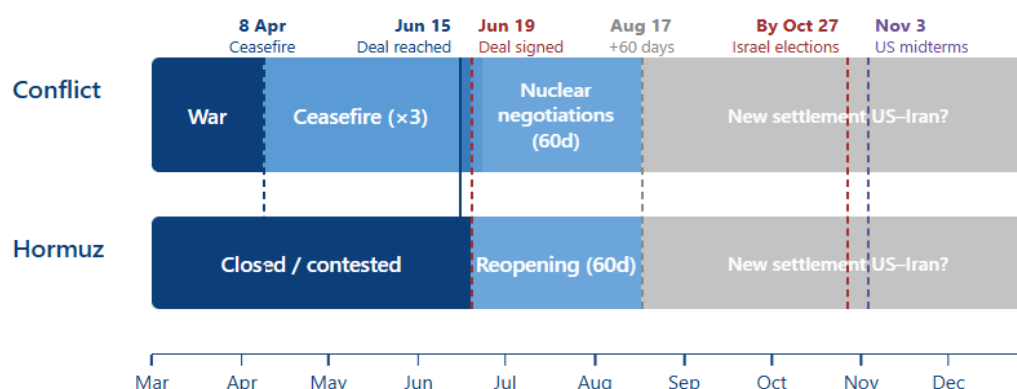
- **For central banks, the risk of a policy mistake is the highest since 2022.** Both the Fed and ECB are expected to deliver one further hike in H2 - the Fed likely in September as core CPI ticks above 3%, the ECB before year-end - before disinflation opens the door to cuts in H2 2027. Both face a time consistency problem as inflation pressure will fade quickly while their data-dependency on backward-looking measures continues to justify tightening well into Q4. Overtightening into an already-fragile household and corporate backdrop remains a plausible - and underpriced - scenario for both banks. But the US retains more growth momentum to digest further tightening, while the Eurozone's growth is already fragile.

- **For capital markets, the signal is rotation and carry, not re-rating.** Equities never priced the conflict as a systemic shock - MSCI US up 8%, MSCI Europe +5% YTD - so there is little upside to unlock on good news but significant room to gap lower on bad. Valuations sit in the 85th percentile with earnings in the 92nd percentile (MSCI World, 20-year history); a 5-10% pullback through summer is plausible. A tilt toward Europe, cyclicals, and energy-sensitive sectors is warranted over a broad melt-up. On credit, EUR IG at 77bps and US IG at 72bps are near all-time tights; the deal protects the carry trade rather than opening new compression. In rates, the short end and belly of European curves offer the most compelling outperformance (35bps front, 20bps belly), while long ends remain vulnerable to fiscal supply - bull steepening, not a parallel rally. In private markets, gains show up in activity - firmer exits, smoother refinancings - not in marks.
- **Relief is the call. It is not the all-clear.** Markets have frontloaded the peace dividend; the economic data has not confirmed it. Inflation will get worse before it gets better. Household and fiscal stress will persist well into 2026. And the deal remains conditional on nuclear negotiations and a multi-front ceasefire with a poor track record. The asymmetry is uncomfortable: upside on good news is modest; downside on bad is fast, disorderly, and correlated. The cost of being slow to react would far exceed the cost of staying cautious.

How does the deal impact our economic outlook?

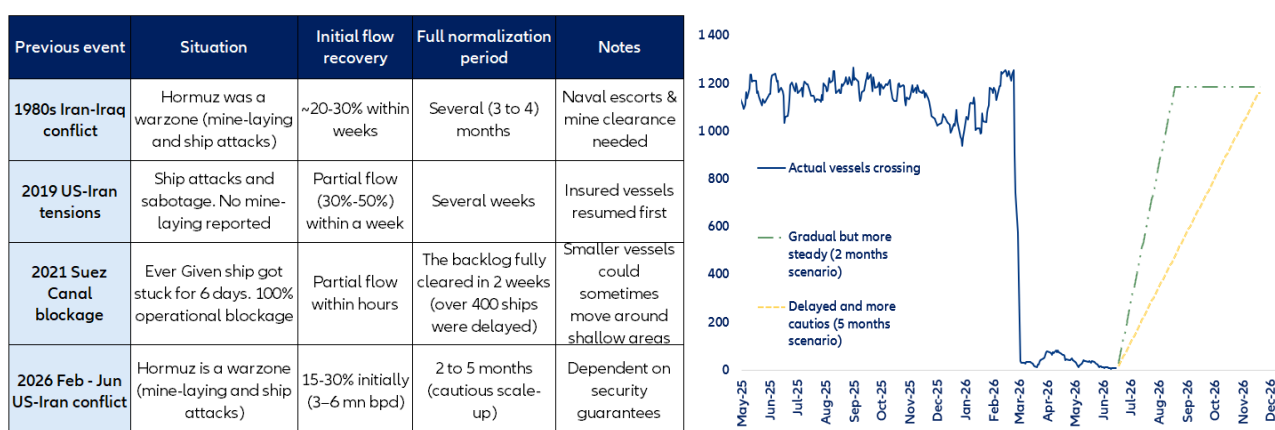
The US and Iran have reached an interim agreement, due to be signed on 19 June, establishing a 60-day ceasefire and paving the way for the reopening of the Strait of Hormuz. After four months of disruption that rattled energy markets and prompted central banks to resume monetary tightening, the announcement marks a significant turning point, reducing upside risks to energy prices and inflation. The agreement is best understood as a sequenced de-escalation framework rather than a final settlement. It establishes a 60-day ceasefire across multiple regional fronts, creating a window for negotiations on Iran's nuclear program and broader geopolitical arrangements. A key provision is the granting of a temporary waiver allowing Iran to sell oil freely during this period, implying a potentially rapid increase in exports relative to current constrained levels, even in the absence of a full sanctions rollback. The deal also envisages a phased approach to sanctions relief, conditional on compliance and progress in negotiations. While the exact sequencing remains unclear, the US position appears to tie any permanent lifting of sanctions to verifiable commitments regarding Iran's nuclear activities. This conditionality introduces a significant implementation risk, as delays or setbacks in negotiations could quickly derail the process. At the same time, the agreement formally incorporates de-escalation across all active fronts, including Lebanon, although participants themselves have identified this dimension as one of the most fragile elements of the arrangement. From an energy-market perspective, the agreement contains several important provisions. The Strait of Hormuz is expected to reopen from 19 June following mine-clearance operations, while in the first hours of the deal announcement President Trump indicated traffic would return "free-of-charge", Iran later signaled it intended to charge "fees for services" for vessels transiting the waterway leaving Tehran with some influence still on the Strait. It is yet unclear what form will these fees take but environmental charges have been floated.

Figure 1: Potential timeline of events until year-end



What needs to be watched for the deal to be sustainable? The sustainability of the agreement will depend on its implementation during the first 60 days. Four indicators will be particularly important to monitor. First, progress on maritime security, notably the pace of mine-clearing operations, which could take from 30 to 50 days to be completed before shipping lanes are deemed fully operational. Second, the speed at which commercial traffic resumes through the Strait of Hormuz (ships in and out), which ultimately depends on the level of confidence in the sector and the security guarantees provided to shipowners. Third, the evolution of war-risk insurance premiums and freight rates, which will provide an early indication of whether shipowners and charterers view the security environment as sufficiently stable to resume regular operations. Finally, progress in negotiations over Iran's nuclear program and the maintenance of ceasefires across other regional fronts, including Lebanon, as these remain the key political conditions underpinning the agreement. Historical experience suggests that confidence is restored more slowly than physical access. Following the 1988 Iran-Iraq ceasefire, commercial traffic required several weeks to start recovering despite the deployment of US naval escorts, while a return to normal operating conditions took more than three months (see Figure 2A). This implies that even under a favorable scenario, the Strait may not return to pre-crisis throughput before Q4 (Figure 2B). Periods of relative calm are therefore likely to be punctuated by short-lived kinetic incidents, with the key test being whether such episodes remain contained. Ultimately, the durability of the agreement will be measured not by the absence of security incidents, but by the ability of the parties to prevent isolated disruptions from escalating into a renewed closure of a maritime corridor that handles roughly one-fifth of global oil demand.

Figure 2A and 2B: Historical benchmarks for maritime traffic recovery in the Middle East; and scenarios for vessel traffic* recovery through the Strait of Hormuz



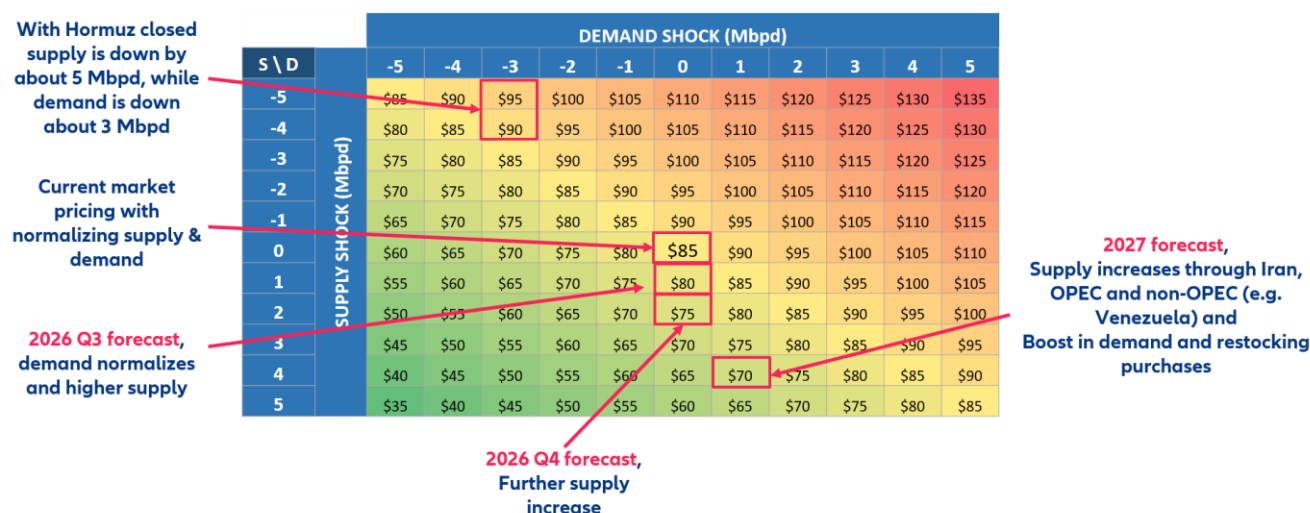
Source: Bloomberg, Allianz Research. *Traffic: number of vessels crossing the Hormuz Strait (7 day rolling total)

Going forward we see a progressive normalization of oil and gas trade flows through year-end rather than an immediate return to pre-conflict conditions. This reflects both existing inventory constraints and the time required to fully restart Gulf production and shipping logistics. In the meantime, partial rerouting of flows, higher exports from the US, Canada, Russia and Latin America, together with strategic reserve releases and moderate demand destruction, would continue to offset a large share of the supply disruption. Most of the effective supply gap would close within three to four months, with roughly 65% of missing supply restored within three months and close to 80% within four months. Crucially, the absence of direct destruction to oil and gas infrastructure prevents the disruption from evolving into a prolonged structural supply shock. We now expect Brent to average 80 USD/bbl in Q3 2026 as the initial supply shock partially unwinds. Currently, the market is absorbing a net disruption of roughly 4-5 mb/d on the supply side while demand destruction of approximately 2-3 mb/d persists, particularly across Asian economies that were hardest hit by the supply shortage and elevated prices. These two forces offset one another in Q3: recovering supply is bearish, but the concurrent demand recovery limits the downside. The net effect is prices stabilizing around 80 USD/bbl as market recalibrate to a world in which the risk of a prolonged Hormuz closure has been removed – for now. By Q4 2026, oil prices should soften further to 75 USD/bbl as supply normalization overshoots the demand recovery. Persian Gulf producers, eager to recapture market share and revenue after

months of shut-ins, are assumed to restore output at a pace slightly ahead of consumption's rebound, tilting the quarterly balance into mild surplus. Natural gas market conditions are also expected to ease for European and Asian buyers and we expect prices to average 43 EUR/MWh in 2026. We also revise our 2027 annual average oil price forecast to 70 USD/bbl - reflecting a structurally more supplied market. Approximately 4 mbpd of incremental production could hit the market : a blend of Iranian output returning to near high levels, OPEC unwinding residual voluntary cuts as the supply emergency passes, and non-OPEC growth from the US and Venezuela. On the demand side, 1 mb/d of incremental consumption offsets part of the supply surge: global activity recovers, Asian industrial demand rebounds, and, critically, large-scale inventory restocking by refiners and strategic reserve managers provides durable support to physical demand. The result is a well-supplied but not a "super glut", anchoring Brent in the high-60s to mid-70s range, with the key risk being larger-than-expected supply increases.

By the time the transportation in the strait of Hormuz (in and out) and resumption of the oil and gas production is full (year-end), mitigating factors will continue to be at play, i.e. demand destruction helping to slow the depletion of global reserves, while fiscal support measures would aim to cushion the inflationary shock. Demand destruction is progressively helping rebalance markets. At present, oil demand destruction is running at around 2.5mbpd, though this largely reflects the immediate disruption effects following the escalation in the Middle East conflict. Assuming a gradual reopening starting in June and normalization of trade flows in the next four months, average demand destruction would moderate to around 0.4mbpd for 2026 as a whole, remaining broadly consistent with a global slowdown rather than a recession scenario.

Figure 3: Brent price forecast based demand and supply shocks



Source: Allianz Research

The good news is that the transmission of the energy shock would not be full which contains the stagflationary flavor of this crisis. The fastest pass-through typically occurs through manufacturing and services input prices as well as energy-related producer prices, with effects peaking within roughly two to six months after the initial shock. Over the following six to twelve months, inflationary pressures progressively broaden as firms pass higher energy and transport costs through supply chains, pushing up intermediate and consumer-goods producer prices as well as industrial and retail selling prices. The most persistent effects are likely to be avoided, i.e. rise of core inflation, consumer-goods prices, inflation expectations and negotiated wages.

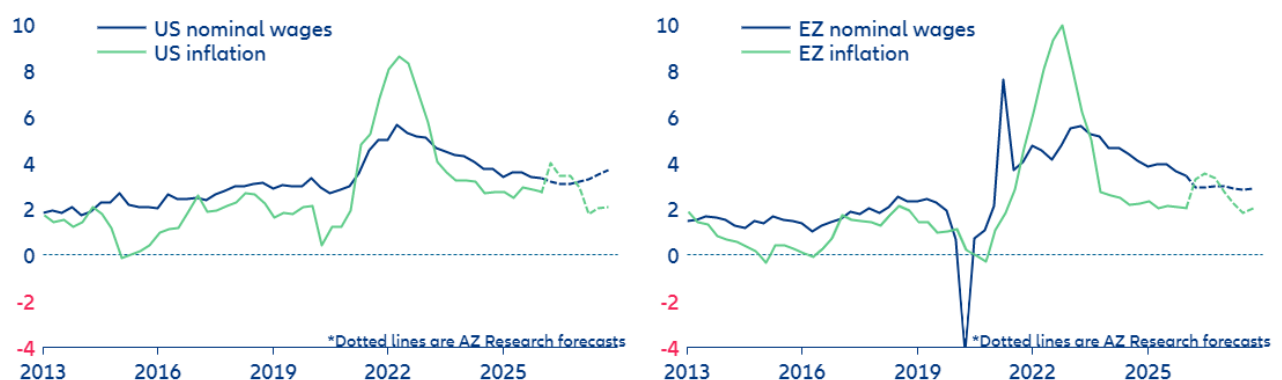
In the US, we continue to expect GDP growth to hold up (+2.0%) thanks to supportive fiscal policy, strong AI-related investment and positive wealth effects. The US being a net energy exporter, its economy benefits from a positive terms of trade shock supporting aggregate income, with the income gains in the mining sector accruing to some households and the government through higher receipts. Nevertheless, the rise in inflation is a net negative for US household, whose aggregate real disposable income has declined in Q2, but fiscal policy is supportive, with tax rebates in H1 2026 providing some partial offset. Furthermore, strong equity valuations encourage households to save less to smoothen their spending on non-energy items. Meanwhile, on the corporate side, capex intentions

remain solid amid strong demand for AI-related business equipment and software. The AI buildout and manufacturing output are relatively immune to high global energy prices thanks to abundant gas supply and low industrial electricity prices, though supply-chain bottlenecks in the global electronics sector is a concern. Recent business surveys indicate that the expansion of the US manufacturing sector is likely to continue in coming months. On the inflation front, we expect headline CPI at +3.3% in 2026. Energy inflation is expected to decline from June. However, higher energy prices relative to their pre-Middle East crisis level, and only a gradual reduction of global supply chains strains should push up core prices. We expect core CPI inflation to tick up above +3% by October, from +2.8% in May. In this environment of increasing core inflation, we expect the Fed to re-affirm its anti-inflation credentials by increasing interest rates by 25bps, in September.

Europe has borne the brunt of the Hormuz shock, with weaker growth and rising inflation as the price to pay for being a net energy importer. Unlike the US and Asia, the Eurozone cannot offset the drag through an AI-driven investment boom - the region simply lacks the large-cap tech champions that are currently powering capital expenditure elsewhere. The energy shock has hit European industry disproportionately hard, feeding into input costs and squeezing margins, as confirmed by the further deterioration in the Eurozone Composite PMI to 47.5 in May, with the services sector - more exposed to confidence effects from the Iran conflict - contracting sharply. Price pressures are building on both input and output sides, reinforcing the case for a more restrictive monetary stance even as growth falters. Germany is a partial exception: fiscal stimulus channeled into defense and infrastructure is providing some near-term support, with Q1 GDP expanding +0.3% q/q (excluding Ireland) on the back of strong exports and higher government spending, and the German Composite PMI holding broadly stable. France, by contrast, saw its Composite PMI collapse to 43.5 in May, its lowest since November 2020.

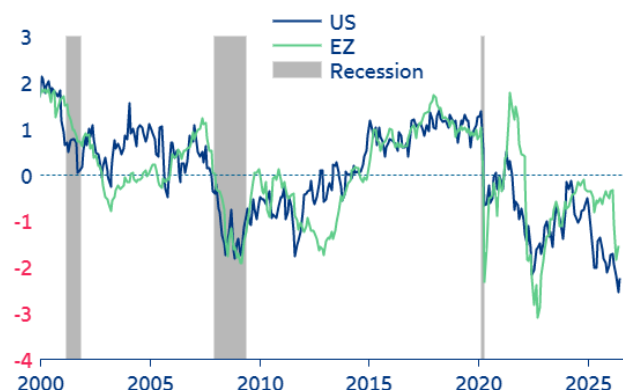
In China, we continue to expect a resilient economy, driven primarily by exports and high-tech manufacturing while domestic demand remains weak amid persistent property-sector stress and subdued consumer confidence. We forecast growth at 4.7% in 2026 and 4.4% in 2027 and maintain our inflation forecast at 1% in 2026 and 1.2% in 2027. As the US-Iran deal will provide a gradual return of oil supply to Asian markets, near-term headwinds to our inflation forecast appear to have eased. However, as conditions normalize, we continue to expect modest inflationary pressure to persist in the near term. We expect the People's Bank of China to remain on hold through 2026 as inflation remain below the 2% policy threshold. While the US-Iran deal will ease inflationary pressures on input prices, the inflationary impact of the return of positive Chinese PPI growth will continue to materialize over the coming quarters. The typical transmission lag between producer and export prices means some modest pass-through remains likely. As a result, we continue to expect Chinese exporters' price competitiveness to slightly erode, while destination markets could see some contained inflationary spillovers. As a leading exporter of manufactured goods, China continues to benefit from strong global demand for clean-energy equipment and AI hardware. However, trade headwinds remain a source of uncertainty. The US has signaled a potential 12.5% tariff under Section 301 targeting forced labor practices to be implemented late July while several additional investigations remain open. Combined with continued RMB appreciation and rising geopolitical tensions, tariff uncertainty could challenge Chinese exporters momentum.

Figure 4: Inflation has surpassed wages leading to a temporary loss in purchasing power



Source: LSEG Datastream, Allianz Research

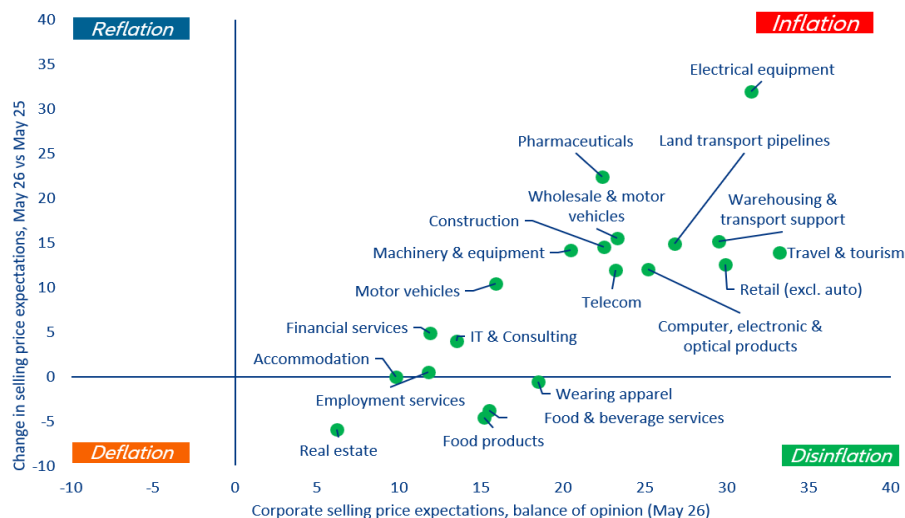
Figure 5: Consumer sentiment has dropped significantly in Europe and the US



Source: LSEG Datastream, Allianz Research

At the corporate level, a gradual reopening is a genuine relief, but pricing power remains the dividing line that a deal break would expose again. The easing in energy prices loosens the squeeze that had reached balance sheets through four channels, namely higher energy costs, logistics disruption, input-cost inflation and weaker demand, with the sectors hit first, transport and petrochemicals where fuel runs 25-40% of operating expenses, the ones that benefit most as crude normalizes. The relief is uneven, set by pass-through capacity: airlines (fuel surcharges and ancillary fees), fertilizer producers and branded pharma can hold price, and semis are cushioned by AI demand, whereas automotive OEMs, computer hardware and generic-drug makers have effectively no pricing power and gain the least margin headroom from cheaper energy. Cash buffers remain solid in aggregate, with listed-company cash falling -3% to EUR36.7trn in 2025 and non-financials more resilient at -1%, though liquidity is concentrating in AI-infrastructure and defense/energy-security names while energy, healthcare and real estate erode. The corporate relief on costs, however, runs into some headwinds on the revenue side: household real purchasing power could remain compressed through most of 2026, and that compression is itself a direct drag on top-line growth and thus on the margin expansion that cheaper energy would otherwise permit. CPI in both the euro area and the US expected to peak around Q4 2026 as energy pass-through is still working its way through utility bills and supply chains. Households are still in a defensive posture: savings rates have edged up, credit-card delinquencies are rising at the margin in the US and eurozone consumer confidence has not recovered to pre-war levels. The practical consequence for corporates is a squeeze on volumes even where pricing power nominally exists. The margin arithmetic therefore looks less clean than the energy line alone implies: input costs are easing but wage bills could remain elevated into 2027, meaning the Q4 2026 peak in CPI marks an inflection point rather than an immediate relief of the operating-cost structure. Again the European exposure is the sharper one: household energy bills carry a larger share of disposable income than in the US, mortgage re-pricing is more immediate in markets such as the UK and the Netherlands where floating-rate structures dominate, and fiscal space to cushion the squeeze through transfers is more constrained after the 2022-23 spending. The caution, consistent with the rest of this note, is that the relief is reversible: should the ceasefire fail and the closure run beyond six months, input-cost inflation that is now only spreading into cyclicals such as pulp and paper and auto-parts would re-intensify, pricing power itself would begin to erode, turning a margin squeeze back into revenue contraction, and Europe would be more exposed than the US throughout given higher energy intensity, greater trade openness and less fiscal room.

Figure 6: Selling price expectations for corporates in the Eurozone, May vs Feb 2026



Sources: Eurostat, Allianz Research

Figure 7 - Short-term pricing power assessment of sectors

Sector	United States	Eurozone
Transport		
Airlines	✓	✓
Shipping	✓	✓
Road	✓	✓
Transport Equipment		
Aviation & Space	✓	✓
Trucks	✓	✓
Shipbuilding	✓	✓
Chemicals		
Petrochemicals	✗	✗
Basic inorganics (fertilizers)	✓	✓
Polymers (plastics)	✓	✓
Specialty chemicals	✓	✓
Pharmaceuticals		
Big Pharma (branded)	✓	✓
Generics	✗	✗
Pulp & Paper		
Graphic paper	✗	✗
Paper packaging & board	✓	✓
Tissue & Hygiene	✓	✓
Machinery & Equipment		
Industrial & robotics	✓	✓
Construction machinery	✓	✓
Agricultural machinery	✓	✓
Construction		
Infrastructure / Civil Engineering	✓	✓
Residential	✗	✗
Automotive		
Vehicles	✗	✗
Parts & accessories	✓	✗
Tires	✗	✗
Electronics & telecom hardware		
Semiconductor	✓	✓
Computer: High/Low-end machine	✓	✗
Smartphone: High/Low-end machine	✓	✗
Household equipment		
Household appliances	✓	✓
Furniture	✗	✗
Textile		
Apparel - luxury/non-luxury	✓	✓
Footwear	✓	✓
Accessories	✓	✓
Retail		
Food	✓	✓
Non-food	✓	✓

✗	No pricing power
✓	Mild pricing power
✓	Strong pricing power

Source: Allianz Research

What is the impact for capital markets?

Middle East de-escalation creates room for outperformance in rates. The short-end and the belly of the curve stand out as the most compelling areas, with the case for steepening and duration strongest there. The repricing of rates was driven by different forces on either side of the Atlantic, which shapes where the most compelling opportunities lie.

In Europe, the repricing was inflation-driven, meaning that a de-escalation scenario and lower energy prices will hit the root cause directly. After last week's ECB hike market still prices one more move into year-end. A durable peace deal plus a lower oil price could give the ECB confidence to stay on hold instead. For now our base case stays one more hike, broadly in line with the market. But the upside is real with ~35bp at the front and up to 20bp vs forwards at the curve belly over the next one to two years

The US offers a more mixed picture. The repricing is only partially inflation-driven, it also reflects renewed US strength and the return of US exceptionalism, which a durable deal could amplify rather than unwind. Front-end holds some outperformance potential of ~25bps as de-escalation may keep the Fed on hold against one hike priced over 12m by markets, especially if AI-driven productivity disinflation starts to show through. Our base case still pencils in one hike in 2026, but risk is tilted to pricing it out again. Belly outperformance on the USD curve is less clear than for the EUR curve, since the growth and exceptionalism legs won't reverse mechanically on lower energy. A significant near-term uncertainty is the Fed's upcoming meeting under incoming Chair Warsh, which could overshadow or offset any peace-deal reaction. Both the ECB and the Fed remain explicitly data-dependent, and inflationary repercussions of the energy price shock are already visible — including early evidence that reflation is broadening beyond energy. That context means neither central bank is likely to move ahead of the data, whatever the geopolitical backdrop. The risk bias deviates however, with the ECB risking a lagged error of commission (hiking too much on stale data), while the Fed risks a deliberate policy shift under new leadership regardless of geopolitics. The fault line is the long end: even on a deal, structural US inflation and fiscal supply leave long-end yields vulnerable, so a curve steepening rather than a parallel rally is the more likely path and the part that could disappoint. Nevertheless, US TIPS look more attractive than EUR linkers, and caution on long-end rates globally remains warranted — even in a scenario of normalizing energy flows — given structural inflation dynamics and elevated upside risks in US and global rates.

In the case of equities de-escalation is helpful, but it is unlikely to create a large new rally because at less than -10% maximum drawdown, equities never priced the conflict as a systemic shock. The June 14 US-Iran deal reduces one of the obvious downside tails, but this is still a partial and fragile geopolitical reversal not a full peace settlement. That distinction matters because the market starting point is already fairly strong (as of June 12 MSCI US was up around 8% since the start of the Iran war, MSCI EM is up around 6.5%, and the MSCI Europe, despite being the relative laggard, is still +5%ytd). In other words, equities have looked through the conflict because earnings, AI leadership and risk appetite have mattered more than geopolitics. The benefit here is relief and not re-rating. Our oil baseline of 80 USD/bbl in Q3, declining to 75 USD/bbl in Q4 and 70 USD/bbl on average in 2027, reflects a gradual supply normalization that removes the inflation and rates scare scenario rather than creating a positive impulse. This should support Europe, cyclicals, transport, airlines and energy-intensive sectors that were carrying a modest war discount, and it lowers the probability of a policy accident. However, at this stage the market is not reversing a war discount, it is simply taking out part of a tail risk that had mostly remained in commodities. On a confirmed deal we would expect consolidation rather than a broad melt-up, with the relief accruing to Europe, cyclicals and the energy-sensitive sectors that carried the discount rather than to the index as a whole. The fault line cuts the other way: because the conflict was never priced as a systemic shock, there is little discount to unwind on good news but plenty of room to gap lower on bad news, leaving the tape asymmetric and headline-driven if the ceasefire frays. This comes against the backdrop of already ambitious market pricing, with valuations in the 85th percentile and earnings growth in the 92nd percentile (MSCI World, 20-year history), leaving little room for upside surprises. In this situation holding exposure, adding strategic risk only very gradually or opportunistically and focusing on rolling hedges at moderate cost offers continues to offer the best risk-reward. From a positioning perspective, the key risk for broader risk assets (including credit) is not just a potential breakdown of the deal, but the second-round effects of earlier shocks on growth, inflation expectations, and rates. Notable downwards revisions could turn earnings revision momentum and pull equities by 5-10% easily through summer. With AI-led

enthusiasm driving equities to new highs, the focus shifts to delivery: any disappointment on monetisation or the burden of sustained capex could trigger renewed volatility. Last weeks' 30% swing in Oracle offered a high-velocity glimpse of how markets can behave on such news — albeit for the market overall it would be far more muted form, as we might see toward end-2025.

For credit the news protects the carry trade, rather than opening a major new spread tightening phase. Credit has been even clearer than equities in saying that this was not priced as a funding shock. EUR IG remains around 77 bps and US IG around 72 bps, both already close to all time tights. Similarly, Euro HY (BB/B) is around 213 bps and has tightened roughly 19 bps since the start of the war. That is not the footprint of a market worried about a major geopolitical credit event. It is the footprint of strong demand for carry, resilient earnings and still open refinancing markets. De-escalation is therefore constructive because it reduces the risk of an oil spike, renewed rate volatility and a confidence shock. But at these spread levels, the most realistic impact is spread support rather than strong compression. On a final resolution we would therefore expect only limited further tightening from already-tight levels, with spreads holding firm rather than compressing meaningfully. The fault line is precisely those tight levels: with spreads close to all-time tights there is almost no cushion, so the risk is heavily skewed to widening if macro or rate concerns return, and HY would feel it first. Inside High Yield, discrimination between more B- and BB-ratings remains at decade highs (2.1x ratio) and indicates some caution in run for carry. IG should remain well bid as macro volatility falls and HY should continue to benefit more from risk appetite and higher rates, but will remain vulnerable as the margin for error if the ceasefire fails or oil risk reappears is low. In this situation we remain cognizant of downside risks but gradually and also selectively embrace opportunities for carry where meeting portfolios have been awaiting more clarity for implementation needs. Carry remains preferred to price appreciation assets in this fragile and again time-consuming outlook. For credit specifically, in a demand led market, the focus is less on fundamentals—which remain broadly average—and more on technicals. Strong inflows continue to anchor spreads at low levels but also increase the risk of a more abrupt reversal than would typically be implied by fundamentals in a normal cycle. Areas we watch are any further divergences within the ratings spectrum, with B/BB segments reaching new highs (<2.5x), as well as emerging signs of fatigue in technology issuance, where bid-cover ratios falling below 3x, new-issue concessions below 20bps and attrition rising above 1/3 could indicate normalization to start.

Lastly, for private markets the interim agreement is likely to improve liquidity and transaction confidence, but it is not a valuation reset. The relevant anchor is that public equities and liquid credit recovered most war-related losses, so they do not have a large visible war discount to reverse, the move is therefore one of mechanics rather than marks. The impact is instead practical and routes through three channels. First, firmer and re-rated public comparables, particularly in energy infrastructure and grid/electrical contracting, where listed names trade at historic EV/EBITDA premiums and a consolidation wave is still building, give Private Equity a clearer exit and valuation reference points, easing exit and IPO discussions that had stalled while sponsors sit on a record backlog of unsold companies and high dry powder. Second, resilient credit markets and the receding oil and rate volatility tail support refinancings, letting sponsors clear debt raises and recaps that were on hold. Third, lower energy tail risk improves the margin outlook for industrials, logistics, consumer services and transport-linked assets, where fuel had run 25-40% of operating costs at the peak. Along these lunes, Private Credit will also benefit from a lower default tail risk but lender spreads, as is the case for liquid credit, are already compressed, with median direct lending spreads having fallen from roughly 700 bps in early 2023 to around 500 bps, so the return impact is likely to be modest. Infrastructure and real estate are likely to get a mixed but generally stabilizing impulse with lower energy volatility helping utilities, tenants and transport assets, and a lower oil inflation path reopens room for the rate cuts that rate-sensitive real estate and residential construction depend on, while some of the energy-security and defense-adjacent premia that re-rated during the conflict may soften at the margin, even as the structural, data center- and electrification-driven grid buildout continues regardless. In practice we would expect the move to show up in activity rather than marks, with deal flow, exits and refinancings picking up gradually rather than valuations re-rating higher.

Pressure points to watch. Every constructive call above is conditional, and the conditionality is the message, this is a half-baked reopening, not a settled peace, and the relief rests on a peace deal, that remains anchored on unresolved negotiations over Iran's nuclear program and a ceasefire on all fronts, including Lebanon, and the track

record of failed de-escalation attempts suggests it can unravel quickly. Because markets never fully priced the war, the move on good news is modest while the room to reverse on bad news is large, so the risk profile is asymmetric and the reaction to any setback would likely be fast and disorderly rather than gradual. In this context, a failure to adhere to the deal, a renewed closure of the Strait, a loss of central bank look-through or a tightening in dollar funding would not arrive in isolation, but would recouple oil, rates, credit and equities at once and turn an orderly normalization into a sharp, correlated drawdown. We therefore treat the base case as constructive but provisional as it warrants close and continuous monitoring of the early warning signals around oil, volatility, credit spreads and long end rates. In short, relief is the call, but it is not the all clear, and the cost of being slow to react would be far greater than the cost of staying cautious.

These assessments are, as always, subject to the disclaimer provided below.

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From kickoff to cash flow – Economic spillovers from the Football World Championship 2026

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In Summary

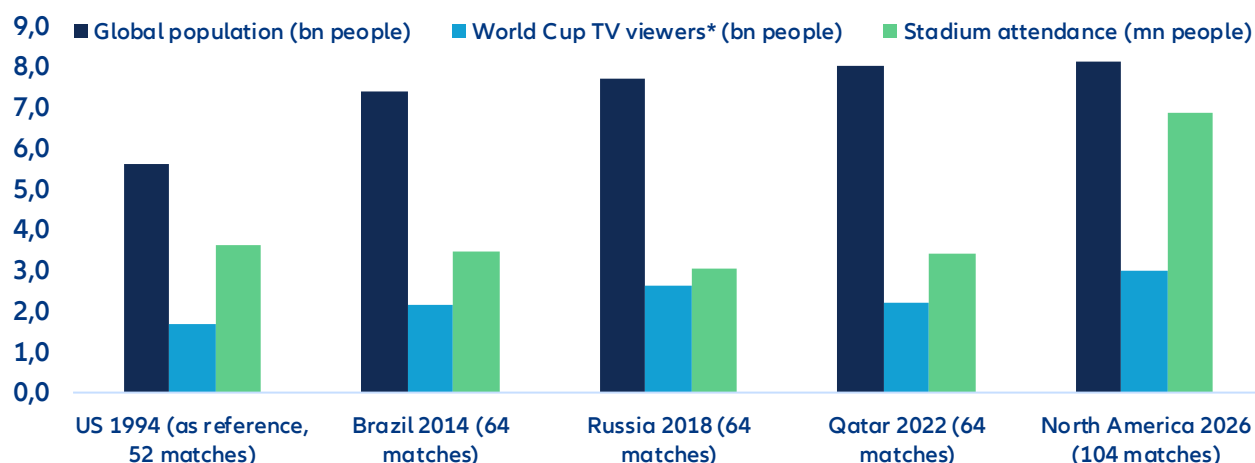
- **The Football World Cup 2026 will be the largest tournament in football history and a structural departure from previous editions, expanding from 32 to 48 teams and from 64 to 104 matches across 16 host cities in the US, Canada and Mexico.** It will test whether a World Cup can successfully evolve from a single-country event into a distributed continental experience without sacrificing atmosphere, attendance or commercial performance. Over a six-week period, it is expected to mobilize approximately 6.5mn attendees, including 2.6mn international visitors, generating an estimated USD9bn in GDP across North America during June–July 2026. For comparison, Taylor Swift's Eras Tour and Beyoncé's Renaissance World Tour, with 149 and 56 shows respectively, generated approximately USD2.1bn and USD579mn in revenue. FIFA projects record commercial revenues of USD13bn for the 2023–2026 cycle but the macroeconomic impact remains more concentrated than transformative, with tourism-related spending accounting for the dominant transmission channel.
- **From a demand perspective, total tourism-related expenditure is expected to reach around USD8bn, split between USD6.8bn in foreign tourism exports and USD1.2 bn in domestic consumption, after accounting for crowding-out effects.** Despite the stricter US visa barrier (refusal rate of 33% on average for non-European qualified nations, 74% for Senegal, 61% for Iran), the US captures the largest share, with an estimated USD5.4bn boost, followed by Mexico (USD1.4bn) and Canada (USD1.2bn). This reflects around 40% international visitors and 60% domestic attendees, each staying an average of 6–10 days and spending between USD180 and USD350 per day depending on the host country. Air travel adds a further USD1.0bn in incremental airline revenues, reinforcing the importance of mobility-linked sectors in overall value creation. Security operations spending will add USD1bn to the economic boost. Most of this spending is government consumption.
- **The distribution of gains will be highly uneven across sectors and geographies.** Lodging and airlines emerge as the clearest winners, supported by peak hotel occupancy rates of 90–95%, with room prices rising by up to +15–20% in selected host cities following the draw phases. Airlines benefit from structurally constrained capacity growth ranging between +0.4% and +2.1% in Q2 2026, supporting strong pricing power on key domestic and international routes. Meanwhile, food & beverage, retail and entertainment industries also stand to gain meaningfully from elevated match-day consumption, particularly in Mexico where football-related social spending is deeply embedded in consumer behavior.
- **However, the macroeconomic impact remains modest relative to the size of host economies, translating into a GDP uplift of approximately USD6.1bn in the US (+0.1pp quarterly growth), USD1.7bn in Mexico (+0.3pp) and USD1.3bn in Canada (+0.2pp).** The event is therefore best characterized as a high-intensity, short-duration demand shock rather than a structural growth driver, with benefits concentrated in tourism-sensitive sectors and constrained by substitution effects, capacity bottlenecks and regulatory frictions. Ultimately, the 2026 World Cup will deliver clear sectoral winners – hotels, airlines and urban tourism ecosystems – while reinforcing the importance of execution, mobility infrastructure and cross-border coordination in shaping the final economic outcome.

Bigger and better? The tournament that outgrew its borders

The Football World Cup 2026 is set to become the largest football tournament ever organized and arguably the most ambitious sporting event in modern history. But will this ultimately lead to stronger economic gains? For the first time, FIFA has expanded the competition from 32 to 48 national teams, increasing the number of matches from 64 to 104 and extending the tournament across three host countries: the US, Canada and Mexico. The competition, which begins in June 2026 and concludes in July, represents a structural break from previous World Cups. Rather than concentrating activity in a single host nation, FIFA has effectively created a continental-scale event stretching across North America, spanning 16 host cities, thousands of kilometers of travel corridors and multiple regulatory jurisdictions. From an economic perspective, the tournament should be viewed less as a sporting event and more as a six-week tourism boost, and a consumption and media platform capable of mobilizing 6.5mn attendees (2.6mn international visitors) and generating USD9.1bn of GDP in June-July across the three nations.

The scale of the tournament is unprecedented. Previous World Cups typically attracted between 3.0-3.4mn stadium spectators. With 104 matches and significantly larger aggregate venue capacity (67,000 people per stadium on average), FIFA 2026 could become the first tournament in history to potentially surpass 7mn cumulative attendees (if all tickets get sold), effectively doubling the scale of fan participation observed in recent editions (Figure 1). The FIFA itself published a 2023–2026 total potential revenue of USD13bn (vs USD6.4bn for Qatar), an increase of USD6.6bn compared to the previous 2019-2022 cycle, driven mainly by higher income from TV broadcasting rights (USD4.3bn total, +USD964mn vs Qatar), ticket sales and hospitality rights (USD3.1bn total, +USD2.6bn) and marketing rights (USD2.7bn total, +USD927mn), with additional smaller gains from licensing and other revenue sources, reflecting strong commercial momentum ahead of the expanded tournament.

Figure 1: FIFA World Cup reach in previous tournaments, vs expectations for 2026



Sources: Reuters LSEG, UN, FIFA, Allianz Research. Note: People watching at least 20 minutes of a match on TV

However, the decentralized nature of this World Cup – spanning multiple countries – creates both opportunities and challenges, with clear winners and losers. The distribution of economic gains will be driven not only by where travelers choose to go, but also by each country's preparedness, infrastructure and capacity to welcome them effectively. One of the most distinctive features of FIFA 2026 is that none of the three host nations needed to build a new generation of World Cup stadiums from scratch. Unlike recent tournaments that required massive upfront infrastructure investment, North America is largely leveraging existing assets. Qatar, in contrast, invested roughly USD220bn ahead of the previous World Cup, largely in long-term infrastructure projects, with USD7–10bn dedicated to build stadiums. This time, all 16 venues were operational before the tournament and require mostly renovations, capacity expansions and FIFA-compliance operational upgrades, with approximately USD2–2.5bn spent in this regard.

This makes FIFA 2026 one of the most capital-efficient World Cups in modern history. However, the primary challenge is not stadium readiness but rather transportation and hospitality logistics. The key execution risk is not stadium readiness but the ability of airports, hotels, transportation networks and border authorities to accommodate millions of visitors moving across 16 cities. Cities such as New York, Los Angeles, Dallas, San Francisco and Miami possess mature tourism ecosystems, extensive hotel inventories and world-class airport infrastructure.

However, congestion risks remain significant. Airports in several host cities already operate near capacity during peak travel periods and the World Cup is expected to create temporary spikes in passenger volumes that will test the resilience of transportation networks.

Canada's preparation efforts have focused more heavily on accommodation and urban mobility. Toronto and Vancouver are highly developed metropolitan areas with strong public transportation systems and excellent reputations for event management. However, both cities are likely to face accommodation pressures. As shown in Table 1, Vancouver (hosting seven matches) has the lowest hotel capacity among all host cities, while Toronto will host six matches with nearly three times the room inventory. This imbalance suggests that Vancouver may experience tighter lodging conditions and greater upward pressure on room rates during the event. Indeed, current hotel pricing already reflects Vancouver's constrained lodging supply. Among all cities, Vancouver has the highest accommodation cost of the tournament, with prices double the amount observed in half of the hosting cities of the list. Toronto, by contrast, benefits from a significantly larger hotel inventory and more extensive transportation infrastructure, which should provide greater capacity to absorb visitor demand, although meaningful pricing pressures are still expected.

Table 1: Comparing hospitality capacity across World Cup 2026 venues

Hosting cities	Hotel rooms capacity	Stadium capacity (# people)	Avg hotel price (USD)
Atlanta	95 000	71 000	\$331
Boston	65 000	65 878	\$611
Dallas	150 000	80 000	\$395
Houston	100 000	72 220	\$236
Kansas City	30 000	76 416	\$344
Los Angeles	103 000	70 240	\$384
Miami	65 000	65 326	\$378
NYC (final match)	140 000	82 500	\$624
Philadelphia	80 000	69 596	\$376
Seattle	40 000	68 740	\$446
San Francisco	150 000	68 500	\$283
Toronto	85 000	45 500	\$496
Vancouver	28 000	54 500	\$785
Guadalajara	65 000	46 232	\$335
Mexico City (open. match)	80 000	87 523	\$509
Monterrey	55 000	53 500	\$538

Sources: FIFA, Kick off adventures, Allianz Research

Mexico's preparation strategy reflects a different set of priorities. Significant investments have been directed toward airport modernization, urban transportation improvements and stadium renovations. Public and private spending associated with the tournament is estimated at roughly USD1.5-2.0bn, including upgrades to the three host venues – Estadio Azteca in Mexico City, Estadio BBVA in Monterrey and Estadio Akron in Guadalajara – as well as investments in airports, roads, public transport systems and surrounding urban infrastructure. While infrastructure quality remains less homogeneous than in the US, Mexico enters the tournament with an important competitive advantage: affordability to international visitors. For many football fans, attending matches in Mexico is materially less expensive than attending equivalent matches in major American cities. This affordability extends beyond accommodation costs to include food, transportation and entertainment expenses, potentially positioning Mexico as the most attractive destination for international supporters operating under budget constraints.

However, lower costs alone do not guarantee proportionally larger economic gains. Part of the spending generated by the tournament is likely to be concentrated in a limited number of host cities and sectors, while import leakages, profit repatriation by international operators and substitution effects may reduce the net domestic impact. Furthermore, accommodation bottlenecks, security concerns and transportation congestion could constrain visitor spending and limit the extent to which tourism demand translates into broader economic activity. As observed in previous host countries, headline visitor numbers and spending estimates often overstate the final contribution to economic growth once these offsetting factors are taken into account.

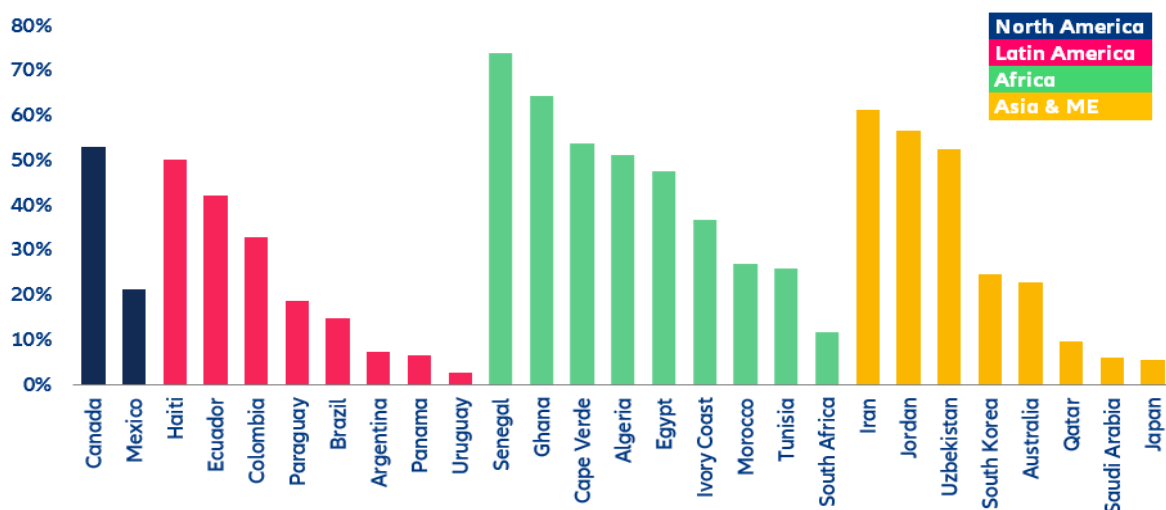
Breaking borders on the pitch, reinforcing them at the gate

While the US is a major global tourism hub – ranking third worldwide in international arrivals with nearly 70mn visitors in 2025 – and will host the largest share of matches in this World Cup, the surge in inbound tourism is not likely to be as strong as one might expect this year. Three factors help explain this: (1) The White House's growing travel ban, (2) differing football cultures across the three host nations and (3) exchange rate dynamics between countries.

One of the most underappreciated risks facing the tournament relates to immigration and visa policy. Eleven of the 16 host cities are located in the US, while Canada and Mexico will host in two and three cities, respectively. The US will stage 78 of the tournament's 104 matches (including the final in NYC), ensuring that the majority of economic activity will naturally accrue to American cities. Yet, the geographic footprint also introduces complexities not observed in previous tournaments. Fans following their national teams across multiple venues may be required to undertake several domestic flights, travel across international borders and navigate significantly different immigration systems. The operational challenge facing FIFA is therefore considerably greater than in Qatar 2022, where all matches were located within a relatively compact geographic area. Success will depend not only on stadium operations but also on the seamless functioning of airports and border control agencies.

Visa-related barriers vary significantly across the three host countries. First, the tourism visa cost difference is noticeable. The US presents the highest friction, with a standard B1/B2 visa fee of USD185 for most non-visa-waiver travelers, alongside stricter screening requirements. Canada offers a lower-cost and more streamlined process, with most visitors paying around CAD100 (USD72) for a tourist visa or only a minimal fee under the eTA system for eligible nationalities. Mexico, by contrast, has the lowest barrier, with tourist entry typically free or included in airfare for air travelers. Beyond the price, entry procedures also differ significantly. Unlike previous World Cups, where entry procedures were often simplified or partially suspended for ticket holders, the 2026 tournament will occur during a period of heightened scrutiny surrounding international mobility. Canada and Mexico generally maintain relatively accessible tourist visa systems for many nationalities. The US, however, presents a more complicated picture. Visa waiting times remain lengthy in several countries, approval rates vary significantly by nationality and border procedures have become increasingly strict under the new Trump Administration. As shown in Figure 2, US State Department data indicates that applicants from 11 of the 48 countries qualified for the World Cup face visa refusal rates exceeding 40%. While these factors may appear secondary from a sporting perspective, they are highly relevant from an economic standpoint because international attendance remains one of the primary drivers of incremental tourism spending.

Figure 2: US visa refusal rates for tourist and business travelers (B-visas only) by nationality in 2025



Sources: US Department of State, Allianz Research. Note: Excluding the 16 European qualified nations

The issue becomes even more important when viewed through a geopolitical lens. Football's fastest-growing fan bases are increasingly located in the Middle East, Africa, Central Asia and certain regions of Latin America but visitors from these regions may perceive travel to the US as administratively burdensome or politically uncertain. In

some cases, specific nationalities face significant travel restrictions or heightened scrutiny during visa processing. Although the Trump administration recently announced that soccer fans from 50 countries subject to an initial USD15,000 visa bond requirement (a deposit) would be exempt – provided they can demonstrate travel to the US for the World Cup with valid match tickets – nationals of Iran and Haiti remain subject to entry restrictions. In addition, travelers from Côte d'Ivoire and Senegal are subject to partial restrictions under the expanded scope of the travel ban. The result may not be a complete absence of international visitors, but rather a redistribution of demand toward Canada and Mexico.

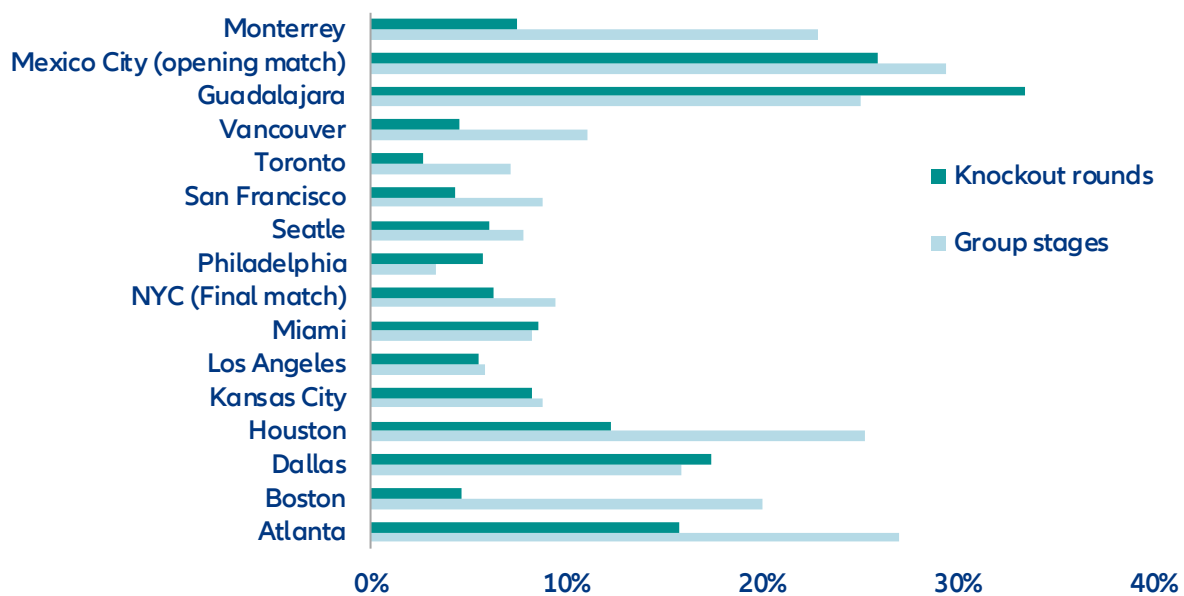
Soccer culture, and how the sport is experienced and consumed, also differs considerably across hosting countries. A defining characteristic of FIFA 2026 is the stark contrast among its hosts. Mexico remains the most football-centric nation, where the sport is deeply embedded in national identity and cultural life. Local demand for matches is expected to be exceptionally strong, and the stadium atmospheres in Mexico City, Guadalajara and Monterrey are likely to rank among the most vibrant of the tournament. The US presents a different profile. Although soccer has experienced significant growth over the past two decades, it remains secondary to the NFL, NBA and college sports in terms of cultural relevance and media attention. Canada occupies an intermediate position, where soccer has benefited from increasing popularity among immigrant communities and younger demographics, but still lacks the historical depth observed in Latin America and Europe. Available inventory tickets suggest that stadium occupancy is unlikely to reach 100% across all matches in the US. Remaining ticket inventory appears to be concentrated in selected group-stage matches, particularly in the US and Canada, while demand for Mexico-hosted matches remains comparatively stronger.

Three countries, three currencies. Exchange rates clearly also have a major influence on travelers' preferences. Exchange rate dynamics are likely to play a meaningful role in shaping travel demand, visitor flows and spending patterns. In recent years, the US dollar has remained relatively strong against most major and emerging-market currencies, increasing the overall cost of travel to US host cities and potentially discouraging price-sensitive international visitors. By contrast, the Canadian dollar has traded at a discount to the USD, enhancing Canada's value proposition for international travelers seeking a similar North American experience at a lower effective cost. Meanwhile, the Mexican peso, despite periods of notable strength driven by robust economic fundamentals and nearshoring investment flows, continues to offer a significantly lower price point for accommodation, food, transportation and entertainment compared with many northern destinations.

Follow the money: The net winners

The lodging sector is positioned to be among the clearest beneficiaries of the tournament, although outcomes will vary considerably by city. New York, Miami, Los Angeles, Dallas and San Francisco possess extensive hotel inventories and significant experience hosting large-scale international events. These cities are likely to absorb demand relatively efficiently, albeit at elevated room rates. In contrast, cities such as Seattle and Vancouver face more meaningful capacity constraints. Limited room inventories is generating substantial price increases during peak periods, benefiting hotel operators but potentially discouraging some visitors. As shown in Figure 3, hotel room prices increased by approximately 15% following the group stage draw, with the three Mexican host cities recording the most pronounced gains (all exceeding +20%). Guadalajara in particular experienced especially strong demand, reflecting its historical position as the most affordable accommodation market among the three Mexican cities. As the final stages of the tournament shift away from Mexico, with Dallas and Atlanta set to host the semifinals, hotel room prices rose by an average of +11% following the knockout round draw.

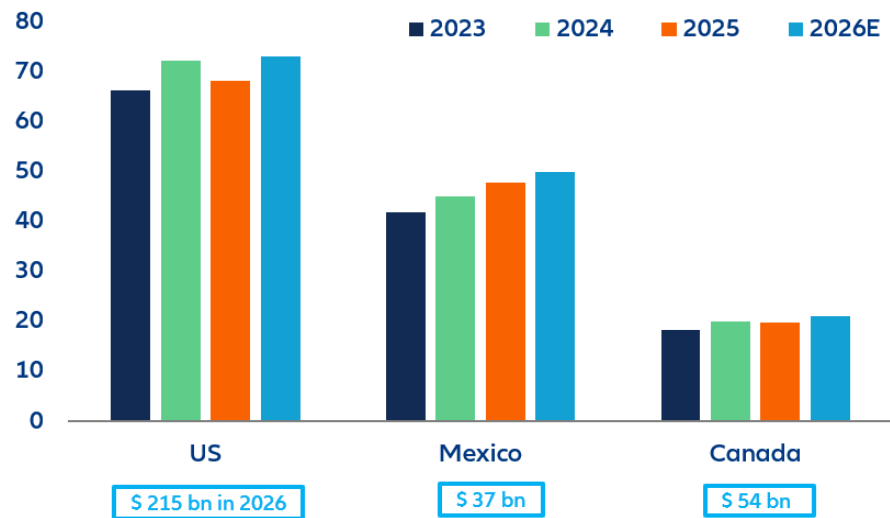
Figure 3: Observed hotel room price change after the draw of



Sources: Lighthouse, Allianz Research.

Overall, the event is expected to provide a meaningful boost to the broader tourism sector in 2026. However, part of the y/y increase – particularly in the US – also reflects a low base in 2025, when traveler confidence was weakened amid uncertainty around Trump-era travel ban policies. Indeed, while international arrivals increased by +6.2% y/y in 2025 in Mexico, they declined by -5.5% in the US (see Figure 4) while were flat in Canada. Against this backdrop, the event could provide some relief to the US tourism sector amid the recent and still ongoing slowdown, with international visitor volumes again down -1.7% y/y in the first quarter of this year. The World Cup could therefore lift 2026’s US international arrivals to approximately 72–74mn visitors, implying a net uplift of +3% to +6% versus a normal baseline year, leading the country to collect around USD215bn of tourism receipts for the entire year.

Figure 4: Tourism evolution by country: number of international visitors (million people) and expected tourism receipts for 2026FY in USD (blue boxes)



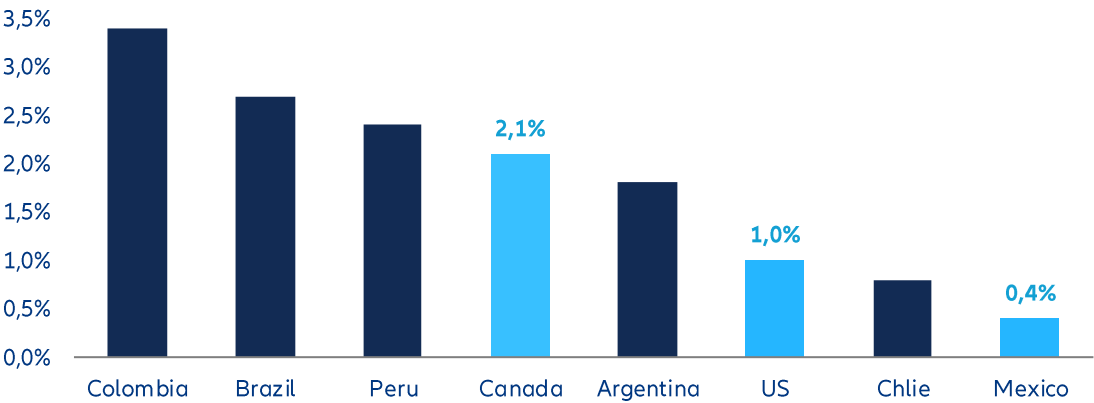
Sources: UN Tourism, Allianz Research.

The food and beverage industry should experience a meaningful boost in parallel, but media and broadcasting companies remain among the largest beneficiaries overall. Stadium consumption, fan festivals, restaurants and bars typically generate strong sales growth during major sporting events. The effect may be particularly

pronounced in Mexico, where football consumption patterns are closely associated with social gatherings and beverage purchases. Nevertheless, the World Cup's greatest economic value has always derived from its global audience rather than in-person attendance. Advertising revenues, sponsorship agreements, streaming subscriptions and broadcasting rights collectively account for a significant share of FIFA's commercial ecosystem. Construction firms, by contrast, are likely to experience only modest incremental gains because most infrastructure spending has already occurred during the preparation phase.

Beyond ground-based tourism activities, airlines also stand out as one of the most direct beneficiaries, turning the period into an opportunity to improve margins after four months of abnormally high jet-fuel prices. The tournament's geographic footprint virtually guarantees elevated demand for both international and domestic flights, particularly as supporters have to move between host cities that are kilometers away. Clearly airlines – alongside hotels – are positioned as the two sectors with the strongest pricing power during the tournament, driven by limited substitution options and surging demand. This is particularly evident on long-haul domestic routes such as Seattle–Miami, Los Angeles–New York, and international ones such as Mexico City–Vancouver, where flight times exceed five hours and alternative modes of transport are effectively non-viable. Since the onset of the Strait of Hormuz conflict, jet fuel prices have averaged roughly twice their 2025 average level. Therefore the World Cup is likely to provide airlines with a temporary opportunity to enhance yields and margins through dynamic pricing, capacity optimization and higher ancillary revenues. Yet, while the event should drive strong demand on selected routes, its impact is expected to be tactical rather than transformational for the industry's overall profitability. Besides the oligopoly nature of the business, the sector's capacity may not be at ideal levels. According to IATA, scheduled passenger seat capacity in Q2 2026 is expected to grow by just +1.0% in the US and +0.4% in Mexico (Figure 5), suggesting that airlines will enter the World Cup with relatively constrained supply. As a result, the tournament is likely to be characterized by strong pricing power rather than significant capacity expansion, supporting higher fares and improved yields. While Canada's capacity growth is projected at a higher 2.1%, its smaller and more concentrated aviation market may still face bottlenecks during peak tournament periods. Overall, the data point to a demand-driven environment where airlines are better positioned to maximize profitability through revenue management and network optimization than through volume growth.

Figure 5: Americas air passenger seat capacity scheduled for Q2 2026 by country, y/y growth rate



Sources: IATA, Allianz Research

A positive but modest GDP boost of USD9.1bn across North America

The World Cup represents a positive yet modest demand shock across North America during the June-July 2026 period, with foreign tourism spending being the most important channel. From a macro-economic perspective, sporting events entail four types of spending which impact GDP growth of the domestic economy: foreign tourism spending, including air travel (counted as services exports in the national accounts); domestic tourism spending (counted as household services consumption); security/public administration spending (mostly government consumption) and infrastructure spending (investment). The economic literature on sporting events points to crowding-out effects in the tourism industry partially offsetting the positive spending impulse: some regular tourists typically avoid host cities, while some locals prefer to stay at home. This substitution effect is typically large for domestic tourism spending. Previous World Cup tend to confirm this: South Africa's only identified about 310,000 foreign tourists who came specifically for the World Cup (vs an annual baseline of approximately 8mn visitors). Russia 2018 tells a similar story: Total tourist volume was similar to the prior year. The tournament shifted the composition of visitors from neighboring countries toward more distant origins; it did not expand the total. "Operations" spending (venue management, broadcasting support, hospitality services, temporary workers etc.) is not a separate GDP component: it ultimately flows in visitors spending (therefore included into household consumption and services exports), security and public services (part of government consumption) and broadcasting services (services exports). We do not count infrastructure spending which has already occurred and was accounted into GDP mostly in prior quarters and years, as our focus is on the June-July period¹. Marketing/broadcasting rights and tickets purchased accruing to Switzerland-based FIFA would not affect GDP of the host countries, but the transfer of income towards Switzerland would be negative for the current account of host countries and create a wedge between GDP (total spending in the domestic economy) and GNI (Gross National Income, i.e. GDP adjusted by the flows of income with foreign countries).

Tourism spending itself is expected to increase by USD8bn in North America, split between USD5.4bn in the US, USD1.4bn in Mexico and USD1.2bn in Canada, largely dominated by foreign spending. Tourism spending is the product of the number of visitors, their average stay and their daily expenditures (on food, local transportation, accommodation, retail purchases etc). The unprecedented scale of the 2026 World Cup should increase the number of visitors and/or the length of the average stay. Table 2A and 2B depicts our forecast of foreign and domestic tourism spending across the three countries, laying out our assumptions on the number of visitors, the average stay, daily expenditure, air travel expenses (accruing to local airlines) and displaced tourism. On average, we expect foreign visitors to spend USD350 per day in the US, USD280 in Canada, and USD180 in Mexico, with daily spending by local visitors smaller. The average stay is expected to be 8-10 days for international visitors and 3-4 for domestic tourists. Air travel accruing to domestic haulers (counted as services exports) is estimated at USD0.8bn for US airlines and USD0.2bn for Mexican and Canadian airlines. Latest estimates suggest that the six-week tournament will attract around 6.5mn visitors over the course of the competition (including 3.8mn in the US), split between 4mn local attendees and 2.6mn international visitors. In all, the aggregate tourism spending (international + domestic) would amount to USD8bn across North America, split between USD6.8bn in tourism exports (foreign tourism spending) and USD1.2bn in household consumption (domestic tourism spending). These numbers are net of displaced tourism spending (crowding out), which are sizeable for domestic tourism. In the US alone, the net total boost of spending is expected to be USD5.4bn (USD4.6bn for foreign tourism and USD0.8bn for domestic tourism), versus USD1.4bn in Mexico and USD1.2bn in Canada.

¹ In contrast, services spending such as flight ticket purchase is recorded into GDP when the flight takes place, not when the ticket is purchased. The impact of infrastructure spending on GDP is, moreover, harder to estimate because of potential positive spillovers effects magnifying the initial impact (fiscal multiplier above 1).

Table 2A: Foreign tourism spending and domestic tourism spending forecasts

Country	Foreign visitors (mn)	Average stay (days)	Daily expenditure (\$)	International air transport travel (\$bn)	Gross foreign spending (\$bn)	Displaced tourism (\$bn)	Net tourism export gain (\$bn)
United States	1.5	10	350	0.8	6.0	-1.4	4.6
Mexico	0.7	9	180	0.2	1.3	-0.1	1.2
Canada	0.4	8	280	0.2	1.1	-0.1	1.0
Total	2.6	—	—		8.4	-1.6	6.8

Table 2B: Domestic tourism spending and domestic tourism spending forecasts

Country	Domestic visitors (mn)	Average stay (days)	Daily expenditure (\$)	Gross domestic spending (\$bn)	Substitution / leakage (\$bn)	Net GDP effect (\$bn)
United States	2.3	4	250	2.3	-1.5	0.8
Mexico	1.0	3	120	0.4	-0.2	0.2
Canada	0.7	3	180	0.4	-0.2	0.2
Total	4.0	—	—	3.1	-1.9	1.2

Sources: FIFA, UN Tourism, various, Allianz Research.

Security operations spending will add to the economic boost. Governments will increase spending on police overtime, border controls, intelligence activities, cybersecurity, emergency medical services and venue security. Most of this spending is government consumption. In total, ballpark estimates put these to an aggregate amount of USD0.7bn in the US, USD0.3bn in Mexico and USD0.1bn in Canada – only accounting for the spending during the June-July period (earlier preparation spending could increase these figures by 20-30%).

In total, the FIFA World Cup is expected to boost GDP over June-July by USD6.1bn in the US (+0.1pp quarterly growth), USD1.7bn in Mexico (+0.3pp) and USD1.3bn in Canada (+0.2pp). Table 3 summarizes the channels of impacts and the total GDP effect across the three host countries for the June-July period. The foreign tourism channel is by far the largest, at USD6.8bn, with domestic tourism (USD1.2bn) and security spending (USD1.1bn) being close to each other. In total, US GDP is expected to receive a boost of USD6.1bn over June-July, Mexico USD1.7bn and Canada USD1.3bn. In terms of quarterly non-annualized GDP growth (mostly falling on Q3), this represents a boost of +0.1pp (+0.4pp annualized) above a counterfactual without FIFA World Cup in the US, +0.3pp (+1.2pp annualized) in Mexico and +0.2pp (+0.8pp annualized) in Canada.

Table 3: Summary of channels of impacts of FIFA World Cup on June-July 2026 GDP

Country	Net foreign tourism (\$bn)	Net domestic tourism (\$bn)	Security spending (\$bn)	Total tournament GDP impulse June-July (\$bn)	Quarterly GDP growth impact (p.p.)
United States	4.6	0.8	0.7	6.1	0.1
Mexico	1.2	0.2	0.3	1.7	0.3
Canada	1.0	0.2	0.1	1.3	0.2
Total	6.8	1.2	1.1	9.1	

Sources: FIFA, various, Allianz Research.

The US is expected to capture the majority of economic benefits simply because it hosts the majority of matches, though relative to its economy the effect will be small. The largest gains will likely be concentrated in transportation, hospitality, food services, entertainment and retail. Cities such as New York, Los Angeles, Dallas and Miami are particularly well positioned due to their combination of international connectivity, large hotel inventories and established tourism ecosystems. Corporate hospitality is also expected to play a major role, particularly for knockout-stage matches where premium seating and sponsorship packages command substantial prices.

Mexico's aggregate economic impact will be smaller in absolute terms but more meaningful relative to the size of its economy and of its tourism sector. In 2025, international overnight visitors nearly reached 48mn, i.e. +6% compared to 2024. Early 2026 indicators also point to continued strength, with international arrivals during the first quarter rising by a similar pace year-on-year and air passenger traffic from the US and Canada remaining on an upward trend. Given the close proximity of the three host countries, the tournament could also encourage significant "nearcation" travel, with fans opting for shorter trips within North America instead of long-haul destinations due to ongoing geopolitical tensions, disruptions to key international transport routes, higher transportation costs and security concerns affecting some long-distance destinations. From a macroeconomic perspective, the tournament is unlikely to have a transformative effect on growth given the size of the Mexican economy and the temporary nature of the event. However, it could contribute around 0.1-0.3pp to GDP growth in 2026 (i.e. between USD2bn and USD6bn) through higher tourism revenues, stronger private consumption and event-related investment.

Canada's gains will likely fall between those of the US and Mexico on a per-visitor basis but remain small in aggregate due to its limited number of host cities. With one of the smallest stadiums (BMO Field in Toronto), only 13 matches out of 104 will take place in Canada, with Toronto and Vancouver representing the country's sole host cities for the tournament. The country's strength lies in attracting relatively affluent travelers willing to pay premium prices for accommodation, transportation and hospitality experiences. Consequently, although Canada hosts only 13% of matches, spending per visitor may compare favorably with other markets.

These assessments are, as always, subject to the disclaimer provided below.

FORWARD-LOOKING STATEMENTS

The statements contained herein may include prospects, statements of future expectations and other forward-looking statements that are based on management's current views and assumptions and involve known and unknown risks and uncertainties. Actual results, performance or events may differ materially from those expressed

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(v) persistency levels, (vi) particularly in the banking business, the extent of credit defaults, (vii) interest rate levels, (viii) currency exchange rates including the EUR/USD exchange rate, (ix) changes in laws and regulations, including tax regulations, (x) the impact of acquisitions, including related integration issues, and reorganization measures,

and (xi) general competitive factors, in each case on a local, regional, national and/or global basis. Many of these factors may be more likely to occur, or more pronounced, as a result of terrorist activities and their consequences.

NO DUTY TO UPDATE

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Emerging markets in a fragmented world: From geography to resilience - the 4Rs framework

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In Summary

- **The Iran war marked the first major oil shock that did not trigger a broad emerging markets sell-off.** Markets repriced countries based on strengths and weaknesses rather than the traditional EM–DM divide. This supports a resilience-based investment framework built around the “4Rs” – Resource Position, Reserve Strength, Rate Credibility and Refinancing Structure – which increasingly explains cross-country differentiation more effectively. Although institutional mandates, benchmarks and trading-desk structures will continue to rely on the EM–DM distinction for the foreseeable future, portfolio construction frameworks that lean primarily on this historical classification risk becoming progressively less relevant.
- **Resource Position, not the EM–DM label, defines the fault line of the Iran shock.** Economies with large import dependencies such as Egypt, Romania, South Korea, Greece and the UK, have faced the strongest repricing pressures, while commodity exporters have benefited from improved terms of trade. Even in a downside scenario with oil prices above USD180/bbl, the pain would be concentrated within the energy-importing cohort.
- **Reserve Strength increasingly separates resilient sovereigns from vulnerable “triple-deficit” economies, irrespective of EM or DM classification.** Since the 2013 taper tantrum, many EMs have rebuilt fiscal discipline, strengthened current-account positions and stabilized debt trajectories, entering the Iran shock with roughly 1pp of GDP more fiscal headroom than at the onset of Covid-19. EM economies now account for roughly 60% of global GDP in PPP terms, up from around 40% in 2000. FX reserve buffers have continued to strengthen across the Middle East, Central Asia and Emerging Europe, while several advanced economies remain mired in persistent fiscal deficits and deteriorating external balances.
- **Rate Credibility has structurally improved across EMs and increasingly resembles DM-style monetary frameworks.** Inflation targeting is now the norm across most major EMs, and EM central banks tightened by an average of 780bps during the post-pandemic cycle versus around 400bps in the DM, frontloading hikes despite weaker growth. The Iran war has confirmed this convergence: no major EM central bank has been forced into emergency hikes, capital controls or disorderly stabilization. EM FX volatility has fallen materially, with several G10 currencies experiencing higher volatility during 2024–25. Outliers, notably Türkiye, Argentina and Nigeria, remain, but the broader convergence trend is intact.
- **Refinancing Structure has fundamentally changed the transmission of external shocks.** Foreign-currency debt shares have fallen by roughly 20–40pps across major countries including Brazil, Mexico, India, Indonesia and several CEE. Deeper domestic institutional investor bases and larger local-currency bond markets have reduced vulnerability to USD and Fed, turning FX depreciation into a macroeconomic adjustment mechanism rather than a solvency trigger. Currently, EM currencies adjusted in an orderly manner with no widespread defaults or emergency IMF interventions; even in a downside scenario of further Fed and ECB tightening, the likely outcome is slower convergence rather than a return to a crisis like the 1990s or early 2010s.
- **The compression of EM risk premia increasingly reflects structural convergence rather than cyclical overvaluation.** The excess spread of EM hard-currency debt over comparable DM credit has largely disappeared on a rating-adjusted basis. Even in a downside Iran escalation scenario, EM hard-currency spreads would likely widen from around 178bps to approximately 235bps and peak near 280bps, materially below the roughly 700bps reached during the Covid-19 shock. A compelling long-term case increasingly lies in EM local-currency debt, where structurally higher real yields continue to generate superior long-term risk-adjusted returns relative to DM fixed income.

The line between EMs and DMs is blurring

Emerging vs. developed is not a label – it is an investment hypothesis about institutions, policy credibility and the distribution of tail risks that is being tested by the fragmentation of the global economy. The traditional EM–DM split has been a shorthand for differences in the strength of institutions, the predictability of policymaking and the degree to which investors can trust rules to be stable over time. These institutional features matter because they shape inflation outcomes, fiscal discipline, market depth and, ultimately, the risk premium investors demand. Importantly, recent market experience has also highlighted that DM does not automatically mean risk-free: volatility in some developed government bond markets has risen materially since 2022, and fiscal uncertainty has become a more visible driver of term premia and curve behavior. In parallel, a structural fragmentation of the global economy, the US-China decoupling, the partial US retreat from its role as anchor of the rules-based order, friendshoring, supply-chain regionalization and growing regulatory divergence across blocs, has begun to redistribute risk away from the EM-DM axis and toward country-specific positioning within a contested international system. The two forces compound each other: DM credibility is eroding from within just as the global architecture that historically rewarded DM positioning is being rewritten from the outside.

Introducing the 4R framework: a resilience-based alternative to EM–DM

Throughout this paper we organize the analysis around four structural dimensions that explain cross-country market segmentation better than the EM–DM taxonomy:

- Resource Position: energy dependence, commodity exposure, vulnerability to terms-of-trade shocks such as the Iran war
- Reserve Strength: fiscal discipline, external balances, FX reserves, exposure to twin or triple deficits
- Rate Credibility: central bank independence, inflation-targeting frameworks, convergence of monetary policy reaction functions
- Refinancing Structure: local-currency funding shares, depth of domestic investor bases, resilience to USD and Fed cycles

The subsequent analysis maps the evidence onto these four dimensions of the “4Rs” framework.

A large and growing share of EM debt trades on fundamentals that increasingly resemble DM-style underwriting like credible central banks, improving external buffers and solid policy frameworks while EM risk dynamics improve. Parts of the DM universe display EM-like stress points through political constraints and fiscal slippage. The fragmenting global economic system reinforces this EM-DM convergence, as several EMs are positioned to benefit from ongoing supply-chain and trade realignments while parts of the DMs see their competitive positions deteriorating. In parallel, dispersion across countries remains high, meaning “EM beta” is less informative than identifying where fundamentals warrant investment-grade-like pricing and where high-yield-like tail risks still dominate.

A conceptual difference appears in EM sovereign debt. EM hard currency government bonds are, by construction, a spread asset because default/restructuring risk is non-zero and must be priced. DM government bonds, in contrast, are – just like EM local currency sovereign bonds – typically treated as the benchmark “risk-free” curve in their own currency. A significant change in a country’s economic and political outlook is first and foremost digested on FX markets. Investors should recognize that term premia can still move meaningfully with inflation uncertainty, fiscal outlook and market functioning, even absent classical default risk. For portfolio construction, the practical takeaway is that investors should separate spread risk from rate, term-premium and FX risk, and to avoid relying on labels when the underlying risk drivers have shifted.

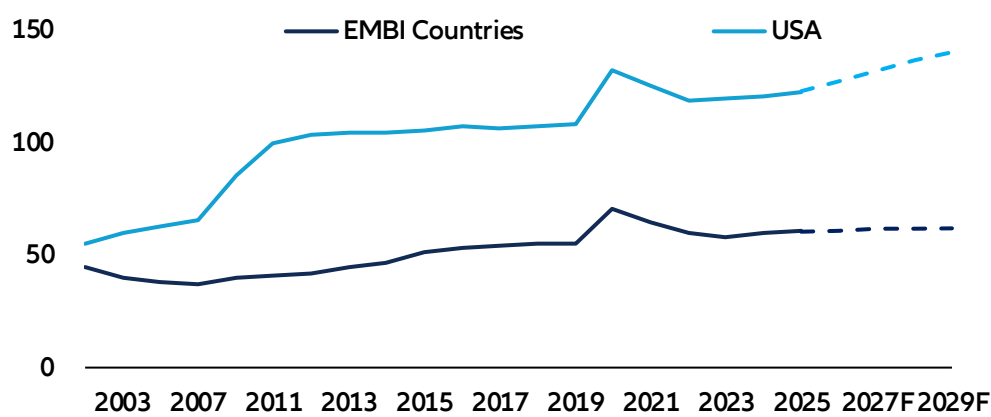
Improved fundamentals: fiscal discipline, external position

This section examines the first two Rs — Resource Position and Reserve Strength. The current shock is the latest data point in a deeper structural shift: the distinction between developed and emerging markets is blurring. Political instability, rising populism and unsustainable public finances were traditionally associated with EM

countries, but they have become descriptions for some of the world's largest and most 'developed' economies in recent years. While each country's political backstory is unique, the common theme is a pronounced deterioration of macroeconomic fundamentals – a dynamic that global fragmentation increasingly reinforces as weakening institutions and fracturing trade relationships feed on each other in a self-reinforcing spiral. Today, glaring fiscal imbalances in the US and other 'advanced' economies point to an inversion of the traditional world order, with little prospect of a meaningful reversal.

In strong contrast to their DM peers, EMs have seen strengthening fundamentals over the past decade. Spurred by the upheaval of the 2013 taper tantrum, which exposed underlying economic imbalances, many EMs have undergone significant rebalancing, strengthening their resilience. Even after the Covid-19 pandemic took hold, many EM policymakers remained fiscally prudent, resulting in primary fiscal balances returning to surplus within just a few years and debt-to-GDP stabilizing at modest levels, improved external positions across EM sovereigns and stronger growth outlooks for EMs compared to DM peers.

Figure 1: General government gross debt, EMBI countries and US, 2003–2029F (% of GDP)

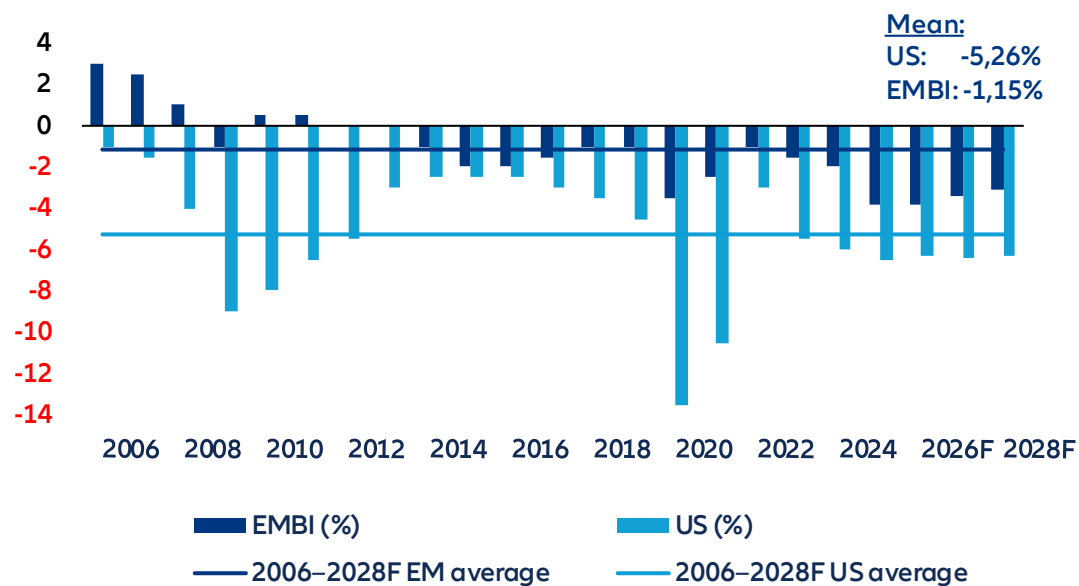


Note: Dashed lines indicate IMF/Bloomberg forecasts.

Sources: IMF, Bloomberg (forecasts), Allianz Research

The Iran war is a live test of Reserve Strength. The early evidence suggests EMs have entered the shock with more headroom (1pp of GDP) than the Covid-19 shock overall. However, this aggregate masks a wide dispersion in fiscal positions, from those that experienced forced reforms in recent years due to shocks and cyclical beneficiaries to those deteriorating. The successive shocks of Covid-19, the Ukraine war and the 2022–23 inflation surge functioned as a stress test that catalyzed reform in two distinct cohorts of improvers. Pakistan, Ghana and Romania were forced into adjustment by near-default conditions and IMF programs, with Argentina also in this group given Milei's dramatic primary surplus delivery in 2024–25. Colombia, Brazil and to a more ambiguous extent Mexico are in the second cohort, having improved through commodity tailwinds and limited tariff exposure. On the deteriorating side, Indonesia, Thailand, Bangladesh, Hungary and Nigeria have each seen pronounced erosion since 2020, driven respectively by subsidy expansion, fiscal stimulus, political transition costs, pre-election spending and oil-revenue volatility.

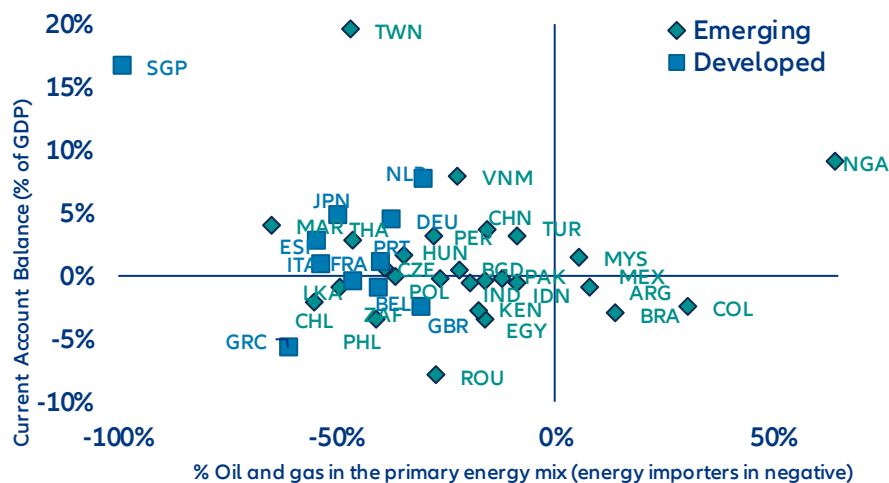
Figure 2: General government fiscal balance, EMBI countries and US, 2006–2028F (% of GDP)



Note: Horizontal lines show period averages (2006–2028F).
Sources: IMF, Bloomberg (forecasts), Allianz Research

The Iran war shock has revived the severity of global imbalances, but the differentiation it has produced is **not** between EM and DM – it runs along Resource Position, separating energy importers from exporters. In prior decades, an external shock of this magnitude would have triggered a near-mechanical sell-off of EM assets against developed-market safe havens, with the EM-DM dividing line acting as the primary axis of risk repricing. The 2026 episode has yet to produce such pattern. Those countries more exposed to the ongoing war are those EMs and DMs with current account deficits and large imports of hydrocarbons, which include countries such as Romania, Egypt, as well as the UK or Greece. If the conflict escalates further, this distinction would sharpen rather than dissolve: in a downside scenario of sustained supply shortages and oil prices spiking above USD180 per barrel, averaging close to USD100 for 2026, the most acute repricing would fall on this energy-importing, current-account-deficit cohort rather than on EMs as a whole. The remaining group would still experience a sell-off, but a materially reduced one, consistent with the broader resilience indicators discussed throughout this section. Even in a severe escalation, in other words, the dividing line would run through energy exposure, not through the conventional EM-DM taxonomy.

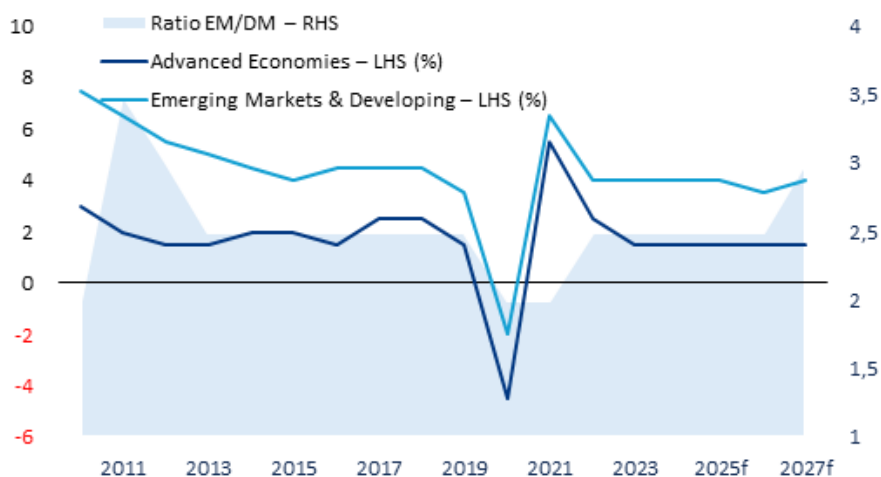
Figure 3: Heterogenous current account imbalances, but EMs better positioned with regards to energy independence than DMs



Sources: LSEG, Allianz Research

EM growth continues to outperform that of DMs, and the relative growth differentials are expected to widen further, continuing to reweight EMs at the center of the global economy. Emerging and developing economies now account for roughly 60% of global GDP in purchasing-power-parity terms, up from approximately 40% in 2000, a 20pps reallocation of global economic weight. The growth trajectory indicates a widening distance between EMs and DMs, causing a further acceleration of the EM catch up towards DM purchasing power levels. In addition, the Iran war shock has not disrupted this trajectory. Hence, the center of gravity of the global economy will continue to move towards the EM bloc in the years to come.

Figure 4: GDP growth, advanced economies and emerging markets & developing markets, 2011–2027F (% y/y)



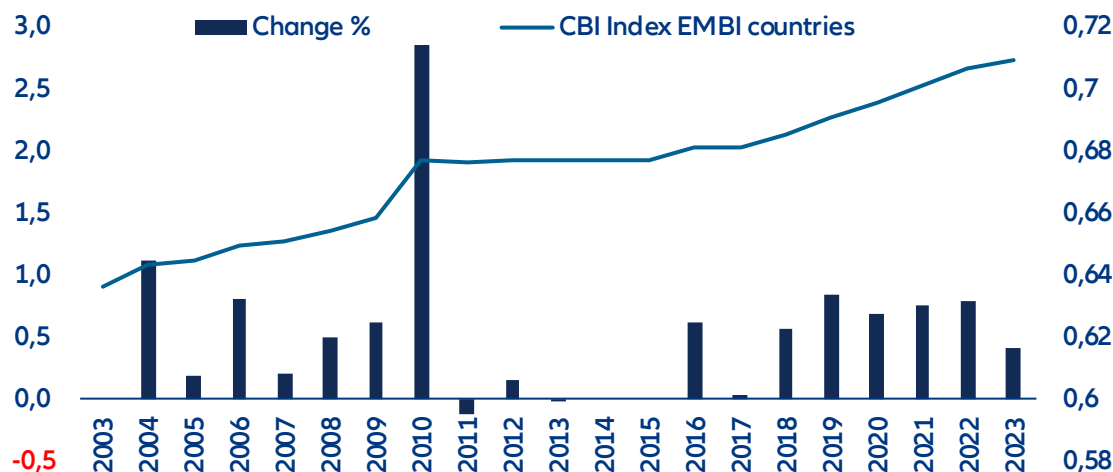
Note: . Shaded area shows the EM/DM growth ratio (RHS).

Sources: IMF, Bloomberg (forecasts), Allianz Research

Monetary policy convergence: evidence and implications

The third R – Rate Credibility – captures three decades of EM progress on central bank independence and policy frameworks, accelerated by the GFC and pandemic shocks, although regions and income levels diverge. As shown in Figure 5, the most pronounced increase in central bank independence among EMBI countries occurred in the aftermath of the GFC, and while the index plateaued temporarily in the subsequent years, it resumed a continuous upward trajectory from 2016 onward. The upper-middle-income group (which captures most of the "core" emerging markets: Brazil, Mexico, South Africa, Malaysia, Thailand etc.) shows the most consistent upward trend, narrowing the gap with high-income countries throughout the 1990s–2010s. While Latin America remains ahead and the closest to DM convergence, MENA (with the exception of Israel) remains the laggard region. Eastern and Central Europe experienced a dramatic catch-up wave tied to EU accession. While low-income and low-middle income countries have experienced the quickest convergence towards DM standards in the last 15 years, the reforms undertaken by West Africa Economic and Monetary Union (WAEMU) countries sharply improved Africa's convergence, though it remains uneven. These reforms include central bank mandates covering everything from board composition, governor appointment rules and limits on lending to the government to financial and budgetary independence, reflecting a structural rather in how EM central banks are designed to operate. Overall convergence is being driven by regional integration (experienced in the EU and WAEMU), external inducements especially tied to IMF-induced reforms, given the conditionality of IMF programs on central bank independence, as well as crisis and shocks that trigger reforms.

Figure 5: Central Bank Independence (CBI) Index for Emerging Market Bond Index (EMBI) countries



Note: The line depicts the Central Bank Independence (CBI) Index, a weighted composite of six sub-dimensions (board independence, policy autonomy, policy objectives, lending restrictions, financial independence, and reporting requirements), scaled from 0 (no independence) to 1 (full independence). Bars show the year-on-year change in percentage points.

Source: CBI Database

Inflation targeting frameworks are the new normal. Twenty years ago, most EM central banks managed exchange rates or monetary aggregates. Today, South Africa, Brazil, Mexico, Chile, Colombia, India, Indonesia, Philippines, Thailand, Türkiye (nominally), Egypt (post-2024 reform), Ghana, Kenya and others run explicit inflation-targeting regimes with published targets, forecasts and forward guidance. The South Africa Reserve Bank is the latest to shift from a 3–6% range to a 3% point target, in itself a convergence move towards DM-style inflation anchoring. EM central banks tightened by an average of 780bps during the post-pandemic cycle, versus around 400bps in advanced economies, a decisive demonstration of improved policy credibility.

The Iran war also serves as a live test of increased EM central bank credibility and EM-DM monetary policy convergence. So far, no major EM central bank has been forced into emergency hikes, capital controls or disorderly stabilization measures, a significant break from the historical pattern of oil shocks triggering balance-of-payments crises across EM economies. The differentiation once again runs through fundamentals rather than the EM-DM label. In a downside scenario of further escalation, this resilience would be tested but not overturned: sustained energy-price pressure would likely prompt the Fed to deliver two further 25bp hikes and the ECB three, tightening global liquidity and raising refinancing costs for sovereigns with larger hard-currency debt loads. Crucially, even this tightening would be milder than the cumulative policy response seen during the Covid-19 and Ukraine-war inflation episodes, and its impact would be more contained. Hard-currency debt now represents a materially smaller share of total EM sovereign debt than in previous shock episodes, limiting the mechanical transmission of tighter global liquidity into EM balance sheets. In other words, the downside scenario would slow the convergence but not reverse it.

Increased orthodox monetary policy has reduced the inflation risk premium embedded in currencies, reducing volatility self-reinforcing the investor base. As inflation-targeting regimes have matured and central-bank independence has been entrenched, the inflation risk premium embedded in EM exchange rates has fallen. Lower pass-through reflects anchored inflation expectations, which in turn reduce the volatility and tail-risk character that historically defined EMFX as an asset class. During 2024/25, EMFX volatility was overtaken by G10 FX moves due to several parallel moves: the yen's performance, Trump's tariffs announcements and EM strength. The 2026 Iran war episode has provided new evidence: EM currencies have adjusted to the oil shock through orderly depreciation rather than crisis-style breaks, EMBI+ spreads have remained contained and no EM central bank has been forced into emergency hikes or capital controls, a stark contrast to the chaotic EM responses of 1973 or 1997.

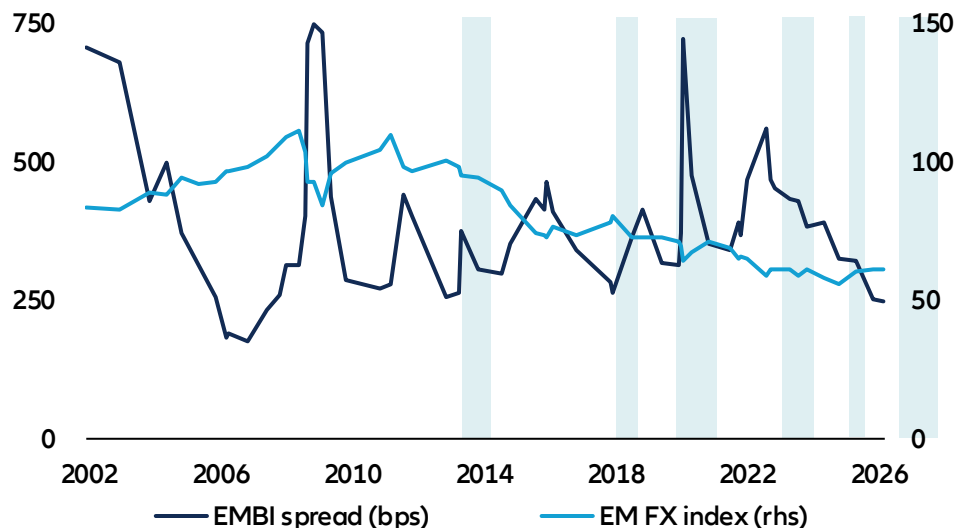
While the convergence has been taking place, a gap remains between EMs and DMs due to more government interference in key countries, market perception, as well as several outliers undermining the average. The convergence thesis is supported by the median EM central bank, but relevant central banks of the EM asset class, such as China or India, operate with higher government influence or within distinct frameworks. Meanwhile, a

meaningful tail of outliers continues to undermine the unweighted average, such as Türkiye (the 2021–23 unorthodox episode under political pressure, with credibility now being rebuilt), Argentina (whose monetary regime debate goes beyond central bank framework questions), and Nigeria (where multiple FX windows and "stealth tightening" remain in place despite Governor Cardoso's reform agenda). Egypt and Pakistan are mid-transition cases where IMF programs have catalyzed reform but track records are short. Notably, Brazil, often cited as a test case given the government pressures on the central bank, has emerged as a convergence success as political pressure did not prevent the September 2024 reversal back into hikes.

Shock absorption and resilience in EMs

This section turns to the fourth R – Refinancing Structure – alongside the broader shock-absorption toolkit. The structural fragility that has historically defined EMs, including exposure to external shocks, dollar funding and commodity volatility, has paradoxically become the foundation of their resilience. Repeated stress events over the past two decades forced EMs to adopt institutional and structural reforms and the evidence is visible in market behavior itself. EM hard-currency (HC) spread widenings firmly recovered after each crisis and show a level of resilience since the global rate hiking cycle triggered a sell-off in 2021–2022, which would have seemed unthinkable just a few years earlier (Figure 6). Even more astonishing is the stability in EM currencies. Previous crises like the GFC in 2008–2009 or the EM sell-off in 2015 triggered major losses in EM currencies, but since then EM currencies show a remarkable stability, as moderate FX losses typically fall short of the EM excess carry. The 2018 EM sell-off, the 2020 Covid-19 shock, the 2022 Ukraine-driven inflation surge and the 2026 Iran war have each tested and reinforced the asset class. Fundamental strengthening as well as larger and deeper shock absorbers like FX reserves levels, higher FX flexibility to absorb and lower level of HC debt are decreasing external vulnerabilities linked to the USD and the Fed.

Figure 6: EM hard-currency spreads and FX as stress indicators and recovery dynamics across major crisis episodes, 2002–2026



Note: The time series approximate the development of EM hard-currency spreads (JP Morgan's flagship EMBI index) and EM FX (JP Morgan's EM currency basket EMCI) around major crisis affecting EM debt, namely the Global Financial Crisis 2008-2009, taper tantrum 2013, the EM sell-offs in 2015 and 2018, Covid-19 in 2020 as well as the global rate hiking cycle in 2021-2022. Intervals between major crisis events are smoothed for readability.

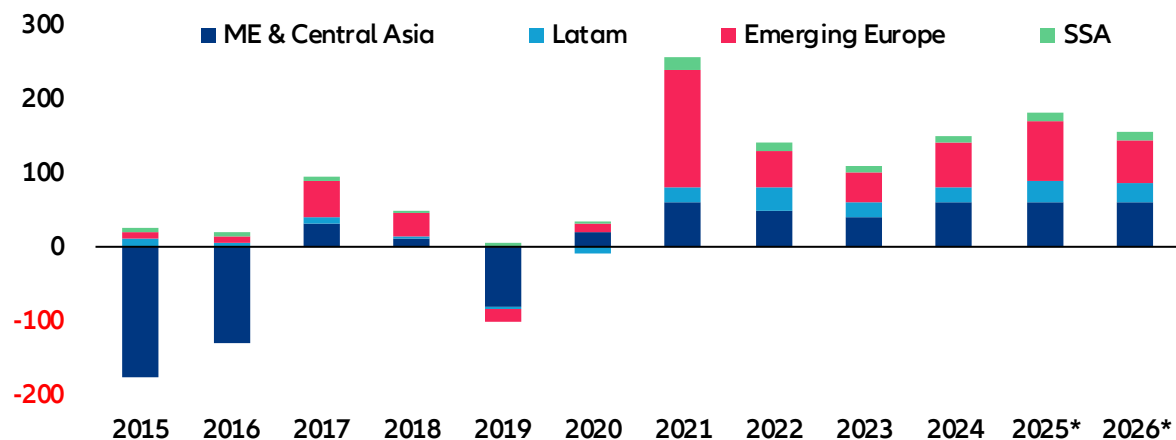
Sources: JPMorgan, Allianz Research

EM FX reserves have proven remarkably resilient over the long term, despite repeated bouts of external stress.

As shown in Figure 7, aggregate reserve accumulation across EM regions turned sharply positive again after the pandemic-driven drawdown in 2019 and 2020, with the Middle East, Central Asia and Emerging Europe contributing to the bulk of inflows since 2021. While the pace of accumulation has moderated from its post-

pandemic peak, the structural trend remains constructive, with forecasts for 2025 and 2026 pointing to continued reserve building across most regions. Sub-Saharan Africa and Latam remain the most vulnerable in terms of reserve adequacy, but even here the direction of travel has improved relative to the acute stress periods of 2015 to 2016 and 2019. Taken together, the data suggests that EM reserve buffers are broadly stronger today than a decade ago, providing a meaningful cushion against external shocks, including the kind of energy-price and financing-condition volatility associated with the current geopolitical environment.

Figure 7: EM FX reserves by region, 2015–2026F (USD bn)

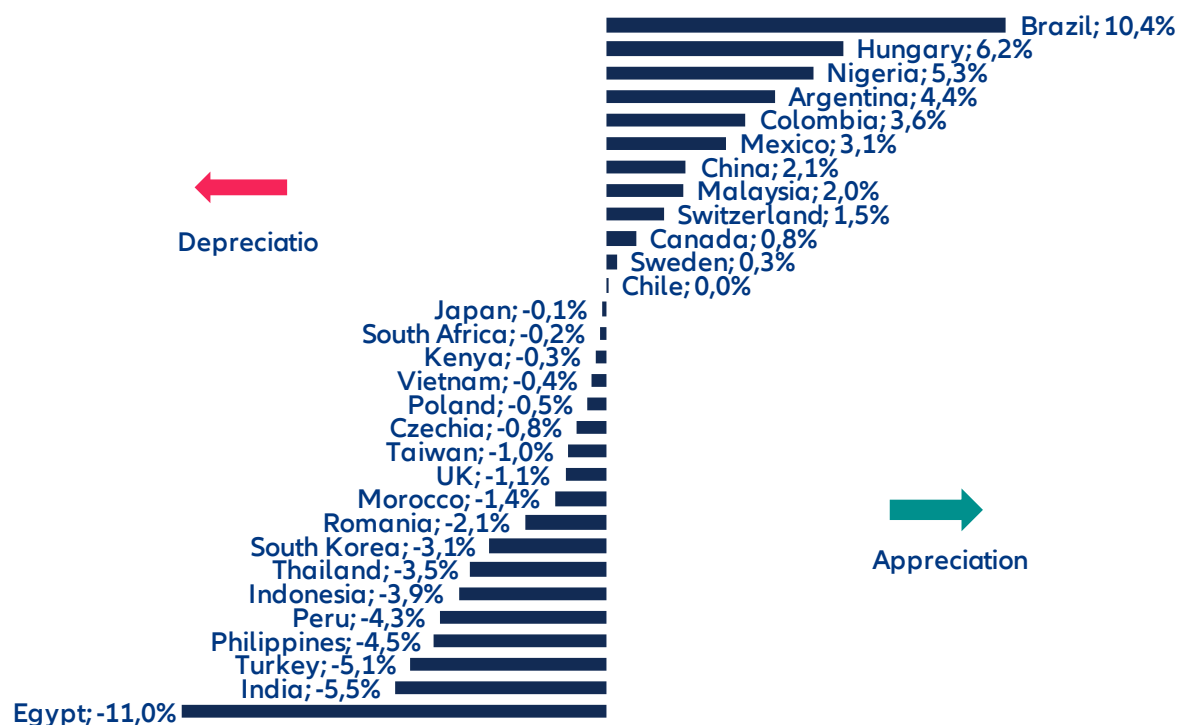


Note: as of April 2025, *indicates forecasts.

Source IMF, Bloomberg (forecasts), Allianz Research

Beyond reserve buffers, the broader shift toward flexible exchange rate regimes has been the second pillar of EM shock absorption, allowing currencies to function as adjustment mechanisms rather than crisis triggers as they often did in prior decades. The ongoing energy-linked shock is a great example. The dividing line between energy importers vs exporters, and exposure to Middle Eastern hydrocarbons, holds true in the currency space as well. The most exposed Asian countries have led the depreciation cycle since the start of the war, given their heavy exposure to Gulf energy imports, among them the South Korean won, together with other energy-importing countries such as Romania, Morocco or Türkiye. On the opposite side, energy exporters, many in the LATAM region, have benefited the most from the shock. Two notable outliers complete the picture: Hungary, where the post-election political transition has overridden the energy-trade signal, and China, whose low oil-import intensity and continued access to discounted Russian energy have largely insulated it from the shock. The Egyptian pound stands out as one of the worst-performing LCU, where pre-existing external imbalances have been compounded by geographic proximity to the conflict. Egypt is, however, also illustrative of a broader structural change: it is one of a cluster of historically crisis-prone EMs – alongside Ghana, Nigeria, Pakistan, Ethiopia and Sri Lanka – that have shifted from managed pegs and multiple exchange-rate windows toward market-determined floats since 2022, in most cases under IMF programs. This matters for the downside scenario. In previous shock episodes, these economies typically defended overvalued pegs until reserves were exhausted, at which point the currency broke and a balance-of-payments crisis followed. With more flexible regimes now in place, a further escalation of the energy shock would still produce a sharp depreciation in this cohort, but as a gradual adjustment rather than a sudden-stop rupture, with the currency absorbing the shock incrementally instead of triggering it. The contrast with the 1997 Asian crisis or Egypt's own pre-2024 experience is the clearest illustration of how regime choice has changed the character of EM shock transmission.

Figure 8: Currency performance 2026 YTD



Sources: LSEG, Allianz Research

Refinancing Structure has been transformed: EM sovereigns have shifted their funding mix from hard currency to local currency over the past two decades, with foreign-currency debt shares falling by 20–40pps across leading EMs. Brazil, Mexico, India, Indonesia, Thailand and the major CEE sovereigns now fund themselves predominantly in domestic currency, supported by deeper domestic institutional investor bases and growing foreign participation in local currency bond markets. The implication is structural: a stronger dollar no longer inflates EM debt burdens to the same extent as it did in 1997, 2013, or still in 2018. Local currency depreciation has become an adjustment mechanism rather than a solvency event which is the single most important reason the Iran war shock has so far been absorbed without disorderly defaults, capital controls or IMF emergency programs.

However, the USD continues to be a strong external influence on EM moves, causing central banks to be ahead of the Fed. Given that the USD anticipates Fed expectations, EM central banks across the board track the Fed. During the 2021–22 inflation surge, EM central banks in Brazil, Mexico, Chile, Czech Republic, Hungary and Poland tightened ahead of the Fed. That is a complete reversal of the historical pattern where EMs reacted defensively to Fed moves. In the subsequent easing cycle, they started to ease only after the Fed's easing cycle began in 2024. During the latest Iran war response, the monetary authorities of South Africa, Egypt and Israel all framed their pauses to monetary easing identically, referring to "inflation expectations anchoring" and "data dependence". No EM central bank has hiked aggressively nor imposed capital controls, in contrast with the 1973 or 1979 oil shocks, when EM responses were disordered and followed devaluations and defaults. This demonstrates the transition towards a converged reaction function: EM central banks now follow their own script anticipating the Fed rather than just acting in reaction to its moves.

What does this mean for EM debt investors?

The 4Rs framework has clear implications for portfolio construction, but the EM–DM taxonomy will not disappear from the investment universe overnight. Benchmark providers, institutional mandates and asset-manager desk structures (EM versus DM credit, EM versus DM rates) remain organized around the historical classification, and these institutional features will continue to shape how capital is allocated for the foreseeable future. With that caveat in mind, improving monetary and fiscal credibility together with an increasing capacity to absorb shocks – captured across all four Rs – reduces the downside of EM debt investments and changes their role in institutional portfolios. These structural improvements carry direct implications for portfolio construction. For risk-

constrained investors – pension funds, insurers, sovereign wealth funds – the historically cautious stance toward EM debt was grounded in real volatility: the "Fragile Five" of 2013, the "Troubled Ten" that followed, serial default cycles and the chronic vulnerability to Fed tightening. That era reflected genuine fragility. But as fiscal credibility improves, monetary frameworks mature and shock-absorption capacity deepens, the risk premium embedded in EM debt increasingly overstates actual portfolio risk. The practical consequence is meaningful: lower volatility and reduced tail risk justify higher EM debt allocations within existing risk budgets, without requiring investors to take on more aggregate portfolio risk. For liability-driven investors in particular, this is not a marginal shift – it represents a structural re-rating of an entire asset class, and portfolios that fail to reflect this new reality may be leaving risk-adjusted return on the table.

Despite the long-term improvement in EM fundamentals and the convergence with DMs, investors in EM hard-currency debt were historically paid a significant premium for "EM risk" over "DM risk". On a like-for-like basis across rating classes, EM hard-currency debt traded at a meaningful spread over comparable developed market bonds. This excess reflected genuine incremental risks, including thinner liquidity, greater sensitivity to dollar strength and commodity cycles, political instability, currency mismatch on sovereign balance sheets and a structurally narrower investor base that demanded compensation for volatility and sudden-stop risk. History vindicated some of that caution. The Tequila, Asian and Russian crises of the 1990s, Argentina's repeated defaults and the 2013 taper tantrum all demonstrated that EM spreads could widen sharply in ways investment-grade DM bonds rarely did. More recently, as convergence has progressed and EM fundamentals have matured, this structural risk premium has compressed (Figure 9). The excess spread of EM over DM credit has now largely disappeared, and relative valuations are broadly fair, with little incremental compensation remaining for investors crossing the EM/DM boundary. The compression of the premium does not, however, imply the disappearance of cyclical risk. In a downside scenario of further escalation in the Iran conflict, we assess that the average EM hard-currency spread would widen from its pre-crisis level of 178bps to around 235bps over 2026, with a peak of close to 280bps during the year. The scale of this move would be modest by historical standards. During the Covid-19 shock, the EMBI Global Diversified spread widened from roughly 300bps to a peak of around 700bps in March 2020, more than doubling in the space of weeks, before retracing over the following months. The downside repricing we envisage for the Iran scenario, by contrast, would represent a widening of around 100bps at its peak. This more contained response suggests that convergence has altered the nature of EM spread risk, rendering it more cyclical and less crisis-prone, without eliminating it entirely.

Figure 9: EM sovereign over US credit pick-up on a rating-adjusted basis



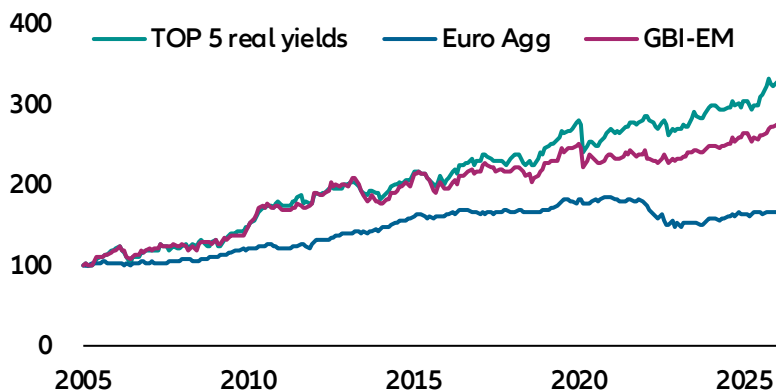
Notes: To calculate the pick-up Emerging Markets hard currency debt investors get for "EM risk" over "DM risk", we compare the spread over US Treasuries for EM Sovereign bonds (JP Morgan EMBIG) with the option-adjusted spread for US Credit (Bloomberg) per rating bucket (AA, A, BBB, BB, B). The pick-up reflects the weighted average of the spread difference based on rating bucket weights in the EMBIG.

Sources: JP Morgan, Allianz Research

An EM risk premium can still be earned when we move our focus from credit to rates. EM local currency bonds offer much higher nominal yields compared to DM bonds. From a DM-based investor's point of view, we have to take FX risk into account. There is a long history of EM currency depreciation against major DM currencies that exceeds 2% p.a. in a long-term average. Nevertheless, exposure to EM local currencies pays off as local excess carry exceeds FX losses in the long-term. The root cause of higher EM performance is the real yield pick-up. After accounting for the inflation embedded in higher nominal EM rates, EM local debt has historically offered real yields

that are substantially higher than what investors can earn in DM markets, where demographics, productivity slowdowns and central bank intervention have compressed real rates. This matters because real yields are not merely an academic construct: empirically, they are among the strongest predictors of long-term fixed income returns, capturing the true compensation investors receive for bearing interest rate risk. For a DM investor, that persistent real yield advantage is the genuine investment case for EM local currency debt. Not only did EM local currency bonds significantly outperform a Euro Aggregate investment over the past 20 years for a EUR-based investors; selecting the highest real yielders only further boosted EM local currency outperformance (Figure 10).

Figure 10: EM local currency debt outperforms DM fixed income for EUR-based investors, higher real rates are the major driver of the outperformance

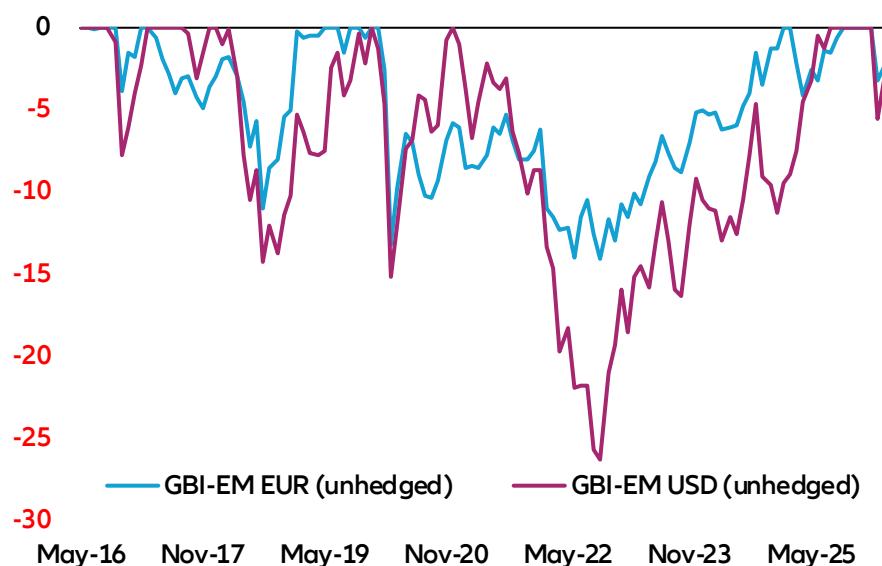


Notes: To analyze the importance of the real yield, we have tested a trading strategy that invests equal-weighted into five issuers offering the highest expected real yields of the GBI-EM investment universe, JP Morgans flagship EM local currency index. At each point in time, the expected real yield is derived by subtracting the inflation forecasts over the next 5 years from the 5yr nominal bond yield.

Sources: LSEG Workspace, Oxford Economics, Allianz Investment Management

While EM local currency FX clearly adds volatility to DM investors, EUR-based investors benefit from mitigation by their home currency. The carry of EM local currency cash is a strong performance contributor for DM-based investors, while FX is without doubt the strongest source of volatility. For a EUR-based investor, FX volatility looks meaningfully different and in some ways more favorable than it does through a USD lens. Because the USD functions as the world's reserve and safe-haven currency, other currencies tend to weaken against it in unison when global risk appetite deteriorates and investors seek shelter. The EUR is more closely tied to risk sentiment than to safe-haven dynamics, giving it a degree of procyclicality that it shares with many EM currencies. The practical implication is that a EUR-based investor holding EM local currency debt experiences meaningfully lower cross-currency volatility than a USD-based peer investor. When EM currencies weaken, the euro tends to weaken too, partially cushioning the FX impact. Figure 11 shows that the EUR's absence of safe-haven status becomes a structural advantage when accessing EM local debt: The investor's base currency aligns more naturally with the asset class, reducing FX volatility and drawdowns that make the same trade more costly for USD-based investors.

Figure 11: GBI-EM drawdown from peak measure in EUR vs. USD (%)



Notes: Drawdown measured as percentage decline from prior peak on a monthly total return basis. EUR unhedged reflects local currency returns converted to EUR without currency hedge. USD unhedged reflects the same index from a USD investor perspective.

Sources: JP Morgan Markets, Allianz Research

The strong progress in EM monetary policy credibility increases the diversification benefits of investing into EM local currency debt for DM investors. EM central banks have moved decisively away from the exchange rate stabilization bias that historically characterized their policy frameworks, adding on average 780bps to policy rates during the post-pandemic tightening cycle compared to just 400bps for advanced economies (IMF, 2024). Crucially, this tightening was front-loaded and aggressive even at the cost of short-term economic pain, signaling a genuine commitment to domestic inflation targeting over growth stabilization. For a globally diversified, DM-based fixed-income investor, a decent EM local currency allocation provided significant diversification benefit during the painful global rates repricing following the Covid-19 and Russia-Ukraine shocks: A sharp EM sell-off driven by rates and FX has been followed by a much faster recovery compared to home currency fixed-income investments.

These assessments are, as always, subject to the disclaimer provided below.

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(v) persistency levels, (vi) particularly in the banking business, the extent of credit defaults, (vii) interest rate levels, (viii) currency exchange rates including the EUR/USD exchange rate, (ix) changes in laws and regulations, including tax regulations, (x) the impact of acquisitions, including related integration issues, and reorganization measures,

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Weed out the laggards: Sustainability as a credit filter

03 June 2026

Allianz Research

Content

Page 3-5

Executive Summary

Page 5

Unsustainable practices can generate financial stress

Page 6-11

Sustainability matters for credit quality and credit risk

Page 12-14

Implications for financial institutions, investors
and businesses

Page 15-17

Appendix: Data and methodology details

Executive Summary



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- **Financial fundamentals remain the primary drivers of credit risk.** Across more than 7,400 companies and 280,000 firm-year observations, profitability (ROA), leverage and size collectively explain the vast majority of variation in credit risk. Sustainability is statistically significant but smaller in magnitude: financial fundamentals carry coefficients 3-7 times larger than the sustainability variable in our global sample. A firm's balance sheet, earnings power and scale determine whether it can service its debt - sustainability adds an incremental but real signal on top of that financial bedrock.
- **Sustainability acts as a credit-risk filter, but its signal concentrates at the bottom of the distribution.** Weaker sustainability is consistently associated with higher default risk. A one-decile improvement in sustainability (roughly +7.5 points on a 0-100 scale) corresponds to an approximately 0.25pp reduction in default probability - a 12-25% relative reduction for firms in the worst default-risk decile. The relationship is non-linear: moving from poor to average sustainability materially improves credit outcomes; moving from average to best-in-class delivers little additional benefit.
- **Environmental performance is the clearest predictor of default risk.** A 10-point improvement in the environmental score is associated with a 0.9-point gain in the Altman Z-score - enough to move borderline firms out of the distress zone. Governance, by contrast, is the key driver of broader credit quality, consistent with financial discipline and transparency underpinning financial strength.
- **European companies lead on sustainability scores and show the greatest sensitivity to ESG factors.** Regional baselines differ sharply: Europe leads (average combined score in the 50s), the US trails (40s, with environmental scores near 30) and Japan scores well on environment (near the high-40s) but weaker on social metrics. Environmental leadership carries the strongest credit premium in emerging markets, where best-in-class issuers benefit from a scarcity premium as standards tighten. In the US, holistic ESG profiles matter more than any single pillar; in Europe, the combined score links to both credit risk and credit quality.
- **The clearest ESG-credit links cluster in sectors exposed to energy transition, operational and reputational risk - notably communication services, consumer staples and energy.** Our sector analysis finds meaningful sustainability-credit relationships in seven of eleven sectors, with the largest effects in these three. Communication/software tends to concentrate financially strong firms that can fund sustainability investment and are rewarded for risk management that encompasses ESG. Consumer staples and energy include carbon-intensive sub-sectors - textiles to oil & gas - where pollution, labor issues, carbon pricing and stranded-asset dynamics can directly hit cash flows, asset values and funding access. The relationship is weaker in materials and IT, reinforcing that sustainability integration in credit requires a sector materiality lens, not a blanket score overlay.

- **These findings are robust across data providers and confirmed by our internal credit assessments.** Replicating the analysis with alternative sustainability scores yields virtually identical results. Using our proprietary internal credit grades, we find that sustainability is already implicitly embedded in analysts' assessments - the sustainability score and environmental and social pillars are all statistically significant predictors of internal grades. Analysts are capturing sustainability-related risks in practice, a more systematic approach could sharpen that signal further.
- **The practical implication is clear: screen out the weakest.** Banks should embed sustainability as a negative screen in credit pricing and due diligence. Credit investors should use it primarily as a downside protection tool. Corporate issuers face a real credit penalty for sitting below the sustainability threshold - and no proportionate reward for exceeding it. The imperative for all market participants is the same: clear the floor. Chasing the ceiling is a marginal bonus.



Unsustainable practices can generate financial stress

Sustainability has moved from a niche concern to a mainstream factor in extra-financial evaluation of corporates over the past decade. The underlying rationale is simple: sustainability issues directly or indirectly influence the fundamentals that eventually determine a company's ability to generate revenues, make a profit or just repay debt. If a company faces environmental fines or a factory is shut down by floods or wildfires, its revenues suffer and costs rise – impeding debt service. If poor labor practices lead to strikes or a consumer boycott, cash flows can dry up. And if weak governance leads to fraud or compliance breaches, a firm can quickly find itself facing legal penalties and loss of investor confidence. All these sustainability-related events hit the company's financial position, translating into traditional credit risk and increasing the probability of default.

Climate change and environmental management are now recognized as material credit factors. Physical risks (e.g. extreme weather, droughts, rising sea levels) can destroy assets and disrupt supply chains¹. For example, in 2019, wildfires in California saddled a major utility with staggering liabilities, contributing to its bankruptcy. Transition risks are also mounting: Governments imposing carbon taxes or banning high-emission activities can strand assets and force costly business model changes². Companies heavily reliant on fossil fuels or with poor pollution controls may face higher costs, asset write-downs and weaker financial ratios, all of which hurt credit quality. On the other hand, firms proactive in cutting emissions or investing in cleaner technology can reduce regulatory risks and operating costs over time, bolstering their credit outlook.

How a company manages its workforce, customers and communities also bears on creditworthiness. Neglecting safety and labor rights, for instance, can lead to workplace accidents or strikes that interrupt production. In the mining industry, fatal accidents not only incur direct costs but can prompt license suspensions. Similarly, poor product safety or customer data breaches can trigger lawsuits and erode a firm's reputation and revenues. By contrast, companies that invest in human capital (e.g. training employees, maintaining good labor relations) and uphold product quality tend to have more resilient operations and loyal customers. Although spending on social initiatives may add costs in the short term, that stability should translate into steadier cash flows and lower credit risk in the long run. Lastly, corporate governance has long been linked to credit outcomes. From Enron's collapse to more recent corporate scandals, history shows that failures were rooted in poor governance. Lax oversight, misaligned incentives or opaque disclosure can mask mounting problems until it is too late. When governance breaks down, firms are prone to accounting fraud, corruption or reckless risk-taking that can rapidly undermine their financial position. For creditors, strong governance is a first line of defense: it means competent management, prudent financial controls and transparency. In practice, governance issues can crystalize into credit events through regulatory penalties, adverse audit findings or sudden leadership crises. Any of these can spook investors and restrict a firm's access to funding.

¹ See our previous report [on adaptation](#)

² See our previous report [on stranded assets](#)



Sustainability matters for credit quality and credit risk

Over the last ten years, a growing body of research has examined whether strong sustainability actually translates into better credit outcomes. The evidence, especially from the most recent studies, generally finds that companies with higher ESG ratings tend to have lower credit risk. On the whole, firms that excel on ESG enjoy tangible credit benefits. They often face lower borrowing costs and fewer nasty surprises. For example, multiple studies of corporate bond markets have found that companies with high ESG scores pay narrower credit spreads than peers, even after controlling for their credit ratings. In other words, bond investors demand a smaller risk premium, viewing high-ESG issuers as safer bets. These firms also exhibit lower volatility and reduced tail risk of extreme losses. The upshot is that a strong ESG profile appears to enhance financial resilience, which lenders reward with cheaper debt capital (see appendix for more references on the sustainability-credit risk nexus).

To further investigate the sustainability-credit risk nexus, we conduct an extensive statistical analysis on a global dataset of companies. Our study covers firms worldwide over the period 1991-2025, using sustainability scores from Refinitiv (see appendix for more details). This dataset spans both developed and emerging markets, with a broad sector representation. We measured credit risk in two ways: (i) the Altman Z-score, a classic metric of default risk based on financial ratios, and (ii) a credit-quality index capturing broader aspects of financial health (primarily profitability, leverage and cash flow metrics). The Altman Z-score was available for a subset of about 2,100 firms (generally larger companies with complete

financial data), while the credit quality index covered a larger sample of 7,400 firms, giving us a comprehensive view of sustainability impacts on creditworthiness. We include firms from North America, Europe, Asia-Pacific and emerging economies. Sustainability scoring coverage has expanded significantly in the past decade; still, not all companies have sustainability ratings or all the inputs for the Altman Z-score. For example, among emerging market companies in our sample, only about 44% have sustainability scores available (as coverage tends to be thinner for smaller EM firms). By contrast, in the US sample almost 67% of firms have sustainability scores, reflecting better disclosure and data coverage in the US. Developed Europe falls in between (around 65% sustainability coverage). These proportions highlight that while sustainability data is now widespread, it is not yet universal. Nevertheless, our total sample comprises over 280,000 firm-year observations, providing a rich basis to detect patterns.

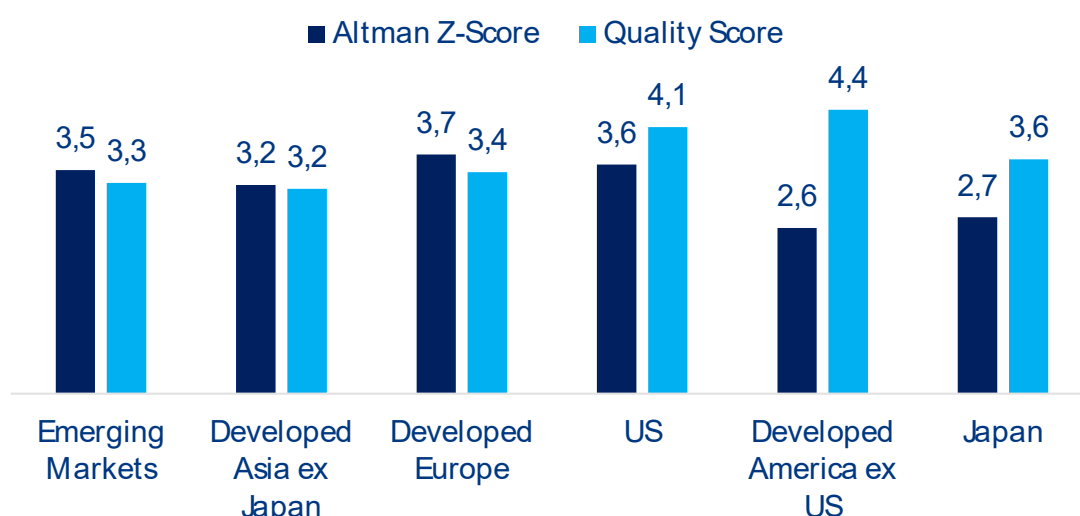
European companies boast the highest sustainability scores (the average combined score in Europe is about 50 out of 100), compared to the low 40s in emerging markets. US companies trail Europe: the average US sustainability score is around 40, with particularly low marks on environmental factors (the average US environmental pillar score is barely 30, reflecting that many American firms score poorly on carbon emissions and resource use relative to global peers). Japan, interestingly, scores well on environmental issues (average E score close to 48) but somewhat lower on social metrics. These regional differences suggest that European

businesses have led on sustainability, especially on social and environmental dimensions, while North American and some Asia-Pacific firms lag in environmental performance. Does strong sustainability in a region correspond with better credit profiles? The Altman Z-score in our sample averages around 3-4 for most regions (see Figure 1), with higher values indicating safer balance sheets³. Developed Europe had a slightly higher mean Z-score (3.7) than the US (3.6) or emerging markets (3.5). These averages put the median company safely out of distress, though the range is wide. We also construct a credit quality index for each firm-year to capture overall financial health. This composite index synthesizes profitability (return on assets) and cash-flow generation (operating cash flow to assets), interest coverage and leverage which are key inputs to credit quality. By design, higher values of this index indicate stronger credit fundamentals (high profits, robust cash flows, less debt). Not surprisingly, developed market firms tend to score better on this quality index on average, given their generally more mature, stable financials.

We find that a strong sustainability score is associated with lower credit risk. We employ panel regression analysis to quantify the relationship between sustainability performance and our credit risk measures⁴. In simple terms, when a company's sustainability score is higher, do we subsequently observe better credit outcomes? To answer this, we ran regressions of Altman Z-scores and the credit quality index on sustainability scores, controlling for other factors. Specifically, we include controls for firm size (larger firms often have more diversified risks)

and profitability (a higher return on assets naturally boosts credit metrics). We also account for fixed effects (essentially controlling for broad differences across sectors, years and regions) to isolate the impact of sustainability. Importantly, sustainability scores were lagged by one year in the models, meaning we related last year's sustainability performance to this year's credit risk indicators. This helps address causality (i.e. ensuring we are capturing sustainability influencing credit, not the other way around, and also to be consistent with the timing of availability of information, as today we know the sustainability score based on the past year's data). First of all, we need to underline that coefficients for fundamentals are significant and larger across all our regressions compared to sustainability measures. This means that financial metrics are the leading drivers of credit risk. However, we find that sustainability also matters to a significant extent: The coefficient on the Environmental Pillar Score implies that if a firm's environmental score rises by 10 points (say from 50 to 60), its Altman Z-score is expected to be about 0.9 higher on average (see Table 1). Such an effect is meaningful: given many firms' Z-scores hover in the 1.5-3.0 range, a nearly 1 point boost could move a firm from the borderline "grey zone" closer to safety. The overall ESG combined score (which integrates all pillars and accounts for controversies) had a positive coefficient of about 0.05. This suggests that, taken as a whole, strong sustainability performance is associated with better credit health (higher Altman Z-score), although among the individual components, the environmental factor is doing most of the work in explaining that link.

Figure 1: Distribution of credit risk and quality scores by regions



Sources: Refinitiv, Allianz Research

³ For context, a Z-score above 2.6 is considered "safe", while below 1.1 is considered "distressed"

⁴ See appendix for methodological details

Table 1: Distribution of credit risk and quality scores by regions

Sustainability measure vs credit risk (Altman Z-score)	Beta coefficient of the sustainability measure	Beta coefficient of fundamentals		
		Profitability	Size	Cashflow Strength
ESG Combined Score	0.05*	0.34***	0.28***	0.14***
E Pillar Score	0.09***	0.27***	0.36***	0.18***
S Pillar Score	-0.01	0.52***	0.42***	0.24***
G Pillar Score	0.02	0.49***	0.38***	0.22***

*, ** and *** denote respectively 10%, 5% and 1% significance level

Sources: Refinitiv, Allianz Research

Sustainability is also linked to broad credit quality.

When looking at credit quality, like for credit risk, fundamentals are more important. Nevertheless, we also find here that the combined ESG score has a positive and statistically significant coefficient, indicating that firms with higher overall sustainability performance tend to exhibit better credit quality. In practical terms, a 10-point increase in the ESG score corresponds to about 0.16 higher on our credit quality index – a modest but non-trivial improvement in underlying financial strength (see Table 2). Interestingly, when we break it down into pillars, we see that the governance pillar shows a positive effect. This suggests that good governance practices (board structure, minority shareholder protection, transparency, etc.) are contributing to better credit fundamentals, which is intuitively consistent with the idea that well-governed firms have more disciplined financial management. At a global level, our findings imply that sustainability matters for credit risk and credit quality, but the relative importance of E, S, and G differs between the two. For default risk (Altman Z), environmental performance is the standout (i.e. companies with stronger environmental stewardship tend to be safer from a credit risk perspective). This resonates with the notion that environmental risks (climate, pollution, regulation) have become financially material and can drive default outcomes if unmanaged. In contrast, social and governance factors did not show up in the default risk regression, possibly because their effects on default are indirect or longer-term (or already reflected in

other control variables like profitability). For credit quality, we see that having a robust overall sustainability profile helps, and there is a hint that governance (and to an extent social) is playing a role.

E matters for EMs while overall sustainability is more important for US and Europe. Looking at different regions⁵, we find that environmental pillar impacts are strongest in emerging markets when it comes to credit quality. In this group, a higher environmental score was associated with a significantly better credit quality index. Emerging market firms that are environmental leaders tend to also have stronger credit fundamentals. One interpretation is that in developing economies, where environmental standards and enforcement are rapidly tightening, firms that proactively manage environmental risks gain a credit edge – perhaps by avoiding fines or attracting sustainability-linked capital. Additionally, emerging market companies with strong sustainability ratings might enjoy a reputational premium that improves investor confidence in their financial stability. Moreover, in emerging markets, the governance pillar had a positive relationship with credit quality, suggesting governance reforms are rewarded in those countries with better perceived creditworthiness. In the US, our models showed that overall sustainability performance had a mild positive relationship with credit quality, but none of the individual E, S, G pillars were statistically significant on their own. This might indicate that US credit investors take a holistic view

⁵ North America (US and other developed Americas), Europe, Asia-Pacific (developed Asia ex-Japan, and Japan) and Emerging Markets.

– i.e. a company has to be well-rounded on sustainability to get credit recognition, and no single pillar being enough. Europe showed a positive link with the overall sustainability performance for both credit risk and credit quality. Europe's long-standing focus on sustainability could mean investors and lenders have integrated these extra-financial criteria in their analysis.

Sustainability matters most for communication services, consumer staples and energy sector credit. Overall we find significant links between sustainability and credit risk/quality for seven sectors out of 11, underlining the broad relevance of extra-financial performance for credit worthiness. However, the influence of sustainability is the largest in communication services & software, consumer staples and energy (see Figure 2). For communication

services and software, our results most likely stem from the fact that the sector includes financially strong companies that spend and invest in sustainability. Meanwhile, both consumer staples and energy sectors include carbon-intensive segments – from textiles to oil & gas – for which sustainability-related risks represent substantial financial risk, too. For instance, pollution risks and labor related-issues could disrupt firms in textiles while carbon prices and stranded assets are clear hurdles for oil & gas companies. At the other end of the spectrum, although relevant, sustainability scores seem less important for credit metrics for firms in the materials and IT sectors.

Table 2: ESG and Credit Quality Index (global panel regression)

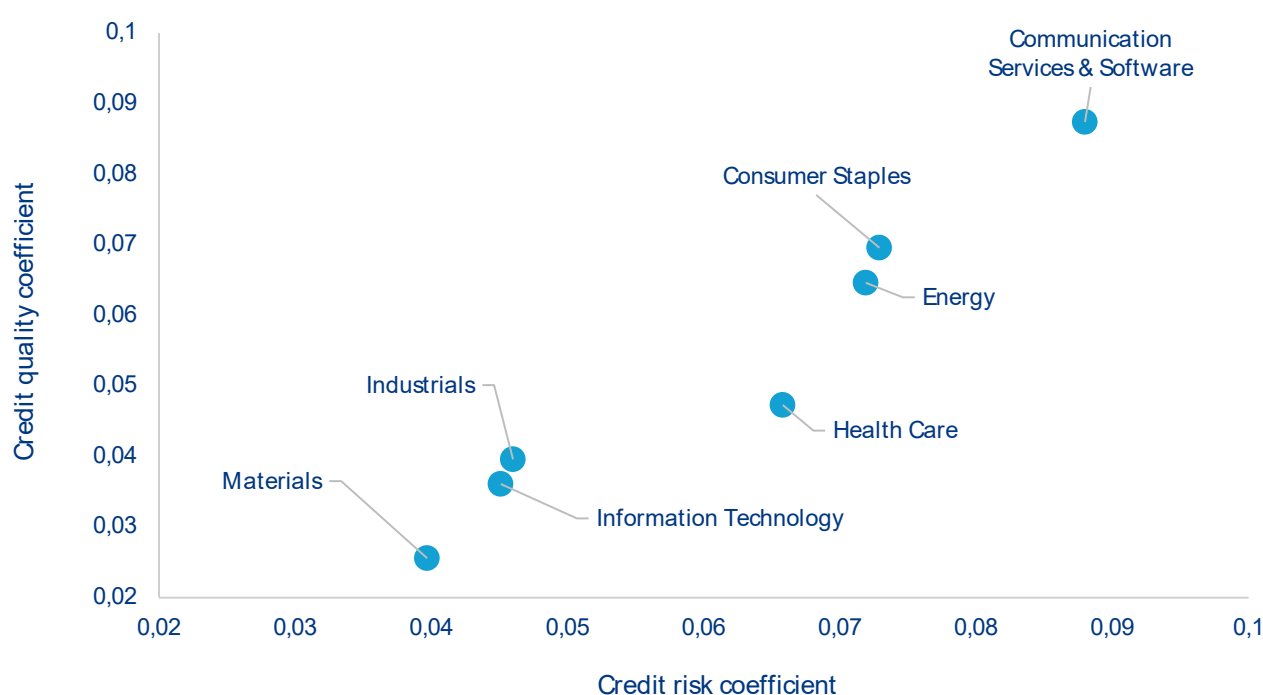
Sustainability measure vs credit quality	Beta coefficient of the sustainability measure	Beta coefficient of fundamentals ⁱ
		Size
ESG Combined Score	0.02***	0.14***
E Pillar Score	0.01	0.14***
S Pillar Score	0.01	0.26***
G Pillar Score	0.01*	0.19***

*, ** and *** denote respectively 10%, 5% and 1% significance level

ⁱ We do not include profitability and cashflow here as they are used to compute credit quality

Sources: Refinitiv, Allianz Research

Figure 2: Sustainability score coefficients for credit risk and credit quality



Sources: Refinitiv & Allianz Research – Note: we only show significant coefficients.

Our findings hold when using MSCI sustainability data instead of Refinitiv, confirming that the sustainability-credit risk nexus is not an artefact of a single data provider.

A recurring concern in the sustainability-credit literature is that results may be sensitive to the choice of rating provider, given well-documented divergences across sustainability scoring methodologies (Berg, Kölbel & Rigobon, 2022). To address this, we replicate our core Altman Z-score regression using MSCI sustainability scores in place of Refinitiv. The MSCI-based sustainability coefficient is positive and highly significant ($\beta = 0.04$, $p < 0.01$ in Table 3), closely mirroring the Refinitiv result ($\beta = 0.05$ in Table 1). All control variables – profitability, size, leverage, cash-flow strength and interest coverage – retain the expected signs and significance. This consistency across two independent data providers strengthens our confidence that the underlying relationship between sustainability performance and credit risk is genuine and not driven by idiosyncratic scoring choices.

Using our proprietary internal credit grades, we confirm that sustainability carries credit-relevant information beyond what traditional financial analysis captures.

Beyond third-party credit-risk proxies such as the Altman Z-score, we test whether the sustainability-credit link holds against our own internal credit grading system, which

runs from 1 (strongest) to 10 (weakest) and incorporates qualitative analyst judgment alongside financial metrics. Table 4 reports the results using Refinitiv sustainability scores as explanatory variables. The sustainability score, environmental pillar and social pillar all exhibit negative and highly significant coefficients ($p < 0.01$), meaning that companies with stronger sustainability performance receive better internal credit grades. The governance pillar, however, is not statistically significant – likely because governance considerations are already explicitly embedded in our internal credit assessment process. While the economic magnitudes are modest – a move from the worst to the best sustainability score corresponds to roughly a quarter-notch improvement on the 1-to-10 scale – the statistical significance validates that sustainability captures credit-relevant information incremental to what traditional financial analysis already reflects. This supports using sustainability as a complementary signal – particularly as a negative screen, as we demonstrate below – rather than as a standalone driver of credit differentiation.

Table 3: Sustainability and Altman Z-score – MSCI data (global panel regression)

Sustainability measure vs credit risk (Altman Z-score)	Beta coefficient of the sustainability measure	Beta coefficient of fundamentals			
		Profitability	Size	Leverage	Cashflow Strength
MSCI Sustainability Score	0.04***	1.29***	0.10***	-1.06***	0.26***

*, ** and *** denote respectively 10%, 5% and 1% significance level

Sources: Refinitiv, Allianz Research

Table 4: Refinitiv sustainability scores and internal credit grades

Sustainability measure vs internal credit grade	Beta coefficient	Significance
Sustainability Score	-0.04	***
Combined Sustainability Score	-0.03	***
Environmental Pillar Score	-0.03	***
Social Pillar Score	-0.04	***
Governance Pillar Score	-0.01	-

Negative coefficients indicate better (lower) credit grades. All regressions control for firm size.

*, ** and *** denote respectively 10%, 5% and 1% significance level. Sources: Refinitiv, Allianz Research

When we split firms into low, medium and high sustainability groups, the credit-risk relationship is concentrated in the bottom tercile – suggesting sustainability is most actionable as a filter to exclude the worst-scoring companies. To test whether the sustainability-credit link is linear or concentrated in specific segments, we estimate a spline regression that allows the slope to differ across sustainability terciles (using MSCI data). The results are striking (Table 5): the coefficient for the low sustainability group is 0.10 and highly significant at the 1% level, implying that among firms with the weakest sustainability profiles, each additional point of improvement is associated with a meaningful increase in the Altman Z-score. By contrast, the coefficients for the medium (0.01) and high (0.02) groups are small and statistically insignificant. In other words, the marginal credit benefit of better sustainability is strongest at the bottom of the distribution and tapers off once firms reach a moderate standard. This non-linearity has a clear practical implication: rather than rewarding top sustainability performers, the data supports using sustainability primarily as a negative screen – flagging or excluding firms in the lowest sustainability tercile, where poor practices are most strongly associated with elevated credit risk.

Translating our results into probability-of-default terms, a one-decile improvement in sustainability corresponds to a reduction in default probability of approximately 0.25pp. To make our findings directly actionable for

credit practitioners, we link the sustainability-to-Altman Z relationship to probability of default (PD) through a two-step estimation. First, from our MSCI regression, a one-decile improvement in sustainability score (approximately 7.5 points) is associated with a 0.29-point increase in the Altman Z-score. Second, using a logit regression of observed PD on winsorised Altman Z-scores ($\beta = -0.0096$, $p < 0.01$), we compute the implied change in default probability. The resulting chain implies that a one-decile sustainability improvement – approximately 7.5 points on the 0–100 sustainability scale, for example moving from a score of 12 to 20 for firms in the bottom decile – translates to approximately a 0.25pp reduction in PD. To put this in perspective, observed default probabilities in our sample range from near zero to around 5%. For a company in the worst decile of default risk, where PDs typically reach 1–2% or higher, a 0.25pp reduction represents a 12–25% relative improvement in creditworthiness – a meaningful shift that can alter risk classification. The effect compounds across large credit portfolios and is substantially larger for the environmental pillar using Refinitiv data: the stronger E-pillar coefficient ($\beta = 0.09$ per point, from Table 1) implies that a 10-point improvement in environmental performance corresponds to an even greater PD reduction – underscoring that environmental risk management is the single most credit-relevant dimension of sustainability.

Table 5: Sustainability tercile effects on Altman Z-score – Spline regression (MSCI data)

Sustainability group	Beta coefficient	Significance
Low (bottom tercile)	0.10	***
Medium (middle tercile)	0.01	-
High (top tercile)	0.02	-

Controls: Profitability, Size, Leverage, Cash-flow Strength, Interest Coverage – all significant at 1%.

Sources: MSCI, Allianz Research



Implications for financial institutions, investors and businesses

Commercial banks should embed the floor; pricing the premium is about conviction. The core finding for commercial banks is specific: Borrowers in the bottom sustainability tercile carry statistically higher default risk, and the 0.25pp PD reduction per sustainability decile is material in credit loss terms at portfolio scale. The practical response is to embed sustainability as a negative screen in credit due diligence: flagging and re-pricing exposure to low-sustainability borrowers, rather than rewarding high sustainability credentials with lower margins across the board. A borrower moving from moderate to strong sustainability does not generate a proportionate reduction in expected credit loss but a borrower in the bottom tercile generates elevated risk that traditional financial analysis may not fully surface. In practice, this integration means incorporating sustainability risk flags into internal ratings and pricing grids, particularly for borrowers in carbon-intensive sectors and those with governance deficiencies.

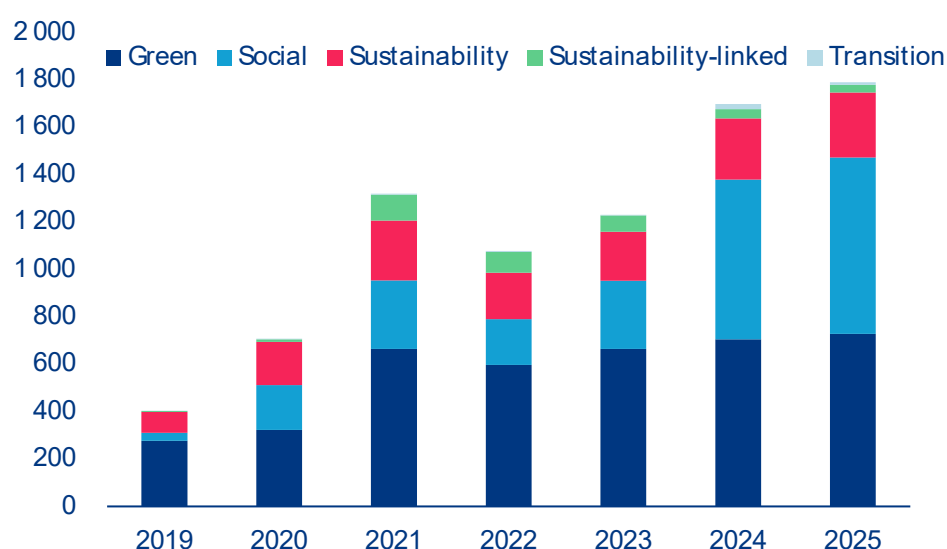
Broad-based credit investors should consider ESG filters as downside protection, not necessarily an alpha generator. For asset managers and credit investors, our findings support sustainability integration primarily as a downside risk mitigation tool. Indeed, the credit signal concentrates in identifying default-prone issuers, not in predicting outperformance by top-rated ones. A portfolio that systematically screens out bottom-tercile sustainability issuers captures the bulk of the credit risk

reduction; however, tilting toward the highest-rated sustainability issuers does not generate proportionate improvement in expected risk-adjusted returns. This reframes the analytical objective: the primary use case is loss avoidance, not return enhancement. This framing is consistent with fund-level evidence that sustainability-tilted credit portfolios have exhibited lower spread volatility and drawdowns during stress periods – precisely because the downside-protection effect is real and concentrated at the weakest issuers. Forward-looking sustainability indicators – transition strategies, green capex trajectories, emissions commitments – are increasingly relevant as transition risk begins to crystallize in credit spreads, particularly in the three sectors where our analysis shows the largest effects. Issuer engagement focused on sustainability risk management in communication services, consumer staples and energy represents a more targeted and defensible application of the credit signal than broad portfolio sustainability tilts. Investors who can distinguish genuine sustainability improvement at the bottom of the distribution from cosmetic score improvements elsewhere will have the most informative signal as the market matures.

Insurers are exposed to sustainability-related credit risk through both underwriting and investment activities and the sector concentration in our findings is directly relevant to where analytical effort should be focused. The credit signal from sustainability is largest in communication services, consumer staples and energy, and weakest in materials and IT. A uniform sustainability overlay applied to a corporate bond portfolio overstates the signal in low-materiality sectors and risks underweighting it in high-materiality ones – concentrating integration on the three high-signal sectors generates a materially better signal-to-noise ratio. The analytical focus for insurers is also shifting from static ESG scores toward forward-looking indicators to distinguish issuers that are credibly managing transition risk from those accumulating deferred exposure. During phases of spread decompression, when idiosyncratic credit risk dominates and return dispersion rises, sustainability-integrated analysis is most valuable precisely because the signal concentrates at the bottom of the distribution: it helps identify which issuers are deteriorating while others hold, rather than providing a uniform quality tilt. As climate change and shifting social expectations reshape sectoral risk profiles over the medium term, sector-level sustainability views should feed directly into strategic allocation decisions – the large dispersion our analysis shows across sectors in how strongly sustainability affects credit risk is itself a portfolio signal.

For corporations, improving sustainability can yield tangible benefits in terms of credit quality and funding costs. Our findings translate into a specific and financially grounded strategic priority for corporate issuers: the credit and financing benefits of sustainability concentrate at the threshold, and the penalty for falling below it is real. Companies with weak sustainability profiles (particularly weak environmental practices and poor governance) face measurably higher borrowing costs, tighter capital market access and elevated default risk. The strategic implication is to prioritize the most material sustainability risks rather than optimizing for the highest possible combined ESG score. Clear, measurable commitments to addressing material environmental and governance risks, transparent reporting and credible transition strategies are the inputs that matter for credit assessment. Companies that credibly clear the sustainability floor benefit from improved credit terms and a broader investor base; those that do not will face systematic pricing penalties as banks and investors formalize negative screen frameworks. The ESG investment that removes a credit penalty has a higher financial return than the ESG investment that earns marginal recognition at the top of a distribution where the signal is weak.

Figure 3: Labelled bond issuance volume (in USD bn)



Sources: Bloomberg. Includes corporate, government, supranational bonds, municipals and securitized products

Figure 4: 12M Normalized OAS

Source: Bloomberg

The expanding market for sustainability-labelled bonds provides investors with targeted exposure to environmental and social objectives, supported by use-of-proceeds and reporting frameworks that enhance transparency and allow for an assessment of issuers' sustainability trajectories. Issuance of GSSS bonds has risen markedly over the last six years, driven largely by issuers with comparatively strong sustainability profiles (see Figure 3). It remains dominated by green bonds, with activity concentrated in the financials and utilities sectors. While corporate green bonds historically traded at a premium ("greenium") due to scarcity amid strong demand, this premium has largely vanished at the aggregate index level as the market has scaled up, though sector-level dispersion persists (see Figure 4). Empirically, green bonds have exhibited lower spread volatility during market widening phases than the broader corporate bond universe, suggesting potentially more resilient risk-return characteristics, provided issuers' sustainability commitments are credible.

Appendix: Data and methodology details

Refinitiv Sustainability Scores – Methodology:

Refinitiv sustainability scores are constructed using a transparent, rules-based methodology that aggregates approximately 180–200 publicly available environmental, social, and governance indicators into standardized measures of corporate sustainability performance. Raw indicators, derived from company disclosures and publicly reported information, are mapped into defined sustainability categories and themes, normalized on an industry-relative basis, and combined using pre-determined, industry-specific weights to produce Environmental, Social, and Governance pillar scores on a 0–100 scale. In addition to the standalone ESG score, Refinitiv calculates an ESG Combined (ESGC) score, which incorporates an overlay for sustainability-related controversies by adjusting the ESG score based on the severity and frequency of documented incidents. Both ESG and ESGC scores are updated as new disclosures and controversy information become available.

Data sources and sample construction: We compiled firm-level data from Refinitiv covering the years 2011 through 2024. The ESG data comes from Refinitiv's ESG database, which provides an ESG Combined Score (a comprehensive rating including environmental, social, governance aspects and deducting for controversies) as well as separate pillar scores for Environmental, Social, and Governance (each scaled 0-100, with higher = better ESG performance). We also collected financial data from Refinitiv for the same firms, including balance sheet and income statement items required to calculate the Altman Z-score and other financial ratios. Our initial universe included over 130,000 companies globally, but many were missing ESG or financial data. We restricted the sample to firm-year observations with non-null values for the variables of interest. The final panel comprises about 280,000 observations on ~7,400 unique firms worldwide. The regional breakdown of firms (by headquarters) was: roughly 18% North America, 20% Developed Europe, 6% Japan, 7% other developed Asia-Pacific, 34% Emerging Markets, and 15% unclassified/unknown region (the latter are cases where region wasn't specified, often smaller frontier market firms). Data coverage varied, with developed markets generally having better ESG disclosure. For instance, 67% of U.S. firm-year observations had an ESG score available, versus ~44% in Emerging Markets, as noted earlier.

Altman Z-score calculation: We used the **Altman Z-score** as a proxy for credit risk or default risk of a firm. The Altman Z-score is a well-known composite financial indicator developed by Edward Altman, which combines five accounting ratios using fixed weights. We applied the version for publicly traded firms (Altman's Z-score for manufacturing, since most of our firms fall in non-financial sectors). The formula is:

$$Z = 1.2 \times \frac{\text{Working Capital}}{\text{Total Assets}} + 1.4 \times \frac{\text{Retained Earnings}}{\text{Total Assets}} + 3.3 \times \frac{\text{EBIT}}{\text{Total Assets}} + 0.6 \times \frac{\text{Market Value of Equity}}{\text{Total Liabilities}} + 1.0 \times \frac{\text{Sales}}{\text{Total Assets}}.$$

This formula produces a single score where higher values indicate a lower probability of bankruptcy. As mentioned, $Z > 2.6$ is usually considered "safe", $Z < 1.1$ is "distressed", and the 1.1–2.6 range is a grey zone. We winsorized extremely high or low Z-scores to avoid undue influence of outliers (e.g., a few observations had negative equity or abnormally large market capitalizations leading to very high Z).

In our dataset, not all firms had the market value data needed (especially if not publicly traded or if data was missing), which is why the Altman Z sample is smaller (~2,100 firms). Notably, financial sector firms (banks, insurance) are excluded from Altman Z analysis since the Z-score isn't designed for them and their balance sheets aren't comparable.

Credit quality composite: To measure credit quality more broadly (and to include more firms), we constructed a composite index reflecting a firm's earnings and cash flow capacity, which are crucial to creditworthiness. Specifically, we took three indicators: Return on Assets (ROA = Net Income / Total Assets), Leverage (Total Debt to Total Assets) and Operating Cash Flow to Total Assets (OCF/TA), which capture profitability and cash generation respectively. We standardized these within each year (to account for economic cycles) and took an average to form a composite score. This "credit quality index" is not a standard metric like Altman Z, but it serves as a general gauge of a firm's fundamental strength to service debt – high values mean the company is profitable and generating cash relative to its asset base (implying it likely can cover interest and reinvest, hence higher credit quality). We included this second measure because credit risk is multifaceted; a firm might have a low Z-score due to high market leverage but still strong cash flows to pay debts, or vice versa. The composite helps capture dimensions the Z-score might miss and greatly expands coverage to smaller firms that don't have market value data.

Regression specifications: We estimated panel regressions of the form:

$$\text{CreditMetric}_{i,t} = \alpha + \beta \times \text{ESGMetric}_{i,t-1} + \gamma \mathbf{X}_{i,t-1} + \text{FE} + \varepsilon_{i,t},$$

where $\text{CreditMetric}_{i,t}$ is either Altman Z or the credit quality index for firm i in year t . $\text{ESGMetric}_{i,t-1}$ is the ESG variable of interest (we ran separate regressions for the Combined ESG Score, ESG Score, Environmental pillar, Social pillar, and Governance pillar) lagged one year. $\mathbf{X}_{i,t-1}$ represents control variables, which included firm size (log of total assets) and profitability (ROA) – both lagged one year as well. “FE” denotes fixed effects; for the Altman Z models, we included sector-by-year fixed effects (SYFE), meaning we account for any shocks or trends affecting a whole sector in a given year (like an oil price spike affecting all oil & gas firms’ Z-scores in 2020, for example). In the credit quality models, we included both sector-year and region-year fixed effects (since that sample was much larger, we could control for regional macro conditions as well). These fixed effects absorb broad macroeconomic or industry effects, isolating the impact of ESG at the firm level. We did not include firm fixed effects because our ESG scores don’t exhibit enough within-firm time variation for many firms (many have short time series), and we aimed to capture between-firm differences as well – effectively assessing if higher-ESG firms have better credit metrics than lower-ESG firms, controlling for size and ROA.

We used robust standard errors clustered at the firm level to account for heteroskedasticity and serial correlation. The statistical significance was assessed at the 10%, 5%, and 1% levels as customary. The regression tables in the main text (Table 1 and 2) summarize the coefficient estimates for the ESG variables and their significance. The R-squared (within) for the Altman Z regressions were around 0.05 (i.e., 5% of the variation in Z within sectors/years was explained by ESG and controls – not surprising as many firm-specific factors drive Z). For the credit quality regressions, the within R-squared was higher (~0.69) because profitability (ROA) is itself part of the dependent variable and explains a lot of its variation, and fixed effects soak up a lot of variance as well.

Supplementary tables: Below we provide a summary comparison of regional results mentioned in the text for the credit quality regressions (coefficients of ESG variables by region), to give a fuller picture:

Table A1. ESG coefficient estimates by region – Credit Quality Index regressions (signs and significance of ESG Combined Score and Pillar effects)

Beta Coefficients of ESG vs Credit Risk	ESG Combined Score	E Pillar Score	S Pillar Score	G Pillar Score
US	positive & significant	positive & significant	positive	positive & significant
Developed Europe	positive & significant	positive & significant	positive & significant	positive & significant
Japan	positive & significant	positive & significant	positive	positive & significant
Developed America ex US	positive & significant	positive & significant	positive	positive
Developed Asia ex Japan	positive & significant	negative	negative	negative
Emerging Markets	negative	positive	positive	positive

Sources: Refinitiv & Allianz Research

References:

Date	Publication Source	Authors	Title	Main Findings
2011	Journal of Banking & Finance	Goss & Roberts	The impact of corporate social responsibility on the cost of bank loans	CSR concerns linked to higher loan spreads and tighter terms; lenders price ESG-related risk, especially for weaker borrowers.
2013	Journal of Business Ethics	Attig et al.	Corporate Social Responsibility and Credit Ratings	Higher CSR performance associated with better credit ratings, robust to endogeneity concerns.
2014	The Financial Review	Oikonomou et al.	The Effects of Corporate Social Performance on the Cost of Corporate Debt and Credit Ratings	Strong social performance linked to lower bond yield spreads and higher assessed credit quality.
2017	Principles for Responsible Investment (PRI)	PRI	Why ESG factors matter in credit risk analysis	ESG affects default risk, downgrade risk, and recovery values rather than just bond pricing.
2020	Banque de France / ACPR	Billio et al.	Inside the ESG ratings: (Dis)agreement and performance	Large ESG rating disagreement implies ESG–credit results depend strongly on provider choice.
2022	Review of Finance	Berg, Kölbel & Rigobon	Aggregate Confusion: The Divergence of ESG Ratings	Low correlation across ESG ratings; measurement noise complicates credit-risk inference.
2022	Journal of Risk Finance	Barth, Hübel & Scholz	ESG and corporate credit spreads	Higher ESG scores associated with lower CDS spreads; ESG shocks are priced in credit markets.
2022	NY Fed Staff Report	Seltzer, Starks & Zhu	Climate Regulatory Risk and Corporate Bonds	Climate regulatory risk causally worsens credit ratings and increases yield spreads.
2022	World Bank	Gratcheva et al.	Credit Worthy: ESG Factors and Sovereign Credit Ratings	Clarifies how ESG factors relate to sovereign creditworthiness; warns against mechanical ESG–rating mapping.
2023	International Review of Economics and Finance	Lian et al.	How does ESG performance affect bond credit spreads?	Better ESG linked to lower bond spreads via reduced financial risk and better transparency.
2025	International Review of Economics and Finance	Chen et al.	Impact of ESG factors on credit ratings	Environmental factors positively associated with European banks' credit ratings.

A photograph showing a group of diverse hands stacked on top of each other, resting on a tree trunk. The hands are of various skin tones and some have accessories like a black wristband and a silver ring. The background is a lush green forest. The text 'Our team' is overlaid on the image, with 'Our' in white and 'team' in yellow.

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Allianz Research encompasses Allianz Group Economic Research and the Economic Research department of Allianz Trade.

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